HEM / HEMK

UTILITY SCALE SOLAR INVERTERS

# Programming and Software Manual

for Power Electronics' staff

SW versions up to 4.6.1

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HEMSW02SI Rev. S

# ABOUT THIS DOCUMENT

### PURPOSE

This manual contains important instructions for the configuration and use of the products HEM and HEMK GEN 2 manufactured by Power Electronics.

Power Electronics reserves the right to modify product features.

### TARGET AUDIENCE

This manual is intended for **Power Electronics' qualified personnel** who will commission, configure and operate the products HEM and HEMK GEN 2 manufactured by Power Electronics.

**Only authorized technical personnel approved by Power Electronics may configure and operate the products.** The instructions assume that the installer is familiar with electrical installation rules and regulations.

## NOTICE

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The manufacturer is not liable for damages, losses, costs or expenses incurred by any user of the product if such damages, losses, costs or expenses result from a failure to comply with the applicable safety instructions or general instructions or operating instructions given by the manufacturer in any of the documents and manuals of the product, including, but not limited to hardware installation, programming and operation, maintenance instructions, handling, or any other. Any damages, losses, costs or expenses resulting from the improper handling, manipulation, modification or operation of the product will be subject to the company's warranty terms.

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EN

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# ACRONYMS

The terms commonly used in the documentation of Power Electronics' products of the *Solar & Storage* division are listed in the table below.

| ACRONYM | MEANING |
|---|---|
| AASS | Auxiliary Services. |
| AC | Alternating Current. |
| AI | Analogue Input. |
| AO | Analogue Output. |
| BESS | Battery Energy Storage System. |
| BMS | Battery Manager Solution. |
| CP | Control Pilot. |
| CPU | Central Processing Unit. |
| dBA | Weighted decibel level. |
| DC | Direct Current. |
| DCL | Discharge Current Limit. |
| DI | Digital Input. |
| DSP | Digital Signal Processor. |
| DO | Digital Output. |
| EMS | Energy Management System. |
| FPGA | Programmable device (Field-Programmable Gate Array). |
| FRU | Field Replaceable Unit. |
| GFDI | Ground Fault Detector Interrupter. |
| GPRS | General Packet Radio Services, a data transmission system. |
| HVAC | Heating, Ventilation, and Air Conditioning. |
| IGBT | Insulated Gate Bipolar Transistor. |
| IMI | Insulation monitoring device. |
| IK | Degree of protection against mechanical impact. |
| IP | Degree of protection against solid/fluid objects. |
| IT | Grid system where the power supply is kept isolated and the electrical equipment system is grounded. |
| LOTOTO | Locking, tagging and verification procedure (Lock Out – Tag Out – Try Out). |
| LVR | Low Voltage Regulation. |
| MCB | Miniature Circuit Breaker. |
| MCCB | Moulded Case Circuit Breaker. |
| MPCS | Multi Power Conversion System. |
| MID | Measuring Instrument Directive. |
| MV | Medium Voltage. This term is used to refer to high voltage in general. |
| OF | Optical Fiber. |

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| ACRONYM | MEANING |
|---|---|
| PCB | Printed Circuit Board. |
| PE | Ground connection (Protective Earth). |
| PI | Proportional and Integral. |
| PLC | Programmable Logic Controller. |
| POI | Point Of Interconnection. |
| PowerComms | Configuration and monitoring interface developed by Power Electronics. |
| PPE | Personal Protection Equipment. |
| PV | Photovoltaic energy. |
| RAL | Color code (Reichsausschuß für Lieferbedingungen und Gütesicherung). |
| RCD | Residual Current Device. |
| RCM | Residual Current Monitor. |
| RFID | Radio Frequency Identification. |
| SOC | State Of Charge – referred to battery. |
| SOH | State Of Health – referred to battery. It compares the actual state of the battery to its initial conditions. It is measured in percentage. |
| STO | Safe Torque Off. |
| SW | Software. |
| THD | Total Harmonic Distortion. |
| TN | Grid system where the power supply is grounded, and the electrical equipment system is brought to the same ground through the neutral connector. |
| TT | Grid system where both the power supply and the electrical devices are connected to the ground via separate connections. |
| UPS | Uninterruptible Power Supply. |
| Vac | Alternating current voltage. |
| Vdc | Direct current voltage. |

# SAFETY SYMBOLS

Always follow safety instructions to prevent accidents and potential hazards from occurring.

In this manual, safety messages are classified as follows:

| | |
|---|---|
|  **WARNING** | Identifies potentially hazardous situations where dangerous voltage may be present which if not avoided could result in minor personal injury, serious injury or death.<br><br>Be extremely careful and follow the instructions to avoid the risk of electrical shocks. |
|  **CAUTION** | Identifies potentially hazardous situations which if not avoided could result in product damage or minor or moderate personal injury.<br><br>Read the message and follow the instructions carefully. |
|  **NOTICE** | Identifies important measures to take in order to prevent damages to the product. |

Other symbols used in this manual for safety messages are the following:

| | |
|---|---|
|  | Hot surface. Be careful and follow the instructions to avoid burns and personal injuries. |
|  | Risk of fire. Be careful and follow the instructions to prevent causing an unintentional fire. |
|  | Energy storage timed discharge. Wait for the indicated time to avoid electrical hazards. |
|  | Caution, risk of hearing damage. Wear hearing protection. |

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# IMPORTANT SAFETY INSTRUCTIONS

**SAVE THESE INSTRUCTIONS**

This manual contains important instructions for models HEM/K and PCSM/K GEN2 that shall be followed during installation and maintenance of the unit.

Read carefully all documentation before manipulating the product and pay special attention to safety recommendations to maximize the performance of this product and to ensure its safe use and installation.



## CAUTION

Read carefully the ***Hardware and Installation Manual***, ***Maintenance Manual*** and all documentation related to the inverter to ensure its safe use and to prevent risks of personal injuries and equipment damage.

Adhere to local and national regulations.

# INTERFACE

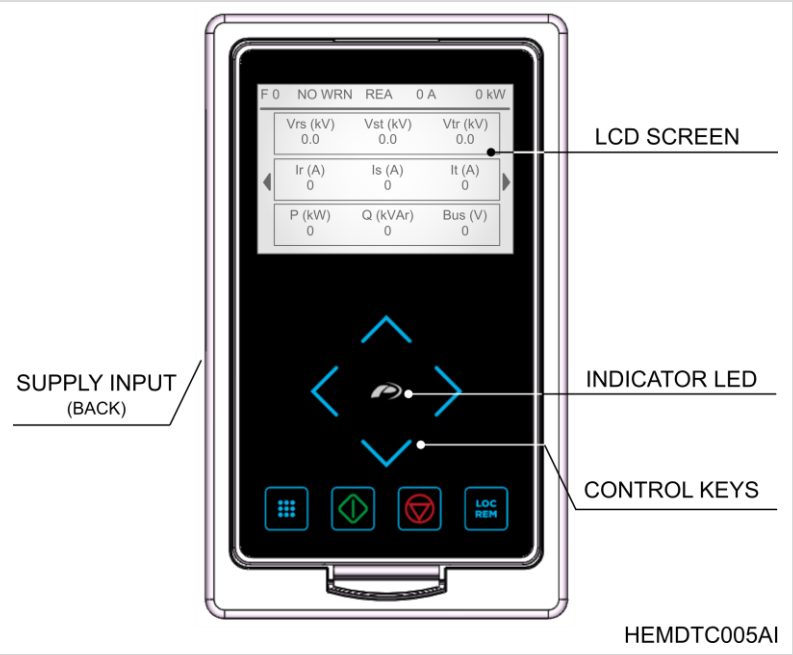
1

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## Display Unit and Control Keypad (Optional)

Power Electronics' external display, available upon request, is a graphic monochromatic 2.8" screen which provides intuitive data representation and easy navigation through parameters. Additionally, it allows storing thousands of customized configurations.

The graphic display is an external unit which must be connected at the main control cabinet's door USB port. It integrates a LED indicator that supplies information about the equipment operational status and eight control keys.




## NOTICE

**Connect the display from the Main Control Cabinet** to the corresponding external port.
Use a with a micro USB male terminal on one end and USB male terminal on the other.

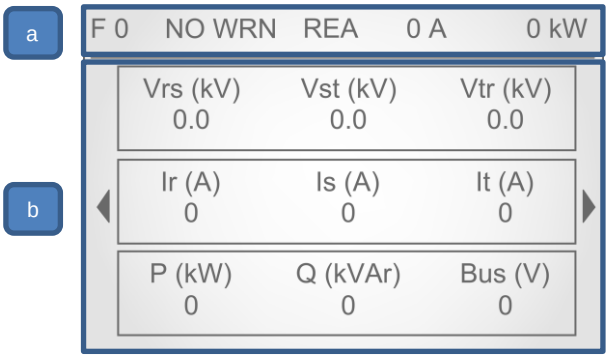
**ONLY use the DISPLAY port to connect the display unit**. Do not connect other USB devices. Otherwise, the port may get permanently damaged and you may cause equipment malfunctioning.

The main screen of the display is divided into two areas:

**a) Status bar:** Shows the main indicators of inverter status.

From left to right (see figure "Main screen"):

- Current fault.
- Current warning.
- Inverter status.
- Output current in Amperes [1].
- Grid power in kW [2].




*Main screen*

**Note:** For the definition of status, warning and faults, please refer to section 3.

**b) Visualization Area:** Shows the home visualization screen and the different menus of the application. If the user has created any customized views, they will be shown in this area as well.

The home visualization screen, or main screen, shows:

- In the first line, the line voltage between phases.
- In the second line, the current per phase.
- In the third, and last, line, information about active power, reactive power and DC bus voltage.

To create customized views, go to: Main menu (*Menú*) → New visualization view.

## Status LED

The status LED shows the inverter status while it is on. It is located in the Power Electronics logo, and will change its color as follows:

- Green: The equipment is RUNNING.
- Red: The equipment has stopped due to a FAULT.
- Yellow: The equipment is in WARNING status.

[1] Modbus Address: 41007. Real value = Modbus value.
[2] Modbus Address: 41008. Real value = Modbus value.

## Control Keypad

The display integrates eight control keys with the following functions:

| | |
|---|---|
|  | Scroll up the menu or change the value of a parameter.<br>Press > to save changes or < to cancel. |
|  | Scroll left the main visualization screen or return to the previous screen in the Menu options. Cancel changes made to a parameter. |
|  | Scroll right the main visualization screen or enter to the different options of the Menu. Save changes made to a parameter. |
|  | Scroll down the menu or change the value of a parameter.<br>Press > to save changes or < to cancel. |
|  | **Menu key.** Press it to enter or exit the menu.<br>It can also be used as editing help during parameter adjustment. To enable it, go to *Menu → Settings → Variables edit mode* and select “Manual precision”. Use the arrows right / left to increase / decrease the digit “ED”. See an example below. |
|  | **Start key.** This key is used to start the equipment when the control has been set to Local. |
|  | **Reset / Stop key.** This key is used to stop the equipment when the control has been set to Local. In the event of a fault, this button can be used to reset the equipment when Local control is enabled. |
|  | Not used. |

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## Adjusting the Scale

Often user must enter high numeric values. To ease this task, the display unit offers the possibility of changing the scale (x1, x10, x100, x1000, x10000) when adjusting a parameter.

1. Ensure “Variables edition mode”, in the SETTINGS menu of the display, is set to “Manual precision”.
2. Enable edit mode by pressing “Menu” key. “EDx1” will appear in the top of the screen.
3. Choose the scale by using the left/right arrow keys (see example).
4. Adjust the digit – according to the selection made on step 2 – by using the up/down arrow keys.

**Example.** Value that will be entered = **1453,2**

| **Parameter value** | **1** | **4** | **5** | **3** | **,** | **2** |
|---|---|---|---|---|---|---|
| **Scale adjustment** | X10000 | X1000 | X100 | X10 | | X1 |

## Changing the Language

It is possible to change the display language (currently English and Spanish are supported). To do so, go to the main menu and follow these simple steps:

**To change from Spanish to English:**

- Main menu (*Menú*) → *Ajustes* → *Idioma* → Scroll and choose “*English*”.

**To change from English to Spanish:**

- Main menu → *Settings* → *Language* → Scroll and choose “*Español*”.

# USER ACCESS LEVEL

2

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Software parameters are divided in groups, depending on their functionality:

- **Visualization and status parameters** cannot be modified and display useful information about the inverter, such as: current and voltage per phase, amount of energy generated by the inverter, fault messages and many more. Please refer to section 4 “Visualization and status parameters” for a detailed list of all parameters and their attributes.
- **Configuration parameters** allow setting and modifying the inverter operating conditions and general parameters such as the current date and user password. Some of these parameters are secured and they can only be accessed by qualified customers o Power Electronics’ personnel. They are explained in section 5, “Description of programming parameters”.
- Parameters from the **service group** are reserved for Power Electronics’ personnel. They allow setting parameters for commissioning and configuring all the functionalities of the inverter. They can be found in section 8, “Description of service group parameters”.

To prevent unintentional modifications that could lead to malfunctioning and even damage to the equipment, access to parameters is password protected and managed by Power Electronics. The following table shows the user modes and access level for each:

| USER MODE [G1.3] | VISIBLE PARAMETERS | | | | |
|---|---|---|---|---|---|
| | Status | Visualization (SV) | Configuration (G) | Service Group (SG) | Debug |
| Customer | All | All | G1.2, G1.3, G1.4 | SG1.4.2 and SG1.4.3 | None |
| Commissioning | All | All | Most configuration parameters, except for those highlighted in coral. | None | None |
| Qualified customer | | | | SG12.1 | |
| Power electronics | All | All | All | All | None |
| Developer | All | All | All | All | All |

**By default, the user mode is Customer**. To change it, go to [G1.4] and enter your password. The corresponding user mode will automatically be set in [G1.3]. If you do not have a password or require assistance, please contact Power Electronics.

To enable Developer and Power Electronics user modes in PowerCommsV, an algorithm that generates a new password every 24 hours is used. The user must log in to PowerCommsV and must be registered as Developer or Power Electronics in the server. It is necessary to select the desired user mode in G1.4-User password, this will configure the parameter G1.3-User automatically.

After 24 hours, the user mode will be downgraded to Customer and the technician must select the user mode in G1.4-User password again.

## NOTICE

Throughout this document, parameters **highlighted in coral** are those reserved to roles “Power Electronics” and “Developer”.

User mode will automatically be reset to the default value (Customer) **only** when the equipment’s power supply is lost or after a software update.

# STATUS, WARNING AND FAULT MESSAGES

3

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There are different type of messages that provide, at all times, information about the inverter. They are the status, warning and fault messages:

- **STATUS** messages provide information about the current state of the inverter (e.g. ready to start, operating…).
- **WARNING** messages provide useful information about what is going on in the inverter (e.g. derating, a module has stopped, heating is active…). They should be checked periodically, and the corresponding actions should be taken (if required).
- A **FAULT** is a condition that, if satisfied, will force the inverter to stop (e.g. an islanding condition has been detected).

Status, warning and fault messages are listed and explained in the following subsections.

Also, the inverter includes three status LEDs in the Main Control Cabinet door which show its operation status:

| | LED ON | LED OFF |
|---|---|---|
| **White LED** | Voltage presence | No voltage presence |
| **Green LED** | Inverter ON | Inverter OFF |
| **Red LED** | Flashing every second: Warning<br>Still: Fault | No faults or warnings |

# Module

## Module Status Messages

These parameters are shown in visualization group SV9.3.

| Code | Name | Description |
|---|---|---|
| 55 | Init | Module is initializing. |
| 22 | Idle | The module waits for 5 s before starting to ensure all DSP parameters have been received. |
| 1 | Stop | Module is stopped. |
| 2 | Softcharge | Module is performing the soft charge. |
| 3 | Ready | The module is ready and it is waiting for the start command. |
| 4 | LCL | LCL filter contactor has been activated and the module is waiting for the feedback to confirm the contactor is closed. Once received, will wait for some time before starting. |
| 5 | Main AC | The main contactor has been activated and the module is waiting for the feedback. Once received, it will enter into "Main AC wait" status. |
| 6 | Main AC wait | Module waits for 100 ms with the contactor closed before starting the injection. |
| 7 | Run | The module is injecting. |
| 8 | nMain AC | When a module stops, the AC contactor is open and keeps closed the LCL filter contactor during 100 ms before opening it. |
| 9 | Discharge | Module is performing the active discharge. |
| 10 | LVRT | An event has occurred in the network. The module waits for a time lapse until the new DSP references are received. If references are not received, the module will stop. |
| 11 | Trackers mode | The module electrically prepared to inject power to the auxiliary services circuit is in trackers mode. |
| 13 | VRT Restart | A voltage drop has occurred and, after an overcurrent fault, the module remains in this state before returning to RUN. |
| 14 | Wait overcurrent | The module stops when an overcurrent is detected. If the current does not return to its range within 500µs, fault F112:OCAC emergency will be triggered. |
| 15 | Wait low Vdc | The module stops on detection of bus voltage below minimum. If the bus voltage does not recover in less than 1ms, fault F109:Low module Vdc will be triggered on the module. |
| 17 | Fault exit OV wait | Once 4 seconds have elapsed since "Softcharge fault", the module will enter in this status and then it will switch to status "Stop". If a run command is not received, the module will stop. |
| 18 | Fault exit wait | Once 4 seconds have elapsed since "Fault", module enters in this status and then switches to "Stop". If no running command is received, the module will stop. |
| 19 | Softcharge fault | Module has failed before completing the soft charge. |
| 20 | Fault | This status has never been used in GEN2. It will never appear. |
| 116 | Module Disabled SC | The module has been disabled. |
| 159 | Master FPGA comms error | Not used. |

**Note**: If a module warning / fault has occurred, the corresponding code will be shown as module status. Refer to section "Module Warning and Fault Messages" for the full list of warning and fault messages.

In addition to the information provided by the display, the electronics of the device can also provide information on the status of the device.

| *Estado* | *LED6* | *LED7* | *LED8* | *LED9* |
|---|---|---|---|---|
| *STOP* | HR | OFF | OFF | OFF |
| *READY* | HR | OFF | BLINK | OFF |
| *RUN* | HR | OFF | ON | OFF |
| *FAULT* | HR | OFF | OFF | BLINK |

HR: Continuous slow flashing.

**Note**: If a remote upgrade is in progress, the different statuses of the procedure will be shown as module status. The status messages of a remote upgrade begin by **RU**_***. It is suggested that SAT save the previous parameter configuration in PowerCommsV and perform the remote update to the appropriate version, initialize the Configuration and Services Group parameters and reload the configuration.

## Module Warning and Fault Messages

All module warnings have an equivalent fault message. Whenever a problem occurs in a module, such module will trigger a warning (stop module) or a fault (stop inverter) depending on its default behaviour.

By default, all warnings are configured to behave as warnings, except for [W109: Low module Vdc]. This will behave as [F109: Low module Vdc].

To configure which warning messages should behave as faults, go to group G5.10.x [1].

The following table shows the list of codes and acronyms for module warnings / faults. They will be described in the next page.

| Code (W / F) | Warning acronym | Fault acronym | Code (W / F) | Warning acronym | Fault acronym |
|---|---|---|---|---|---|
| 65 | OC | ACOVCRR | 97 | SMKD | SMKD |
| 66 | UV | UNBVDC | 98 | IDCD | MIDCD |
| 67 | DOC | DCOVCRR | 99 | IACD | MIACD |
| 68 | FB | FBCNT | 101 | DCF | DCFS |
| 69 | POV | DCOVVP | 102 | UC | IUNB |
| 70 | NOV | DCOVVN | 103 | TIGBT | TIGBT |
| 71 | DR1P | DESR1P | 104 | TAMB | TAMB |
| 72 | DR2P | DESR2P | 105 | PWUNB | PWUNB |
| 73 | DR2N | DESR2N | 106 | BUSUN | BUSUN |
| 74 | DR3N | DESR3N | 107 | NRIDC | NRIDC |
| 75 | DS1P | DESS1P | 108 | HBMX | HBMX |
| 76 | DS2P | DESS2P | 109 | LMVDC | LMVDC |
| 77 | DS2N | DESS2N | 110 | BACF | BACF |
| 78 | DS3N | DESS3N | 111 | RSV | RSV |
| 79 | DT1P | DEST1P | 112 | OCACE | OCACE |
| 80 | DT2P | DEST2P | 113 | CCHK | CCHK |
| 81 | DT2N | DEST2N | 114 | MDFN | MDFN |
| 82 | DT3N | DEST3N | 115 | MDSN | MDSN |
| 83 | DMLT | DESMLT | 116 | MDSC | MDSC |
| 84 | COMS | COMMS | 117 | CMTHR | CMTHR |
| 85 | DUNK | DSTUNK | 118 | TDRFT | TDRFT |
| 86 | FBDC | DCCNT | 119 | DR1N | DESR1N |
| 87 | FBSC | ACSCC | 120 | DR3P | DESR3P |
| 88 | FBAC | ACCNT | 121 | DS1N | DESS1N |
| 89 | FBLC | LCLFB | 122 | DS3P | DESS3P |
| 90 | TIGH | TIGHW | 123 | DT1N | DEST1N |
| 91 | GUV | GUNBDC | 124 | DT3P | DEST3P |
| 92 | DIS | DISFLT | 125 | HCD | HCD |
| 93 | TLCL | THLCL | 126 | IGBTD | IGBTD |
| 94 | NBSC | NBSC | 127 | EstTIGBT | EstTIGBT |
| 95 | PS | PW15V | | | |
| 96 | TAMH | TAMHW | | | |

[1] Please notice this group requires elevated privileges. Go to G1.4 to introduce the password or contact Power Electronics for assistance.

The following table shows all available warning codes for a module and their equivalent fault, along with underlining probable causes and suggested actions to be taken.

## NOTICE

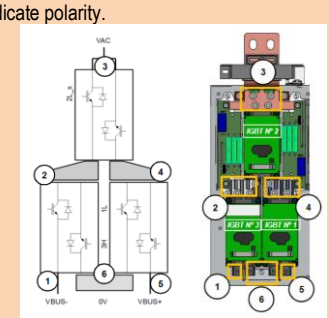
The actions described in the following table must be carried out under safety conditions and wearing the appropriate PPE equipment. Especially, during those troubleshooting actions that require voltage presence in the equipment.

Always follow local regulations or standards regarding safety and health as well as internal application procedures.

| Warning | Fault | Description or probable cause | Actions |
|---|---|---|---|
| **W65:AC overcurrent** | **F65:Mod. AC overcurrent** | If this fault is triggered, it could be caused by 9 independent faults in the module FPGA:<br>- Event or overcurrent in any of the three phases.<br>- Blown AC fuse.<br>- Current sensor cable (black or white) has no contact with the control board in any of the three phases.<br>- If it does not appear recurrently, it can be caused by a network events, regulation faults, fault in the control of the inverter or AC overcurrent due to a DC disconnection.<br>- Optical fiber defect.<br>- Defect in the ribbon cables of the IGBTs.<br>- Voltage derating curve and protection curves intersect. This happens when the configuration is not correct. Contact with front office.<br><br>This fault can also appear due to a defect in the hall converter board.<br><br>This occurs when the module current measurement exceeds the threshold set in SG10.2.1-Overcurr prot low and SG10.2.4-Overcurr prot high depending on the bus voltage. If individual modules fail, the W65 is produced in the inverter and the F65 is produced in the module. If all modules fail, the F65:Mod. Sobrecorriente AC is produced in the inverter. | Check wiring, including optical fiber.<br>If the fault persists, please contact Power Electronics.<br>**Extra information:**<br>- Check configuration SG10.2.1-Overcurr prot low and SG10.2.4-Overcurr prot high.<br>If the fault does not appear recurrently:<br>- Download the SD card and blackbox to analyze the causes of the fault.<br>If the fault is repetitive:<br>- Check that the AC fuses are not blown.<br>- Check flat cables and their connections.<br>- Check hall sensors and wiring.<br>- Check the leds of power supplies of the E973 module board (FPGA board). Led1 = -15V, Led3 = 5V and Led5 = +15V. If any is off, disconnect one by one the flat cables and the current measurements to locate the element that is knocking down the source and repair it.<br>- Check that the Optical fiber is in good state. It must be flush with the connector and without cracks.<br>- Check that the ribbon cables and its connections are in good state.<br>- Replace the AC and DC measurement cables of the FGPA (J4 y J6 in AC, J3 in DC. This cables are located in the upper part of the board, at the left). If once this change is done F67:Mod. DC overcurrent appears, then the problem is in the hall converter board. If the F65 does not disappear, exchange FPGA for the FPGA of another module and check if it disappears.<br>- Check the current in SV9.4-Current R and SV9.7-DC current. |
| **W66: Unbal. Voltage** | **F66:Mod. Unbal DC bus** | There is an unbalance (threshold is set G5.10.4-Unbal bus threshold, its default value should be 100 V) between the high side and low side of the DC bus.<br>This fault can block the start of the module.<br>$\|V(P) - V(N)\| > Threshold$<br>Possible causes:<br>- Damage in the AC contactor.<br>- IGBT clamped to earth, damaged or short-circuited.<br>- Defect in the isolating Nomex laminate.<br>- Bad crimping of AC contactor pins.<br>- Damaged snubber board.<br>- Incorrect reading on FPGA board.<br>- Incorrect reading on the balance board.<br>- Grounding of the balance board incorrectly connected.<br>- Measurement wiring on module FPGA incorrectly crimped.<br>- Measurement wiring on the balance board incorrectly crimped.<br>- AC contactor damaged.<br>- DC or AC fuses of the modules blown.<br>- Invalid module FPGA board version.<br>- Damaged module FPGA board. | Never rearm the equipment in case of this failure.<br>Contact Power Electronics.<br>Check P and N voltages in SV9.2-Selected module with the inverter stopped. Ensure voltages are balanced in all modules and below the value set in G5.10.4-Unbal bus threshold).<br>Verify DC voltages with a multimeter and if unbalance really exists, check IGBTs and E936AA board. Do not measure in the measurement board directly.<br>Increase G5.10.4-Unbal bus threshold) to 500Vdc if the fault is triggered with the modules ON. If the fault persists, change FPGA board.<br><br>**Extra information:**<br>- Remove blackbox from modules and SD. Check whether one of the contactors could be opened during operation.<br>- After this type of fault, a complete review of the power stages (IGBT, buses...) must be carried out.<br>- If nothing is broken, check the version of the module boards. The E973CD version is not valid.<br>- Clean the AC contactors. If the fault does not disappear, replace them. |

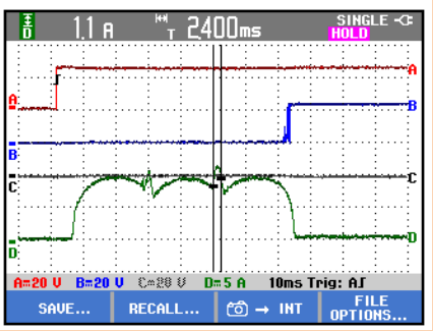
| Warning | Fault | Description or probable cause | Actions |
|---|---|---|---|
| **W67:DC overcurrent** | **F67:Mod. DC overcurrent** | Failure caused when the DC overcurrent threshold (G5.10.14-Unbal bus threshold) is exceeded, during 10 us.<br>It could be caused by:<br>- Overcurrent in the DC bus.<br>- Current sensor cable (black) disconnected from the control board in any of the three phases.<br>- Disconnected current sensor cable (white) in any of the three phases.<br>- Defective DC current sensor cable (grey) on the module FPGA converter board.<br>- Converter board in bad condition.<br>- Defect in the current hall sensor.<br>- Defect in the fiber.<br>- Defect in the module FPGA board.<br>- Damage in the module bus.<br>- Damage in the IGBTs.<br>Specific faults can be caused by network events or inverter regulation and control.<br>When this fault is triggered, F26 will be automatically triggered. | Check wiring.<br>If the fault was triggered by a module, change converter board 4 – 20mA hall DC.<br>If fault persist, please contact Power Electronics.<br><br>**Extra information:**<br>If it is a sporadic failure, it may be due to an event on the network.<br>- Extract SD and blackbox<br>In case of repeated failure<br>- Determine which module generates the fault (SG9-Logger modules).<br>- Check the Idc measurement of the module in question and the measurement circuit. Do not measure in the measurement board directly.<br>- Check the 4-20mA converter board of the current sensor<br>- Check the status of the Hall sensor.<br>- Check the condition of the fiber.<br>- Check for sulphated FPGA components.<br>- Check for possible damage in the bus.<br>- Check the IGBTs. |
| **W68:Contactor feedback** | **F68:Mod. Contactor feedback** | This fault should not appear because it has been disabled. | Contact Power Electronics. |
| **W69:DC Overvoltage (P)** | **F69:Mod. DC Overvoltage (P)** | Overvoltage (>755 V) present in the DC positive side.<br><br>Failure caused when the positive semibus voltage exceeds 755 V.<br><br>Possible causes:<br>- Defective IGBTs.<br>- Defect in the ribbon cables.<br>- Defect in the balance board.<br>- Defect in the module FPGA board.<br>- Defect in the buses.<br>- Defect in any power stage. | If the fault was triggered by a stopped module (W69:DC Overvoltage (P)), check snubbers and IGBT (power stage).<br>If the fault occurs frequently, contact Power Electronics.<br><br>**Extra information:**<br>- Determine which module(s) are the cause(s)<br>- Measure the voltage of both semibuses and check if it coincides with the inverter measurement.<br>If the measure is real:<br>- Extract SD and Blackbox.<br>- Check the power amplifier.<br>- Check balancing boards.<br>- Check bus status.<br>- Check ribbon cables.<br>- Check IGBTs.<br>- Check SV9.8-DC voltage (P).<br>If the measurement is not real (inverter measurement >755V and much lower tester measurement):<br>- Check calibration of DC voltage readings.<br>- Check measurement circuit. Do not measure in the measurement board directly.<br>- Change module board. |
| **W70:DC Overvoltage (N)** | **F70:Mod. DC Overvoltage (N)** | Overvoltage (>755 V) present in the DC negative side of the bus.<br><br>Possible causes:<br>- Defective IGBTs.<br>- Defect in the ribbon cables.<br>- Defect in the balance board.<br>- Defect in the FPGA<br>- Defect in the buses.<br>- Defect in any power stage. | If the fault occurs frequently, contact Power Electronics.<br><br>**Extra information:**<br>- Determine which module(s) are the cause(s)<br>- Measure the voltage of both semibuses and check if it coincides with the inverter measurement.<br>- If the fault was triggered by an stopped module (W69:DC Overvoltage (P)) check the snubbers and the IGBTs of the module.<br>If the measure is real:<br>- Extract SD and blackbox<br>- Check the power amplifier<br>- Check balancing boards<br>- Check bus status<br>- Check ribbon cables.<br>- Check IGBTs.<br>- Check SV9.8-DC voltage (P).<br>If the measurement is not real (inverter measurement >755V and much lower tester measurement):<br>- Check calibration of DC voltage readings and the measurement circuit. Do not measure in the measurement board directly.<br>- Change module board. |

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<table>
<tr><th>Warning</th><th>Fault</th><th>Description or probable cause</th><th>Actions</th></tr>
<tr><td>W71: Desat R1(P)</td><td>F71:Mod. Desat R1(P)</td><td rowspan="11">These faults are caused by a desaturation in the corresponding IGBT driver.<br><br>Faults caused by a loss of power to the gate drive that controls the IGBT that caused the fault.<br><br>Note: These faults are complementary to W119/F119:Mod. Desat R1(N) - W124/F124W124:Mod. Desat T3(P)<br><br>Possible causes:<br>- Defective or shunt in the IGBTs.<br>- FPGA defect.<br>- Incorrect voltage on module FPGA board, 24 Vdc (TD -Test Point ±15 Vdc).<br>- Damaged or loose ribbon cables.<br>- Defective 24 Vdc power supply.<br>- Loose connections on 24 Vdc power supply harting.<br>- Shunt in the DC bus.</td><td rowspan="11">If the fault occurs frequently, the gate drive may be damaged. In this case, contact Power Electronics.<br><br>After this fault, do not start the module again. If possible, isolate the power stage. The IGBT might be damaged. Check image “Distribution of transistors in the power stage”.<br><br>In case of doubt, please, contact your superior.<br><br>Extra information:<br><br>- Review the power stage<br>- Check the power supplies (15V and -15V) of the module board<br>- Check flat cables<br>- If the fault is repetitive or has caused damage, remove blackbox and SD and contact Power Electronics.<br>- R, S and T indicate which IGBT has failed, P and N indicate polarity.<br>  <br><table><tr><th>Nº</th><th>Description</th></tr><tr><td>1</td><td>Negative bus.</td></tr><tr><td>2</td><td>IGBTs interconnection.</td></tr><tr><td>3</td><td>AC phase.</td></tr><tr><td>4</td><td>IGBTs interconnection.</td></tr><tr><td>5</td><td>Positive bus.</td></tr><tr><td>6</td><td>Middle bus.</td></tr></table><br>Note: In the event of these faults, it may be advisable to replace the power stage.</td></tr>
<tr><td>W72: Desat R2(P)</td><td>F72:Mod. Desat R2(P)</td></tr>
<tr><td>W73: Desat R2(N)</td><td>F73:Mod. Desat R2(N)</td></tr>
<tr><td>W74: Desat R3(N)</td><td>F74:Mod. Desat R3(N)</td></tr>
<tr><td>W75:Desat S1(P)</td><td>F75:Mod. Desat S1(P)</td></tr>
<tr><td>W76:Desat S2(P)</td><td>F76:Mod. Desat S2(P)</td></tr>
<tr><td>W77:Desat S2(N)</td><td>F77:Mod. Desat S2(N)</td></tr>
<tr><td>W78:Desat S3(N)</td><td>F78:Mod. Desat S3(N)</td></tr>
<tr><td>W79:Desat T1(P)</td><td>F79:Mod. Desat T1(P)</td></tr>
<tr><td>W80:Desat T2(P)</td><td>F80:Mod. Desat T2(P)</td></tr>
<tr><td>W81:Desat T2(N)</td><td>F81:Mod. Desat T2(N)</td></tr>
<tr><td>W82:Desat T3(N)</td><td>F82:Mod. Desat T3(N)</td></tr>
<tr><td>W83:Multiple desat</td><td>F83:Mod. Multiple desat</td><td>Failure caused by desaturation in more than one IGBT controller at the same time.<br><br>Possible causes:<br>- IGBT or gatedrive defect.<br>- Defect in the FPGA.<br>- Incorrect module FPGA board 24 Vdc supply voltage.<br>- Damaged or incorrectly connected ribbon cables.<br>- Failure in the 15 V gatedrive power supply from the module FPGA board.</td><td>Never rearm the equipment in case of this failure. Contact Power Electronics.<br><br>Extra information:<br><br>- F83:Mod. Multiple desat will appear when there are at least 6 desaturations in the module.<br>- Check for possible gatedrive shunts to ground.<br>- Check the diode of all IGBTs.<br>- Measure continuity of all AC fuses for blown fuses.<br>- Analyse blackbox and SD.<br>- There is no supervision of desaturations between modules at a higher level by the DSP<br>- The recommended actions are the same as for simple desaturation failures.</td></tr>
</table>

| Warning | Fault | Description or probable cause | Actions |
|---|---|---|---|
| **W84:Comms** | **F84:Mod. Comms** | Fault caused when the central FPGA does not receive frames from the module FPGA in a defined period of time. The fault is deactivated at startup, it is activated when the first frame is received.<br><br>Possible causes:<br>- Badly crimped fiber, in bad condition or with excessive bends.<br>- Module FPGA board defect.<br>- Incorrect module FPGA board 24 Vdc supply voltage.<br>- Central board voltage lower than 24 V.<br>- Defect in the central control board P3G-E977XX.<br>- Dirt on connectors. | Check the state of communication fibers (connector crimping, bend angles, flush with the connector and without cracks…).<br><br>- If the fiber is in good condition, check the power supply and the condition of the FPGA boards of both the module and the control unit.<br>- Use a fiber tester to measure the power reaching from one end to the other.<br>- Update the SW version.<br>- From version 2.0.0 check for lost frames through SV12.1-FPGA modules. Received frame values must be greater than 99%. |
| **W85:Mod. Desat unkwn** | **F85:Mod. Desat unkwn** | This fault should not appear because it has been disabled. | |
| **W86:Mod. DC contactor** | **F86:Mod. DC contactor** | | |
| **W87:Mod. ACSC contactor** | **F87:Mod. ACSC contactor** | | |
| **W88:Mod. AC contactor** | **F88:Mod. AC contactor** | Fault caused when the module sends the command to activate the AC contactors and they have not closed within 1s. It makes the check between the order that is sent to the contactor is the same as the state the contactor is in.<br>The module has n attemps ('n' defined in parameter SG1.2.4-Modules retries) before the fault is triggered.<br><br>Possible causes:<br>- Defect in the voltage sources.<br>- Defect in the wiring involved in the manoeuvre of the contactors.<br>- Defect in the backfeed wiring.<br>- Defect in the harting.<br>- Defect in the auxiliary contactor of the main contactor.<br>- Defect in the contactor coil.<br>- Defect in the relay of the digital output of the module FPGA board.<br>- Defect in the 24 Vdc terminal bridge.<br>- Dirty AC contactors. | Check wiring. |
| | | **Extra information:**<br><br>- Check output voltage of 24 V sources (25.5 V and 26 V) and their proper connection through the jump.<br>- Check the wiring of the contactors operation. Check the harting pins, the contactor connection, the 24 V relay connection and the digital output of the control board.<br>- Check feedback and continuity on the contactor auxiliary contacts when the contactor is closed.<br>- Check contactor coil. It may be burned out or not receiving power. If they contact and then come out, the feedback is not working. All wiring should be checked. If a coil is damaged, the voltage sources may trip due to overcurrent.<br>- The jumper of the 24 V terminal block must be correctly installed.<br>- Clean the AC contactors and check if the fault appears again.<br><br>The contactors used in HEM v2.0 are the L5396A with type C+ coils instead of the L5396 (type C). The use of this contactor with type C+ coils leads to the following considerations to be taken into account:<br><br>- It is necessary to jumper the protection and distribution modules 3 by 3 to prevent them from limiting the fuses when the contactors are operated (modular 24 Vdc power distribution module).<br>- The extended range coil installed in the L6587 contactor is the L6608. The equipment in which the extended range coils are to be mounted must have the E973CE module FPGA or higher. | |

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| Warning | Fault | Description or probable cause | Actions |
|---|---|---|---|
| **W89:Mod. LCL contactor** | **F89:LCL contactor feedback** | Fault triggered when the LCL contactor is activated but it does not respond.<br><br>It may be caused by the backfeed of the harmonic filter capacitor contactors or harmonic filter fuses connected in series with each other.<br><br>Possible causes:<br>- Filter C connectors blocked.<br>- Defective contactor or power supply.<br>- Defect in the backfeed wiring.<br>- Defective digital input or output.<br>- Fuse blown. | Fault produced by a discordance of more than 500 ms between the activation command and the feedback.<br><br>- Check the continuity of the power supply wiring of the contactors and the feedbacks.<br>- Check digital input and output of the central board at J40. Check SV4.1.7-Input Status, second bit from the left (0 = no continuity and X = continuity).<br>- Check if all three contactors are actuated at the same time.<br>- Measure the continuity of the three filter fuses, this measurement must be done without voltage.<br>- Measure the command (red), backlight (blue) and current consumption (green). The time measured under ideal conditions is 70ms.<br>  <br>- Measure the 24 V of the relay sources.<br>- In case of obtaining something different, the contactor status should be checked. |
| **W90:Mod. IGBTs tº HW** | **F90:Mod. IGBTs tº HW** | This fault occurs when the temperature of any IGBT overcomes the temperature configured in G5.7.1.1-IGBTs temp thres flt, or when an IGBT temperature sensor is disconnected.<br><br>Possible causes:<br>- Defect in the NTC sensor of the IGBT.<br>- Defect in the NTC sensor cable.<br>- Defect in the FPGA of the module.<br>- Defect in the IGBT hardware.<br>- Fans with paste instead of graphite.<br>- Defect in the module fan.<br>- Clogged radiator. | Check the reading of the current sensors from the display. If any of them is -30, the sensor is disconnected. If all values are normal, an overtemperature has occurred while the equipment was running.<br><br>**Extra information:**<br><br>- Check IGBT temperature measurement in SV9.18-Temperature R1 to SV9.26-Temperature T3.<br>- If any temperature is out of range, check the NTC connection.<br>- Swap wiring with another NTC wiring to verify that the problem is not with the FPGA.<br>- Replace the IGBT.<br>- Replace FPGA.<br><br>If the temperature measurement is consistent, overheating has occurred while the equipment was in operation:<br>- Check the configuration of the fans (supply / extract).<br>- Check the condition of the fans.<br>- Tighten the screws that hold the IGBTs to the heatsink.<br>- Check for graphite instead of paste.<br>- Check the radiator for clogging. |

| Warning | Fault | Description or probable cause | Actions |
|---|---|---|---|
| **W91:Mod. Great unbal DC** | **F91:Mod. Great unbal DC** | DC bus unbalanced more than 500 V.<br>Check descriptions or probable causes of the F106:Vbus unbalanced and the F66:Mod. Unbal DC bus. | Check the status of all module IGBTs.<br>**Extra information:**<br>- Determine which module(s) are the cause(s)<br>- Measure the voltage of both semibuses and check if it coincides with the inverter measurement.<br>- Check the status of the module's IGBTs.<br>If the measure is real:<br>- Extract SD and blackbox<br>- Check the power amplifier<br>- Check balancing boards<br>- Check bus status<br>If the measure is not real:<br>- Check measurement circuit. Do not measure directly in the measurement board.<br>- Check FPGA board of the module.<br>- Check balance board.<br>- Check wiring. |
| **W92:Mod. Discharge fault** | **F92:Mod. Discharge fault** | Occurs when the module FPGA attempts to discharge the module bus and voltage does not drop below 50 V in 40 s.<br>Possible causes:<br>- Wrong FPGA readout of the module. | Verify all DU contactors are open, because if the equipment remains connected to the field, the modules will not be able to discharge the bus.<br>Check that at least two modules are enabled. |
| **W93:Mod. Thermal LCL** | **F93:Mod. Thermal LCL** | Thermal LCL contactor of a module inductance has been opened.<br>Inductances in HEM / HEMK inverters are shared between pairs of modules. Thus, if a module triggers this fault (F93:Mod. Thermal LCL), its paired module will detect the inductance overheating and trigger F117:Mod. Common thermal. Both modules will stop.<br>Possible causes:<br>- Problems in the hot air extraction.<br>- Problems in the cyclone system.<br>- Incorrectly configured ventilation parameters.<br>- Bad positioning of the ducts that cool the chokes.<br>- Incorrectly crimped connector.<br>- Damaged thermal. | Check fan status.<br>Check thermal wiring.<br>Consult Power Electronics.<br>**Extra information:**<br>- Check continuity on the feedback of the module FPGA board connector, connectors J34.1 and J35.3.<br>- If there is a power difference between the connector terminals, it is essential to check the connector located under the module and the thermal. When checking the thermal, check that there is no resin or sulphation on the terminal.<br>- Check the settings of the ventilation of the inductors (G11-Fans).<br>- Check the operation of the extraction fans of the inductance in question.<br>- Check the operation of the cyclones.<br>- Verify the correct positioning of the ducts that focus the air to the inductances. |
| **W94:Mod. No bus softcharge** | **F94:Mod. No bus softcharge** | This fault should not appear because it has been disabled. | Contact Power Electronics. |

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| Warning | Fault | Description or probable cause | Actions |
|---|---|---|---|
| **W95:Mod. Power 15V** | **F95:Mod. Power 15V** | Internal power supply of the board does not provide the adequate voltage.<br>It may mean that the IGBT's gate drive is not receiving the 15V supply from the FPGA board.<br><br>Possible causes:<br>- Incorrectly calibrated voltage power supply.<br>- Defect in the module's 24 Vdc power supply wiring.<br>- Defect in the power supply regulator.<br>- Defect in the ribbon cables of the IGBTs.<br>- Defect in the IGBTs or in the drive.<br>- Defect in the FPGA board.<br>- Damaged hall sensor or current converter board. | Check the status of the 24V power supply.<br>Check flat cables wiring.<br>Exchange flat cables.<br>Change IGBTs of the affected phase.<br><br>**Extra information:**<br>- Measure the 24 V voltage of the failed module and at the output of the power supplies.<br>- Regulate the voltage at the output of the sources in case it is less than 25.5 V.<br>- Check if the combs have been reformed in the 24 V electronic distributor.<br><br>  <br><br>- If the fault persists and the voltages in the 15V source of the module board are not correct, check if any element (Halls, Gate drives...) is short and this knocks down the power supply.<br>- If everything is correct, remove blackbox from the module and the SD and contact Power Electronics. |
| **W96:Mod. Amb tº HW** | **F96:Mod. Amb tº HW** | The temperature of the module's electronic board has overcome the level defined in G5.7.2.1-ModuleTempThresFlt.<br><br>Possible causes:<br>- Wrong calibration of the module FPGA board calibration defect.<br>- Defect in the FPGA board.<br>- Defect in the air inlet.<br>- Defect in the air ducts.<br>- Defect in the air outlet. | Check fan status.<br><br>**Extra information:**<br>- Check temperature measurement in SV9.13-Ambient temperature.<br>- If the measurement is consistent and above the level defined in G5.7.2.1-ModuleTempThresFlt, check extraction fans.<br>- Check if there is any obstruction in the fan tubes or in the heatsink.<br>- Check if the ventilation tubes coming from the cyclone are well positioned.<br>- Compare the temperature of the modules to see if it is a problem in a particular module or in the air intake.<br><br>If the measure is not consistent<br>- Change module board |
| **W97:Mod. Smoke detector** | **F97:Smoke detector** | This fault should not appear because it has been disabled. | - |

| Warning | Fault | Description or probable cause | Actions |
|---|---|---|---|
| **W98:Mod. Idc derivation** | **F98:Mod Idc derivation** | The module has detected a derivation in $I_{DC}$.<br>High DC current has been measured when the module is stopped.<br>Fault produced when the module measures DC current higher than 100 A, at standstill.<br><br>Possible causes:<br>- Defect in DC current measurement wiring.<br>- Defect in the DC hall converter board.<br>- Defect in the module's FPGA board.<br>- Defect in the current hall sensor.<br>- Short circuit in the bus. | Never rearm the equipment in case of this failure. Contact Power Electronics.<br>This fault has an additional action associated by default. Thus, it is included in the list of special faults (G5.10.1-Special faults).<br>Change DC HALL boards. Check wiring.<br>Set G5.10.8-Unb Pac Pdc threshold = 100 A. and verify if the fault is minimized.<br>Then change the current HALL or FPGA and set back to 50 A.<br><br>**Extra information:**<br><br>- Check the Idc measurement wiring of the module that caused the fault.<br>- Download blackbox from modules and SD. If the bus voltage remains constant during the event, the fault is in the current measurement, otherwise it is in the power circuit.<br><br>If the fault id in the power circuit:<br>- Check for continuity between positive-negative, positive-ground, negative-ground busbars. If there is a short-circuit, a complete check of the power stages (IGBT, buses, etc.) and busbars must be carried out.<br><br>If the fault is in the measurement circuit:<br>- Check DC current hall sensor, sensor converter board, sensor wiring, module board. Exchange these components with those of another module to check for proper operation. |
| **W99:Mod. Iac derivation** | **F99:Mod Iac derivation** | The module has detected a derivation in $I_{AC}$.<br>High AC current when the module is stopped.<br>This fault only shows up if it happens when performing the self-diagnosis.<br><br>Fault produced when the module measures AC current higher than 50 A, at standstill.<br><br>Possible causes:<br>- Defect in AC current measurement wiring.<br>- Defect in the AC hall converter board.<br>- Defect in the module's FPGA board.<br>- Defect in the current hall sensor. | Never rearm the equipment in case of this failure. Contact Power Electronics.<br>Change AC HALL boards. Check wiring.<br>Set G5.10.10-Deriv. Iac threshold = 100 A and verify if the fault is minimized. Then change the current HALL or FPGA and set back to 50 A.<br><br>**Extra information:**<br><br>- Remove blackbox from modules and SD.<br>- After this type of failure, a complete review of the power stages (IGBT, buses...) must be carried out.<br>- Check the Iac measurement wiring of the module that caused the fault.<br>- Check the module of the FPGA.<br>- Set SG1.3.3-Module stop meas = Sí and check in the module which of the three phases is measuring incorrectly. |

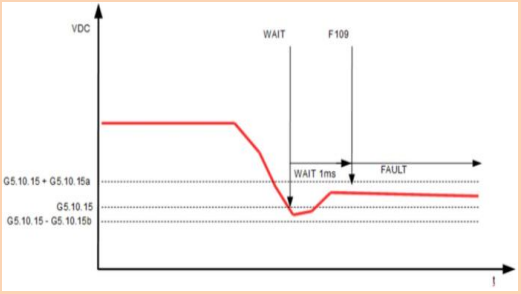
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| Warning | Fault | Description or probable cause | Actions |
|---|---|---|---|
| **W101:Mod. DC Fuse** | **F101:Mod. DC Fuse** | Module DC voltage measure is below the average value of all modules.<br>The difference between the two must be greater than 200 V to trigger the fault.<br><br>Possible causes:<br>- Fuse blown.<br>- Defect in the balance converter board.<br>- Defect in the module FPGA board.<br>- Defect in the measurement wiring.<br>- Defect in calibration. | Check DC fuses.<br>Check DC voltage measurement circuit of the module and the bus.<br><br>**Extra information:**<br><br>- Check the DC fuses of the module<br>- Check the measurement in SV9-Modules information and SV2-Photovoltaic gen and measure with a tester the voltage of both the internal bus of the module that fails and that of the common bus. And verify that the difference is correct.<br>- Check for current and voltage unbalance. If the unbalance is only voltage unbalance, the problem is in the measuring circuit. Do not measure in the measurement board directly.<br><br>If the failure is not real:<br>- Check voltage calibration.<br>- Check module measurement circuit.<br>- Check DC fuses.<br>- Check DC measurement circuit of the module.<br>- Check balance board, FPGA and wiring.<br>- Check DC measurement circuit of the bus; if it is damaged the measure could oscillate and trigger F101:Mod. DC Fuse, accompanied usually by a general F13:No modules.<br>- Remove Blackbox and SD and contact Power Electronics. |
| **W102:Unbalance current** | **F102:Unbalance current** | A module output current unbalance greater than 30% has been detected for 5 seconds.<br><br>Possible causes:<br>- Defect in hall sensor measurement wiring.<br>- Defect in the hall converter boards.<br>- Defect in the module FPGA board.<br>- Blown AC fuses.<br>- Incorrect tightening torques on the AC bus.<br>- Defect in the IGBTs or gate drives. | Check AC fuses.<br>Should the fault persist, please contact Power Electronics.<br><br>**Extra information:**<br><br>- Check the configuration of G5.5-Unbalanced.<br>- Measure output currents and check for unbalance.<br><br>If the unbalance is real<br>- Check the condition of the AC fuses.<br>- Measure AC voltages.<br><br>If it is not real<br>- Check status of AC measurement circuit (wiring, halls...). Do not measure in the measurement board directly. |
| **W103:IGBTs temperature** | **F103:IGBTs temperature** | The temperature of one or more IGBTs has overcome the maximum threshold.<br>Module fan is not working properly. It may have several causes:<br>- PWM cable pins contact is bad.<br>- Fan has no power supply, has stopped working or it is rotating in the opposite direction.<br>- Fan wiring incorrectly crimped.<br>- Module harting wiring incorrectly crimped.<br>- Module radiator clogged.<br>- Defective NTC sensor on IGBT.<br>- Incorrect torque on IGBT.<br>- Defect on the graphite.<br>- Damaged IGBT or gate drive.<br>- Heatsink defect.<br>- Incorrect module FPGA board reading.<br>- Damaged module FPGA board.<br>- Incorrectly crimped FPGA wiring.<br>Failure occurring when a module IGTB exceeds the temperature protection threshold configured in SG10.2.5-Igbts temp prot. | Check fan status.<br>Check NTC cables, one of them may not be making proper contact.<br><br>**Extra information:**<br><br>- Check the temperature of the IGBT of the module in question (SV9-Modules information). Check if this is a generalized problem in all IGBTs.<br>- If the fault is only in one IGBT, check the tightening torque of the IGBT plates.<br>- If the fan is rotating in the opposite direction, exchange the phases.<br>- Check the function of the exhaust ventilation, its connections and if it is clogged.<br>- If the problem persists, replace power stage. |

| Warning | Fault | Description or probable cause | Actions |
|---|---|---|---|
| **W104:Ambient temperature** | **F104:Ambient temperature** | Ambient temperature is over the maximum threshold configured in G5.7.2.1-ModuleTempThresFlt. It is set as 75ºC by default. This temperature is measured by de NTC sensor fo the module's board.<br><br>Possible causes:<br>- Incorrectly calibration of the module FPGA board.<br>- Damage in the module FPGA board.<br>- Defect in the ventilation system.<br>- Defect in the ventilation ducts. | Check fan status.<br><br>**Extra information:**<br><br>- Check temperature measurement in SV9<br><br>If the measurement is consistent and above G5.7.2.1-ModuleTempThresFlt.<br>- Check extraction fans of the module.<br><br>If the measure is not consistent<br>- Change module board |
| **W105:Pac/Pdc unbalanced** | **F105:Pac/Pdc unbalanced** | There is an unbalance between Pac and Pdc values. They differ a percentage defined in G5.10-Special protections (30% by default).<br><br>This fault may be triggered because injected power is too low and the Idc measure of a module is not good; causing the unbalance between Pac and Pdc.<br><br>Possible causes:<br>- Module FPGA board wiring defect.<br>- Damage in the module FPGA board.<br>- Damage in the DC board or its wiring.<br>- Damage in the AC hall board or its wiring. | Never rearm the equipment in case of this failure. Contact Power Electronics.<br>This fault has an additional action associated by default. Thus, it is included in the list of special faults (G5.10.1-Special faults).<br>Check FPGA board cable.<br>Check power, voltage and current readings in SV9.2-Selected module.<br>Change DC and AC HALL boards. Check wiring.<br>Set G5.10.8-Unb Pac Pdc threshold = 30%. If the fault persists, change FPGA board.<br><br>**Extra information:**<br><br>- Remove blackbox from modules and SD.<br>- After this type of fault, a complete review of the power stages (IGBT, buses...) must be carried out.<br>- If everything is correct, start the equipment by self-diagnosis.<br><br>If the failure is repetitive and there is no damage:<br><br>- Check the configuration.<br>- Review the measurement of Pdc (SV2-Photovoltaic gen) and Pac (SV3-Insulation monitoring).<br>- Check the current of the modules is SV9 – Modules information.<br>- Contact Power Electronics. |
| **W106:Vbus unbalanced** | **F106:Vbus unbalanced** | There is an unbalance between the high side and low side of the DC bus:<br>$\|V(P) - V(N)\| > 200V$<br>This fault can appear if, G5.10.13-Thr Activ Flt Idref is enabled. The default value is 200 V with a trigger time of 1 ms.<br>If N modules trigger this fault, the inverter will stop and get disconnected from the network.<br><br>Possible causes:<br>- IGBT shunted to ground or shorted.<br>- Defect in the IGBTs.<br>- Defect in the nomex insulation laminate.<br>- Defect in the capacitor.<br>- Badly crimped contactor pins.<br>- Snubber board defect.<br>- Defect in the FPGA reading.<br>- Bad reading of the E936XX balance board.<br>- Bad grounding of the balance board. It can be due to a wrong connection of the intermediate bus cable or a wrong connection or damage in the radiator.<br>- Badly crimped measurement wiring in the balance board or in the module FPGA board.<br>- Defective AC contactor.<br>- Blown AC or DC fuses of modules.<br>- Damaged module FPGA board or invalid version. | Never rearm the equipment in case of this failure. Contact Power Electronics.<br>This fault has an additional action associated by default. Thus, it is included in the list of special faults (G5.10.1-Special faults).<br>Check P and N voltages in SV9.2-Selected module with the inverter stopped. Ensure voltages are balanced in all modules and below 200V.<br>Verify DC voltages with a multimeter and if unbalance really exists, check IGBTs and E936XX board.<br>If the fault persists, change FPGA board.<br><br>**Extra information:**<br><br>- Remove blackbox from modules and SD.<br>- After this type of fault, a complete review of the power stages (IGBT, buses...) must be carried out.<br>- If nothing is broken, check the version of the module boards. The E973CD version is not valid. |

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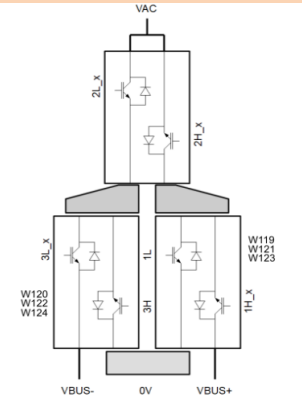
| Warning | Fault | Description or probable cause | Actions |
|---|---|---|---|
| **W107:Iac not reached** | **F107:Iac not reached** | Module FPGA will trigger this fault if the output current does not reach, as minimum, 50% of the reference current, having overcome the threshold set in G5.10.13-Thr Activ Flt Idref.<br><br>This fault may indicate there is not enough bus voltage, or that the 3 AC fuses have blown.<br>The fault is activated by threshold in G5.10.13-Thr Activ Flt Idref. Once the Id exceeds this threshold, if the Id is less than 50% of the Idref for 100ms, the fault will jump.<br><br>Possible causes:<br>- Low power in the DC field.<br>- DC fuses blown.<br>- Defective DC transformer hall sensor.<br>- Defect in the wiring of the DC hall sensor.<br>- Defect in the module's FPGA board.<br>- Excessively sensitive protections. | Never rearm the equipment in case of this failure. Contact Power Electronics.<br><br>Bus voltage may be low, or all three AC fuses are blown.<br>Check wiring and current sensors placement.<br>Check whether the bus voltage is sufficient and the DC fuses are in good condition.<br>- Check DC bus insulation.<br>- Change DC HALL boards, check wiring.<br>- Change the FPGA board if the fault persists.<br>- Check modules' status, Vbus and inverter status.<br>- Check through PowerCommsV the DC current measurement and compare it with the rest of the modules, traps, hall sensors, wiring and FPGA. If the reading is not correct, replace the affected material.<br>- If the measurements are correct, it is possible that this is due to a transient and over-sensitive protection. In this case, increase the threshold in G5.10.13-Umb Activ Fal Idref.<br><br>**Extra information:**<br><br>- Remove blackboxes from modules and SD.<br>- After this type of failure a complete review of the power stages (IGBTs, buses...) must be carried out. |
| **W108:Half bus Vmax** | **F108:Half bus Vmax** | F69:Mod. DC Overvoltage (P) was triggered and module was stopped at 755 V, but bus voltage remains unstable. This fault will be triggered if an unbalance greater than 800 V appears in the positive.<br><br>Possible causes:<br>- IGBT shunted to ground, damaged or shorted.<br>- Defective Nomex insulation laminate.<br>- Damaged capacitor.<br>- Damaged snubber board.<br>- Faulty reading of balance board or module FPGA board.<br>- Bad grounding of the balance board.<br>- Incorrectly crimped measurement wiring on balance board or module FPGA board. | Never rearm the equipment in case of this failure. Contact Power Electronics.<br>This fault has an additional action associated by default. Thus, it is included in the list of special faults (G5.10.1-Special faults).<br>Check module hardware.<br>**Extra information:**<br><br>- Check the voltages P and N on SV9.2-Selected module with the inverter stopped. Make sure that the voltages are balanced on all modules.<br>- Remove blackbox from modules and SD.<br>- After this type of fault, a complete review of the power stages (IGBT, buses...) must be carried out.<br>- Determine which module(s) are the cause(s).<br>- Measure the voltage of both semibuses and check if it coincides with the inverter measurement.<br><br>Verify the DC voltages using a multimeter, if the unbalance is real:<br>- Check the IGBT and the E936XX board.<br>- Extract SD and blackbox.<br>- Check the power amplifier.<br>- Check balancing boards.<br>- Check bus status.<br><br>If the measurement is not real (inverter measurement >800 and much lower tester measurement):<br>- Check measurement circuit. Do not measure in the measurement board directly.<br>- Change module board.<br>- Contact Power Electronics.<br><br>**Note:** If nothing is broken, check the version of the module boards. The E973CD version is not valid. |

<table>
<tr><th>Warning</th><th>Fault</th><th>Description or probable cause</th><th>Actions</th></tr>
<tr><td rowspan="2">W109:Low module Vdc</td><td rowspan="2">F109:Low module Vdc</td><td>A start command has been received but bus voltage is insufficient to start and maintain operation.<br><br>There could be a bus short circuit.<br><br>This fault occurs when there is a difference between the positive and negative bus greater than the threshold set at G5.10.15a-Low mod Vdc High and G5.10.15b-Low mod Vdc Low.<br><br>Possible causes:<br>- Defect in the wiring of the voltage measurement and module FPGA board.<br>- Incorrect reading of the voltage measurement and module FPGA board.<br>- Defect in the routing of the bus measurement cable to the module FPGA board.<br>- Incorrectly crimped wiring on FPGA and balance boards.<br>- Badly wired AC measurement board. Do not measure directly on the board.<br>- DC fuses of the module blown or damaged.<br>- Faulty P3G board.<br>- Short circuit on bus or busbar.<br>- Wrong tightening torque on hardware.<br><br>Note: In versions prior to HEM_1.2.1, the value of this parameter was the only condition to activate fault F109. The need to distinguish bus oscillations from a real bus short circuit was identified.</td><td>Never rearm the equipment in case of this failure. Contact Power Electronics.<br>Check wiring.<br>This fault has an additional action associated by default. Thus, it is included in the list of special faults (G5.10.1-Special faults).<br><br>- Download and analyze the blackbox, the SD and the parametrization from the modules.<br>- Check the wiring.<br>- After this type of failure a complete review of the power stages (IGBTs, BUSES...) must be carried out.<br>- If nothing is broken, check the version of the module boards. The E973CD version is not valid.<br><br>If the fault appears after a storm, it may be due to an asymmetrical dip at low power causing bus oscillations. Configure the following parameters as:<br>- G5.10.15-Low module Vdc = 500 V (less sensibility)<br>- G2.4.1-Activation level = 30 %<br>- G2.4.2-Deactivation level = 30 %<br>- G2.4.6-Available Q backup = 0 %<br><br>The fault is activated depending on these three criteria:<br>- If it stays between the thresholds of G5.10.15 + G5.10.15a and G5.10.15, it waits 1 ms and the Vdc does not recover, the fault is triggered.<br>- If the voltage rises again above G5.10.15 in less than 1 ms, it switches back.<br>- If the voltage drops below the threshold of G5.10.15 - G5.10.15b, the fault F109 appears instantaneously.<br>If the fault is activated due to sensibility, lower the parameter G5.10.15 50 V.<br>If the fault is generalized in all the inverters of the plant, Configure the following parameters as:<br>- SG1.2.a1 - VRT Mode = Mode 2.<br>- SG1.2.a2 - Sudden V drop = 0.08.<br>- SG1.2.a3 - Sudden f drop = 0.100.<br>- G5.10.15a - Low mod Vdc High = 50 V<br>- G5.10.15b - Low mod Vdc Low = 100 V</td></tr>
<tr><td colspan="2">Parameters related to this fault and value they should have:<br><br>• Low Vdc: can be configured through G5.10.15 - Low module Vdc = V nominal * 1,4142 – 350 (Vac peak).<br>• Low Vdcth2 + G5.10.15ª - Low mod Vdc High – l = low Vdc + 50 V.<br>• Low Vdcth3 - G5.10.15b - Low Vdc mod Low = low Vdc – 100 V.<br><br> 

<br><br>Note: From version 2.0.1: all parameters are already set by default to the appropriate value.</td></tr>
<tr><td>W110:Blown AC fuse</td><td>F110:Blown AC fuse</td><td>This fault should not appear because it has been disabled.</td><td></td></tr>
<tr><td>W111:Reserved</td><td>F111:Reserved</td><td>Reserved for future use.<br>Could appear during the initialization of the DSP or when the system detects a wrong configuration of the module.</td><td>Reserved for future use.</td></tr>
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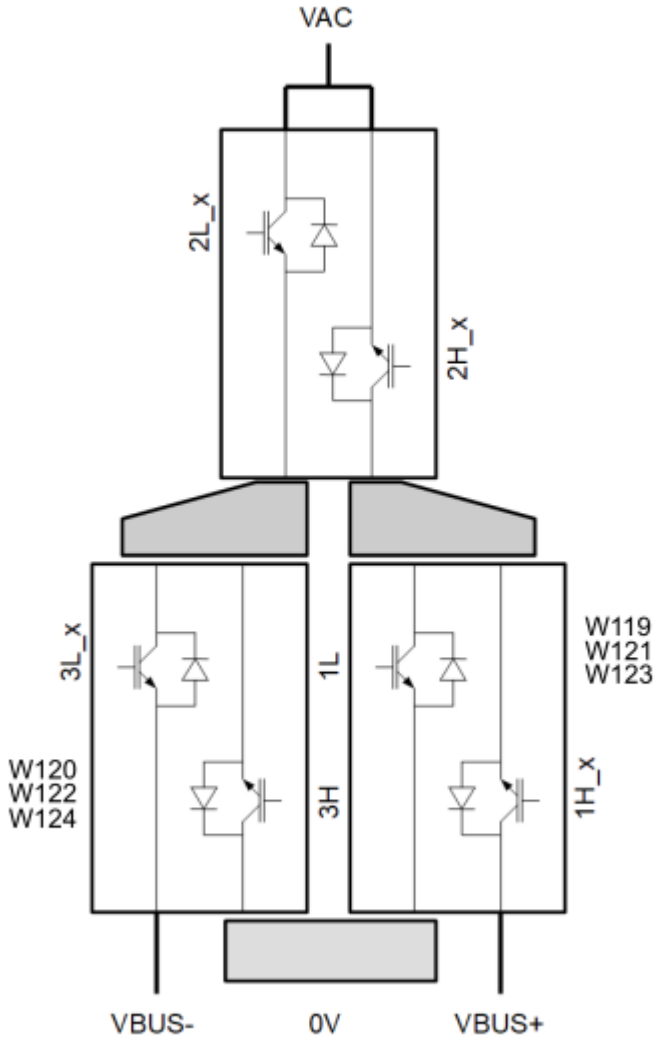
| Warning | Fault | Description or probable cause | Actions |
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| **W112:OCAC emergency** | **F112:OCAC emergency** | An overcurrent has been detected in AC and it has not been stinguished after 500 µs (this threshold is set internally).<br><br>Possible causes:<br>- Defect in the parameterization.<br>- Reading on the IGBTs.<br>- Ribbon cable damaged or badly connected.<br>- Voltage on FPGA lower than 24 V.<br>- Fiber connector incorrectly crimped on FPGA board (E973XX) or central board (E997XX).<br>- Bad fiber or connectors.<br>- Defect in hall sensor.<br>- Badly calibrated or wired AC measurement board (Do not measure directly in this board).<br>- Defect in FPGA board.<br>- Defect in converter board.<br>- Defect in central control board P3G -E977XX. | Upgrade Software to the latest version.<br>In case you already have a higher SW version updated:<br><br>- Remove the Blackbox from the modules.<br>- Perform a complete review of the power modules (IGBT, buses, ribbon cables…).<br>- Check the module logger, identify which module has failed due to an increased current in its phases and check the current halls, wiring and converter board.<br>- Check that the calibration of the central AC voltage measurement board does not exceed 1.5%, because above this value the switching may not be correct. Do not measure in this board directly.<br>- Check that the fiber in the connector is flush, polished and has no cracks. Measure the power from one end to the other with the fiber tester.<br>- Check that the parameters G8.2.10-Kp Vdc night and G8.2.11-Ki Vdc night have other value than 0.<br>- From software version 2.0.0, check that the frame rate is higher than 98% in SV12.1-FPGA modules. If this measurement is correct, check power supplies and the status of the central board and FPGA.<br>Never reset the equipment if this fault is activated.<br>Contact Power Electronics.<br><br>There is a hardware failure in the equipment.<br>This fault has an additional action associated by default. Thus, it is included in the list of special faults (G5.10.1-Special faults). |
| **W113:Coms checksum** | **F113:Coms checksum** | An error has been detected in communication between the module x and central control. There are some failed optical fiber communication frames.<br><br>If TX or RX fiber is removed, fault F84:Mod. Comms will be triggered.<br><br>Possible causes:<br>- Information overload in the microprocessor. | Check the status of communication fibers.<br><br>From software version 2.0.0 registers for checking the percentage of correct frames have been added.<br><br>**Extra information:**<br><br>- Check optical fiber cabling.<br>- Measure the optical power (if possible) of fiber cables.<br>- Problem with the frame.<br>- If F84:Mod. Comms is also triggered; the FPGA may not be receiving data from one of the modules.<br>- If the fault is repetitive, contact Power Electronics. |
| **W114:Module Fans** | **F114:Module Fans** | Feedback has not been received from the module fan.<br><br>Possible causes:<br>- Defect in the fan or FPGA power supply.<br>- Damaged fan capacitor.<br>- Defect in the digital output of the FPGA board.<br>- Defect in the fan enable relay.<br>- Faulty power supply to the fan enable relay.<br>- Badly wired fan feedback relay.<br>- Defect in the fan. | Check module fans wiring.<br><br>- Check the fan power supply, it must provide 230 V and there should be a balanced consumption in all modules.<br>- Jumper the digital fan activation output on the FPGA board. If it does not activate, check that the relay located next to the board receives 24 V.<br>- Measure the capacitance of the capacitor connected to the fan.<br>- Check the fan feedback circuit.<br><br>After checking wiring, provide supply again to the fans and see if they turn on.<br>If the fans are still not operating, they must be extracted for replacement / repair. Contact Power Electronics for assistance. |
| **W115:Module sensor** | **F115:Module sensor** | One, or more, of the three current sensors of the module are not measuring properly, there is an offset error.<br><br>Possible causes:<br>- Defect in current measurement wiring.<br>- Defect in the AC hall converter board.<br>- Defect in the hall current sensor.<br>- Damage to the module's FPGA. | Check wiring and connections of hall effect sensors in all modules.<br>- Check 4-20 mA conversion board.<br>- Configure SG1.3.3-Module stop meas to see which phase of the module is measuring incorrectly.<br>- Check the Iac measurement wiring of the module causing the fault.<br>- Replace AC hall boards.<br>- Check wiring.<br>- Check module FPGA. |

| Warning | Fault | Description or probable cause | Actions |
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| **W116:Module Disabled SC** | **F116:Module Disabled SC** | The module has been disabled by the DSP. It could be caused due to a fault during soft charge or because the module has failed too many times, depending on the number of times configured in SG1.2.4-Modules retries and SG1.2.8-Mod. Start fault.<br><br>Possible causes:<br>- Fault on the module. | If the module has failed too many times, re-enable it by using G2.4.5-Reset retries counter or by stopping and starting the equipment.<br>If there was a fault during soft charge, the module can be restarted by disabling and enabling it again.<br>If this happens, the module is disabled and cannot be re-enabled unless the selector is removed or 30 minutes elapse.<br><br>**Extra information:**<br>- Analyze which module fails and why it fails. The reason for the failure can be seen in SG9-Module Logger<br>- Tackling the problem that causes the failure of modules. Generally due to W88: AC contactors |
| **W117:Mod. Common thermal** | **F117:Mod. Common thermal** | Inductances in HEM / HEMK inverters are shared between pairs of modules.<br>If the inductance in a module overheats, that module will trigger F93:Mod. Thermal LCL and its paired module will trigger this fault F117:Mod. Common thermal. Both modules will stop.<br><br>Possible causes:<br>- Defect in the wiring of the inductance thermal sensor.<br>- Defect in the inductance thermal sensor.<br>- Bad contact between the inductance busbars. | See which module triggered the fault F93:Mod. Thermal LCL and check inductance in its paired module.<br>Check position, ventilation and status of the coil.<br>If the fault is frequent, check wiring and the thermal sensor.<br><br>- Verify that the technical characteristics of the inductance thermal sensor match the máximum value of the temperature.<br>- This failure occurs when the module with which it shares inductance, reports a W93:Thermal LCL. |
| **W118:Thermal drift** | **F118:Thermal drift** | Reserved for future use.<br>This fault will be implemented in future versions. It will be triggered if an anomalous behaviour is detected in an IGBT which could cause a breakdown. To fix it, the defective IGBT must be replaced (check the SD log to find out which). | Reserved for future use. |
| **W119:Desat R1(N)** | **F119:Desat R1(N)** | These faults are caused by a desaturation in the corresponding IGBT driver.<br><br>Other possible cause is a loss of power to the gate drive that controls the diode which caused the fault.<br><br>Possible causes:<br>- Defect or shunt in an IGBT.<br>- Defect in the module's FPGA board.<br>- Incorrect gate drive voltage must be ±15 Vdc.<br>- Incorrect voltage of the FPGA board of the module. It should be 24 Vdc (±15 Vdc should be measured at TD - Test Point).<br>- Damaged or loose ribbon cables.<br>- Defect in the 24 Vdc power supply.<br>- Loose connections in the 24 Vdc power supply harness.<br>- Shunts in the DC bus.<br><br>**Note**: These faults are complementary to W71/F71:Desat R1(P) – W82/F82:Desat T3(N). | If the fault occurs frequently, the gate drive may be damaged. In this case, contact Power Electronics.<br><br>After this fault, do not start the module again. If possible, isolate the power stage until the affected power amplifier is checked. The IGBT might be damaged. Check image "Distribution of transistors in the power stage".<br><br>In case of doubt, please, contact your superior.<br><br>**Extra information:**<br>- Check the power stage.<br>- R, S and T indicate which IGBT has failed; P and N indicate polarity.<br>- Check if the power stage needs to be replaced.<br>- Check the power supplies (5 V, 15 V and -15 V) of the module board<br>- Check flat cables<br>- If the fault is repetitive or has caused damage, download blackbox, parametrization and SD and contact Power Electronics.<br>  |
| **W120:Desat R3(P)** | **F120:Desat R3(P)** | | |
| **W121: Desat S1(N)** | **F121:Desat S1(N)** | | |
| **W122: Desat S3(P)** | **F122:Desat S3(P)** | | |
| **W123: Desat T1(N)** | **F123:Desat T1(N)** | | |
| **W124: Desat T3(P)** | **F124:Desat T3(P)** | | |

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| Warning | Fault | Description or probable cause | Actions |
|---|---|---|---|
| **W125: Homopolar I detect** | **F125. Homopolar I detect** | This fault appears when the homopolar current detection calculated by the module is greater than 300A for more than 100us. This warning appears when the value defined by parameter G5.5.9-Module Homopolar I is exceeded.<br><br>Possible causes:<br>- Defective power stages.<br>- Defective DC or AC fuses.<br>- Defect in the AC current hall sensor.<br>- Defect in the module's FPGA.<br>- Defect in the wiring. | If there is a homopolar current detection, there could be a short circuit in the module.<br>Change the module and inspect the power stage looking for damaged components.<br><br>Download blackbox, parametrization and SD and contact Power Electronics.<br>For software version 1.7.0:<br>- If parameter SG2.2.1-Config FPGA 1 = 4000 causes an overflow and the threshold is low.<br><br>For software version 1.7.1:<br>- Parameter SG2.2.1-Config FPGA 1 disables the fault.<br><br>For software version 2.2.1 (recommended):<br>- The parameter SG2.2.1-Config FPGA 1 appears as disabled. |
| **W126: Temperature Deviation IGBTs** | **F126:Temperature Deviation IGBTs** | This fault appears if a module exceeds the "IGBT deviation" of G3.3.1.3-TªIGBTsDeviationMax or G3.3.4 in prior versions to 2.0.<br>It is the maximum difference between the highest IGBT temperature and the average of the highest IGBTs temperatures per module.<br>By default, it is set at 15 ºC, the fault appears if this temperature is compared with respect the average temperature.<br>This fault also appears if the temperature set in G3.3.1.2-Initial Tª IGBTs (initial point for IGBT temperature derating) is exceeded.<br><br>Possible causes:<br>- Defective ventilation.<br>- Lack of thermal paste or graphite.<br>- Wrong tightening torque.<br>- Obstruction in radiator or air inlet.<br>- Defect in the cable or in the IGBT of the NTC sensor. | Inspect the power stage.<br>- Check ventilation.<br>- Check the use of thermal paste or graphite for better thermal transfer.<br>- Check for correct tightening torques.<br>- Check for possible obstruction in the stage radiator or air inlet.<br>- Check continuity and connectors of NTC cables.<br>- Check NTC sensor resistance.<br>- Swap NTC sensor channels in FPGA. |
| **W127: EstTIGBT** | **F127:EstDeltaTªIGBTmod fault** | Protection against temperature deviations in IGBTS<br>Depending on the current of each module, a ΔT_IGBT is calculated to realize a protection. It is displayed when any IGBT on any module increases its temperature above the calculated temperature of the IGBT. In the case that ΔT_IGBT is higher than G5.7.1.4-EstDeltaTªIGBTfault it will jump the fault by deactivating it.<br><br>Possible causes:<br>- Wrong configuration of the SG10.1.3b – Power Stage parameter.<br>- Wrong intake temperature reading.<br>- Defect in the ventilation.<br>- Lack of thermal paste or graphite.<br>- Wrong torque.<br>- Obstruction in radiator or air intake.<br>- Defect in the cable of the NTC sensor or IGBT. | Check the module and identify if there is any obstruction in the air intake to the module, if the fan is working the module should be visually inspected.<br>- Check if the fan is working properly.<br>- Check the type of stage configured in SG10.1.3b-Power Stage and ensure that it is in accordance with the IGBTs installed.<br>- Check that the intake temperature reading is correct. |

The following figure shows the distribution of transistors in the power stage. It is useful to identify which block must be replaced, according to the desaturation fault (W71/F71:Desat R1(P) – W82/F82:Desat T3(N) or W119/F119:Desat R1(N) – W124/F124:Desat T3(P)) that was triggered. *x* represents the phase (R, S or T).




*Distribution of transistors in the power stage*

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# Inverter

## Inverter Status Messages

The table below summarizes all possible inverter status and their acronyms.

| Code | Status | Name | Description |
|---|---|---|---|
| 0 | PUP | Power up | The start command has been received and the inverter takes 5 seconds to stabilize the system (communications, measurements, configuration, etc.). |
| 1 | INIT | Initialization | Basic conditions are being checked. |
| 2 | OFF | Off | The inverter is stopped waiting for the start command which can be activated with the START button of the display unit when the inverter is in local mode or with the ON/OFF switch when the local mode is disabled. |
| 3 | PCHG | Precharge | The inverter is doing the soft charge. |
| 4 | REA | Ready | The soft charge has been completed and the inverter is waiting for the command to start. |
| 5 | WAIT | Wait | The start command has been received, starting conditions are fulfilled and the inverter is waiting for the delay time (during which starting conditions must be maintained) to elapse. |
| 6 | ON | On | The inverter is operating normally. |
| 7 | STOP | Stop | The inverter is stopping. |
| 8 | DISC | Discharge | The capacitors on the AC and DC sides are discharging (discharge time is 1 minute). Please wait until discharge has been completed before performing any actions in the unit. |
| 9 | FLT | Fault | There is a fault in the inverter. See fault code FXX and refer to section 3 for troubleshooting. |
| 10 | LVRT | LVRT | Algorithm Low Voltage Ride Through is running. |
| 11 | OVRT | OVRT | Algorithm Over Voltage Ride Through is running. |
| 12 | NGHT | Night | The inverter has detected night conditions. |
| 13 | NDCO | Night dc off | The inverter has detected it is night-time already, DC side is opened, but it is not performing heating or operating in night mode. |
| 14 | STB | Standby | The inverter is waiting for the conditions to perform the soft charge to be fulfilled. |
| 15 | HVPL | HVPL | High Voltage phase lost. |
| 16 | TRCK | Trackers | Trackers mode is running. |
| 17 | PRON | Pre-on | The inverter is in a status before “on” status, where after performing the soft charge, it checks the synchronization with the grid. This status can last 4 seconds and during the process the references are zero. |
| 18 | DIAG | Diagnosis | Extended self-diagnosis, in which the soft charge is done rectificating and then discharging each module sequentially. If this process works well, then the equipment will start. |
| 19 | LCON | Filter contactors | LC filter contactor has been activated and the inverter is waiting for the feedback to confirm the contactor is closed. |
| 20 | PRMG | Premagnetization | Medium voltage transformer premagnetization is in progress. The MV switchgear is open. The inverter synchronizes the voltage in the MV transformer with the grid voltage to reduce the inrush current. Once synchronized, the inverter will close the MV switchgear and the inverter will get connected to the grid. |
| 23 | ANGH | Autonight | Indicates that the inverter switches to the autonight state. The "NIGHT" state is used to make a soft transition from the "on" state to the "NIGHT" state. |
| 24 | ECON | Econmode | Indicates that the inverter has activated the Econmode functionality in order to save MV transformer losses (inverter with MV switchgear open and DU closed) |

## Inverter Warning Messages

The following table details all possible warning messages. If there are no warnings, the message "**NO WRN**" will appear.

**Note:** In case a 4-digit value is shown, it must be deciphered using an algorithm. Contact Power Electronics for assistance.

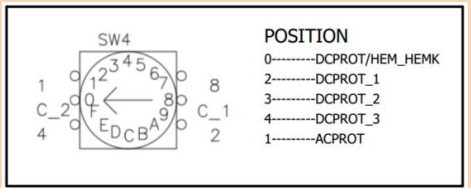
| Warning | Acronym | Name | Description |
|---|---|---|---|
| W1 | HVAC | High input voltage | The inverter input voltage is reaching a dangerous level. Its value is above the value set in the protections (G5.2-Running conditions). |
| W2 | LVAC | Low input voltage | The inverter input voltage is reaching a dangerous level. Its value is below the value set in the protections (G5.1-General). |
| W3 | HFRQ | High frequency | The input frequency is reaching a dangerous level. Its value is above the value set in the protections (G5.4-Algorithms). |
| W4 | LFRQ | Low frequency | The input frequency is reaching a dangerous level. Its value is below the value set in the protections (G5.5-Unbalanced). |
| W5 | NSC | No start conditions | In STANDBY status, it indicates that there are no AC conditions to perform the bus soft charge.<br>In START status, indicates that nor AC neither DC start conditions are fulfilled. |
| W6 | NOMD | No modules | Heating is not working because no modules are available. |
| W7 | PLIM | P limit | P limit has been reached. |
| W8 | QLIM | Q limit | Q limit has been reached. |
| W9 | ISLIM | Is limit | Is limit has been reached. |
| W10 | IDCLIM | Idc limit | Idc limit has been reached. |
| W11 | DVDC | Derating DC bus V | A derating due to high voltage in the PV field DC bus is active. |
| W12 | HEAT | Heating | Heating is active. |
| W13 | NIGHT | Night mode | Night mode is active. |
| W14 | LP | Low P | Active power is lower than the minimum AC power set on G2.2.2. |
| W15 | LVDC | Low Vdc | DC voltage is lower than the minimum DC voltage set on G2.2.5. |
| W16 | DIBT | Derating IGBT T. | Inverter is limiting power due to high internal temperature (IGBT) or high external temperature. |
| W17 | PPF | Power Per Frequency | The Power Per Frequency algorithm is running. Power is reduced due to a high frequency. |
| W18 | DELTAT | Modules delta T exceeded | Protection to prevent overheating of the module if the air admission is blocked and not enough air reaches the module area. This point involves checking the cyclone filter of the inverter and the module air intake.<br>Protection is the calculation of the average module temperature value (measurement of the PCB temperature of each module). The internal ambient temperature of the module (the one taken directly from the temperature measurement of the PCB of each module) can be found in the following records: Module 1: 45011, Module 2: 45011+32.<br>If the difference between the average module temperature and the intake temperature is above the value of G5.7.2.3 then the warning W18 will appear. |
| W19 | HVDC | High Vdc | DC voltage is more than 1500V. A fault will be triggered if the delay time set in [G10.5.2] elapses. |

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| Warning | Acronym | Name | Description |
|---|---|---|---|
| W20 | EstTIGBT | EstDeltaTªIGBT mod warn | Warning about the temperature estimation limit of the IGBT and its protection. Depending on the current of each module, a ΔT_IGBT is calculated to realize a protection. It is displayed when any IGBT on any module increases its temperature above the calculated temperature of the IGBT. If ΔT_IGBT is greater than G5.7.1.3 then W20 will be triggered. |
| W21 | TRKNM | Trackers no modules | The module electrically prepared to inject power to the auxiliary services circuit is not available and trackers mode cannot be activated. |
| W22 | TRKLV | Trackers low Vac out | The voltage output of the module in trackers mode is lower than the configured value (only configurable by Power Electronics' personnel). |
| W23 | TRKCC | Trackers close contactor | The order to close auxiliary supply circuit contactors has been received, but there is a problem with the feedback. |
| W24 | TRKOC | Trackers open contactor | Similar to W23, but the order is to open contactors. |
| W25 | TRKBF | Trackers blown fuse | Blown fuse detected in the auxiliary supply circuit for trackers mode. |
| W26 | LCFB | LC feedback | LC contactors are not operating adequately in the stop procedure. |
| W27 | QFV | Curve Q vs V | Grid support reactive power curve ongoing. |
| W28 | PFV | Curve P vs V | Grid support active power curve ongoing. |
| W29 | VDCDS | Vdc discordance | Discordance between bus voltage and PV voltage (it will only be triggered when the DU is closed, in that case they should be practically the same, but they are not). |
| W30 | EMDCC | Econmode DC | NEC 2014 configuration it indicates, DU is going to close at night without PV voltage in 30 seconds.<br><br>**Note**: If Econmode is available, control will try to close DU during the night after a LVRT/OVRT event. |
| W31 | EMACC | Econmode AC | NEC2017 configuration and Econmode available, if DU opens after a LVRT/OVRT event, control will close the MV switchgear without pre-magnetization (in 30 seconds).<br><br>This warning will not come out if the parameter G4.12.6 is disabled. |
| W32 | EMDCA | Econmode DC Open | When the inverter is in Econmode state, it will try to connect MV switchgear following the pre-magnetization process. DU must be closed for that.<br><br>This warning will be triggered when DU is open. Then, the pre-magnetization process cannot continue.<br><br>Check F192 in order to understand next steps. |
| W33 | EMSEL | Econmode Selector | In the ECONMODE state, if the SELECTOR is OFF and DU is not open then W33 will appear.<br><br>**Note 1**: NEC 2014 cannot be in this situation as the 24Vdc digital output controlled by the control board is removed as soon as the SELECTOR is set to OFF.<br><br>**Note 2**: NEC 2017 can occur this event. The holding coil of the NEC 2017 is fed directly from 24Vdc so there is no control to open it as during the ECONMODE the inverter does not have the 230Vac to effectively open the DU. |
| W34 | LIMDU | DU Overc Lim Power | It will appear when a current reduction is carried out in the modules of the NEC2017 equipment because the DC cabinet current IDC limit has been exceeded depending on the inlet temperature. It can also appear when the current limit through the DC switch disconnectors (Q8 and Q9) is exceeded.<br><br>These limits are defined through the parameters of the subgroup G3.6-DU limits. |
| W35 | FORDOB | Force Double | More than one modulation switch per minute has been detected (from single to double). |

| Warning | Acronym | Name | Description |
|---|---|---|---|
| W36 | DAT | Derating admission T. | A derating due to admission temperature is being carried out. |
| W37 | DF | Derating frequency | A derating by frequency is being carried out. |
| W38 | NOPPS | No PPS Received | The PPS (pulse per second; a signal used to synchronize clocks) has not been received. |
| W40 | OVP | AC overvoltage | AC overvoltage protection has tripped (surge arrester P1, P2, P3). Check the status of the hardware and the backfeed wiring up until the control board. |
| W41 | RTC | RTC | The RTC is not configured. Set the time in G1. |
| W42 | MSWM | Modules version | There are one or more modules with different software versions. Check the version of each module in SV9. |
| W43 | VSWM | uP DSP version | Software versions are not compatible. Contact Power Electronics. |
| W44 | NOSD | SD not present | No SD card detected. Verify that it is inserted and properly placed. |
| W45 | BMI | Battery module isolated | One or more modules of the PCS has been isolated due to a fault. |
| W46 | DUIUN | DU I unbalance | There is a current unbalance in the DU channels. By default, the threshold is set at 25% during 15 s but these values can be adjusted in parameters G5.5.6 and G5.5.7.<br>**Note:** This warning only applies if the inverter type is PCS. |
| W47 | SPRLD | Spring loaded | Some of the circuit breaker triggers are not loaded. This warning will only show up when the DU is configured with NEC 2017. Check wiring and breakers |
| W48 | DUVDC | DU Vdc measure error | Reserved for future use in HEM NEC 2017 inverters. |
| W49 | PPC_C | PPC Communications | When power limiting is activated according to the parameters set in the communication timeout with the PPC (G6.3.1-Home refs time), this warning will appear. |
| W50 | DUCOM | Comms lost | There is a problem in the Disconnecting Unit communications. |
| W51 | CHISL | Channel isolated | DU channel isolated because it is derived, feedback has failed, or it does not measure voltage. |
| W52 | INSA | Algorithm on | Derived channels detection algorithm is running. Once finished, the warning will automatically disappear. |
| W53 | DUOF | Fb open error | Problem with feedback from one of the DU channels in an opening event. |
| W54 | DUCF | Fb close error | Problem with feedback from one of the DU channels in a closing event. |
| W55 | OVDCP | DC overvoltage | DC overvoltage problem. Check the status of the Disconnecting Unit and J7 connector digital input. |
| W56 | ISFLT | Isolation fault | Warning not used currently. |
| W57 | CHCOM | Acq comms error | Communications problem between the Disconnecting Unit and acquisition channels. |
| W58 | NOCHN | No ch available | Indicates that there are no available channels. Similar to fault 50, but this warning automatically disappears if any channels become available. |
| W59 | GRIDIN | Isolat flt AC grid | Indicates that the insulation fault has occurred in the AC side. |
| W60 | DUBSY | Processing cont | Contactors processing event ongoing. Disappears once finished. |
| W61 | CHSWM | Acq version | DU acquisition channels with different versions have been detected. |

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| Warning | Acronym | Name | Description |
|---|---|---|---|
| W62 | PCHM | Poles config | This warning will be triggered when the number of channels in the positive pole of grounding contactors (in *GFDI+IMI ext* mode) differs from those detected in the negative pole. |
| W63 | CHCD | Current detected | A current has been detected during a DU channels opening event. |
| W64 | CHMW | Channel miswired | One or more acquisition channels does not measure any voltage, whether because the measurement is not connected or because it was connected with the wrong polarity. |
| W100 | DCOMS | Display comms | Communication between the display and the control board has been lost. |
| W130 | ISODCCM | DC insulation monitor comms | Communication with the DC insulation monitor cannot be stablished. Check the RS485 cable or, if your inverter does not include a DC insulation monitor, make sure G9.1.1-Type is not set to "Power Electronics".<br>- Check the communication between the DC cabinet and the cyclone.<br>- Check that the software version of the DU is different from 0.<br>- Check the good condition of the fans in G11.2.6-Imp fan 1 status and G11.2.7-Imp fan 2 status.<br>- Ensure continuity in IMI and cyclone. The grid must be connected all along the RS485 network and at one point to the DC cabinet.<br>- Check the connection of the resistor at the beginning and at the end of the line using jumpers.<br>- Check the correct connection of the IMI and the DU:<br>IMI: A-> White, B-> Green, GND->Yellow.<br>DU: A-> Green, B-> White, GND->Yellow.<br>- Carry out tests with the inverter stopped or running. If more warnings appear while running, use an external cable to check if the problem is noise.<br>- If parameter G6.1.8-Modbus error CRC shows a value of less than 2%, consult G6.1.9-Modbus OK frames = 0, taking into account that when it reaches 65535 it returns to 0.<br>- Check the connection of the line resistors.<br>- Check DC connections and wiring.<br>- Check status of IMI E840XX board.<br>- Check status of resistance measurement board E843XX.<br><br>Configure the IMI board:<br>  |
| W131 | INSDC | Insulation DC | The DC insulation monitor has detected an insulation resistance below the threshold set in G9.1.4-Warning threshold. |
| W132 | IRTHDC | Ins resistor high DC | The DC insulation monitor has detected a higher insulation resistance than the admissible. |
| W135 | ISOACCM | AC insulation monitor comms | Communication with the AC insulation monitor cannot be stablished. Check the RS485 cable or, if your inverter does not include an AC insulation monitor, make sure G9.2.1-Type is not set to "Power Electronics". |
| W136 | INSAC | Insulation AC | The AC insulation monitor has detected an insulation resistance below the threshold set in G9.2.4-Warning threshold. |
| W137 | IRTHAC | Ins resistor high AC | The AC insulation monitor has detected a higher insulation resistance than the admissible. |
| W140 | FNIMCM | Imp fans comms | Communication with the impulsion fans (cyclone) cannot be stablished, communication fault. See G11.2.5-Error type and G11.2.17-Wrn Power Reduction. |
| W141 | FNIMER | Impulsion fans | One (or both, in HEM inverters) of the impulsion fans (cyclone) has reported an error. See G11.2.5-Error type and G11.2.17-Wrn Power Reduction. |

| Warning | Acronym | Name | Description |
|---|---|---|---|
| W142 | FNXCER | Extrac. Fans control | Feedback from extraction fans in the low voltage side has not been received. |
| W143 | FNXMER | Extrac. Fans MV | Feedback from extraction fans in the medium voltage side has not been received. |
| W146 | APIDON | Antipid on | The AntiPID is an optional device on the inverter that serves to regenerate the plates on single-pole inverters. This device can only be activated when there are night conditions for the time defined in G4.13.3-Time Aft Night Cond under night conditions. After this defined time the Antipid device is energised and W146:Antipid on will appear. |
| W147 | BBON | Blackbox On | Blackbox download operation has started. |
| W148 | DUFB1 | DU FB1 error | **These warnings only apply to HEM inverters.** The DC cabinet has two DC switch disconnectors (Q8 and Q9) for connecting to the PV panels. If the order and feedback of any of them does not match, the corresponding warning will be triggered (W148:DU FB1 error for switch disconnector Q8 and W149:DU FB2 error for switch disconnector Q9). |
| W149 | DUFB2 | DU FB2 error | |
| W150 | BFPSNC | BFPS no cond | A command to close the BFPS has been received but could not be executed because there is DC voltage still present. |
| W151 | PWBFPS | BFPS power | The maximum power supported by the BFPS (parameter G12.1.1-Max Power) has been exceeded during the time set in G12.1.1-Max Power. |
| W152 | IBFPS | BFPS current | The maximum current supported by the BFPS (parameter G12.1.3-Max I) has been exceeded during the time set in G12.1.4-Delay Max I fault. |
| W153 | HVBFPS | BFPS high V | Voltage is higher than the threshold defined in G12.1.5-Max Voltage fault. |
| W154 | LVBFPS | BFPS low V | Voltage is lower than the threshold defined in G12.1.6-Min Voltage fault. |
| W155 | CNBFPS | BFPS contactor | An order of closing the BFPS has been received, but the feedback has not been recieved. Check the contactor and feedback. |
| W156 | BFBFPS | BFPS fusible | After closing BFPS contactor, the measured voltage is still below 200V. |
| W157 | BFPSCM | BFPS comms | Communication with the BFPS cannot be stablished. Check the RS485 cable or, if your inverter does not include BFPS, make sure G12.2-Enable = "N". |
| W160 | DCDC KO | DCDC KO | This warning is related to the Econmode function and the pre-magnetization of the transformer.<br>In G4.12-Econmode we can find more information about these features.<br><br>If the Econmode option is available, during the night the MV switchgear will be open.<br>At the morning start, the transformer will be pre-magnetized, and for this purpose it is necessary to have power from the PV field through the internal DCDC.<br>If this DCDC is not available, it will not be able to perform the pre-magnetization (since the Econmode does not have enough power to close a module by itself), and the MV switchgear will close without premagnetizing the transformer.<br>This will generate the W160:DCDC KO. |
| W161 | TRAUXTHR | Trans Aux Therm | Indicates that the thermal sensor of the AASS transformer has detected an overtemperature.<br>It only appears when G5.11.11-Transf Aux Thermal = Warning.<br>To avoid false positives, F63:Ground connection is replaced by W161:Trans Aux Therm during the transition to autonight mode in equipment with double GFDI. |
| W162 | SDCRRP | SD corrupted | SD card is corrupted and it is not possible to write in it. When this warning is triggered, the SD card is automatically formatted.<br><br>Possible causes:<br>- It is not possible to write in the SD card.<br>- SD card is corrupted.<br>- Defect in the P3G E977XX central control board.<br>- Incorrect software version.<br><br>**Note:** Applies from software versions 2.2.3 and 5.0.0.0 for the isolation monitor. |

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| Warning | Acronym | Name | Description |
|---|---|---|---|
| W163 | ETLCL | Exc temp LCL box | Is triggered when the thermal sensor detects an overtemperature in the cabinet where the LCL filter is placed.<br>This warning will not be shown if G4.6.1-Enable = External or if SG1.1.19-OverT LCL box error is configured as fault or disabled. |
| W164 | AEMO_C | Communications AEMO | Communications between the PPC and the inverter have been lost. |
| W165 | IANC | Insulation alg NC | Conditions for the insulation algorithm are not met.<br><br>**Note:** this warning, in versions prior to 4.0.1, used to be fault [F63: Ground connection]. |
| W166 | NTPNOK | NTP Server KO | It is not possible to stablish connection with the NTP server. |
| W167 | NEWFASTLOG | New fast log | A new fast logger has been created. |
| W215 | LCFUSDVDT | LC fuse - dV/dt | A higher than expected dV/dt voltage ripple has been detected.<br><br>If [G5.14.1-Configuration] = Warning, this warning will be triggered. If [G5.14.1-Configuration] = Fault, [F215:LC fuse - dV/dt] will be triggered. |
| W216 | LCFUSDIDT | LC fuse - dI/dt | A higher than expected dI/dt voltage ripple has been detected.<br><br>If [G5.14.1-Configuration] = Warning, this warning will be triggered. If [G5.14.1-Configuration] = Fault, [F216:LC fuse - dI/dt] will be triggered. |

**Note**: See figures "HEM ventilation layout" and "HEMK ventilation layout" for details about ventilation elements and sides.

## Inverter Fault Messages & Troubleshooting

A fault in the inverter will force it to stop.

The table below shows all possible faults, along with the probable causes of each fault and recommended actions to fix them.

If you need further assistance, please contact Power Electronics' technical department.

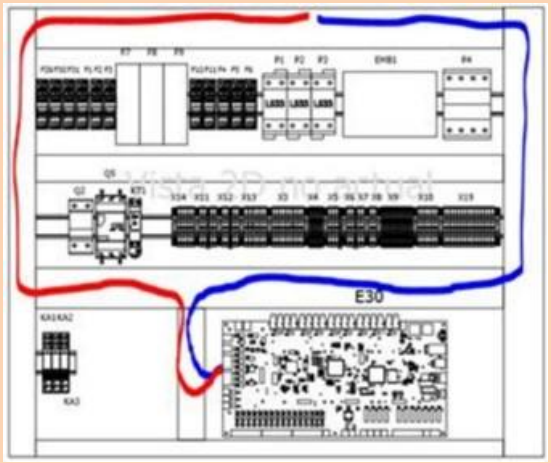
### NOTICE

The actions described in the following table must be carried out under safety conditions and wearing the appropriate PPE equipment. Especially during those troubleshooting actions that require voltage presence in the equipment.

Always follow local regulations or standards regarding safety and health as well as internal application procedures.

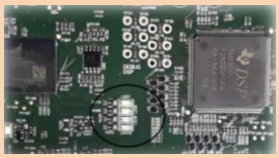
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| Acronym | Fault | Description or probable cause | Actions |
|---|---|---|---|
| **WD** | **F1: Watchdog** | A fault in the DSP has occurred.<br>Possible causes:<br>- Can occur in a controlled manner after a remote update.<br>- Defective software update.<br>- Defect in the central control board P3G - E977XX. | - Disconnect for ten seconds and reconnect the auxiliary services supply.<br>- Reinstall the microcontroller and initialize the Service Group.<br>- Reinstall DSP using Remote Upgrade. Reset control board for 10 s.<br>- If it occurs from time to time without update SW report to Power Electronics by attaching SD registers file.<br>- Replace the central control board P3G -E977XX.<br>- If the fault persists, request Power Electronics' technical assistance. |
| **HWVBS** | **F2: HW Vbus** | DC overvoltage detected. Threshold: 1520 Vdc.<br>Possible causes:<br>- Problem in the wiring.<br>- Defective converter board.<br>- Defect in the central control board P3G - E977XX. | - Check the voltage with an external voltmeter and the displayed voltage in the inverter through the SV2.1-DC voltage bus parameter.<br>- Compare the displayed voltage in the inverter through the SV2.1-DC voltage bus parameter with the voltage of the modules.<br>- Check the DC voltage measurement fuses.<br>- Do not measure in the measurement board directly. |
| **SFTCHR** | **F3: Softcharge** | Instant failure: Wrong configuration (central AC without DU…). | - Check module configuration. In almost every equipment it should be:<br>Soft charge = Central AC.<br>DC module contactor = NO<br>Alternative soft charge = NO.<br>- If the fault persists, request Power Electronics' technical assistance. |
| | | Fault at 60 s: Timeout, one of the soft charge processes did not end successfully or when there is a configuration problem.<br>Possible causes:<br>- Error in the DC voltage reading displayed on the SV2.1-DC voltage bus screen.<br>- DC voltage is not enough to be able to start the module.<br>- Malfunction of fuses or contactors.<br>- Malfunction of the measuring board of the bus (E984). Do not measure directly on this board under any circumstance. | - Check the fuses (F19:ACSC no voltage).<br>- Check contactors and feedback (F18:ACSC feeback).<br>- Check calibration of the voltage reading. Do not measure on the board directly.<br>- Check soft charge configuration and the required conditions for the desired soft charge.<br>- Force the contactor closure from the board relay (do not use a screwdriver directly). Check that the reached bus voltage is Vac*√2 and it must be <70. Otherwise, there is a derivation or some kind of HW problem. Measure the bus voltage that is reached during the process.<br>- If the bus voltage does not increase, there is a shunt or short-circuit in the equipment; check wiring, modules, resistors, rectifier and contactor.<br>- If the DC voltage of the modules is lower than the displayed by the inverter through SV2.1-DC voltage bus, set SG6.3.0.1 Gain I dc ch 1 adj.<br>- Request technical assistance. |

<table>
<tr><th>Acronym</th><th>Fault</th><th>Description or probable cause</th><th>Actions</th></tr>
<tr><td>DISCHR</td><td>F4: Discharge</td><td>Fault at 50 s. Timeout, one of the modules has not discharged correctly (< 50 Vdc)</td><td>Check voltage in all modules using a voltmeter and compare it with the displayed by communications to identify which module (or modules) did not discharge correctly.<br>If a module is repeatedly triggering this fault, contact Power Electronics.<br><br>Extra information:<br>- Upgrade to the lastest Software Version.<br>- Check that there are at least three modules enabled.<br>- Check fiber communications.<br>- Check the electronic power supply of each module.<br>- Check the DC fuse of each module.<br>- Check the FPGA board of each module.<br>- Check the IGBTs and wiring.<br>It is known that depending on the number of modules available, the time is not enough. It is not important because the autoreset clears the fault and the second attempt ends the download.</td></tr>
<tr><td rowspan="2">HGHVAC</td><td rowspan="2">F5: High Vac</td><td>High AC voltage fault.<br>Five programmable levels with five programmable delays.<br><br>Possible causes:<br>- Overvoltage in the AC network.<br>- Incorrect AC voltage reading.<br>- Check wiring and connectors.<br>- Defect in the central control board P3G - E977XX.</td><td>- Check the voltage with an external voltmeter and the displayed voltage in the inverter.<br>- Check the AC voltage measurement fuses.<br>- Check wiring and connectors.<br>- Check G5.2-High AC voltage settings.<br>- Request technical assistance.<br>- Download the blackbox, the SD card and the parametrization of the equipment.</td></tr>
<tr><td colspan="2">Extra information:<br>This fault occurs when the grid voltage exceeds the thresholds configured in G5.2-High AC voltage. Both HEM and HEMK use the low measurement to manage the mains protections.<br>- Check G5.2-High AC voltage configuration. It must be in accordance with the applicable regulations and the configuration table per plant.<br>- Check if the fault has occurred in several units of the same plant (fault review or SD). In the affirmative case, the protections have acted correctly and we have disconnected in accordance with the network regulations.<br>If the problem persists:<br>- Measure with an oscilloscope the Vac measurements at the output in the capacitors of the LC filter in HEM.<br>- Observe if the measurement is stable with the equipment running / with the equipment stopped. If the measurement oscillates, improve the routing of the measurement cables.<br>- Check the status of the board E981XX (E21 in the scheme). Check the voltage at J10, if there is no network it is because there is an offset of 12 mA (3.7 Vdc = 0 Vac). The reading range of the board is from 4 mA to 20 mA. Do not measure in the measurement board (E981XX) directly.<br>  <br>- Revision of the Vac during the last 5 days in the SD. Limit the range of daily variation and take corrective measures if necessary (lower tapping, etc...).<br><br>Note: If the equipment stays in INIT or the Vac measures 0 it is necessary to check the wiring of the Vac measurements.</td></tr>
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| Acronym | Fault | Description or probable cause | Actions |
|---|---|---|---|
| **LWVAC** | **F6: Low Vac** | Low AC voltage fault.<br>5 programmable levels with 5 programmable delays.<br><br>Possible causes:<br>- Defective calibration.<br>- Wiring problems.<br>- AC measuring board (E981) damaged. Do not measure on this board directly. | - Check the voltage with an external voltmeter and the displayed voltage in the inverter.<br>- Check the AC voltage measurement fuses.<br>- Check wiring and connectors (defect in the crimping, proximity to power cables...).<br>- Check G5.1-Low AC voltage setting.<br>- Request technical assistance.<br>- Applies same information shown in F5:High Vac.<br>- Check the voltage at the primary fuses of the SSAA transformer, located in the AC switchboard or in the LC filter switchboard. Confirm a correct reading.<br>- Check the correct operation of the AC measuring board by exchanging it for a new one. Do not measure on this board directly.<br>- Check calibration. |
| **HGHFRE** | **F7: High frequency** | High frequency fault. It occurs when the network frequency exceeds the thresholds set in G5.4-Alta F ac.<br>There are 5 programmable levels with 5 programmable delays.<br><br>Possible causes:<br>- Overfrequency in the AC network.<br>- Incorrect AC voltage reading.<br>- Defect in the wiring or connectors.<br>- Defect in the central control board P3G - E977XX. | - G5.4-Alta F ac configuration review. Must be in accordance with applicable regulations and the plant configuration table issued by Power Electronics.<br>- Analyze if the fault has occurred in several units of the same plant (fault review or SD). In the affirmative case, the protections have acted correctly and we have disconnected according to the network regulations.<br>- Check if the circuit breaker has been opened and opened by Ig, disable the protection (protections configurable directly on the AC circuit breaker).<br>- Apply all actions described in F5:High Vac.<br>- Check the status of the E981XX board, in diagram E21. Check the voltage at J10, if there is no network it is because there is an offset of 12 mA (3.7 Vdc = 0 Vac). The reading range of the board is from 4 mA to 20 mA. Do not measure in the measurement board (E981XX) directly.<br>- Check the AC voltage measurement fuses.<br>- Check wiring and connectors.<br>- Check G5.4-Alta F ac settings.<br>- Request technical assistance.<br><br>**Note**: If the equipment stays in INIT or the Vac measures 0 it is necessary to check the wiring of the Vac measurements. |
| **LWFRE** | **F8: Low frequency** | Low frequency fault. It occurs when the network frequency exceeds the thresholds set in G5.3-Baja F ac.<br>There are 5 programmable levels with 5 programmable delays. | - G5.3-Baja F ac configuration review. Must be in accordance with applicable regulations and the plant configuration table issued by Power Electronics.<br>- Analyze if the fault has occurred in multiple pieces of equipment in the same plant (fault review or SD). In the affirmative case the protections have acted correctly and we have disconnected according to the network regulations.<br>- Check if the circuit breaker has been opened and opened by Ig, disable the protection (protections configurable directly on the AC circuit breaker).<br><br>- Check the AC voltage measurement fuses.<br>- Check wiring and connectors.<br>- Check G5.3-Baja F ac settings.<br>- Request technical assistance.<br><br>**Note**: If the equipment stays in INIT or the Vac measures 0 it is necessary to check the wiring of the Vac measurements. |

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<table>
<tr><th>Acronym</th><th>Fault</th><th>Description or probable cause</th><th>Actions</th></tr>
<tr><td>FPGACMS</td><td>F9: FPGA comms</td><td>Very fast fault. 2 frames to the central FPGA are wrong. It is due to a communications problem between DSP and central FPGA.<br><br>Possible causes:<br>- Error during the SW charging.<br>- Electronics blocked.<br>- Defect in the central control board P3G - E977XX.</td><td>- Disconnect and reconnect the auxiliary services supply.<br>- Check fiber optics wiring.<br>Check the FPGA LEDs. Led7 and Led5 must be on.<br><table><tr><th>LED7</th><th>LED5</th><th>LED8</th><th>LED10</th></tr><tr><td>OFF</td><td>OFF</td><td>OFF</td><td>OFF</td></tr><tr><td>ON</td><td>X</td><td>X</td><td>OFF</td></tr><tr><td>X</td><td>ON</td><td>X</td><td>OFF</td></tr><tr><td>ON</td><td>X</td><td>ON</td><td>OFF</td></tr><tr><td>X</td><td>X</td><td>X</td><td>BLINK</td></tr></table>- Reload the SW (DSP, Micro and central FPGA) of the official version.<br>- If the fault persists, replace the central control board.<br>- If the fault persists, request Power Electronics technical assistance.</td></tr>
<tr><td>MSTFPGA</td><td>F10: Master FPGA</td><td>The central FPGA has not been configured correctly. Check that the 3 joined leds are turned on. It is due to a communications problem between DSP and central FPGA.<br><br>Possible causes:<br>- Error during the SW charging.<br>- Defect in the central control board P3G - E977XX.</td><td>- Disconnect and reconnect the auxiliary services supply.<br>- Check fiber optics wiring.<br>- Check the FPGA LEDs. Led7 and Led5 must be on.<br>  <br>- If LEDs 5, 7 and 8 are on, it indicates that all three states are occurring at the same time (configuration has been received, frames are being received and these frames are wrong).<br><table><tr><th>Status</th><th>LED 7</th><th>LED 5</th><th>LED 8</th><th>LED 10</th></tr><tr><td>The FPGA is not receiving information from the DSP.</td><td>OFF</td><td>OFF</td><td>OFF</td><td>OFF</td></tr><tr><td>The FPGA is receiving information from the DSP.</td><td>ON</td><td>X</td><td>X</td><td>OFF</td></tr><tr><td>The FPGA has received, at least once, the configuration of the number of modules</td><td>X</td><td>ON</td><td>X</td><td>OFF</td></tr><tr><td>The FPGA is receiving erroneous information from the checksum.</td><td>ON</td><td>X</td><td>ON</td><td>OFF</td></tr><tr><td>There is one module wrong at least</td><td>X</td><td>X</td><td>X</td><td>BLINK</td></tr></table>X indicates no influence on the LED.<br><br>- Reload the SW (DSP, Micro and central FPGA) of the official version. Check the installed software versions. If the FPGA has a version equal to or higher than 2.0.0 installed, the module must have a version equal to or higher than 1.9.0 installed.<br>- If the fault persists, replace the central control board.<br>If the fault persists, request Power Electronics technical assistance.</td></tr>
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| Acronym | Fault | Description or probable cause | Actions |
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| **ACANTISL** | **F11: Active anti island** | The anti-islanding algorithm (in mode "Active") has detected an island condition by a variation in the measured frequency.<br><br>This algorithm works by frequency difference between the grid wave and the generated wave. If the frequency difference between load and inverter is 0, the loads are balanced and the inverter is in island condition. Normally the trigger time is between 0.5-1 seconds.<br>The anti-islanding algorithm in passive mode has detected an island condition due to minimum frequency gradient variations (when disconnected from the grid the frequency drops very fast). Normally the tripping time is 0.5 seconds.<br><br>Possible causes:<br>- Power cut caused by the electricity company.<br>- The AC circuit breaker is open. | - Verify with the company if power was actually cut.<br>- Check if the plant requires the algorithm to be activated, otherwise it requires the algorithm to be deactivated.<br>- If it does not require it, check that it is configured with the default values.<br>- If fault persists, check the plant configuration and, if necessary, consult with Power Electronics attaching the SD file |
| **VDCMISMA** | **F12:Vdc mismatch** | Normally, this fault appears after W29:Vdc discordance.<br><br>It can appear when the DC cabinet is conducting and the P3G board detects a difference between Vbus (SV2.1-DC voltage bus) and Vpv (SV2.2-DC voltage PV) greater than 10 % during 5 s. Both values (voltage threshold and time) are defined in G5.11.9-Vpv-Vdc mismat. thres and in G5.11.10-Vpv - Vdc mismat. dlay parameters. The voltage difference normally happens when the module makes the pre-charge of the bus and closes the DC switch disconnector.<br><br>If it is necessary, set the values of SG6-Calibration according with the readings of the parameters SV2.1-DC voltage bus and SV2.2-DC voltage PV.<br>When this fault is triggered, all modules stop working and a disconnection is carried out in the DC cabinet. | - Check the last module failure to identify the problem and take necessary action.<br>- Check wiring, usually associated with W51:Channel isolated, W53:Fb open error or W54:Fb close error warnings.<br>- Check voltage sensors and fuses. Do not measure on the measurement board directly.<br>- If the Vpv measurements and the measurements on the Vbus electronics differ by more than 500 V, the calibrations should be checked. Do not measure on the measurement board directly.<br>- Check the DC cabinet input voltages through parameters SV10.1-Selected channel and SV10.3-Measured voltage.<br>- Replace central control board P3G -E977XX.<br>- Set G2.2.3-P min delay = 20 s and G5.5.3- Delay unbalance I = 30 s<br>- Reprogram the operation of the DC cabinet (number of channels or G10.1- DU Type).<br>- If it is not possible to calibrate the DC cabinet (SV2.2-DC voltage PV) or if this fault appears continuously, set parameters G5.11.9-Vpv-Vdc mismat. thres = 50 Vdc and G5.11.10-Vpv - Vdc mismat. dlay = 5 seg.<br>- Check the voltage of the bus (SV2.1-DC voltage bus). If voltage calibration is necessary, set SG6.3.0.1-Gain Vdc ch1 adj ≠ 1.00.<br>- In NEC2017 inverters, check that neither W148:DU FB2 error nor W149:DU FB2 error have been triggered . If one DC switch disconnector does not close when the inverter starts, the boards will have a wrong reading and the fault will be triggered.<br>- Check the configuration of the parameter G5.11.8-Fb DC breaker fault.<br><br>Calibrate the operation of the DC cabinet in SG6.1-DU calibration:<br><br>- Minimum: select the values one by one by checking the absence of voltage (Vdc = 0 V) using a calibrated multimeter on each string.<br>- Maximum: select the values one by one by checking the exact voltage (Vdc) using a calibrated multimeter on each string. |

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| **NOMOD** | **F13: No modules** | All modules have stopped.<br>It occurs when all modules fail, and it has not been for W65: AC overcurrent .<br><br>Possible causes:<br>- Any module is active.<br>- Defect in the AC Hall board.<br>- Defect in the wiring.<br>- Wrong drive select configuration.<br>- Another fault has occurred in all modules and has disabled them. | - Check the last module fault to identify the problem and take the necessary actions.<br>- Change AC HALL boards and the wiring.<br>- Review the reason for module downtime (SG9-Logger modules or in SD).<br><br>If this fault appears in the status of a module, configure the following parameters:<br><br>- G2.2.3 P min delay = 20 s.<br>- G5.5.3 Delay unbalance I = 30 s. |
| **DRVSLT** | **F14: Drive select** | Wrong *drive select* configuration.<br><br>Possible causes:<br>- Defect in the central control board P3G - E977XX.<br>- Wrong configuration of the drive select. | Contact Power Electronics.<br>- Ensure the software package installed is complete and compatible with the inverter.<br>- Check SG10-Drive select configuration.<br>- Initialize SG10.1.2-Initialise SG10.1.2-Forcing by default.<br>- Initialize SG1.1.4-Initialise G.Serv.<br>- Initialise G1.1-Initialise parameters.<br>- Load microprocessor and DSP software.<br><br>**Extra information**:<br><br>- Check the drive select and force default configuration.<br>- Initialization of all parameters is recommended. |
| **SYNC** | **F15: Synchronization** | Wrong inverter configuration. This fault indicates that the inverter is measuring AC voltage, but it is configured to work without grid (parameter SG4.13-Grid = 'NO').<br>This fault is due to an incorrect configuration in SG4-Test mode.<br><br>Possible causes:<br>- Incorrect reading of the network frequency.<br>- Measuring AC voltage board has its phases incorrectly connected. | Contact Power Electronics.<br>If the grid is available, SG4.13-Grid should be set to 'YES'.<br><br>**Extra information:**<br><br>- Initialize service group SG1.1.4-Init service group.<br>- In addition, an initialization of all parameters is recommended. |
| **DCMSW** | **F16: DC miswired** | There is a problem with the DC measurement from central control.<br>The DU measurement is more than 10% of voltage. The DU is closed.<br>It occurs when the bus voltage is less than 100 V during 30 s.<br><br>Possible causes:<br>- Defect in the DC voltage converter board of the bus.<br>- Defect in the DC voltage converter board of the DC cabinet busbar.<br>- Defect in the central board of the DU.<br>- Defect in the wiring.<br>- Fuse blown.<br>- Defect in the central control board P3G - E977XX.<br>- Malfunction of the DC contactors in NEC2014 inverters.<br>- Malfunction of the DC switch disconnectors in NEC2017 inverters. | Check that the DC cabinet is closed.<br>Check the central DC measurement.<br>The main cause is that ithe poles are inverted (positive with negative).<br>- Check the measurements on SV2.1-DC voltage bus and SV2.2-DC voltage PV and verify that there is no difference greater than 100 V.<br>- Measure the bus and busbar of the DC cabinet with a multimeter to verify the voltage. Do not measure on the measurement board directly.<br>- In NEC2014 check DC contactors.<br>- In NEC2017 check DC switch disconnectors.<br>Check wiring and fuses. |

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| **DUFB** | **F17: DU feedback** | The DC cabinet is not closing when required. It can be triggered by a fault in the order or in the cabinet.<br><br>Possible causes:<br>- RS485 communications failure.<br>- Defect in the digital output of the DU central board.<br>- Defect in the timer relay.<br>- Defect in the contactor coil.<br>- Defect in the power supply of the contactor coil.<br>- Defect in the auxiliary contactor. | Check communications.<br>Check manually if the DU closes and opens correctly (no voltage test).<br>The digital output of the DC cabinet must be 24 V.<br><br>**Extra information:**<br><br>The central control sends the closing command and DC cabinet does not close.<br><br>- Check the status of the RS485 communications (jumpers, erroneous frames in G6.1-Modbus, continuity of the grid, connection to the DC cabinet) and fiber (erroneous frames in SV12.3-DU).<br>- Check the ground connection of the grid.<br><br>NEC2014:<br>- Check the auxiliary contact of contactor K202 and DC cabinet central board J8A3 (close circuit breaker by referring to all contactors).<br>- Check the timer relay if its LEDs change (red, green).<br>- Check if the equipment has done the reform of the time relay delay for closing the DC cabinet.<br>- Set SG8.13-Status of DU inputs for Bit 7, circuit breaker 1 closed.<br><br>NEC2017:<br>- Check the backfeed on the DC cabinet main board with circuit breakers 1 and 2 (J8A3 and J8A7) closed.<br>- Check time relay KT1.<br>- Set SG8.13-DU input status for Bit 4 and Bit 7, circuit breakers 1 and 2 closed respectively. |
| **ACSCFB** | **F18: ACSC feedback** | Feedback from the soft charge contactor does not match the order during 5 s.<br><br>Possible causes:<br>- High temperature.<br>- Fault in the thermistor installed in series in the resistor.<br>- Fault in the 24 V power supply of the auxiliary contact.<br>- Fault in the relay of the central board.<br>- Faulty connection on the central board.<br>- Fault in the contactor connection. | Check contactor wiring and feedback energizing it from the main board contact, check that 230 Vdc is supplied to the coil and that the auxiliary contact has continuity.<br><br>**Extra information:**<br><br>Fault caused by the feedback of the contactor of the central AC soft load.<br>- Check wiring<br>- Check if it is due to a thermal reason that makes the thermal of the soft load jump |
| **ACSCNV** | **F19: ACSC no voltage** | The central soft charge contactor is closed, but buses are not charging at the expected rate (50% of the target value in 4s). The bus voltage must reach (√2*Vac) / 2.<br><br>Possible causes:<br>- Fault in the fuses of the soft load.<br>- Shunt in module.<br>- DC module fuses blown.<br>- Defect in the bridge rectifier.<br>- Shunt in the resistance of the soft load.<br>- Defect in the K4 contactor of the soft load.<br>- Defect in the measurement wiring of the soft load.<br>- Defect in the wiring of the DC measurement boards. Do not measure on the boards directly.<br>- Defect in the fuses of the soft load. | Measure voltage 4 s after closing the contactor. In equipment with DC/DC equipped with drive select this time is higher than 4 s.<br>Measure soft charge auxiliary voltage. It must be 690 V.<br>Check soft charge fuses, diodes and wiring.<br><br>**Extra information:**<br><br>- Check soft load maneuver<br>- Check soft charge fuses.<br>- Force the Soft Charge contactor and check that bus voltage is reached in 3 minutes. √2*Vac must be reached. Otherwise, there is a bypass or some kind of HW problem. It is recommended to force the contactor closure from the board's relay, wear the appropriate PPE and do not use a screwdriver directly.<br>- Check if the DC fuses of the modules are blown.<br>- Check if the voltage coming from the secondary of the SSAA transformer is cancelled at any point.<br>- Check if there is any shunt in the DC busbar or in the module. |
| **MODFLT** | **F20: Module fault** | This fault should not appear because it has been disabled. | |
| **DUSTCK** | **F21: DU stucked** | This fault should not appear because it has been disabled. | |

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| **MVF** | **F22: Medium V fuse** | A medium voltage fuse has blown.<br><br>Possible causes:<br>- Defect in the MV switchgear fuses.<br>- Defect in the AC voltage measuring board or on its wiring. Do not measure on the board directly. | See group G5.9-HV Phase lost for information about HVPL algorithm.<br>Check medium voltage inputs.<br>Contact your electrical company.<br>Contact Power Electronics if the problem persists.<br>Measure on the connectors of the voltage presence indicator located in the switchgear (the values of each grounded phase should be around 100 Vac). |
| **UNBVAC** | **F23: Unbalanced Vac** | There is an abnormal AC unbalanced voltage. It is configured in group G5.5.4-Unbalance V. It occurs when there is a 30% imbalance between phases.<br><br>Possible causes:<br>- Incorrect calibration or defect in the AC voltage measurement board or on its wiring. Do not measure on this board directly. | Check the description or probable causes of the F41:GFDI fuse<br>Check inverter's voltage measurement shown in the display.<br>Check input voltage.<br>Check inverter measurement wires and fuses.<br><br>**Extra information:**<br><br>Check if the unbalance is real. If it is real, warn the customer and leave the equipment stopped with the circuit breaker/cell open. If it is not real, check Vac measurement, wiring and calibration through the SG6.2-AC Calibration parameter. |
| **HVDC** | **F24: High Vdc** | There is an overvoltage (>1400 V) in the DC bus. The fault is delayed according to the value of G5.10.2-Special protections for voltages below 1509 V. For higher voltages, fault is instantaneous.<br><br>Possible causes:<br>- Incorrect calibration or defect in the AC voltage measurement board or on its wiring. Do not measure on this board directly.<br>- Defect in the central control board P3G - E977XX. | Check inverter's voltage measurement shown in the display.<br>Check input voltage.<br>Check inverter measurement wires and fuses.<br><br>**Extra information:**<br><br>Check if the bus measurement is real. If the real voltage is greater than 1500 Vdc, disconnect the equipment from the PV and communicate to the customer (equipment out of specification). If not real, check Vdc measurement, wiring and calibration through the SG6.3-DC Calibration parameter.<br>Compare the AC voltage of the equipment with the voltage of the modules.<br>Check the wiring and the fuses of the voltage measuring equipment.<br>Verify the voltage measures shown in SV2.1-DC voltage bus using a multimeter. |
| **LVDC** | **F25: Low Vdc** | Voltage is too low ($< V_{ac}\sqrt{2} - G5.10.16(Low\ DSP\ Vdc)$). Fault is delayed 1 ms (not configurable).<br><br>Default fails when Vdc drops 300 V (G5.10.16-Low DSP Vdc) from Vac peak measured for 3 ms (G5.10.17-Low DSP Vdc delay). This fault will not be triggered if no modules are found during the night mode.<br><br>Possible causes:<br>- DC short circuit.<br>- Incorrectly adjusted protections.<br>- Shunt to earth.<br>- Defect in the AC voltage measurement board or on its wiring. Do not measure on this board directly.<br>- Defect in the DC voltage measurement board or on its wiring. Do not measure on this board directly. | Never rearm the equipment in case of this failure. This fault could be triggered by a short-circuit in the DC bus.<br>Contact Power Electronics.<br><br>- Check G5.10.16-Low DSP Vdc settings.<br>- Check the general DC measurement board, connected to the main board, and its wiring. Do not measure on the measurement board directly.<br>- Check the general AC measurement board, connected to the central board, and its wiring. Do not measure on the measurement board directly.<br>- Download the blackbox, the SD card and the parameterization from PowerCommsV.<br>- Check if there is a short-circuit in the DC part of the power stages.<br>- If the equipment have an earthed pole, check for possible earth shunts in the stage, insulators, bushings or in the photovoltaic field.<br>- Extra information:<br><br>**Extra information:**<br>If no faults are found in the installation, this may be due to a transient. After consultation with back office, adjust the following parameters.<br>- G5.10.16-Baja Vdc DSP = 400 V.<br>- Increased up to 10 ms (G.5.10.17-Low DSP Vdc delay). |
| **MEF** | **F26: Modules emergency fault** | **Note: This fault has been disabled in HEM_1.2.1.** Instead, the inverter will trigger the corresponding fault to each situation and stop. | This fault is disabled, it will never be triggered. |

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| **MSF** | **F27: Modules start fault** | The **first N** modules (N is set in SG1.2.8-Mod. Start fault) have failed consecutively during start. If **any of the first N** modules starts correctly, this fault will not be triggered.<br><br>Possible causes:<br>- Incorrect calibration.<br>- Bad network conditions.<br>- Faulty switching to perform soft charging. | Check network conditions.<br>Try restarting the modules one by one.<br>Check the setting of SG1.2.8-Mod. Start fault.<br><br>**Extra information:**<br>- If this happens after 10 s running, load a new DSP.<br>- Check that the Vac and Vdc measures are correct.<br>- Check that SG4.1-Force start conditions = No.<br>- Check 23 Vdc sources.<br>- Review the module logger and tackle the problem based on the bugs that are occurring. |
| **PSANTISL** | **F28: Passive anti island** | The anti-islanding algorithm (in mode "Passive") has detected an island condition.<br>Fault produced when the frequency changes abruptly in a certain time.<br><br>Possible causes:<br>- Loss of energy supply.<br>- Defect in the voltage measurement board. Do not measure in this board directly.<br>- Fault in the wiring that connects the voltage measurement board to the central board.<br>- AC grid event in the customer side. | Verify with the company if power was actually cut.<br>If this fault is triggered continuously, it could be a false fault. Check the plant configuration and, if necessary, consult with Power Electronics.<br><br>**Extra information**:<br>- Check the Anti-island settings (G5.6.1-Mode, G5.6.5-ROCOF (Hz/s) and G5.6.6-Fault delay). It must be configured according to the electrical requirements.<br>- Check if the fault has occurred in more equipment in the plant simultaneously. In a positive case, an event has occurred in the network. If not, check the Vac and frequency measurements through the Vac measurement board. Do not measure in this board directly. |
| **HWIDCDIF** | **F29:HW Idc Dif** | A current derivation has been detected on the panels side.<br><br>Possible causes:<br>- DC fuses blown.<br>- Defect in the contactors (NEC2014 equipment). | Contact Power Electronics.<br>- Check the equipment and verify if the derivation occurred inside.<br>- Parameter initialization is recommended. |
| **MVCMS** | **F30: MV comms** | Failure in communication with the medium voltage measurement board.<br>Fault produced when communications with the medium voltage board fail. | Check communication fibers and voltage supply.<br>If it happens in a HEMK:<br>- SG1.2.22-MV board enable Must be at NO. If the parameter does not appear, configure the equipment as HEM, disable it and reconfigure it as HEMK.<br><br>If it happens in a HEM:<br>- Checking the medium-voltage board. The supply voltage can't be lower than 24 V.<br>- Check the voltage supplied to the main control board, it can be lower than 24 V.<br>- Check the SW version of the MT board.<br>- Defect in the MV board.<br>- Defect in the expansor board.<br>- Check the crimping and the wiring of the fiber optic between central control and MT board. Check the status of the fiber.<br>- Use a fiber tester to measure the power reaching from one end to the other. |
| **DIAGTIM** | **F31: Selfdiagnosis Fault** | A fault has been triggered while performing the auto diagnosis process, e.g.: There is no DU or the bus loading time is excessive.<br>The self-test has not been completed within the defined timeout (300 s).<br><br>Possible causes:<br>- Blown fuses.<br>- Defect in the contactors.<br>- Defect in the soft charge circuit.<br>- Wrong bus voltage measurement. Do not measure in the measurement board directly.<br>- W5:No start conditions appears during the self-diagnosis. | - Check the fuses of the soft charge.<br>- Check contactors of the soft charge and the feedback.<br>- Check the soft load circuit (rectifier, resistors, wiring).<br>- Check that the voltage measurements of each bus are correct. Do not measure in the measurement board directly.<br>- Check that the equipment does not display the warning W5:No start conditions during the diagnostic process.<br>If the fault persists, request Power Electronics' technical assistance.<br><br>**Extra information:**<br>- Check the soft charge operation (contactor, fuses, rectifier...).<br>- Update Software (version higher than 1.4.0). |

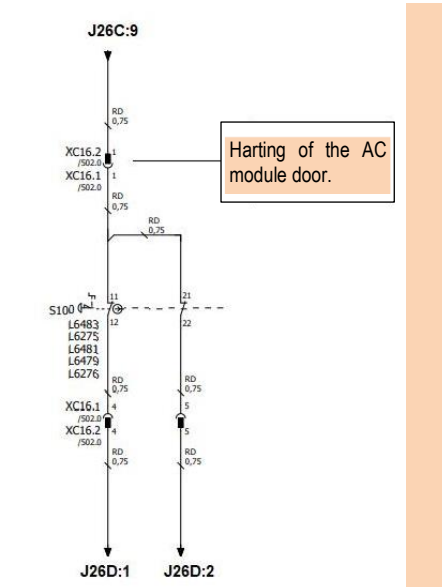
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| Acronym | Fault | Description or probable cause | Actions |
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| **ERMDDIAG** | **F32: Error Mod Selfdiagnosis** | An error occurred while the self-test was running, and the self-test is being performed after a special failure.<br><br>Possible causes:<br>- Fault in the module.<br>- Fault in the soft charge contactor.<br>- Defect in soft charge power supply.<br>- Defect in the soft charge auxiliary contactor.<br>- Defect in the thermal sensor of the soft charge. | - Check Fault History and ensure there is not a short circuit present, nor any hazards that could develop in major events, such as a fire.<br>- Download Blackbox, SD card and parameterization from PowerCommsV.<br>- Check the fuses of the soft load.<br>- Check the contactor and feedback of the soft charge.<br>- Check the soft load circuit (rectifier, resistors and wiring).<br>- Check that the measurements of the bus voltage are correct. Do not measure in the measurement boards directly.<br>- Check that the equipment does not display the warning W5:No start conditions during the diagnostic process.<br>- Review the module logger, must be "Stop – Download - Stop".<br>- In case of failure, check for possible problem of the module (IGBT, snubber board, ribbon cables, FPGA...).<br>If the fault persists, request Power Electronics' technical assistance. |
| **UNREC** | **F33: Unable Reconnect** | The inverter could not restart after the last grid event.<br><br>After a grid event, the inverter will attempt to restart. If the inverter fails to restart repeatedly after a certain a time (defined internally), F33:Unable Reconnect will be triggered.<br><br>Possible causes:<br>Centralized in one equipment:<br>- Wrong AC voltage readings. Do not measure in the measurement board directly.<br>- Wrong AC current readings. Do not measure in the measurement board directly.<br>- Defect in the main control board.<br>Generalized in the plant:<br>- Wrong configuration of the LVRT algorithm (SG1.2.a-VRT Fault Avoid, G4.2-LVRT, G4.3-OVRT). | - Contact Power Electronics.<br>- Check whether the fault is plant-wide or only centralized in one equipment.<br>- Check the configurations of the dip algorithms.<br>- Check the number of modules configured in SG1.2a4-Var Quick n, default n=3.<br>- Download the blackbox, SD card and parametrization from PowerCommsV to report the problem to the R&D department and ask for instructions to extend the range of service group parameters SG1.2.24-En. Undue LVRT.<br>- Check the configurations of the dip algorithms. The reconnection time should be less than 60 – 80 ms.<br>- Request and check the oscillograms of the event.<br>- The number of dips detected by the equipment is shown in SV11.7.2-LVRT number, SV11.7.3-Last LVRT Depth and SV11.7.4-Last LVRT duration. |
| **HOMDIFLV** | **F34:Hom Diff LV** | A zero-sequence current (earth leakage) has been detected on the low voltage side of the transformer.<br><br>If the current exceeds the threshold defined in parameter **G5.9b.1.5-Differential protec** during the time set in **G5.9b.1.5b-Differential time**, this fault will be triggered.<br><br>Possible causes:<br>- Incorrect parameterization.<br>- Faulty SW load.<br>- Faulty auxiliary transformer wiring.<br><br>**Note**:<br>Protection disabled by default. Must not be enabled in field. Associated with SW load. | - Verify if there really is a leak (check cable insulation in the low and high sides of the transformer).<br>- Initialize parameters, check all parameters to verify that are configured correctly.<br>- Contact Power Electronics. |
| **MVPRE** | **F35: MV Premag** | There was a problem during MV transformer premagnetization.<br><br>If the fault is quick (less than a second after attempting to start premagnetization), it indicates that initial conditions to start premagnetization are not met.<br><br>If the fault is slow (up to 100 s after attempting to start premagnetization), it means the process started but could not be completed within the set timeout due to a problem.<br><br>Possible causes: | Contact Power Electronics.<br><br>- Initialize parameters.<br>- Check initial conditions for premagnetization are met (position of the MV switchgear, MV measure board configuration, and bus voltage must be over $\sqrt{2}\,Vin$).<br><br>**Extra information:**<br><br>- Check group SV4.5.9-Premagn. Status and report its status to Power Electronics.<br>- Add the modbus 41199 status register in the SD card. |

| Acronym | Fault | Description or probable cause | Actions |
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| | | - Low power in the field, 30 kW must be guaranteed.<br>- Fault in the wiring from the cubicle to the HV-E980 board.<br>- Defect in the wiring linking the E983 converter board to the E991 voltage reading board. Do not measure on the measurement board directly.<br>- Defect in the fiber linking the E980 board to the P3G board.<br>- Defect in the high voltage measurement board E991. Do not measure on this board directly<br>- Defect in the HV-E980 cell signal board.<br>- Defect in the converter board E983, placed on P3G.<br>- Defect in the main control board P3G - E977XX.<br>- Switchgear mechanism damaged. | - Check the previous conditions of the protection switchgears, earthing and communications with the MV measuring board. |
| **OVOLIRLV** | **F36:Ovc Inrush LV** | One of the current thresholds (overload, overcurrent or input inrush current) has been overcome in the low voltage side of the transfomer.<br>Possible causes:<br>- Incorrectly adjusted protections.<br>- Defect in the wiring that connects the toroid to the P3G central board.<br>- Defect in the 3500/5 current toroid.<br>- Defect in the central control board P3G - E977XX. | - Check settings of G5.9b.1.1-Overcurr threshold to G5.9b.1.3b-Inrush timeout and verify the configuration is correct.<br>- If the problem persists, there might be a hardware failure. Check the output on the low voltage side of the equipment.<br>- Check the current measurement equipment in low voltage.<br>Contact Power Electronics. |
| **OVOLIRMV** | **F37:Ovc Inrush MV** | One of the current thresholds (overload, overcurrent or input inrush current) has been overcome in the medium voltage side of the transfomer.<br>Possible causes:<br>- Incorrectly adjusted protections.<br>- Defect in the low voltage measurement wiring.<br>- Defect in the 100/1 current toroid.<br>- Defect in the converter board E983.<br>- Defect in the central control board P3G - E977XX. | - Check settings of G5.9b.2.1-Overcurr threshold to G5.9b.2.3b-Inrush timeout and verify the configuration is correct. These parameters are configured by default.<br>- Should the problem persist, there might be a hardware failure. Check the output on the medium voltage side of the equipment.<br>- Check the current measurement equipment in low voltage.<br>Contact Power Electronics. |
| **CUROFFLV** | **F38:CurrentOff LV** | A current measure has been detected in the low voltage side of the transformer while the equipment is not running.<br>Possible causes:<br>- Incorrectly adjusted protections.<br>- Wrong calibration of AC currents at low voltage.<br>- Defect in the wiring that connects the toroid to the P3G central board.<br>- Defect in the 3500/5 current toroids.<br>- Defect in the central control board P3G - E977XX. | Check measurement circuit. Do not measure in the measurement board directly.<br>Check the configuration of the G5.9b.1.4b-I off timeout and G5.9b.1.4-I off thresh parameters; these parameters are disabled by default.<br>Contact Power Electronics. |
| **CUROFFMV** | **F39:CurrentOff MV** | A current measure has been detected in the medium voltage side of the transformer while the equipment is not running.<br>Possible causes:<br>- Incorrectly adjusted protections.<br>- Wrong calibration of AC current measurement board. Do not measure on this board directly.<br>- Defect in the wiring that connects the toroid to the P3G central board.<br>- Defect in the 100/1 current toroid.<br>- Defect in the converter board E983.<br>- Defect in the central control board P3G - E977XX. | Check measurement circuit. Do not measure in the measurement board directly.<br>Check the configuration of the G5.9b.1.4b-I off timeout and G5.9b.1.4-I off thresh parameters; these parameters are disabled by default.<br>Contact Power Electronics. |

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<tr><th>Acronym</th><th>Fault</th><th>Description or probable cause</th><th>Actions</th></tr>
<tr><td>OVTMPACM</td><td>F40: AC mod overtemperature</td><td>Internal NTC temperature over the value defined in G5.7.4.1-Main control board.<br>Fault occurring when the temperature in the AC cabinet exceeds the threshold set in G5.7.4.1-Main control board.<br><br>Possible causes:<br>- Fault in ventilation relays KA3, KA4 and KA5 for AC cabinet ventilation.<br>- Fan failure.</td><td>Check temperature. If it is out of specification, request technical assistance.<br>Make sure there is nothing obstructing the cooling fans. Verify they rotate correctly.<br>Verify ambient conditions are proper for the equipment<br><br>Extra information:<br><br>- Check temperature reading in internal display group (SV4.1.4-Internal tº), if incorrect, change control board.<br>- Check fault configuration (G5.7.4.1-Main control board).<br>- Check if the air inlets are obstructed.<br>- Use a thermographic camera to check the condition of the drawer components and replace if any are excessively hot.<br>- On the AC box, check the distance between the plate and the central control board, if the turrets are short add one more.</td></tr>
<tr><td rowspan="2">GFDIFS</td><td rowspan="2">F41: GFDI fuse / DU power lost</td><td>Blown GFDI fuse or fuse holder open.<br>Occurs when one of the circuit breakers does not open by removing the selector or leaving the equipment isolated from the field.</td><td>Check J10 pin in the Disconnecting Unit board or digital input 5 of the uP in case SG1.1.9-GFDI fuse input = ‘YES’<br>Check circuit breaker backwatches.</td></tr>
<tr><td colspan="2">Extra information:<br>Fault occurring when the GFDI fuse blows. Check fuses. This fault activates the module self-diagnosis procedure.<br><br>The cause could lie on the photovoltaic field if the fault appears only when the DC switch disconnectors have been closed and the equipment starts supplying current. If the fault appears always at the same hour, there is trackers mode in the field and the value of the RISO is correct, it can be due to a damage on the cables of the client.<br><br>Check if fuse is blown:<br>- Check if the field is derived. For example, damaged wiring.<br>• If the equipment has isolation monitor, check the isolation measurement in SV3.1.2-Resistor.<br>- Check if equipment is derived. For example, deterioration in busbars insulators, ground wires shunted to any busbars, DC input wires shunted to the chassis or to the wire of the edge covers, rectifier diode of the DC/DC board shunted to chassis, shunt in the AASS transformer, wet or dusty busbars, connectors in bad condition…<br>• Perform continuity measurement between positive-negative with the multimeter.<br>• Perform a self-diagnosis to see if there is an unbalanced stage.<br>• Measure positive-ground and negative ground and check that the values are correct.<br>• Check voltage balancing.<br>• Check AC and DC fuses.<br>• Check that the phase voltages are balanced by measuring the phase-to-phase and phase-to-earth voltages with an AC multimeter.<br>- Check for possible damages, shunt to ground or humidity on the RFI boards.<br>- Check power stages. For example: An IGBT could be shunted to the chassis.<br>- Check for possible shunts in L (module output), C and L filters.<br><br>If fuse is not blown:<br>- Check the wiring of the fuse backwatch.<br>- Check SV4.1.7 bit 8 (from left to right). Open the fuse and check whether bit 8 is set to 0. Close the fuse and check that bit 8 is set to X. If this does not change correctly, check feedback wiring.<br>- Check J10 in the DC cabinet board. Check the input 5 of the microprocessor.<br>- Contact Power Electronics.<br><br>Note: If the equipment has GFDI and IMI, disable the RFI filter on the grounded pole.</td></tr>
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| DCCNTLCK | F42: DC contactor locked | Locked contactor in a DU opening process.<br><br>Possible causes:<br>- Contactor blocked.<br>- Circuit breaker blocked.<br>- Keys in lock position.<br>- Back feed incorrectly wired.<br>- Bad voltage calibration on bus side when carrying out soft charging. | Change the channel in SG8.8-Acq selected and check in SG8.11-Acq Status Check which channel, or channels, is reporting that the contactor is open but according to feedback it is closed. |
| | | **Extra information:**<br><br>Occurs when one of the DC switch disconnectors does not open by removing the selector or leaving the equipment isolated from the field.<br>This fault can occur if the DC switch disconnectors are set to "Lock" instead of "Release".<br>- Disable and enable DC cabinet.<br>- Check the correct configuration of the DC cabinet.<br>- Check circuit breaker feedback.<br>- Update to the latest software version. | |
| EMRGSTP | F43: Emergency stop | It is produced when the stop pushbutton has been pushed. Check the status of the stop pushbutton in the door of the main control cabinet.<br><br>Possible causes:<br>- Emergency stop has been activated<br>- Wrong connection of the wiring of the emergency stop circuit.<br>- Fault in camera-push buttons.<br>- Fault in the digital input of the P3G central board. | - Check the pushbutton status at the main control cabinet door.<br>- Check if the client has sent the order from the SCADA.<br>- Check on the central P3G board the J26 RETURN – J26D1 – J26D2 connectors.<br>- Check on the central P3G board the 24Vdc – COMMON J26C9, XC16.1 and XC16.2 connectors.<br>- Check the harting of the XC16.1 and XC16.2 connectors.<br><br>  |
| DRVSLT | F44: Drive select | Drive select parameters out of range.<br>The main board does not have the Drive Select properly programmed.<br><br>Possible causes:<br>- Incorrect software version.<br>- Defect in the parameterization.<br>- Defect in the initialization of the software. | Check values and adjust them correctly in SG10-Drive select.<br>In addition, an initialization of all parameters is recommended.<br>Ensure the software package installed is complete and compatible with the inverter.<br>Check software versions in PowerCommsV. |

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<tr><th>Acronym</th><th>Fault</th><th>Description or probable cause</th><th>Actions</th></tr>
<tr><td rowspan="2">INSL</td><td rowspan="2">F45: Insulation</td><td>IMI has detected an insulation problem.<br>Possible causes:<br>- Bad insulation of the PV field.<br>- Defect in the Ethernet router/switch of the AC cabinet.<br>- Wrong configuration of the IMI.<br>- Failure in the RS485 connections.<br>- Bad insulation of the power stage.<br>- Failure in the RFI filters (AC and DC).<br>- Failure in the 690 V wiring or in the measuring wires (AC and DC).<br>- Failure in the auxiliary transformer.<br>- Failure in L or C filters.<br>- Isolator in bad conditions or contaminated.<br>- Rubber bushings with humidity or dust.<br>- Connectors in bad conditions.<br>- Failure in the power supply units.<br>- Failure in the ground of the measuring board (do not measure in this board directly).<br>- Failure in the high voltage transformer.<br>- Wrong wiring in the feedback signal.<br>Failure can be reset by qualified customer.</td><td>Check if the DU has any isolated string. If so, check the insulation of those strings.<br><br>Enables self-diagnosis of modules.<br>Fault produced when the isolation monitor measures an impedance value lower than the one configured in G9.1.2-Fault threshold (isolation monitor DC) or G9.2.2-Fault threshold (isolation monitor AC).<br>If the disconnectors or contactors in the DC cabinet are disconnect positive and negative, they are opened and the internal insulation measurement is carried out. Then, they are closed and the insulation of the complete busbar is measured.<br>Check the following page for extra information.</td></tr>
<tr><td colspan="2">Extra information:<br><br>To obtain stable conditions, configure G9.3.3- Delta V max as 30 V. Check that the isolation monitor is enabled through G9.1.1a-Activation = Yes.<br><br>Verify if the insulation measurement is real by checking with a multimeter the impedance between positive/negative and ground. If the insulation fault is real, the cause of the derivation must be sought.<br><br>If the shunt is real:<br>- If the fuse of the GFDI blows frequently:<br>• Check for ground wires shunted to busbars or phases. This can occur when the voltage of the busbars is unbalanced during the soft charge.<br>• Check for possible deterioration in busbars insulators.<br>• Check that the DC cabinet input wires are not in contact with the chassis or with the wire of the edge covers.<br>• Check that the rectifier diode of the DC/DC board is not shunted to the chassis.<br>• Check for possible damages, shunts to ground or humidity on RFI boards (AC and DC).<br>• The cause could lie on the photovoltaic field if the fault appears only when the DC switch disconnectors have been closed and the equipment starts supplying current. If the fault appears always at the same hour, there is trackers mode in the field and the value of the RISO is correct, it can be due to a damage on the cables of the client.<br>- Isolate the field device (open strings) and repeat the measurement with the insulation monitor (open GFDI in case of grounded devices).<br>• If there is an insulation fault (impedance value lower than 50 kΩ), the shunt is in our equipment (IGBTs shunted to ground, cables touching chassis...).<br>• Check that the wiring from the insulation monitor board to the E977 control board is correct.<br>• If there is no longer an insulation fault, the tap is in the field. Close string to string and perform insulation measurement with each string.<br>• Check that there is no rust on the NTC probe.<br>• Check that the measuring boards are earthed (do not measure in these boards directly) and that the measuring cables are in good condition.<br>- If the equipment has isolated poles:<br>• The fault could appear if there is an additional consumption due to a component installed by the customer. For example, the installation of a three-phase transformer to supply any other component in the AC cabinet. The isolation monitor could work properly before starting the equipment but one minute after commutation the fault could apperar.<br><br>In case there is no real derivation and the F17:DU feedback is triggered:<br>- In facilities with high capacitance or where the RISO is high, increase G9.3.2-Check period to 900 s (600 s for versions prior to 5.0.0) and G10.3-Isolation meas tout to 600 s.<br>- In equipment with GFDI, configure G9.3.1-Algorithm execution = Before run and G10.2-Working mode = GFDI + IMI.<br>- If this fault appears and disappears after the VDC is stabilized in a high value or it only appears in the morning, the RISO could have a problem, increase the starting voltage of the inverter.<br>- Program the autotest to be carried on every 24 h through G9.1.7- Self test exec period = 1440 min.<br>- If there are communication losses frequently:<br>• Check the lost frames in G6.1.8- Modbus error CRC, the result must be between 1-2% every 100.<br>• Update the version of the insulation monitor if there are sudden losses of the RISO.<br>- If the equipment is grounded:<br>• Check the maneuver of contactors that isolate the negative from ground.</td></tr>
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| Acronym | Fault | Description or probable cause | Actions |
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| | | • Check the connection of the insulation monitor's boards.<br>• If the measurement performed by the equipment gives a high value (>50 kΩ) when the negative is earthed and a low value (<5 kΩ) when isolated, check the resistance board of the insulation monitor (change if necessary).<br>- If the equipment has isolated poles:<br>• Check the connection of the board and configure G9.3.1-Algorithm execution=Stable conditions.<br>**Note:** From software version 5.0.0, the accuracy and stability of the measurement will be improved. | |
| **DTFLT** | **F46: Data fault** | Out of range parameters have been detected.<br><br>Possible causes:<br>- The system has been started incorrectly.<br>- Microcontroller overload due to excessive communication information.<br>- Communication fault between inverter and SCADA.<br>- Hardware problem in the Ethernet network.<br>- Duplicate MAC addresses.<br>- Problem localized in a specific IP.<br>- Hardware problem in the central board. | Contact Power Electronics.<br><br>**Extra information:**<br><br>This is due to a communication problem with the data memory which may be due to the microcontroller being saturated.<br>Disconnect the Ethernet cable and check if it jumps.<br>If it happens sporadically and blocks the equipment, it is necessary to remove the control voltage and contact Power Electronics.<br>Swap the central board with the board of another inverter to rule out HW problems.<br><br>If the fault is going to be reset:<br><br>- There is nothing to be done. There was a parameter out of range and the microcontroller has put it in range.<br><br>If the fault occurs with the equipment running and sporadically, the load on the microprocessor must be reduced:<br>- Update the software version.<br>- Disable temperature and humidity sensor (SG1.1.17-Humidity/Temp sensor).<br>- Disable the writing in the SD board.<br><br>If the F46:Data fault stops jumping, check the sensor and reset it correctly.<br>- Capture (with wireshark) communications frames to see the communications between inverter and SCADA.<br>- Check that switch-type peripherals do not have shunts.<br>- Check the MAC address of all network devices to rule out duplicates.<br>- Exchange IPs with other equipment to rule out a failure linked to a specific IP. |
| **WTCHDG** | **F47: Watchdog uP** | A fault in the microcontroller has occurred.<br><br>Possible causes:<br>- The system has been started incorrectly.<br>- Microcontroller overload due to excessive communication information.<br>- Communication fault between inverter and SCADA.<br>- Hardware problem in the Ethernet network.<br>- Duplicate MAC addresses.<br>- Problem localized in a specific IP.<br>- Hardware problem in the central board. | Disconnect and reconnect the auxiliary services supply. If the fault persists, request Power Electronics' technical assistance.<br><br>**Extra information:**<br><br>It is associated to SW update processes.<br>- Reset the fault. If it persists, remove auxiliary power.<br>- Remove power supply, disable Weidmuller Switch and connect LAN cables to the router.<br>- Avoid going through the router. Direct cable to P3G, **removing mesh at one point.**<br>- The Ethernet cable (with metal terminals) that comes from the Skid, if it is wired directly to the P3G, it is necessary to remove the shield from one of the connectors at a single point.<br>- Configure G6.2.7-Sockets per IP = 3 or 4 sockets.<br>- **The scada must make requests for packages of no more than 50 records and polling times of 100ms.**<br><br>If it occurs while the equipment is running, change central control board.<br><br>If the fault persists, contact Power Electronics. |

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| **INTCMS** | **F48: Internal comms** | Internal communication problem.<br>Possible causes:<br>- Problem in loading the software.<br>- Defect in the central board. | Disconnect and reconnect the auxiliary services supply.<br>**Extra information:**<br>Micro and DSP versions are not compatible. It occurs in the SW update process. Load software and initialize parameters. If the fault persists, replace the central board.<br>In case it does not occur during a SW upgrade, notify Power Electronics. |
| **IMISLFT** | **F49: IMI autotest** | Periodic insulation monitoring devices auto test has finished with errors.<br>The IMI tests itself to check that its hardware is operational. There are internal relays that allow it to have measurements in a known range and if these are outside this range, the IMI sends the fault to the main micro.<br>Possible causes:<br>- Problem loading software.<br>- Board blocked.<br>- IMI damaged.<br>- Defective RS485 bus communications. | Check DU connections.<br>If the fault persists, replace the IMI.<br>- Load software.<br>- Check voltages 1100-1107.<br>- Make a restart by removing the 24 V power supply.<br>Rule out communications problems on the R485 bus:<br>- Verify that the wire mesh is grounded in the P2G cabinet, and that there are no shunts.<br>- Check continuity of the cable mesh, and that each connection point (central control, IMI, etc.) has only one input and one output. Avoid Y-connections.<br>- Verify that the end-of-line resistance jumper are placed at the start (P2G board) and end (DC cabinet board) of the RS485 bus.<br>If communication problems in the RS485 bus have been ruled out, change insulation monitor boards. |
| **NODUCNL** | **F50:No DU channels** | All DU channels have been isolated, and it is not possible to establish a connection with the plant. This fault is disabled.<br>Possible causes:<br>- Wrong connection or contact.<br>- Defect in the DC cabinet measuring boards (voltage and current). Do not make any measurement directly in these boards. | Check all connections.<br>Reset all modules.<br>- The red LED of the central board of the DC cabinet blinks twice, the green LED once. Reset DU and channel status. Resetting channel status may be enough.<br>- Reset P3G control board.<br>- Check the RS485 connections from the P3G board to the boards of the DC cabinet, to the cyclonic fan and to the isolation monitor.<br>- Check the measuring boards (voltage and current). Do not make any measurement directly in these boards.<br>- Check the configuration of the master and slave boards and restart them.<br>- Check the configuration of the channels.<br>If the problem persists, contact Power Electronics. |
| **PPCC** | **F51: PPC comms** | PPC and inverter communication problem during the time configured in G6.3.2-Comms fault time.<br>Possible causes:<br>- Errors in parameterization.<br>- Wrong wiring.<br>- Defective connectors.<br>- Defective switches, routers, repeaters and network power supplies.<br>- Ethernet port of the central control board (P3G) damaged.<br>- Ethernet port of the PPC (E769XX) damaged.<br>- Failure in the customer communications. | Disconnect and reconnect the PPC. Check PPC connections.<br>**Extra information:**<br>- Review Ethernet communications wiring.<br>- Check defective switches, routers, repeaters and network power supplies.<br>Check the communications configuration (both PPC and inverter).<br>- In case of installing one PPC, it is recommended to set the IP address in G6.3.6-PPC IP Filter.<br>- In case of installing two or more PPCs, it is necessary to set the IP address in G6.3.6-PPC IP Filter.<br>- Check MAC and IP addresses, ports and other communication between equipment and PPC.<br>- Ping the PPC and inverters to ensure good communication.<br>- Check for loops in switches or routers. |

| Acronym | Fault | Description or probable cause | Actions |
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| SCDUFB | F52: Softcharge DU feedback | Softcharge timeout has expired and no feedback was received.<br>This fault is disabled.<br>This failure occurs when the DU soft charge contactors (PCS equipment) are ordered to close and the contactors do not close. | Check wiring (relay and feedback).<br>**Extra information:**<br>- Check DU soft charge on/off and backwatch wiring.<br>- Force without DC voltage the contactors to close on the corresponding digital output of the DU board. If they close, start the soft charge manoeuvre and check whether the activation relay closes. If by forcing the contactors, they do not close and the wiring is OK, change contactors.<br>- Verify the contactors of the soft charge (positive and negative). |
| EARTHL | F53:GFDI current leak | GFDI current has exceeded the threshold defined in G5.8.2-Current threshold during the time set in G5.8.3-Trip time.<br>Enables self-diagnosis of modules.<br>Possible causes:<br>- Defect in the insulation of the photovoltaic field.<br>- Defect in the insulation of the equipment.<br>- Defect in the hall reading board. | Check strings insulation.<br>**Extra information:**<br>- If the project does not have a current sensor, disable the protection.<br>- Check threshold settings (G5.8.2-Current threshold and G5.8.3-Trip time).<br>- Check sensor measurement wiring<br>**Note**: It should only be enabled for projects that require it and with versions equal to or greater than 1.5.0. This bug can only be manually reset with advanced client key or higher. Once the fault has been reset, before starting the control performs the self-diagnostic operation of the module. |
| DUCNOV | F54: DU channel overcurrent | Overcurrent in one of the DU channels. The threshold is defined in G5.11.3-DU overcurrent.<br>Possible causes:<br>- Wrong setting of the overcurrent threshold in G5.11.3-DU overcurrent.<br>- Wiring fault.<br>- DU fuse blown.<br>- Defect in DU board connections to the channels.<br>- Defect in the hall sensor.<br>- Defect in the DU board. | Check DU fuses.<br>Check connections between the DU and the strings.<br>**Extra information:**<br>- Check the threshold set.<br>- Check the display group of the DU that, with the equipment stopped, no channel measures current.<br>- If any channel is measuring current while stationary, the wiring of that channel should be checked.<br>- Check for any hot spots on busbars caused by faulty torque.<br>If the current measurement is real:<br>- Check the state of the fuses of the rest of the channels.<br>- Check that the tightening torque of all the joints in that channel is adequate.<br>**Note:** If any channel has a faulty reading, the channels below the faulty channel can be remotely disabled through G10.4-Number of DU channels, G10.4b-Channels DU meas 1 and G10.4c-Channels DU meas 2. |
| EXT/OVGR | F55: External fault/OVGR | It is an external fault (it can be required by some countries like Japan).<br>Digital input 6 of the microcontroller is open.<br>There is a problem with the external protections associated with this DI6, on the board corresponds to J26C5 - J26C6, see diagram.<br>This fault can be configured from the switchgear or from the client. The external OVGR relay has tripped.<br>Possible causes:<br>- Bad configuration.<br>- Faulty wiring.<br>- External fault sent by the user.<br>- Defect in the overvoltage protection. | In case the equipment is not using it, disable it in SG1.1.13-Ext fault/OVGR in.<br>Check the external protections associated with this input.<br>Check the overvoltage protections associated with this input.<br>Check that configuration SG1.1.13-Ext fault/OVGR in. (NO, NC, Flank) is correct.<br>If the project defines this fault:<br>- Check with the client if this digital input has been activated and why.<br>- Check the wiring.<br>Check external OVGR relay.<br>If the inverter fails, the software will activate a J31A output wired to a test terminal block.<br>Contact Power Electronics. |

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| Acronym | Fault | Description or probable cause | Actions |
|---|---|---|---|
| **REMRSTP** | **F56: Remote emergency stop** | A remote emergency stop command has been received through Modbus communications (43200).<br><br>Possible causes:<br>- Command sent via SCADA or PowerCommsV. | Verify with your plant controller to determine if the emergency stop was triggered by a technician or by a plant operator. Verify if the plant can be restarted.<br><br>**Extra information:**<br><br>- Check that the SCADA has given the order.<br>- Check the SD.<br>- Check communication frames with wireshark.<br>- Contact Power Electronics.<br>- See section 'Remote Comunications Failure'. |
| **DUMWR** | **F57: DU channel miswired** | A command to close the DU has been received, but one or more channels is measuring negative voltage.<br>Fails when any configured DU measurement channel measures below -50 V.<br><br>Possible causes:<br>- Defect in the calibration of the voltage measurement.<br>- Defect in the voltage measurement board. Do not measure on this board directly.<br>- Connection of the exchanged GFDI board. | Contact Power Electronics.<br>Check the connector in the DU board. If it is interchanged with the GFDI reading, connect it properly and the fault should disappear.<br><br>**Extra information:**<br><br>- Review DU configuration. That all channels with measurement and none without measurement are enabled.<br>- Display the measurement of each of the configured channels (SV10-DU measures). Detect the channel that is triggering the fault and check the wiring.<br>- Configure the voltages. |
| **SWCMM** | **F58: Control SW mismatch** | The microcontroller and DSP software versions are not compatible.<br><br>Possible causes:<br>- Different microprocessor and DSP software versions.<br>- Problems during software loading.<br>- Corrupted software.<br>- Voltage measurement calibration defect.<br>- Defect in the central control board P3G - E977XX. | Check and update, if necessary, the software version.<br><br>In case of not occurring in a SW update, contact Power Electronics.<br><br>**Extra information:**<br><br>- Review SW versions.<br>- Upgrade, following the correct sequence, to a compatible version. |
| **MODSWI** | **F59: Module SW mismatch** | Not all modules have the same software version.<br><br>Possible causes:<br>- Incorrect software version.<br>- Problems while loading the software.<br>- Defect in the fibers.<br>- Defect in the module's FPGA board. | Check the software version of each module in SV9.17-SW version. Please contact Power Electronics to assist you with your software update.<br><br>**Extra information:**<br><br>- Check that all modules have the same version in SV9.17-Sw version.<br>- If any is different, upgrade all modules to the same version.<br>- If despite upgrading to the same version, you still see that someone has a different version, check fiber optic cabling.<br>- Upgrade to a compatible version. |
| **MODHWI** | **F60: Module HW mismatch** | Not all modules have the same electronic board revision.<br><br>Possible causes:<br>- Module boards with different hardware.<br>- Problems during software loading on the module board.<br>- Defect in the communication fibers.<br>- Defect in the module FPGA board.<br>- Defect in the central control board P3G - E977XX. | Disable the modules with mismatching boards to keep the inverter running and Contact Power Electronics. |

| Acronym | Fault | Description or probable cause | Actions |
|---|---|---|---|
| | | **Extra information:**<br><br>Fault produced when the HW versions of the module do not coincide. It can be checked in SV9.16-Hw version. Must occur during a SW update, otherwise:<br>- Check SV9.16-Hw version for all modules.<br>- Check module board version.<br>- Check that the frame rate sent to the FPGA is greater than 98%, refer to SV12.1-FPGA Modules in software versions 2.0.0 and higher.<br>- Check that the frame rate between the DSP and the microprocessor is greater than 100%, see SV12.2-DSP-uC in software versions 2.0.0 and higher.<br>- If the frames are correct and all modules have the same hardware, check via SG9-Logger modules which module is faulty. Then manually load the FPGA software of this module.<br>- If there are communication problems with the module or with the DC cabinet, check that the fibers are crimped correctly, have no cracks or 105° angles.<br>- If communication between the microprocessor and the DSP fails, check the power supply to the host board.<br>- If everything is correct notify to Power Electronics. | |
| **DUSD** | **F61: Smoke detector** | The DU smoke detector has been activated.<br><br>Activation parameter is G5.10.11-Smoke detector DU.<br><br>**Note**: the smoke detection functionality is disabled. | In the case of false faults, check the status of the digital input at the DU.<br><br>**Extra information:**<br><br>- Check G5.10.11-Smoke detector DU. The Fault must be disabled.<br>- If once disabled, it is still triggered, contact Power Electronics. |
| **DUCOM** | **F62: DU comms** | Communication between DU and central control has been lost.<br><br>Possible causes:<br>- Defect in the connector of the cyclone fan.<br>- Defect in the connector of the isolation monitor.<br>- Defect in the central board connector.<br>- Defect in the connector of the DU board. | Check wiring.<br>In case of false faults, verify that RS485 communications between DU and AC module are noise free.<br><br>**Extra information:**<br><br>- Check frame counter 485 (G6.1.8-Modbus error CRC).<br>- Check the 485 (line terminations, earth meshes).<br>- Check DU supply voltage.<br>- Check the connector of the cyclone fan and the connector of the isolation monitor.<br>For further assistance, contact Power Electronics. |
| **GRNDCN** | **F63: Ground connection** | The ground contactor is closed but the insulator detector detects that the pole is isolated.<br>To connect the negative to earth, the 1500 V contactor is powered, this being NA, it closes and withstands the voltage peak. The 1000 V contactor, being NC, is not powered and remains closed, thus connecting the negative to earth.<br><br>To disconnect the negative to earth, the 1000 V contactor is powered and, as it is NC, it opens, and then the 1500 V contactor is not powered and, as it is NO, it opens, thus disconnecting the negative to earth.<br><br>To avoid false positives, in SW branch 2.4.x, F63 was replaced by W161:Trans Aux Therm during the transition to autonight mode in equipment with double GFDI.<br><br>Possible causes:<br>- Defect in contactors K51 and K52.<br>- Defect in load relays K60, K51A and K52A.<br>- Defect in 1500 V and 1000 V contactors.<br>- GFDI blown.<br>- Insulation monitor fault due to a relay incorrectly wired.<br>- Incorrect IMI measurement process.<br><br>**Note:** this fault, in SW version 4.0.1, was replaced by W165 (Insulation alg NC). | Check ground contactor.<br>Check insulation monitor.<br><br>**Extra information:**<br><br>- Check contactor status K51A and K52A.<br>- Check the relays K60, K51A and K52A.<br>- Check the GFDI fuse (it has been seen that this fault occurred because the GFDI was blown but placed upside down in the fuse holder).<br>- Check that the insulation monitor relay output is connected to the normally closed contact.<br>- Check if the measurement process of the IMI is carried out correctly.<br>- There could be a problem in the communications through the RS485 bus. Check the quantity of lost frames in the G6.1.8- Modbus error CRC parameter. If the number is >3% approximately, it may be caused by a bus and/or main microcontroller saturation. Improve internal communications (microcontroller – DC cabinet – ISOPRO) and check that the microcontroller does not have heavy workloads related to external communications. |

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| Acronym | Fault | Description or probable cause | Actions |
|---|---|---|---|
| **INVMAC** | **F64: Invalid MAC** | Invalid MAC address (00-00-00-00-00-00).<br>Possible causes:<br>- Wrong configuration on the MAC address. | - Check the configuration of G6.2-Ethernet.<br>- If it is necessary, change MAC address in G6.2.5-Dirección MAC. To modify MAC address it is necessary to introduce the password in SG1.1.2-MAC protection key.<br>- Check for duplicate MAC addresses after updating the software.<br>If you require assistance, contact Power Electronics. |
| **DCOMS** | **F100: Display comms** | Communications between the display and the control board are lost.<br>Possible causes:<br>- Power supply contact failure.<br>- Failure during SW update.<br>- Fault in wiring.<br>- Defect in the GDU display software.<br>- Defect in the display. | This fault will automatically disappear once communications are restored. If it is triggered frequently, it is possible that the wiring cable used is not correct and introduces noise into the communications module. In this case, it is recommended to use a category 6 screened cable.<br>**Extra information:**<br>- Check display cable and connectors.<br>- Change display. |
| **INDTHR** | **F140: Inductance thermal** | Feedback has not been received from the LCL filter inductance thermal.<br>This thermal is the one located near the inductance, along with the capacitors.<br>Failure caused by the opening of the common inductance thermal contactor, located next to the filter capacitors.<br>Possible causes:<br>- Inadequate cooling.<br>- Incorrectly crimped connector.<br>- Thermal defect.<br>- High resistance in the connection. | Check position, ventilation and status of the coil.<br>If the fault is frequent, check wiring.<br>**Extra information:**<br>- Check fans.<br>- Check thermal wiring.<br>- Check the continuity in connector J26B4 (J26B2 + 24 Vdc) of the backfeed to the central board. Check if there is a power difference between the pins of the connector, in this case check the connector located on the choke and the thermal connector, making sure that there is no resin or sulphuration on the terminal.<br>- Take pictures with thermographic camera to the whole cabinet to check if the ventilation is correct in the inductor.<br>- If the equipment does not have the inductance mounted, a bridge must be made in the thermal back warning, in the central control board. |
| **MVTT** | **F141:MV transformer Tº** | Temperature in the medium voltage transformer (PT100 readings) is too high.<br>Possible causes:<br>- Inadequate cooling of cyclone or exhaust fan.<br>- Clogged filters.<br>- Incorrectly connected PT100 probe or inappropriate height.<br>- Incorrect parameter setting.<br>- Clogged ducting.<br>- Bad connection between ducts. | Check settings in G5.7.3.1-MVTransformrThreshld and G5.7.3.2-MV transformr delay.<br>Check ventilation airflow. |
| | | **Extra information:**<br>Fault produced when the configurable threshold is exceeded in the PT100 temperature measurements or when one of the measures is -40ºC.<br>- Check PT100 and check that the measurements are correct.<br>- Clean cyclonic filter and its evacuation.<br>- Check the connection of the PT100 and verify the height.<br>- Check the settings of G5.7.6 and G5.7.7.<br>- Check in the SD the measurements of the temperature measurements. Make sure that there are no spurious and check that the temperature reaches the threshold (defect 125ºC). If the threshold is reached, check the ventilation (fans and ducts). If there are spurious contact with Power Electronics. | |

| Acronym | Fault | Description or probable cause | Actions |
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| **AFLV** | **F142: Air flow LV** | Airflow temperature in the low voltage side [2] is too high.<br><br>Failure caused when the air pressure measurement exceeds the threshold set in G5.11.6-LV manometer fault for the time set in G5.11.7-LV manometer delay.<br><br>Possible causes:<br>- Defect in pressure gauge wiring to P3G E977 board.<br>- Defect in pressure gauge tube.<br>- Defect in pressure gauge board E997.<br>- Obstruction in the ventilation ducts.<br>- Defect in the cyclone. | Check settings in G5.11.6-LV manometer fault and G5.11.7-LV manometer delay.<br>Check air inlets and outlets for possible obstructions.<br>Check ventilation airflow in the low voltage side.<br><br>**Extra information:**<br>- Check configuration.<br>- Check SV4.1.12-LV manometer, with the fan running we should obtain a measurement of 3600 and with the fan stopped 2000.<br>- Check the state of the pressure gauge and its wiring to the P3G board.<br>- Check the condition of the manometer.<br>- Check that there are no obstructions in the air inlets and outlets.<br>- Check air flow on LV side.<br>- Check configuration of G5.11.6-LV manometer fault and G5.11.7-LV manometer delay.<br>- Check the operation of the fans.<br>- If the measurement is not consistent (excessively high or negative) check the wiring. |
| **AFMV** | **F143: Air flow MV** | Temperature in the medium voltage side [2] airflow is too high.<br><br>Failure caused when the air pressure measurement exceeds the threshold set in G5.11.4- MV manometer fault for the time set in G5.11.5-MV manometer delay.<br><br>Possible causes:<br>- Defect in the wiring of the pressure gauge to the MV E980 board.<br>- Defect in pressure gauge tube.<br>- Defect in pressure gauge board E997.<br>- Obstruction in the ventilation ducts.<br>- Defect in the cyclone. | Check settings in G5.11.4-MV manometer fault and G5.11.5-MV manometer delay.<br>Check air inlets and outlets for possible obstructions.<br>Check ventilation airflow in the medium voltage side.<br><br>**Extra information**:<br>- Check the configuration of G5.11.4-MV manometer fault and G5.11.5-MV manometer delay, with the fan running we should obtain a measurement of 3600 and with the fan stopped 2000.Check the condition of the manometer.<br>- Check the condition of the pressure gauge and its wiring to the MV board.<br>- Check the configuration of the manometer in SV4.1.13-MV manometer = 0<br>- Check that there are no obstructions in the air inlets or outlets.<br>- Check air flow on high voltage side.<br>- Check settings in G5.11.4-MV manometer fault and G5.11.5-MV manometer delay.<br>- Check the operation of the fans.<br>- If the measurement is not consistent (excessively high or negative) check the wiring.<br><br>**Note:** This only applies to HEM equipment. |
| **FNIMER** | **F144: Impulsion fans** | One of the impulsion fans (cyclone) [2] has reported an error.<br><br>Possible causes:<br>- Power supply does not reach 230 V.<br>- Error frames greater than 1%.<br>- End of line jumper disconnected.<br>- Defect in the cyclone communication board.<br>- Fan blocked.<br>- Defect in the fan power supply harting.<br>- Defect in the communication harting. | Verify AC input voltage for the ventilation is 230 V.<br>Verify fans are not obstructed and rotate smoothly.<br><br>**Extra information:**<br>- Check RS-485 communication on both cyclones. Error frames should be less than 1%.<br>- Ensure that the ground loop has continuity, is grounded only at the last point of the DU and is connected to the end-of-line resistor.<br>- Check the information given by the fan G11.2.5-Error type.<br>- Reeplace fan.<br>- Contact Power Electronics and transmit the error code.<br><br>**Note:** If the LC filter fails, voltage peaks can be coupled through the AASS transformer and damage the cyclone as well as other peripherals. |

[2] See figures "**HEM ventilation layout**" and "**HEMK ventilation layout**" for details about ventilation elements and sides

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| Acronym | Fault | Description or probable cause | Actions |
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| **NXCER** | **F145: Extrac. Fans control** | Feedback from the low voltage extraction fans [2] has not been received.<br>In HEM units the backfeed reaches the DU ventilation and in HEMK it reaches the DU ventilation and the exhaust ventilation of the AC module.<br><br>Possible causes:<br>- Defect in the P3G E977XX main control board.<br>- Defect in the exhaust ventilation.<br>- Defect in the backlight wiring. | Check low voltage fans supply and that the status of fault output is "closed".<br>If fans work properly, check feedback wiring.<br><br>**Extra information:**<br>- Check wiring.<br>- Check fan power supply.<br>- Check fan condition.<br>- Check the connections in the J26A1 and J26A2 in the P3G board. |
| **FNXMER** | **F146: Extrac. Fans MV** | Feedback from the medium voltage extraction fans [2] has not been received.<br><br>Possible causes:<br>- Defect in the MV E980 board.<br>- Defect in the wiring of the exhaust fan.<br>- Defect in the fan.<br>- Defect in the wiring of the backlight. | Check medium voltage fans supply and that the status of fault output is "closed".<br>If fans work properly, check feedback wiring.<br><br>**Extra information:**<br>- Check the connecting wiring of the fans (red color).<br>- Check fan power supply (blue and black colors).<br>- Check terminals J15.7 and J15.8 on the MV E980 board.<br>- Check fan condition. |
| **FNIMCM** | **F147: Impulsion comms** | Communication with the impulsion fans (cyclones) [2] cannot be stablished.<br>It is a failure caused by a loss of communications between the micro and the drive fans.<br><br>Possible causes:<br>- Fault in the AC power supply.<br>- Defect in the RS485 network.<br>- Defect in the port or in the fan electronics. | Check operation of the other bus devices (DU, insulation monitor) and verify terminating resistors are only in the central DU.<br>Check the RS485 cable, end-of-grid resistors, mesh continuity from start to finish, and only with earth connection at the end of the grid.<br>Connect a RS485 cable directly between central control and the fan to verify if they communicate. Replace fan.<br><br>**Extra information:**<br>Check the information given by the fan G11.2.5-Error type.<br>Check the power supply at the engine and at the fan.<br>Set G11.2.2-Module = Fixed.<br>Communicate the fan with a computer to check if the problem is the fan.<br>If the problem persists, contact Power Electronics. |
| **MVTTC** | **F148: MV Critical T° Trans** | The temperature threshold of the medium voltage transformer (defined in G5.7.3.3-MVTransTcriticThld) has been overcome.<br><br>Possible causes:<br>- Defective ventilation.<br>- Defective wiring.<br>- Defect in the signal of the PT100 board.<br>- Terminal block or terminals incorrectly connected.<br>- Damaged LCL filter. | Verify the MV switchgear protection has been activated. If the switchgear has not opened, check the temperature in the MV transformer cabinet and contact Power Electronics.<br><br>**Extra information:**<br>- Check transformer temperature measurements in SV4.5.3-PT100 (1), SV4.5.4-PT100 (2) and SV4.5.5-PT100 (3).<br><br>If the measurement is not consistent:<br>- Review communications with the MT board, F30:MV comms must appear.<br>- Check PT100 wiring.<br><br>If the measure is consistent:<br>- Check the routing of the ventilation ducts.<br>- Check the operation of the supply fans (power supply, speed, etc.).<br>- Check transformer terminal block (check connection and ferrules).<br>- Check LCL filter. |
| **DU_TEMP_EX** | **F149:DU T° exceeded** | The protection can be activated in G5.7.4.2-DU central board When the temperature read in SV4.1.4b-DU Internal t° is higher than G5.7.4.2-DU central board, F149:DU T° exceeded will be activated. | This protection is disabled by default and should not be enabled.<br><br>If this fault appears, contact Power Electronics. |

<table>
<tr><th>Acronym</th><th>Fault</th><th>Description or probable cause</th><th>Actions</th></tr>
<tr><td>CPMVO</td><td>F150: MV Cell prot open</td><td>Protection contactor of the medium voltage cell is open.<br><br>Possible causes:<br>- Defect in the backlight.<br>- Defective wiring or bad contact.<br>- Defect in the 24 V HV input board.<br>- Incorrectly set protection device.<br>- Defect in the power supply of the electronics (24 Vdc) in the AC cabinet.</td><td>Visually check the status of the cell and verify it matches with the settings of SG12-MV Cell service.<br>Contact Power Electronics.<br>Fault caused when the media cell's feedback indicates that the media cell has been opened. This fault has to jump if the cell is opened by hand or by itself. Never skip this fault if the cell has been opened from the control.<br><br>Extra information:<br><br>- Check cell status in SV4-Int. Visualization.<br>- Review cell configuration.<br>- Check the wiring of the backstop.<br>- Extract SD.<br>- Check the protection on the switchgear (toroids).</td></tr>
<tr><td>SDMV</td><td>F151: Smoke detector MV</td><td>After a software update, this fault was disabled<br>Note: The activation parameter, G5.10.19-Smoke detector MV, was deleted in HEM_1.3.0. Therefore, this fault will never be triggered.</td><td>Contact Power Electronics.<br>This fault it will never be triggered.</td></tr>
<tr><td>TRAUXTHR</td><td>F152:Transf Aux Thermal</td><td>When the thermal circuit breaker that protects the auxiliary transformer triggers its protection, it sends a digital signal to the main control board.<br><br>This fault will be shown if G5.11.11- Transf Aux Thermal = Fault.</td><td>Contact Power Electronics.</td></tr>
<tr><td>FBBK1</td><td>F153:Fb DC Breaker 1</td><td>This fault appears when there is a discrepancy between the command and the reset of circuit breaker 1 in NEC2017.<br>This fault can occur if the device has the DC disconnect switch on.</td><td rowspan="2">The DC cabinet has two DC switch disconnectors (Q8 and Q9) for connection to boards. If the command and the feedback of any of them do not match, the corresponding warning will be triggered ( W148:DU FB1 error for switch disconnector Q8 and W149:DU FB2 error for switch disconnector Q9).<br><br>Adjusting G5.11.8-Fb DC breaker fault = Warning instead of Fault could trigger the fault F12:Vdc mismatch.<br>- Check the power supply (230 V) of the circuit breaker.<br>- Check the status of the switch disconnectors in SG8.13-DU input status and SV13.3.1 Digital Inputs.<br>- Check the position of the circuit breaker key.<br>- Check the continuity of the backlight wiring.<br>- Check if the DC disconnect button is activated.<br>- Check power supply (24 V) to the minimum coil.<br>- The motor that loads the spring must be powered at 230 V, although in some projects it is powered with 24 V.<br>- Check the timing of the minimum coil power supply relays.<br>- Check that the DU configuration is set in NEC2017.<br>- There is a fault in the command of the circuit breaker. It is possible to operate one manually if the others are energized. To do this safely, the DC switch disconnectors of the DC cabinet must be opened and the bus must be discharged.<br>- Check the status of the switch disconnectors in SG8.13-DU input status and SV13.3.1-Digital Inputs. For further information, check the parameters description.<br><br><table><tr><th>Status</th><th>Switch Disconnector 1 (bits 7, 6 and 5)</th><th>Switch Disconnector 2 (bits 4, 3 and 2)</th><th>Overvoltage (bit 1)</th><th>Aux (bit 0)</th></tr><tr><td>Inverter OFF</td><td>X0X</td><td>X0X</td><td>X</td><td>0</td></tr><tr><td>Inverter RUN</td><td>0X0</td><td>0X0</td><td>X</td><td>0</td></tr><tr><td>Fault</td><td>X0X</td><td>0X0</td><td>X</td><td>0</td></tr></table></td></tr>
<tr><td>FBBK2</td><td>F154:Fb DC Breaker 2</td><td>This fault appears when there is a discrepancy between the command and the reset of circuit breaker 2 in NEC2017.<br>This fault can occur if the device has the DC disconnect switch on.</td></tr>
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| Acronym | Fault | Description or probable cause | Actions |
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| **PRDID** | **F155:Product id** | The product identifier read in the drive select is not valid.<br>Normally the F44:Drive select should jump first.<br><br>Possible causes:<br>- Wrong drive select configuration.<br>- Fault in the central control board P3G E977XX. | - Check drive select configuration.<br>- Initialize drive select by default.<br><br>Verify the central control board.<br>Consult Power Electronics |
| **DUINVC** | **F156:DU Inverse current** | When the negative current of a channel in the DC cabinet (DU) exceeds G5.11.3a-Inv current prot, the fault occurs and the inverter stops (but does not open the DU).<br><br>The fault can appear in all inverter states.<br><br>Possible causes:<br>- Defective wiring or hall sensor.<br>- Defect in the DU current board.<br>- Current flow to the PV array. There could be short circuits or power injection from a DC/DC.<br><br>**Note:** In the case of PCS inverters or DU being NEC 2017, the fault will be disabled. | Investigate if the inverse (negative) currents are real and is the responsibility of the customer's photovoltaic field.<br>Check wiring for external short circuits.<br>Check if the measurement boards are correct.<br><br>If the device has Bus Plus, it is necessary to configure G5.11.3a-Reverse I-Prot. I in such a way that the DC/DC power is prevented from going into the PV field. |
| **AEMO_C** | **F157:Communications AEMO** | This fault is triggered when the maximum number of consecutive losses allowed (configurable parameter) is reached through AEMO. | Check the inverter configuration and that the IP is correct, as well as the heartbeat counter ([G6.3.7-PPC Heartbeat Reg]). If it increases, that means configuration is correct at communications level and it could be an sporadic fault related to HW.<br><br>If it does not increase<br>- Ensure IP filtering is not enabled, this could cause a problem with access permissions.<br>- Check wiring and verify communications at software level.<br>- Check that RJ45 connectors of the client's network are in good state and properly connected. If any defect is found, inform the client so they fix it.<br><br>If the counter increases, but irregularly in time, check the communications at HW level with the SCADA.<br><br>In case of sporadic losses, there could be a problem with the communications switch where the client's cable is connected.<br><br>To rule out problems related to interferences, check that all cables are screened.<br><br>If after all the previous checks the problems still has not been found, ask the client to check the continuity of the RS485 wiring. |
| **ETLCL** | **F159:Exc temp LCL box** | Is triggered when the thermal sensor detects an overtemperature in the cabinet where the LCL filter is placed.<br><br>This fault will not be shown if G4.6.1-Enable = External or if SG1.1.19-OverT LCL box error is configured as warning or disabled. | Contact Power Electronics. |

| Acronym | Fault | Description or probable cause | Actions |
|---|---|---|---|
| **HOMDIFLV** | **F181:Hom Diff MV** | A zero-sequence current (earth leakage) has been detected on the medium voltage side of the transformer.<br>If the current exceeds the threshold defined in parameter **G5.9b.2.5-Differential protec** during the time set in **G5.9b.2.5b-Differential time**, this fault will be triggered.<br>Fault related to MT protections that currently should not be enabled.<br><br>Possible causes:<br>- MV protections enabled.<br>- Fault in the wiring.<br>- In HEM equipment, defect in the MV 100/1 toroids or in its orientation.<br>- Defect in the E983 converter board. | Verify if there really is a leak (check cable insulation in the low and high sides of the transformer).<br>Initialize the drive select by default.<br>Contact Power Electronics.<br><br>**Extra information:** See note below. |
| **CRBALVMV** | **F182:Curr Bal LVMV** | A current unbalance has been detected between the low and medium voltage sides (measures do not match).<br>If the current exceeds the threshold defined in parameter **G5.9b.3-LV-MV I balance** during the time set in **G5.9b.3b-LV-MV I balance T**, this fault shall be triggered.<br>It might mean there is a derivation in the transformer.<br>Fault related to MT protections that currently should not be enabled.<br><br>Possible causes:<br>- MV protections enabled.<br>- Problem in the balance of real voltages and currents.<br>- Fault in the wiring.<br>- In HEM equipment, defect in the MV 100/1 toroids or in its orientation.<br>- In HEMK equipment, defect in the LV 3500/5 toroids or in its orientation.<br>- Defect in the central control board P3G - E977XX.<br>- Defect in the converter board E983. | - Verify if there really is a derivation and, if the problem persists, check the measurement circuit. Do not measure in the measurement board directly.<br>- Check the MV transformer.<br>Contact Power Electronics.<br><br>**Extra information:** See note below. |
| **NEGPWLV** | **F183:Neg Pow LV** | Active power in the low voltage side of the transformer has overcome the threshold set in **G5.9b.1.7-Negative P** during the time adjusted in **G5.9b.1.7b-Neg P timeout.**<br>Fault related to MT protections that currently should not be enabled.<br><br>Possible causes:<br>- Defective wiring.<br>- In HEMK equipment, defect in the LV 3500/5 toroids or in its orientation.<br>- Defect in the central control board P3G - E977XX. | - This fault indicates there is a problem in the equipment.<br>- Check wiring.<br>- In HEMK equipment, defect in the LV 3500/5 toroids or in its orientation.<br>- Check central control board P3G -E977XX.<br>- Check it and, if the fault persists, contact Power Electronics.<br><br>**Extra information:** See note below. |
| **NEGPWMV** | **F184:Neg Pow MV** | Reserved for future use.<br>This fault will be similar to **F183:Neg Pow LV**, it will be triggered if negative power is detected in the low voltage side of the transformer. Activation conditions will be set in **G5.9b.2.7-Negative P** and the period for the activation of the fault is defined in **G5.9b.2.7b-Neg P timeout.** Fault related to MT protections that currently should not be enabled.<br><br>Possible causes:<br>- Defective wiring.<br>- In HEM equipment, defect in the MV 100/1 toroids or in its orientation.<br>- Defect in the E983 converter board.<br>- MV protections enabled. | Reserved for future use.<br><br>**Extra information:** See note below. |

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| Acronym | Fault | Description or probable cause | Actions |
|---|---|---|---|
| **CRUNBALV** | **F185:Cur Unbl LV** | The unbalance threshold between phases in the low voltage side of the transformer has been overcome.<br>Activation conditions will be set in **G5.9b.1.6-I unbalance** and the period for the activation of the fault is defined in **G5.9b.1.6b-I unbalance tim.**<br>This fault usually appears whenever a grid event occurs, voltage unbalance causes current unbalance.<br>Fault related to MT protections that currently should not be enabled.<br><br>Possible causes:<br>- Defective wiring.<br>- MV protections enabled.<br>- In HEMK equipment, defect in the LV 3500/5 toroids or in its orientation.<br>- Defect in the central control board P3G - E977XX. | Wait until the fault disappears. Should the fault persist, contact Power Electronics.<br><br>- Ensure a grid event occurred. If there have been no grid events and the problem persist, check the measurement circuit.<br>- Verify the correct installation of the bolts and nuts that hold the toroid.<br><br>**Extra information:** See note below. |
| **CRUNBAMV** | **F186:Cur Unbl MV** | Similar to **F185:Cur Unbl LV**, but for currents on the medium voltage side of the transfomer.<br>Activation conditions will be set in **G5.9b.2.6-I unbalance** and the period for the activation of the fault is defined in **G5.9b.2.6b-I unbalance tim = 1000 ms.**<br>Fallo relacionado con protecciones MT que actualmente no deben estar habilitadas.<br><br>Possible causes:<br>- Defective wiring.<br>- MV protections enabled.<br>- In HEM equipment, defect in the MV 100/1 toroids or in its orientation.<br>- In HEMK equipment, defect in the LV 3500/5 toroids or in its orientation.<br>- Defect in the central control board P3G - E977XX. | Wait until the fault disappears. Should the fault persist, contact Power Electronics.<br><br>- Ensure a grid event occurred. If there have been no grid events and the problem persists, check the measurement circuit. Do not measure in the measurement board directly.<br>- Check wiring.<br>- In HEM equipment, check the MV 100/1 toroids and its orientation.<br>- In HEMK equipment, check the LV 3500/5 toroids and its orientation.<br>- Check the converter board E983.<br>- Check central control board P3G -E977XX.<br>**Extra information:** See note below. |
| **CRBLDSPMD** | **F187:Cur Bal DSPMD** | Current measurement made by central control does not match the sum of measurements of all modules.<br>The threshold between sums of current measured by the modules and the total current measured on the low side of MV has been exceeded (this measurement is done by the central control).<br><br>Activation conditions will be set in **G5.5.8-Mod-DSP I balance** and the period for the activation of the fault is defined in **G5.5.8b-Mod-DSP I balan tim.**<br><br>There may be a hardware problem in the equipment of a measurement failure.<br>Fallo relacionado con protecciones MT que actualmente no deben estar habilitadas.<br><br>Possible causes:<br>- Defective wiring.<br>- MV protections enabled.<br>- In HEMK equipment, defect in the LV 3500/5 toroids or in its orientation.<br>- Defect in the central control board P3G - E977XX. | - Check the equipment and measurement circuit. Do not measure in the measurement board directly.<br>- Check wiring.<br>- In HEMK equipment, check the LV 3500/5 toroids and its orientation.<br>- Check the converter board E983.<br>- Check central control board P3G -E977XX.<br>Contact Power Electronics.<br><br>**Extra information:** See note below. |

| Acronym | Fault | Description or probable cause | Actions |
|---|---|---|---|
| **PWBALLVMV** | **F188:Pow Bal LVMV** | Reserved for future use.<br><br>This fault will be implemented in future versions. It will be triggered whenever a power unbalance is detected between the low voltage and medium voltage sides (threshold and timeout to be implemented too). It may indicate a possible derivation in the transformer and, if it appears, customer must contact PE.<br>Fallo relacionado con protecciones MT que actualmente no deben estar habilitadas. | Reserved for future use.<br><br>**Extra information:** See note below. |
| **THDIMV** | **F189:THDi MV** | The threshold of harmonic distortion (THDi), **G5.9b.5-MV THDi protection**, has been overcome during the timeout set in **G5.9b.5b-MV THDi prot. Time**.<br><br>Fault produced by more than 500ms between the activation command and the feedback.<br>Fault related to MT protections that are not enabled currently.<br><br>Possible causes:<br>- Defective wiring.<br>- MV protections enabled.<br>- In HEM equipment, defect in the wiring of the MV 100/1 toroids.<br>- In HEMK equipment, defect in the wiring of the LV 3500/5 toroids.<br>- Defect in the high voltage toroidals or in its orientation.<br>- Defect in the E983 converter board.<br>- Defect in the LC filter.<br>- Failure in the module switching. | - Measure order, and current consumption. The time measured under ideal conditions is 70 ms.<br>- Measure 24 V from relay sources.<br>- In case of obtaining something different, the state of the contactor should be checked.<br>- Check that the voltage THDv is below 5% and that the current THDi is less than 3% at 80% load or higher.<br>- If the current waveform is greater than 6% under the above conditions, check the switching of the individual modules and the capacitor filter.<br>- Individually start up each module and check that they all have a correct THDi.<br><br>Check the equipment. Should the problem persist, contact Power Electronics.<br><br>**Extra information:** See note below. |
| **CPVF** | **F190:Comb prot Volt Frq** | One of the protection thresholds set in group G5.4b-Volt/freq Combined has been overcome.<br><br>This protection will be activated if the voltage exceeds the value V1 (**G5.4b.1.2-V1**) and the frequency is greater than f1 (**G5.4b.1.3-f1**) during the time t1 (**G5.4b.1.-t1**).<br><br>Possible causes:<br>- See F5:High Vac, F6:Low Vac, F7:High frequency, F8:Low frequency. | Follow the actions depending on the threshold that has been overcome:<br>- For high AC voltage: see actions for fault F5:High Vac.<br>- For low AC voltage: see actions for fault F6:Low Vac.<br>- For high frequency: see actions for fault F7:High frequency.<br>- For low frequency: see actions for fault F8:Low frequency.<br><br>Contact Power Electronics if you require assistance.<br><br>**Extra information**:<br>- Check the configuration of G5.4b-Volt/freq Combined.<br>- Extract SD to analyze which of the protections jumps.<br>- Act as for F5: High Vac, F6:Low Vac, F7:High frequency and F8:Low frequency. |
| **CLBV** | **F191: Critical Low Battery V** | Voltage in any of the Buses is too low ($V_{DC} < V_{DC\,Min} + G5.10.16 * 0.5$).<br><br>Similar to F25: Low Vdc, but this one be triggered while the equipment is in READY status. It happens due to a SCADA wrong operation.<br><br>**Note:** This fault only applies when the inverter is configured as MPCS. | - Check SCADA operation.<br>- Check wiring.<br>- Check DC cabinet measure board. Do not measure on this board directly.<br>- Check DC cabinet main board.<br>- Check batteries.<br><br>Contact Power Electronics. |

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| Acronym | Fault | Description or probable cause | Actions |
|---|---|---|---|
| **ECOM** | **F192: Econmode** | This fault is related to the Econmode function and the pre-magnetization of the transformer.<br>In G4.12-Econmode we can find more information about these features.<br><br>If Econmode is available and activated, during the night, the MV switchgear is open. DU will be normally close and the inverter will be waiting sunrise to start up.<br><br>During sunrise it is necessary to close the MV switchgear to start the inverter, and if Econmode is available and connected, inverter will close MV switchgear after a pre-magnetizaction process.<br>If DU is open (and not closed as it should be), MV switchgear can not close, W32:Econmode DC Open will be triggered.<br><br>Later on, there are two situations depending on the DU type:<br><br>**NEC2014:** To close the MV switchgear without pre-magnetization:<br><br>- During night, DU closes automatically.<br>- During day conditions, inverter cannot close the DU since there would be inrush current to the DC bus (MV swichgear is open, so AC softwcharge does not work). After 10 min without closing MV switchgear, will appear F192:Econmode failure.<br><br>**NEC2017:** If MV switchgear is open and the DU opens after a grid event, then we cannot close the DU because Econmode has not enough power to close DC switch disconnectors. Then, during sunrise and day conditions are activated, MV switchgear will close without pre-magnetization.<br><br>If the new parameter G4.12.6-Clos Witho Prm Allow, to force closing without pre-magnetization, is disabled, then F192:Econmode will be triggered. | During those 10 minutes in NEC2014 or when F192:Econmode is triggered, our technician or the SCADA can exit from this state using parameter SG12.1-Action Gen. Switch.<br>In these cases the DC cabinet will be closed without pre-magnetization. Parameter G4.12.6-Clos Witho Prm Allow could be disabled.<br><br>- Check that the DC cabinet is closed.<br>- Check DC cabinet electronics.<br>- Check the feedback light of DC switches.<br><br>Contact Power Electronics. |
| **DCDISFLT** | **F193:DC Disconnect Fault** | The self-diagnosis module takes into account if the DC switch is actually in the open state.<br>For that reason, the inverter will always perform a first deep discharge (as low as 250 Vdc) to make sure that we are disconnected from the solar photovoltaic energy. Afterwards, the functionality will remain the same by performing the discharge sequence on the rest of the modules.<br>If the deep discharge lasts more than 300 s, then the F193:DC Disconnect Fault is triggered.<br>This functionality can be activated by the parameter SG11.4-Self-diagnosis enable deep discharge.<br><br>Possible causes:<br><br>- Defective contactors in NEC2014.<br>- Defect in the circuit breakers in NEC2017.<br>- Defect in the module that performs the discharge.<br>- Defect in the module's DC measurement board. Do not measure on this board directly. | The circuit breaker did not open and failed. Check the circuit breaker and the wiring to it.<br><br>- Check that there is no closed contactor in NEC2014.<br>- Check that the circuit breakers are open in NEC2017.<br>- Check that the module has been discharged correctly.<br>- Check the DC voltage measurement of the module that has discharged. Do not measure on the measurement board directly. |

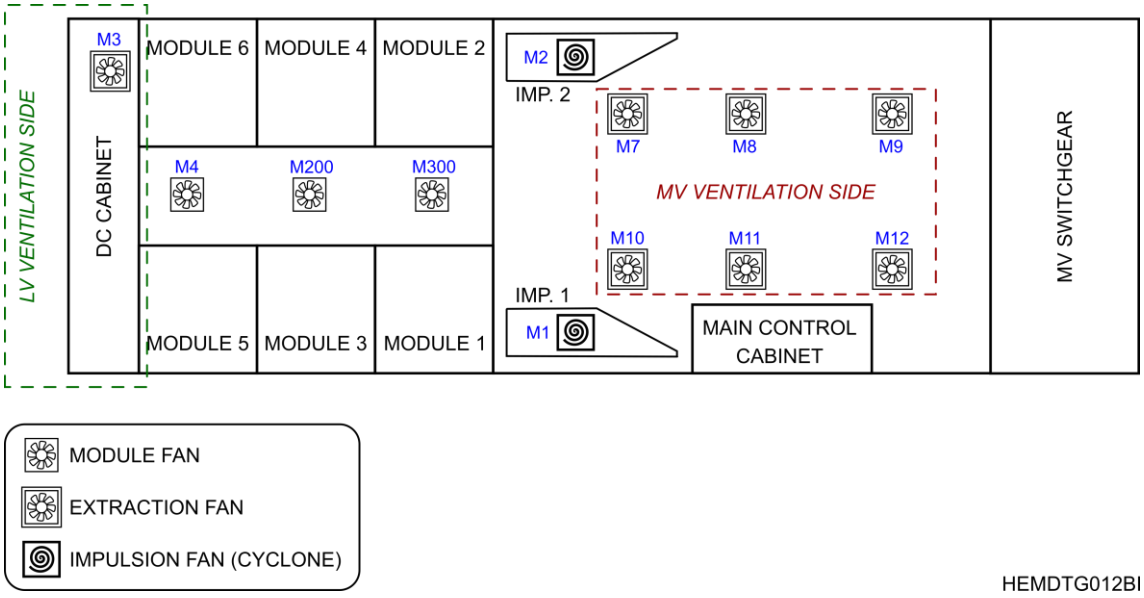
| Acronym | Fault | Description or probable cause | Actions |
|---|---|---|---|
| NEG_I_DU | F194:DU Neg current exceeded | When SV10.6-Current Sum (DU Current) is below - G5.11.3e1-DUtotal neg cur DU total negative current protection for more than G5.11.3e2-DUtotal neg cur del.delay. F194 will be trigger upon these conditions.<br><br>Possible causes:<br>- Defective wiring.<br>- Defective hall sensor.<br>- Defect in the DC cabinet current board.<br>- Current flow to the PV array. There could be short circuits or power injection from a DC/DC. | Check current measurement board and wiring. Do not measure in this board directly. |
| LCFUSBLOWN | F195: LC fuse blown | When SV10.6-Current Sum DU current sum is greater than internal configuration parameter DSP 5, F195 will be triggered.<br><br>Internal configuration parameter DSP 5 is SG2.1.5-Config DSP 5.<br><br>Possible causes:<br>- Defective wiring.<br>- Defective hall sensor.<br>- Defective in the current measure board of the DC cabinet. Do not measure on this board directly. | Contact with Power Electronics. |
| TANTR | F196: Timeout autonight trans | When the inverter is in AUTONIGHT state for more than 5 min.<br><br>Possible causes:<br>- Wrong configuration. | Extract log from SD card.<br>Contact with Power Electronics. |
| HWVACHIGH | F197:HW Vac High | Similar to [F5: High Vac] with the difference that voltage levels are fixed (configured internally) to protect the hardware. | See [F5: High Vac] |
| HWVACLOW | F198:HW Vac Low | Similar to [F6: Low Vac] with the difference that voltage levels are fixed (configured internally) to protect the hardware. | See [F6: Low Vac] |
| SIMPMOD | F200: Simple modulation | More than one modulation switch per minute was detected (from single to double).<br><br>Possible causes:<br>- Fluctuations in bus voltage, caused by clouds.<br>- Fluctuations in the power of the equipment.<br>- Unstable AC voltage.<br>- Reactive power fluctuation in the equipment.<br>- Defect in the power stages.<br>- SW versions older than 2.2.3 | Fault is self-resetting. The count starts again.<br>From version 2.5.0, a new measuring DC filter has been added and it requires to set these parameters:<br><br>- SG1.3.a7-DSP Change Thresh= 50 V.<br>- SG1.3.a8-FPGA change threshold = 70 V.<br>- Check the power stages. |
| LCFUSDVDT | F215:LC fuse - dV/dt | A higher than expected dV/dt voltage ripple has been detected.<br><br>This fault is configured in [G5.14.1-Configuration]. By default, it is disabled. If the protection is enabled, it can be configured as a fault (it will trip F215) or warning (it will trip W215). | If it is enabled and it is triggered, the configuration of [G5.14.1-Configuration] must be reviewed and set to its default value.<br><br>Once this is done, confirm the anomalous behavior with a Blackbox analysis or a current and voltage measurement. If the excess ripple is true, the main coil should be replaced. If the reason is resonance, the following steps should be analyzed with the Quality Department. |
| LCFUSDIDT | F216:LC fuse - dI/dt | A higher than expected dI/dt voltage ripple has been detected.<br><br>This fault is configured in [G5.14.1-Configuration]. By default, it is disabled. If the protection is enabled, it can be configured as a fault (it will trip F216) or warning (it will trip W216). | If it is enabled and it is triggered, the configuration of [G5.14.1-Configuration] must be reviewed and set to its default value.<br><br>Once this is done, confirm the anomalous behavior with a Blackbox analysis or a current and voltage measurement. If the excess ripple is true, the main coil should be replaced. If the reason is resonance, the following steps should be analyzed with the Quality Department. |

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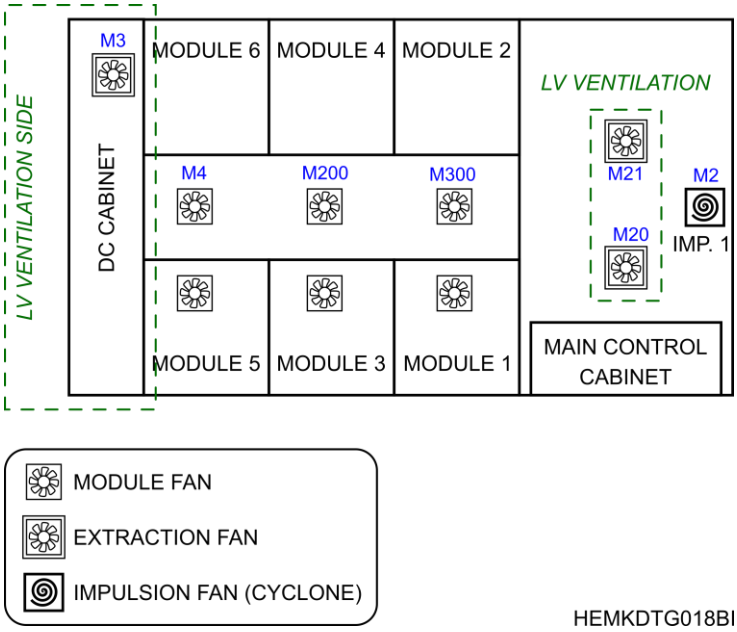
**Note**: Faults F181, F182, F183, F184, F185, F186, F187, F188 and F189 are related to the MV protections that currently must not be enabled.

If any of them jumps, check that it is disabled and reset the fault. If the fault persists, the control voltage must be removed.

The following figures show the ventilation layout for HEM and HEMK inverters. Please notice in HEM inverters, we distinguish between “medium voltage ventilation side” and “low voltage ventilation side”; this does not apply to HEMK inverters, for which we will only talk about “low voltage ventilation side”.




*HEM ventilation layout*




*HEMK ventilation layout*

**Note**: Both for HEM and HEMK:

- Module fans are connected to their module’s FPGA.
- Fans in the DC cabinet are connected in series and are wired to the central control board J26A:2).
- Impulsion fans communicate with the central control directly, through the communication bus.

Besides, in HEM inverters, fans on the medium voltage side (M10-M15) are connected in series to the medium control board (J15:8).

## Special Faults

As mentioned before, fault F26 has been disabled. Likewise, the concept “emergency fault” disappears.

Therefore, from version HEM_1.2.1 we will only talk about **special faults** and these will be the faults (either module or inverter) that are included in G5.10.x:

Whenever any of the faults defined as special is triggered, one of the following actions will be taken (depending on the setting of **G5.10.2-Special protections**):

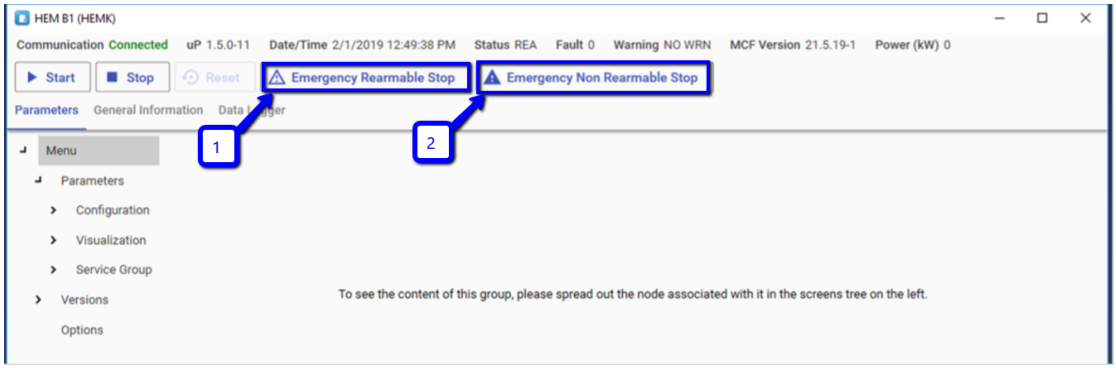
- If G5.10.2 = Fault ➔ Stop inverter.
- If G5.10.2 = Emergency stop ➔ Stop inverter and open DC.
- If G5.10.2 = Circuit breaker ➔ Stop inverter and fully isolate the equipment (open DC and AC circuit breaker).

**Note:** Whenever an inverter trips due to special fault, the Blackbox and SD card should be downloaded and analyzed, to see if there is any type of short circuit in DC. The equipment must be checked to find out what caused this short circuit, normally in the FRU.

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## Remote Communications Failures

These features will be available to users with role of Power Electronics employee (SAT) or developer. The remote emergency stop can be activated via modbus registers or via the PowerCommsV application, as shown in the picture below.




1. Remote resettable emergency stop: Stops the inverter and opens the DU generating the F56 fault simulation. Simulates a 'non-special' fault. By means of modbus registers, this is done by writing a "1" to address 43200.
2. Non-Rearmable Emergency Stop: Opens the circuit breaker, opens the DU and stops the inverter generating an F56 fault. Simulates a special fault. When this option is executed:

- The program will ask "Attention: Do you really want to make a non-resettable emergency stop of the inverter?
- If yes, the password "1492" will be sent to the modbus address 43199
- Write a "2" in modbus address 43200
- Read the confirmation password "1571" at modbus address 43199
- After 1 second of the writing of "2" the equipment will lose communications and will require an operator intervention to restart

# VISUALIZATION AND STATUS PARAMETERS

4

It is possible to check at any time the solar inverter status and information related with its dynamic parameters. All parameters to visualize and monitor the operation status of the inverter are grouped and listed in the menu “Visualization”. They will be described in this section.

## Group SV1: Grid Visualization

This group shows information about voltage and current per phase, as well as the power injected to the grid, power factor and grid frequency.

| Screen | Units | Description |
|---|---|---|
| **SV1.1-Grid voltage RS = 0.0 kV** | kV | Shows the grid voltages in medium voltage phases L1, L2, L3 in HEM inverters. |
| **SV1.2-Grid voltage ST = 0.0 kV** | kV | |
| **SV1.3-Grid voltage TR = 0.0 kV** | kV | |
| **SV1.1-Grid voltage RS = 0.0 V** | V | Shows the grid voltages in low voltage phases L1, L2, L3 in HEMK inverters. |
| **SV1.2-Grid voltage ST = 0.0 V** | V | |
| **SV1.3-Grid voltage TR = 0.0 V** | V | |
| **SV1.1b-Grid voltage RS (LV) = 0 V** | V | Shows the grid voltages in low voltage phases L1, L2, L3.<br>**Note:** These parameters are only shown in HEM inverters. |
| **SV1.2b-Grid voltage ST (LV) = 0 V** | V | |
| **SV1.3b-Grid voltage TR (LV) = 0 V** | V | |
| **SV1.4-Grid current I1 = 0.0 A** | A | Shows the currents injected to the grid in medium voltage phases L1, L2, L3 in HEM inverters. |
| **SV1.5-Grid current I2 = 0.0 A** | A | |
| **SV1.6-Grid current I3 = 0.0 A** | A | |
| **SV1.4-Grid current I1 = 0.0 A** | A | Shows the currents injected to the grid in medium voltage phases L1, L2, L3 in HEMK inverters. |
| **SV1.5-Grid current I2 = 0.0 A** | A | |
| **SV1.6-Grid current I3 = 0.0 A** | A | |
| **SV1.4b-Grid current I1 (LV) = 0 A** | | Shows the currents injected to the grid in low voltage phases L1, L2, L3.<br>**Note:** These parameters are only shown in HEM inverters. |
| **SV1.5b-Grid current I2 (LV) = 0 A** | | |
| **SV1.6b-Grid current I3 (LV) = 0 A** | | |
| **SV1.7-P = 0 kW** | kW | Shows the power injected to the grid. |
| **SV1.8-Q = 0 kVAr** | kVAr | |
| **SV1.9-S = 0 kVA** | kVA | |

| Screen | Units | Description |
|---|---|---|
| **SV1.10-Cos phi = 0.000** | NONE | Shows the power factor. |
| **SV1.11-Grid frequency = 0.00 Hz** | Hz | Shows the grid frequency. |
| **SV1.11b-Abs Val Grid freq = 0.00 Hz** | HZ | Shows the grid frequency always in positive value. |
| **SV1.12-HVPL level = 0** | NONE | Shows the current measurement of the HV blown fuse detection algorithm. |
| **SV1.13-POI V estimated = 0 V** | V | Shows the value of the estimated voltage in the POI (point of interconnection) from the inverter input currents and the impedance of the line to the POI (parameters from group G8.3-Grid). |

## Group SV2: Photovoltaic Generator

This group displays data related to DC voltage and input power and current.

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| Screen | Units | Description |
|---|---|---|
| **SV2.1-DC voltage bus = 0 V** | V | Shows the DC voltage of the bus. |
| **SV2.2-DC voltage PV = 0.0 V** | V | Shows the DC voltage of the photovoltaic field (voltage measured in the Disconnecting Unit).<br>**Note:** This parameter is not shown in STATCOM. |
| **SV2.3-DC voltage last linp = 0 V** | V | Shows the DC voltage the last time the inverter started. |
| **SV2.4-Batt voltage bus = 0 V** | V | Shows the DC voltage of the batteries (voltage measured in the Disconnecting Unit).<br>**Note:** This parameter is only visible for PCS inverters. |
| **SV2.5-DC input power = 0 kW** | kW | Shows the power supplied by the photovoltaic generator.<br><br>DC input power is calculated as the sum of currents in the Disconnecting Unit multiplied by the measured voltage in the DU. |
| **SV2.5a-BESS DC power input= 0 kW** | kW | Displays the input power of BESS DC.<br><br>**Note**: Appears when HEM inverter type is selected with BUS PLUS mode activated and with DU current measures SG10.1.13-DU, SG10.1.4a-Bus Plus. |
| **SV2.6-DC input current = 0 A** | A | Shows the total of the measured currents in the inverter modules (direct current provided by the photovoltaic plant).<br><br>**Note**: The current measurement of the Disconnecting Unit channels is not included by default. If the customer has contracted this option, the measured values in SV10 group. |
| **SV2.7-DC voltage control = 0 V** | V | Shows DC control voltage. This is, the DC measure that is used in voltage algorithms, obtained from applying a filter to the DC voltage of the bus. |

# Group SV3: Insulation monitoring

From this group, the user can check the status of insulation protection and parameters related to the isolation auto test.

**Note:** subgroup SV3.1 will only be displayed if [G9.1.1] = Power Electronics. Similarly, subgroup SV3.2 is only shown when [G9.2.1] = Power Electronics.

## Subgroup SV3.1: DC

<table>
<tr><th>Screen</th><th>Units</th><th>Description</th></tr>
<tr><td>SV3.1.0-Internal status = 0.Inter meas guard time</td><td>NONE</td><td>Shows the status of DC insulation protection. Possible values are shown in the table below:
<table>
<tr><th>OPT.</th><th>STATUS</th><th>OPT.</th><th>STATUS</th></tr>
<tr><td>0</td><td>Inter meas guard time</td><td>13</td><td>Pos pulse 5</td></tr>
<tr><td>1</td><td>Power up start</td><td>14</td><td>Neg pulse 1</td></tr>
<tr><td>2</td><td>Power up auto test initial</td><td>15</td><td>Neg pulse 2</td></tr>
<tr><td>3</td><td>Power up auto test step 1</td><td>16</td><td>Neg pulse 3</td></tr>
<tr><td>4</td><td>Power up auto test step 2</td><td>17</td><td>Neg pulse 4</td></tr>
<tr><td>5</td><td>Power up auto test step 3</td><td>18</td><td>Neg pulse 5</td></tr>
<tr><td>6</td><td>Power up risk state</td><td>19</td><td>Rp calc</td></tr>
<tr><td>7</td><td>Start</td><td>20</td><td>Auto test initial</td></tr>
<tr><td>8</td><td>Grid connect</td><td>21</td><td>Auto test step 1</td></tr>
<tr><td>9</td><td>Pos pulse 1</td><td>22</td><td>Auto test step 2</td></tr>
<tr><td>10</td><td>Pos pulse 2</td><td>23</td><td>Auto test step 3</td></tr>
<tr><td>11</td><td>Pos pulse 3</td><td>24</td><td>Risk state</td></tr>
<tr><td>12</td><td>Pos pulse 4</td><td>25</td><td>Idle</td></tr>
</table></td></tr>
<tr><td>SV3.1.0b-Status bits = 00000</td><td>NONE</td><td>Insulation monitor status. Each bit (from left to right) represents one status (0: not active, X: active)
<table>
<tr><th>BIT</th><th>STATUS</th></tr>
<tr><td>1</td><td>Fault</td></tr>
<tr><td>2</td><td>Warning</td></tr>
<tr><td>3</td><td>Self test failed</td></tr>
<tr><td>4</td><td>Resistance over 2MOhms</td></tr>
<tr><td>5</td><td>Class B auto test failed</td></tr>
</table></td></tr>
<tr><td>SV3.1.1-Status = Idle</td><td>NONE</td><td>Shows the status of insulation protection, according to the following table:
<table>
<tr><th>OPTION</th><th>STATUS</th></tr>
<tr><td>0</td><td>Idle</td></tr>
<tr><td>1</td><td>Inter measurement guard</td></tr>
<tr><td>2</td><td>Measurement in progress</td></tr>
<tr><td>3</td><td>Self-test in progress</td></tr>
<tr><td>4</td><td>Risk addressed state.<br>The isolation monitor change to this status when the autotest fail.<br>It will remain in this status until a hard reset is executed.</td></tr>
</table></td></tr>
<tr><td>SV3.1.2-Resistor = 65535 Ohms</td><td>OHMS</td><td>Shows the measured resistance to earth.<br><br>The value of this parameter is shown as a 32-bits DWord, two 16-bits words (MSB and LSB).<br>The high byte (MSB) corresponds with the lowest Modbus address and the low byte (LSB) corresponds with the highest Modbus address. Thus, to correctly read the value, perform the calculation:<br>V = MSB * 2^16 + LSB</td></tr>
<tr><td>SV3.1.3-Meas started time = 0 s</td><td>s</td><td>Shows the time that has elapsed since the last resistance measure.</td></tr>
</table>

| Screen | Units | Description |
|---|---|---|
| **SV3.1.4-Selftest started time = 0 s** | s | Shows the elapsed time from the auto test start. |
| **SV3.1.5-Selftest exec count = 0 min** | min | Shows the number of minutes remaining for the next auto-test. |
| **SV3.1.6-Diff last measur Vdc = 0 V** | V | Shows the difference between the last Vdc measurements. |

## Subgroup SV3.2: AC

<table>
<tr><th>Screen</th><th>Units</th><th>Description</th></tr>
<tr><td>SV3.2.0-Internal status = 0.Inter meas guard time</td><td>NONE</td><td>Shows the status of AC isolation protection. Possible values are shown in the table below:
<table>
<tr><th>OPT.</th><th>STATUS</th><th>OPT.</th><th>STATUS</th></tr>
<tr><td>0</td><td>Inter meas guard time</td><td>13</td><td>Pos pulse 5</td></tr>
<tr><td>1</td><td>Power up start</td><td>14</td><td>Neg pulse 1</td></tr>
<tr><td>2</td><td>Power up auto test initial</td><td>15</td><td>Neg pulse 2</td></tr>
<tr><td>3</td><td>Power up auto test step 1</td><td>16</td><td>Neg pulse 3</td></tr>
<tr><td>4</td><td>Power up auto test step 2</td><td>17</td><td>Neg pulse 4</td></tr>
<tr><td>5</td><td>Power up auto test step 3</td><td>18</td><td>Neg pulse 5</td></tr>
<tr><td>6</td><td>Power up risk state</td><td>19</td><td>Rp calc</td></tr>
<tr><td>7</td><td>Start</td><td>20</td><td>Auto test initial</td></tr>
<tr><td>8</td><td>Grid connect</td><td>21</td><td>Auto test step 1</td></tr>
<tr><td>9</td><td>Pos pulse 1</td><td>22</td><td>Auto test step 2</td></tr>
<tr><td>10</td><td>Pos pulse 2</td><td>23</td><td>Auto test step 3</td></tr>
<tr><td>11</td><td>Pos pulse 3</td><td>24</td><td>Risk state</td></tr>
<tr><td>12</td><td>Pos pulse 4</td><td>25</td><td>Idle</td></tr>
</table></td></tr>
<tr><td>SV3.2.0b-Status bits = 00000</td><td>NONE</td><td>Insulation monitor status. Each bit (from left to right) represents one status (0: not active, X: active)
<table>
<tr><th>BIT</th><th>STATUS</th></tr>
<tr><td>1</td><td>Fault</td></tr>
<tr><td>2</td><td>Warning</td></tr>
<tr><td>3</td><td>Self test failed</td></tr>
<tr><td>4</td><td>Resistance over 2MOhms</td></tr>
<tr><td>5</td><td>Class B auto test failed</td></tr>
</table></td></tr>
<tr><td>SV3.2.1-Status = Idle</td><td>NONE</td><td>Shows the status of isolation protection according to the following table:
<table>
<tr><th>OPTION</th><th>STATUS</th></tr>
<tr><td>0</td><td>Idle</td></tr>
<tr><td>1</td><td>Inter measurement guard</td></tr>
<tr><td>2</td><td>Measurement in progress</td></tr>
<tr><td>3</td><td>Self-test in progress</td></tr>
<tr><td>4</td><td>Risk addressed state</td></tr>
</table></td></tr>
<tr><td>SV3.2.2-Resistor = 65535 Ohms</td><td>OHMS</td><td>Shows the measured resistance to earth.<br><br>The value of this parameter is shown as a 32-bits DWord, two 16-bits words (MSB and LSB). The high byte (MSB) corresponds with the lowest Modbus address and the low byte (LSB) corresponds with the highest Modbus address. Thus, to correctly read the value, perform the calculation:<br>V = MSB * 2^16 + LSB</td></tr>
<tr><td>SV3.2.3-Meas started time = 0 s</td><td>s</td><td>Shows the time that has elapsed since the last resistance measure.</td></tr>
<tr><td>SV3.2.4-Selftest started time = 0 s</td><td>s</td><td>Shows the elapsed time from the auto test start.</td></tr>
<tr><td>SV3.2.5-Selftest exec count = 0 min</td><td>min</td><td>Shows the number of minutes remaining for the next auto-test.</td></tr>
</table>

EN

# Group SV4: Internal visualization

This group shows the inverter status, whether there are start conditions, drive select information and parameters related to medium voltage control.

## Subgroup SV4.1: Status

| Screen | Units | Description |
|---|---|---|
| **SV4.1.1-Current fault = 0** | NONE | Shows the current fault code in case there is any. To consult fault codes and actions to be taken in case any of them occur, please refer to section 3.<br><br>The coding of the fault is given by a decimal number. After making all the required conversions, first to hexadecimal and then to binary, a sixteen bits register is generated. This register defines the code of the fault and in which module has it occurred. From right to left, the ten first bits (LSB) define the code of the fault and the following six bits (MSB) represent the module.<br>**Note**: if the MSB represents 0, it means that is an inverter fault.<br><br>**Example:**<br>Decimal Modbus value 1127 = Binary value 0000010001100111➔MSB = 1; LSB =103➔ Module number 1 has triggered F103: TIGBT (IGBTs temperature).<br><br>**Note:** if there was more than one fault at the same moment, they will be shown cyclically, one after another, until they are resolved. |
| **SV4.1.1b-Current fault module = 0** | NONE | When a module fault results in a system fault, this parameter shows the number of the module that has produced the original fault. If several modules have produced the fault, it will show the last module that reported it. |
| **SV4.1.2-Current warning = NO WRN** | NONE | Shows the current warning code in case there is any. The full list of warnings and their corresponding code is shown in section 3.<br><br>The procedure to decode and interpret warning messages is the same as the one described in [SV4.1.1-Current fault]. |
| **SV4.1.3-Current status = PUP** | NONE | Shows the current status.<br><br>The value read at the Modbus register is equivalent to the status code. Refer to section 3 for further details about status messages. Status messages for each module are displayed in parameter [SV9.3-Status].<br><br>**Examples**:<br>• Modbus value 4 = REA (Ready).<br>• Modbus value 7 = STOP. |
| **SV4.1.4-Internal tº = 35 ºC** | ºC | Shows the AC module internal temperature. |
| **SV4.1.4b-DU Internal tº = 35 ºC** | ºC | It shows the temperature measured by the DU electronics. |
| **SV4.1.5-Max IGBTs tº = 0 ºC** | ºC | Shows the maximum IGBTs temperature. |
| **SV4.1.6-Max modules tº = 0 ºC** | ºC | Shows the maximum modules internal temperature. |
| **SV4.1.7-Input status = 0000000000** | NONE | Shows the status of the central control board inputs.<br>Each bit (from left to right) represents an input (0 = Not active; 1 or X = Active).<br>**BIT** / **INPUT**<br>1 Auxiliary services transformer thermal<br>2 LCL Contactor feedback<br>3 Inductance thermal<br>4 Not used currently<br>5 AC overvoltage protection<br>6 Selector<br>7 IMI<br>8 GFDI fuse<br>9 External fault<br>10 Hygrostat |

<table>
<tr><th>Screen</th><th>Units</th><th>Description</th></tr>
<tr><td>SV4.1.7b-STO status = 00</td><td>NONE</td><td>Shows the status of the emergency pushbuttons connected to the main control board.<br><table><tr><th>BIT</th><th>STATE</th></tr><tr><td>1</td><td>Emergency pushbutton 1 status</td></tr><tr><td>2</td><td>Emergency pushbutton 2 status</td></tr></table></td></tr>
<tr><td>SV4.1.8-Output status = 00000</td><td>NONE</td><td>Shows the status of outputs.<br>Each bit (from left to right) represents an output (0 = Not active; 1 or X = Active).<br><table><tr><th>BIT</th><th>OUTPUT</th></tr><tr><td>1</td><td>Not used currently</td></tr><tr><td>2</td><td>Not used currently</td></tr><tr><td>3</td><td>Not used currently</td></tr><tr><td>4</td><td>Fault / warning LED</td></tr><tr><td>5</td><td>General fault (AC circuit breaker)</td></tr></table></td></tr>
<tr><td>SV4.1.9-Admission tº = 0.0 ºC</td><td>ºC</td><td>Shows the temperature measured by the sensor located at the air inlet in the AC module.</td></tr>
<tr><td>SV4.1.10-Normative = UL 1741</td><td>NONE</td><td>Shows the current normative.</td></tr>
<tr><td>SV4.1.11-Internal humidity = 50 %</td><td>%</td><td>Shows the Main Control Cabinet module internal humidity.</td></tr>
<tr><td>SV4.1.12-LV manometer = 0</td><td>NONE</td><td>Shows air pressure in the low voltage (LV) ventilation zone.<br>The LV manometer is located next to the cyclone 1.</td></tr>
<tr><td>SV4.1.13-MV manometer = 0</td><td>NONE</td><td>Shows air pressure in the medium voltage (MV) ventilation zone.<br>The MV manometer is located next to the cyclone 2.<br>Note: This parameter only applies to HEM inverters.</td></tr>
<tr><td>SV4.1.14-Power source 1 = 0.00 V</td><td>V</td><td rowspan="2">The central control board has two inputs where power supply can be connected (terminals J8A1 and J8B1).<br><br>These parameters show the voltage measured on the supply and auxiliary transformer cabinet.<br><br>Note: Currently only one supply is connected. Thus, one of these parameters should always display a measurement close to 24V.</td></tr>
<tr><td>SV4.1.15-Power source 2 = 0.00 V</td><td>V</td></tr>
</table>

## Subgroup SV4.2: Available power

| Screen | Units | Description |
|---|---|---|
| **SV4.2.1-P limit inst = 100.00 %** | % | Shows the current active power available in PCS inverters. |
| **SV4.2.2-Q limit inst = 100.00 %** | % | Shows the current reactive power available in PCS inverters. |
| **SV4.2.3-P limit active = No** | NONE | Shows that the inverter has activated a power reduction (derating).<br><br>**Note:** The reduction of the admission temperature is not considered.<br>**Note**: In order to work, some derating limit has to be set to according to the limits defined in G3.1. |
| **SV4.2.3a- P limit Reduction = G3.1.1-P limit %** | % | Power limit value P according to the limits defined in G3.1. |
| **SV4.2.3b- Q limit Reduction = G3.1.2-Q limit %** | % | Power limit value Q according to the limits defined in G3.1. |
| **SV4.2.3c- S limit Reduction = G3.1.4-S limit %** | % | Limit value of power S according to the limits defined in G3.1. |

## Subgroup SV4.3: Start conditions

| Screen | Units | Description |
|---|---|---|
| **SV4.3.1-Start conditions all = No** | NONE | Shows if there are start conditions. |
| **SV4.3.2-Start conditions V = No** | NONE | Shows if there are AC voltage start conditions. |
| **SV4.3.3-Start conditions f = No** | NONE | Shows if there are frequency start conditions. |
| **SV4.3.4-Start conditions Vdc = No** | NONE | Shows if there are DC voltage start conditions. |
| **SV4.3.5-Start cond MV cell = No** | NONE | Shows if there are AC MV cell start conditions (circuit breaker and disconnector closed, earth open).<br>**Note:** This parameter only applies to HEM inverters. |
| **SV4.3.6-Start conditions P = No** | NONE | Shows if there are power start conditions. |
| **SV4.3.7-Start conditions Bbox = No** | NONE | Shows how the Blackbox download has been configured in SG3.7a-AutoBbox download. |

## Subgroup SV4.4: Drive select

### SV4.4.1: General

| Screen | Units | Description |
|---|---|---|
| **SV4.4.1.1-Inverter type = Solar inverter** | NONE | Shows the application of the inverter (Set in SG10.1.4).<br>**Solar Inverter**: inverter for a photovoltaic field<br>**PCSK**: inverter for batteries<br>**Statcom**: Statcom<br>**M-PCS:** Multi batteries |
| **SV4.4.1.1a-Bus Plus= 0** | NONE | Displays whether Bus Plus mode has been selected. |
| **SV4.4.1.1b-Size = Normal** | NONE | Shows the hardware version of the inverter.<br>**OPTION** / **HARDWARE**<br>Normal: HEM Non-repowered inverter V1.<br>PLUS: PLUS repowered inverter V2.<br>1500: PLUS with overcurrent and overvoltage protections, adjustable for 1500 V.<br>P1112 AB: Power stage with ABB IGBTs. |
| **SV4.4.1.2-Nominal AC voltage = 690 V** | V | Shows the rated AC voltage. |
| **SV4.4.1.3-Nominal S = 510.00 kVA** | kVA | Shows the rated apparent power. |
| **SV4.4.1.4-Number of modules = 0** | NONE | Shows the number of modules. |
| **SV4.4.1.5-Number of MPPs = 0** | NONE | Shows the number of MPPs. |
| **SV4.4.1.6-Nominal bus voltage = 1500 V** | V | Shows the rated DC voltage. |
| **SV4.4.1.7-Max AC current = 429 A** | A | Shows the AC current limit. |
| **SV4.4.1.8-Max DC current = 535 A** | A | Shows the DC current limit. |
| **SV4.4.1.9-DC mod contactors = No** | NONE | Shows if there are DC contactors in the modules. |
| **SV4.4.1.10-DU curr measures = No** | NONE | Shows if the equipment has DC current measurements or not. |

### SV4.4.2: Protections

| Screen | Units | Description |
|---|---|---|
| SV4.4.2.1-Overcurr prot low = 950 A | A | Shows the low overcurrent protection threshold. |
| SV4.4.2.2-Overcurr prot low = 830 A | A | Shows the middle overcurrent protection threshold. |
| SV4.4.2.3-Overcurr prot mid 2 = 625 A | A | Shows the second middle overcurrent protection threshold. |
| SV4.4.2.4-Overcurr prot high = 560 A | A | Shows the high overcurrent protection threshold. |

### SV4.4.3: Limits

| Screen | Units | Description |
|---|---|---|
| SV4.4.3.1-Inductive I limit = 357 A | A | Shows the inductive reactive current limit. |
| SV4.4.3.2-Capacitive I limit = 407 A | A | Shows the capacitive reactive current limit. |
| SV4.4.3.3-Max P limit = 100.00 % | % | Shows the maximum P value. This value can be adjusted in SG10.3.3. |
| SV4.4.3.4-Max Q limit = 100.00 % | % | Shows the maximum Q value. This value can be adjusted in SG10.3.4. |
| SV4.4.3.5-Max S limit = 100.00 % | % | Shows the maximum S value. This value can be adjusted in SG10.3.5. |

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### SV4.4.4: Transformer

**Note**: These parameters only apply to HEM inverters.

| Screen | Units | Description |
|---|---|---|
| SV4.4.4.1-Cell type = V | V | Shows the cell protection type. |
| SV4.4.4.2-MV Voltage = 34500 V | V | Shows the MV transformer voltage in volts. |
| SV4.4.4.3-Pn = 3000 kW | kW | Shows the MV transformer active power in kilowatts. |
| SV4.4.4.4-Xcc = 8.00 % | % | Shows the MV transformer short-circuit reactance in percentage. |
| SV4.4.4.5-Pfe = 8.0 kW | kW | Shows the MV transformer no load losses. |
| SV4.4.4.6-Pcu = 25.0 kW | kW | Shows the MV transformer copper losses at full-load current. |
| SV4.4.4.7-Test Frequency = 60 Hz | Hz | Shows the MV transformer frequency in hertz. |
| SV4.4.4.8-Open current = 0.410 % | % | Shows the MV transformer open current in percentage. |

### SV4.4.5: Filter

| Screen | Units | Description |
|---|---|---|
| SV4.4.5.1-R (mOhm) = 5.00 | NONE | Shows the LCL filter resistance in ohms. |
| SV4.4.5.2-L (uH) = 140 | NONE | Shows the modular inductance in microhenries. |
| SV4.4.5.3-C (uF) = 61.50 | NONE | Shows the LCL filter capacitance in microfarads. |
| SV4.4.5.4-Lc (uH) = 2.70 | NONE | Shows the LCL filter inductance in microhenries. |

## Subgroup SV4.5: MV Control

**Note**: These parameters only apply to HEM inverters.

<table>
<tr><th>Screen</th><th>Units</th><th>Description</th></tr>
<tr><td>SV4.5.1-Output status = 00000</td><td>NONE</td><td>Shows the status of MV board outputs.<br>Each bit (from left to right) represents an output (0 = Not active; 1 or X = Active).<br><br><table><tr><th>BIT</th><th>STATUS</th></tr><tr><td>1</td><td>O1: MV cell close (J9).</td></tr><tr><td>2</td><td>O2: MV cell open (J10).</td></tr><tr><td>3</td><td>O3: Auxiliary digital output 1 (J11). Fans in the medium voltage transformer area.</td></tr><tr><td>4</td><td>O4: Auxiliary digital output 2 (J12). Not used currently.</td></tr><tr><td>5</td><td>O5: Auxiliary digital output 3 (J13). Not used currently.</td></tr></table></td></tr>
<tr><td>SV4.5.2-Input status = 00000000000000</td><td>NONE</td><td>Shows the status of MV board inputs.<br>Each bit (from left to right) represents an input (0 = Not active; 1 or X = Active).<br><br><table><tr><th>BIT</th><th>STATUS</th><th>BIT</th><th>STATUS</th></tr><tr><td>1</td><td>I1: Cell 1 open (J14:2) Not used currently.</td><td>8</td><td>I8: Cell protection close (J15:2)</td></tr><tr><td>2</td><td>I2: Cell 1 closed (J14:3) Not used currently.</td><td>9</td><td>I9: Cell protection earth close (J15:3)</td></tr><tr><td>3</td><td>I3: Cell 1 earth (J14:4) Not used currently.</td><td>10</td><td>I10: Cell protection earth open (J15:4)</td></tr><tr><td>4</td><td>I4: Cell protection open (52) (J14:6)</td><td>11</td><td>I11: OV protection (J15:6) Not used currently.</td></tr><tr><td>5</td><td>I5: Cell protection close (52) (J14:7)</td><td>12</td><td>I12: Fan feedback (J15:8) Not used currently.</td></tr><tr><td>6</td><td>I6: Spare input (J14:8)</td><td>13</td><td>I13: MV smoke detector feedback (J16:1)</td></tr><tr><td>7</td><td>I7: Cell protection open (J15:1)</td><td>14</td><td>I14: Spare input (J16:3)</td></tr></table><br>Note: Inputs 1, 2 and 3 are not used currently because V cells are being used.</td></tr>
<tr><td>SV4.5.3-PT100 (1) = 0 ºC</td><td>ºC</td><td>Shows the MV transformer PT100 probe temperature 1.</td></tr>
<tr><td>SV4.5.4-PT100 (2) = 0 ºC</td><td>ºC</td><td>Shows the MV transformer PT100 probe temperature 2.</td></tr>
<tr><td>SV4.5.5-PT100 (3) = 0 ºC</td><td>ºC</td><td>Shows the MV transformer PT100 probe temperature 3.</td></tr>
<tr><td>SV4.5.6-Status Gen. Switch = Unknown</td><td>NONE</td><td rowspan="3">Based on the status of the inputs and outputs (SV4.5.1 and 4.5.2) it is possible to determine the status of the MV switch disconnector, the MV earth switch and the MV automatic interrupter. They are shown in parameters SV4.5.6, SV4.5.7 and SV4.5.8 respectively.<br><br><table><tr><th>OPTION</th><th>STATUS</th></tr><tr><td>0</td><td>Unknown. The status cannot be determined.</td></tr><tr><td>1</td><td>Open</td></tr><tr><td>2</td><td>Closed</td></tr></table></td></tr>
<tr><td>SV4.5.7-Status Earth Switch = Unknown</td><td>NONE</td></tr>
<tr><td>SV4.5.8-Status Auto Int. = Unknown</td><td>NONE</td></tr>
<tr><td>SV4.5.9-Premagn. Status = Off</td><td>NONE</td><td>Shows the status of premagnetization:<br><br><table><tr><th>OPTION</th><th>STATUS</th></tr><tr><td>0</td><td>Off</td></tr><tr><td>1</td><td>Checks MV cell conditions</td></tr><tr><td>2</td><td>Checks frequency conditions.</td></tr><tr><td>3</td><td>Checks voltage conditions.</td></tr><tr><td>4</td><td>Closes LC contactor</td></tr><tr><td>5</td><td>Starts 2 modules</td></tr><tr><td>6</td><td>Increases voltage to reference</td></tr><tr><td>7</td><td>Checks voltages</td></tr><tr><td>8</td><td>Closes MV switch</td></tr><tr><td>9</td><td>Done</td></tr><tr><td>10</td><td>Failed</td></tr></table></td></tr>
</table>

## Subgroup SV4.6: Modulation

| Screen | Units | Description |
|---|---|---|
| **SV4.6.1-Current modulation = Double** | NONE | Shows the current modulation mode. |
| **SV4.6.2-Current modul changes = 0** | NONE | Shows the number of changes in modulation that have taken since the first change from single to double. |
| **SV4.6.3-Forced Double signal = 0** | NONE | Shows the reason for the sudden modulation change from single to double.<br><br>The reasons may be the following:<br>- A LVRT (low voltage ride through) disturbance is detected.<br>- Maximum reactive power level (Q) in single signal is exceeded.<br>- An unbalance in the DSP is detected.<br>- A peak of active power (P) or reactive power (Q) is detected.<br>- An unbalance in the FPGA is detected.<br>- Starting a new module.<br>- An excessive number of changes in modulation is detected.<br>- The limit of the single signal is reached. |
| **SV4.6.4-Commutation frequency = 0 Hz** | NONE | Shows the commutation frequency. |

EN

# Group SV5: Counters

This group displays the historical information about energy, run time (in days, months and years) and grid connections.

Please notice that, due to space limitations of the display, groups SV5.1 and SV5.2 will only show data from one module at a time. These counters are network connection counters, when the modular AC contactors close they start counting and the IGBTs start switching. To view data from several modules at a time, use Modbus communication and refer to section 7 of this manual.

**Note:**

The value of energy counters is shown as a 32-bits DWord, two 16-bits words (MSB and LSB).

The high byte (MSB) corresponds with the lowest Modbus address and the low byte (LSB) corresponds with the highest Modbus address. Thus, to correctly read the value, perform the calculation:

*V = MSB * 2^16 + LSB*

**Example**: Parameter [SV5.1.2-Daily energy]. The MSB corresponds to Modbus address 41202 and the LSB to address 41203. Thus:

*Daily energy = [Value read in 41202] * 2^16 + [Value read in 41203]*

## Subgroup SV5.1: Active energy

| Screen | Units | Description |
|---|---|---|
| **SV5.1.1-Select module = All** | NONE | Allows selecting all or a specific module to visualize data related to active energy. |
| **SV5.1.2-Daily energy = 0 kWh** | kWh | Shows the amount of active energy, in kilowatt-hour (kWh), generated in a day. |
| **SV5.1.3-Yesterday energy = 0 kWh** | kWh | Shows the amount of active energy, in kWh, generated in the day before the present. |
| **SV5.1.4-Month energy = 0 kWh** | kWh | Shows the amount of active energy, in kWh, generated in a month. |
| **SV5.1.5-Last month energy = 0 kWh** | kWh | Shows the amount of active energy, in kWh, generated in the month before the present. |
| **SV5.1.6-Total energy = 0 kWh** | kWh | Shows the total amount of energy, in kWh, generated since commissioning. |
| **SV5.1.7-Consum daily energy = 0 kWh** | kWh | Shows the amount of active energy, in kWh, consumed in a day. |
| **SV5.1.8-Consum total energy = 0 kWh** | kWh | Shows the amount of active energy, in kWh, consumed since commissioning. |

## Subgroup SV5.2: Reactive energy

| Screen | Units | Description |
|---|---|---|
| **SV5.2.1-Select module = All** | NONE | Allows selecting all or a specific module to visualize data related to reactive energy. |
| **SV5.2.2-Daily reactive = 0 kVArh** | kVArh | Shows the amount of reactive energy, in kilo volt amperes reactive hours (kVArh), generated in a day |
| **SV5.2.3-Yesterday reactive = 0 kVArh** | kVArh | Shows the amount of reactive energy, in kVArh, generated in the day before the present. |
| **SV5.2.4-Month reactive = 0 kVArh** | kVArh | Shows the amount of reactive energy, in kVArh, generated in a month. |

| Screen | Units | Description |
|---|---|---|
| SV5.2.5-Last month reactive = 0 kVArh | kVArh | Shows the amount of reactive energy, in kVArh, generated in the month before the present. |
| SV5.2.6-Total reactive = 0 kVArh | kVArh | Shows the amount of reactive energy, in kVArh, generated since commissioning. |

## Subgroup SV5.3: Run time

| Screen | Units | Function |
|---|---|---|
| SV5.3.1-Total run years = 0 | NONE | Total number of years the inverter has been running. |
| SV5.3.2-Total run days = 0 | NONE | Total number of days the inverter has been running. |
| SV5.3.3-Total run hours = 0 | NONE | Hours in total that the inverter has been running. |
| SV5.3.4-Partial run years = 0 | NONE | Partial run time years. |
| SV5.3.5-Partial run days = 0 | NONE | Partial run time days. |
| SV5.3.6-Partial run hours = 0 | NONE | Partial run time hours. |

EN

## Subgroup SV5.4: Grid connections

| Screen | Units | Function |
|---|---|---|
| SV5.4.1-Total grid conns = 0 | NONE | Total number of grid connections.<br>This parameter is increased in 1 every time that the inverter connects to the grid (ideally, it woud be once every day). To reset it, use parameter SG7.3. |
| SV5.4.2-Partial grid conns = 0 | NONE | Partial number of grid connections.<br>This parameter counts the number of connections to the grid since the last time SG7.4 was reset. |

## Subgroup SV5.5: MV Operations

| Screen | Units | Description |
|---|---|---|
| SV5.5.1-Loaded Closures = 0 | NONE | MV disconnector closure counter with MV Switchgear pre-magnetization. |
| SV5.5.2-Unloaded Closures = 0 | NONE | MV disconnector closure counter without MV Switchgear pre-magnetization. |
| SV5.5.3-Openings = 0 | NONE | MV Switchgear disconnector openings counter. |

## Subgroup SV5.6: Special faults

Counters for special faults.

The failure counting system continues to count special failures, even if the inverter is in fault state, but it only counts failures if they have not occurred before. Once the inverter is reset, faults will be counted again, even if they occurred before the reset.

In case of requiring to reset the special fault counters, parameter [SG7.12-Rst special faults] has been enabled to do so.

| Screen | Units | Description |
|---|---|---|
| SV5.6.1-Emergency stops = 0 | NONE | Emergency stop counter. If the emergency stop is not in special faults, SV5.6.1 will not be incremented even if the fault occurs. |
| SV5.6.2-Other special faults = 0 | NONE | Counter of other special faults. |

# Group SV6: Faults

## Subgroup SV6.1: Fault history

This group shows the last 10 faults that have been triggered.

| Screen | Description | Function |
|---|---|---|
| SV6.1.1-Last fault = No faults | Register number 1 of fault history | Displays a list of the last ten inverter faults, in chronological order.<br><br>The most recent fault is shown first and stored in parameter SV6.1.1-Last fault. Whenever a new fault occurs, it will be placed in the top of the list and older fault messages will drop a position (the oldest fault message will be discarded and registry SV6.1.19-First fault replaced with the value that was stored in SV6.1.17-Second fault).<br><br>The inverter is reset by pressing the STOP-RESET button on the display or using an external reset if it is available. Certain failures may rearm automatically using the Auto Reset.<br><br>Please refer to section 3 for details about faults and troubleshooting. |
| SV6.1.2-Date = 01/01/2000 00:00 | Date and time of the register number 1 of fault history | |
| SV6.1.3-Ninth fault = No faults | Register number 2 of fault history | |
| SV6.1.4-Date = 01/01/2000 00:00 | Date and time of the register number 2 of fault history | |
| SV6.1.5-Eighth fault = No faults | Register number 3 of fault history | |
| SV6.1.6-Date = 01/01/2000 00:00 | Date and time of the register number 3 of fault history | |
| SV6.1.7-Seventh fault = No faults | Register number 4 of fault history | |
| SV6.1.8-Date = 01/01/2000 00:00 | Date and time of the register number 4 of fault history | |
| SV6.1.9-Sixth fault = No faults | Register number 5 of fault history | |
| SV6.1.10-Date = 01/01/2000 00:00 | Date and time of the register number 5 of fault history | |
| SV6.1.11-Fifth fault = No faults | register number 6 of fault history | |
| SV6.1.12-Date = 01/01/2000 00:00 | Date and time of the register number 6 of fault history | |
| SV6.1.13-Fourth fault = No faults | Register number 7 of fault history | |
| SV6.1.14-Date = 01/01/2000 00:00 | Date and time of the register number 7 of fault history | |
| SV6.1.15-Third fault = No faults | Register number 8 of fault history | |
| SV6.1.16-Date = 01/01/2000 00:00 | Date and time of the register number 8 of fault history | |
| SV6.1.17-Second fault = No faults | Register number 9 of fault history | |
| SV6.1.18-Date = 01/01/2000 00:00 | Date and time of the register number 9 of fault history | |
| SV6.1.19-First fault = No faults | Register number 10 of fault history | |
| SV6.1.20-Date = 01/01/2000 00:00 | Date and time of the register number 10 of fault history | |

**Note:** to read correctly the date and time real values, perform the following operations:

Date real value:
- Day = *(Modbus value AND 0xF800) / 2^11*
- Month = *(Modbus value AND 0x0780) / 2^7*
- Year = *(Modbus value AND 0x007F)*

Time real value:
- Hours = *(Modbus value AND 0xFF00) / 2^8*
- Minutes = *(Modbus value AND 0x00FF)*

**Note:** Only the last 10 faults are shown in the display, but the COMPLETE fault history is stored in the SD card, in case user needs to consult it later. Notice that the SD stores the fault + when it occurred, but nothing is recorded once the fault disappears.

## Subgroup SV6.2: Fault status

These parameters show the status of each fault (0 = Not active; X = Active) in groups of sixteen faults (being the lowest number the bit to the upmost right).

| Screen | Units | Description |
|---|---|---|
| SV6.2.1-Fault 1 to 16 = 0000000000000000 | None | Parameter [SV6.2.1-Fault 1 to 16] displays the status of faults 1 to 16.<br>Parameter [SV6.2.2-Fault 17 to 32] displays the status of faults 17 to 32.<br>And so on, until [SV6.2.19-Fault 289 to 299], which shows the status of faults 289 to 299. |
| SV6.2.2-Fault 17 to 32 = 0000000000000000 | | |
| SV6.2.3-Fault 33 to 48 = 0000000000000000 | | |
| SV6.2.4-Fault 49 to 64 = 0000000000000000 | | |
| SV6.2.5-Fault 65 to 80 = 0000000000000000 | | |
| SV6.2.6-Fault 81 to 96 = 0000000000000000 | | |
| SV6.2.7-Fault 97 to 112 = 0000000000000000 | | |
| SV6.2.8-Fault 113 to 128 = 0000000000000000 | | |
| SV6.2.9-Fault 129 to 144 = 0000000000000000 | | |
| SV6.2.10-Fault 145 to 160 = 0000000000000000 | | |
| SV6.2.11-Fault 161 to 176 = 0000000000000000 | | |
| SV6.2.12-Fault 177 to 192 = 0000000000000000 | | |
| SV6.2.13-Fault 193 to 208 = 0000000000000000 | | |
| SV6.2.14-Fault 209 to 224 = 0000000000000000 | | |
| SV6.2.15-Fault 225 to 240 = 0000000000000000 | | |
| SV6.2.16-Fault 241 to 256 = 0000000000000000 | | |
| SV6.2.17-Fault 257 to 272 = 0000000000000000 | | |
| SV6.2.18-Fault 273 to 288 = 0000000000000000 | | |
| SV6.2.19-Fault 289 to 299 = 00000000000 | | |

EN

# Group SV7: Warnings

## Subgroup SV7.1: Warning history

This group shows the last 10 warnings that have been detected by the inverter.

| Screen | Description | Function |
|---|---|---|
| SV7.1.1-Last warning = 0 | Register number 1 of the warning history | Displays a list of the last ten warnings, in chronological order.<br><br>When a warning occurs, it will be displayed on the status bar flashing with the current state. It will remain being displayed until the problem is fixed.<br><br>The most recent warning is shown first and stored in parameter SV7.1.1-Last warning. Whenever a new fault occurs, it will be placed in the top of the list and older fault messages will drop a position (the oldest fault message will be discarded and registry SV7.1.19-First warning replaced with the value that was stored in SV7.1.17-Second warning).<br><br>The list of all possible warnings and their codes is shown in section 3.<br><br>**Note**: check the instructions to read correctly the date and time real values in SV6.1: Fault history. |
| SV7.1.2-Date = 01/01/2000 00:00 | Date and time of the register number 1 of warning history | |
| SV7.1.3-Ninth warning = 0 | Register number 2 of the warning history | |
| SV7.1.4-Date = 01/01/2000 00:00 | Date and time of the register number 2 of warning history | |
| SV7.1.5-Eighth warning = 0 | Register number 3 of the warning history | |
| SV7.1.6-Date = 01/01/2000 00:00 | Date and time of the register number 3 of warning history | |
| SV7.1.7-Seventh warning = 0 | Register number 4 of the warning history | |
| SV7.1.8-Date = 01/01/2000 00:00 | Date and time of the register number 4 of warning history | |
| SV7.1.9-Sixth warning = 0 | Register number 5 of the warning history | |
| SV7.1.10-Date = 01/01/2000 00:00 | Date and time of the register number 5 of warning history | |
| SV7.1.11-Fifth warning = 0 | Register number 6 of the warning history | |
| SV7.1.12-Date = 01/01/2000 00:00 | Date and time of the register number 6 of warning history | |
| SV7.1.13-Fourth warning = 0 | Register number 7 of the warning history | |
| SV7.1.14-Date = 01/01/2000 00:00 | Date and time of the register number 7 of warning history | |
| SV7.1.15-Third warning = 0 | Register number 8 of the warning history | |
| SV7.1.16-Date = 01/01/2000 00:00 | Date and time of the register number 8 of warning history | |
| SV7.1.17-Second warning = 0 | Register number 9 of the warning history | |
| SV7.1.18-Date = 01/01/2000 00:00 | Date and time of the register number 9 of warning history | |
| SV7.1.19-First warning = 0 | Register number 10 of the warning history | |
| SV7.1.20-Date = 01/01/2000 00:00 | Date and time of the register number 10 of warning history | |

**Note:** Same as with faults, the last 10 warnings are shown in the display, but the COMPLETE warning history is stored in the SD card, in case user needs to consult it later. Notice that the SD stores the warning + when it occured, but nothing is recorded once the warning disappears.

## Subgroup SV7.2: Warning status

These parameters show the status of each warning (0 = Not active; X = Active) in groups of sixteen warnings (being the lowest number the bit to the upmost right).

| Screen | Units | Description |
|---|---|---|
| SV7.2.1-Warning 1 to 16 = 0000000000000000 | None | Parameter [SV7.2.1-Warning status 1 to 16] displays the status of warnings 1 to 16.<br>Parameter [SV7.2.2-Warning status 17 to 32] displays the status of warnings 17 to 32.<br>And so on, until [SV7.2.19-Warning status 289 to 299], which shows the status of warnings 289 to 299. |
| SV7.2.2-Warning 17 to 32 = 0000000000000000 | | |
| SV7.2.3-Warning 33 to 48 = 0000000000000000 | | |
| SV7.2.4-Warning 49 to 64 = 0000000000000000 | | |
| SV7.2.5-Warning 65 to 80 = 0000000000000000 | | |
| SV7.2.6-Warning 81 to 96 = 0000000000000000 | | |
| SV7.2.7-Warning 97 to 112 = 0000000000000000 | | |
| SV7.2.8-Warning 113 to 128 = 0000000000000000 | | |
| SV7.2.9-Warning 129 to 144 = 0000000000000000 | | |
| SV7.2.10-Warning 145 to 160 = 0000000000000000 | | |
| SV7.2.11-Warning 161 to 176 = 0000000000000000 | | |
| SV7.2.12-Warning 177 to 192 = 0000000000000000 | | |
| SV7.2.13-Warning 193 to 208 = 0000000000000000 | | |
| SV7.2.14-Warning 209 to 224 = 0000000000000000 | | |
| SV7.2.15-Warning 225 to 240 = 0000000000000000 | | |
| SV7.2.16-Warning 241 to 256 = 0000000000000000 | | |
| SV7.2.17-Warning 257 to 272 = 0000000000000000 | | |
| SV7.2.18-Warning 273 to 288 = 0000000000000000 | | |
| SV7.2.19-Warning 289 to 299 = 00000000000 | | |
| SV7.2.20-Deratings register = 0000 | None | Shows whether a derating is taking place. Each bit corresponds to a derating and it will show 0 = Not active or 1 = Active. The bit 0 is the one to the upmost right.<br><br>W11:Derating DC bus V = bit 0<br>W16:Derating IGBT T. = bit 1<br>W36: Derating admission T. = bit 2<br>W37:Derating frequency = bit 3<br><br>**Note:** this parameter will be shown when the derating cause monitoring is enabled internally [SG1.1.22-Cause derat. Monitor = Yes. |

**Others:**

| Screen | Units | Description |
|---|---|---|
| SV7.3-Warnings/Faults counter = 0 | NONE | Faults and warnings counter.<br>**Note:** this parameter will be shown if the counter has been enabled internally (SG1.1.21-F/W bit display = Yes). |

# Group SV8: Events History

This group shows the last 10 events that have been detected by the inverter.

<table>
<tr><th>Screen</th><th>Description</th><th>Function</th></tr>
<tr><td>SV8.1-Last event = No event</td><td>Register number 1 of the event history</td><td rowspan="21">Displays a list of the last ten events, in chronological order.<br><br>When an event occurs, it will be displayed on the status bar flashing with the current state. It will remain being displayed until the problem is fixed.<br><br>The most recent event is shown first and stored in parameter SV8.1-Last event. Whenever a new event occurs, it will be placed in the top of the list and older event messages will drop a position (the oldest event message will be discarded and registry SV8.10-Date replaced with the value that was stored in SV8.9-Sixth event).<br><br>The following table shows all possible events and their codes:<br><br><table><tr><th>COD</th><th>EVENT</th></tr><tr><td>0</td><td>No event</td></tr><tr><td>1</td><td>Emergency stop</td></tr><tr><td>2</td><td>Remote emergency stop</td></tr></table></td></tr>
<tr><td>SV8.2-Date = 01/01/2000 00:00</td><td>Date and time of the register number 1 of event history</td></tr>
<tr><td>SV8.3-Ninth event = No event</td><td>Register number 2 of the event history</td></tr>
<tr><td>SV8.4-Date = 01/01/2000 00:00</td><td>Date and time of the register number 2 of event history</td></tr>
<tr><td>SV8.5-Eighth event = No event</td><td>Register number 3 of the event history</td></tr>
<tr><td>SV8.6-Date = 01/01/2000 00:00</td><td>Date and time of the register number 3 of event history</td></tr>
<tr><td>SV8.7-Seventh event = No event</td><td>Register number 4 of the event history</td></tr>
<tr><td>SV8.8-Date = 01/01/2000 00:00</td><td>Date and time of the register number 4 of event history</td></tr>
<tr><td>SV8.9-Sixth event = No event</td><td>Register number 5 of the event history</td></tr>
<tr><td>SV8.10-Date = 01/01/2000 00:00</td><td>Date and time of the register number 5 of event history</td></tr>
<tr><td>SV8.11-Fifth event = No event</td><td>Register number 6 of the event history</td></tr>
<tr><td>SV8.12-Date = 01/01/2000 00:00</td><td>Date and time of the register number 6 of event history</td></tr>
<tr><td>SV8.13-Fourth event = No event</td><td>Register number 7 of the event history</td></tr>
<tr><td>SV8.14-Date = 01/01/2000 00:00</td><td>Date and time of the register number 7 of event history</td></tr>
<tr><td>SV8.15-Third event = No event</td><td>Register number 8 of the event history</td></tr>
<tr><td>SV8.16-Date = 01/01/2000 00:00</td><td>Date and time of the register number 8 of event history</td></tr>
<tr><td>SV8.17-Second event = No event</td><td>Register number 9 of the event history</td></tr>
<tr><td>SV8.18-Date = 01/01/2000 00:00</td><td>Date and time of the register number 9 of event history</td></tr>
<tr><td>SV8.19-First event = No event</td><td>Register number 10 of the event history</td></tr>
<tr><td>SV8.20-Date = 01/01/2000 00:00</td><td>Date and time of the register number 10 of event history</td></tr>
</table>

# Group SV9: Modules Information

This groups shows information about modules.

Subgroup SV9.1 shows general information: how many and which modules are running, and the number of the next module to be started.

More detailed information is shown in subgroup SV9.2, where the user can select a module and all its related information (status, values of current and temperature, etc.) will be displayed.

## Subgroup SV9.1: General information

| Screen | Units | Description |
|---|---|---|
| **SV9.1.1-Running modules = 000000** | NONE | Shows which modules are running (0 = stopped, 1= running).<br>Each bit represents a module, Module 1 is the first from the right. |
| **SV9.1.2-Num running modules = 0** | NONE | Total number of modules running. |
| **SV9.1.3-Next starting module = Module 1** | NONE | Shows the number of the module that will start next. |
| **SV9.1.4-Modules start stop = 000000** | NONE | Register for internal use. |
| **SV9.1.5-Modules enable reg = 000000** | NONE | Register for internal use. |
| **SV9.1.6-Num Mod Started DSP = 0** | NONE | Number of modules started on DSP.<br>**Note**: this parameter is only displayed for users with role Developer. |
| **SV9.1.7-Hw modules version = 0000000000** | NONE | Shows the FPGA version of each module. Each bit (from right to left) represents a module.<br><br>The number of bits displayed corresponds to the number of modules in the product (SV4.4.1.4). If the product has 4 modules, 4 bits will be displayed in this parameter, and so on. The value of the bit indicates the model of FPGA of each module:<br>• 0 = FPGA 35T.<br>• 1 = FPGA 50T. |

## Information about the selected module

Parameters SV9.2 to SV9.26 show information related to the module selected in [SV9.2-Selected module].

If no module has been selected yet, information from the first running module will be displayed.

| Screen | Units | Description |
|---|---|---|
| **SV9.2-Selected module = Module 1** | NONE | Allows selecting a module for which its related information is displayed. |
| **SV9.3-Status = 0** | NONE | Shows the module status.<br>**Note**: Refer to section 3 for the list of module statuses. |
| **SV9.4-Current R = 0.0 A** | A | Show the value of module current per phase (L1, L2 and L3, respectively).<br><br>**Note:** In these three parameters, if **SG1.3.3 = Yes,** the real measured value will be shown, even when the module is stopped. If, on the other hand, **SG1.3.3 = No,** they will always show 0.0. This is useful whenever there is a problem in the current measure, to identify in which phase is the failure. |
| **SV9.5-Current S = 0.0 A** | A | |
| **SV9.6-Current T = 0.0 A** | A | |
| **SV9.7-DC current = 0.0 A** | A | Shows the value of input DC current of the module. |
| **SV9.8-DC voltage (P) = 0 V** | V | In the 3-level power stage, the DC bus is divided in two halves.<br><br>**SV9.8** shows the positive DC voltage value ($V_{BUS+}$ – 0V) and **SV9.9** shows the negative DC voltage value (0V – $V_{BUS-}$). |
| **SV9.9-DC voltage (N) = 0 V** | V | |
| **SV9.10-DC voltage = 0 V** | V | Shows the medium DC voltage value. |
| **SV9.11-P = 0.0 kW** | kW | Shows the value of active power. |

<table>
<tr><th>Screen</th><th>Units</th><th>Description</th></tr>
<tr><td>SV9.12-Q = 0.0 kVAr</td><td>kVAr</td><td>Shows the value of reactive power.</td></tr>
<tr><td>SV9.13-Ambient temperature = 0 ºC</td><td>ºC</td><td>Shows the ambient temperature in Celsius.</td></tr>
<tr><td>SV9.14-Max temperature = 0 ºC</td><td>ºC</td><td>Shows the maximum IGBT temperature of the selected module.</td></tr>
<tr><td>SV9.15-I/O status = 000000000000</td><td>NONE</td><td>Allows knowing the status of contactors (AC main, AC softcharge, LCL and DC switch), fans and thermal LCL.<br><br>Each bit (from right to left) represents an input or output, as described in the following table.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>1</td><td>O: Fan</td><td>Shows the status of the fan contactor<br>(0: Open, 1: Closed).</td></tr>
<tr><td>2</td><td>O: AC main</td><td>Shows the status of the main AC contactor<br>(0: Open, 1: Closed).</td></tr>
<tr><td>3</td><td>O: AC softcharge</td><td>Shows the status of the AC soft charge contactor<br>(0: Open, 1: Closed).</td></tr>
<tr><td>4</td><td>O: LCL</td><td>Shows the status of the LCL filter contactor<br>(0: Open, 1: Closed).</td></tr>
<tr><td>5</td><td>O: DC switch</td><td>Shows the status of the DC contactor<br>(0: Open, 1: Closed).</td></tr>
<tr><td>6</td><td>Reserved</td><td>-</td></tr>
<tr><td>7</td><td>I: Thermal LCL</td><td>Shows the status of the LCL filter contactor<br>(0: Open, 1: Closed)<br><br>If it is open, it indicates that there is overheating in the coil.</td></tr>
<tr><td>8</td><td>I: Fan</td><td>Shows the fan contactor feedback.<br>(0: Open, 1: Closed).</td></tr>
<tr><td>9</td><td>I: AC main</td><td>Shows the AC main contactor feedback.<br>(0: Open, 1: Closed).</td></tr>
<tr><td>10</td><td>I: AC softcharge</td><td>Shows the AC soft charge contactor feedback.<br>(0: Open, 1: Closed).</td></tr>
<tr><td>11</td><td>I: LCL</td><td>Shows the LCL filter contactor feedback.<br>(0: Open, 1: Closed).</td></tr>
<tr><td>12</td><td>I: DC switch</td><td>Shows the DC contactor feedback.<br>(0: Open, 1: Closed).</td></tr>
</table></td></tr>
<tr><td>SV9.16-Hw version = 35T</td><td>NONE</td><td>Shows the hardware version (hexadecimal value) of the FPGA (35T or 50T).</td></tr>
<tr><td>SV9.17-Sw version = 0.0.0-0</td><td>NONE</td><td>Shows the firmware version (hexadecimal value).<br><br>The value of this parameter is shown as a 32-bits DWord, two 16-bits words (MSB and LSB). The high byte (MSB) corresponds with the lowest Modbus address and the low byte (LSB) corresponds with the highest Modbus address. Thus, to correctly read the value, perform the calculation:<br>V = MSB * 2^16 + LSB</td></tr>
<tr><td>SV9.18-Temperature R1 = 0 ºC</td><td>ºC</td><td>Displays the temperature, in Celsius, of IGBT R1.</td></tr>
<tr><td>SV9.19-Temperature R2 = 0 ºC</td><td>ºC</td><td>Displays the temperature, in Celsius, of IGBT R2.</td></tr>
<tr><td>SV9.20-Temperature R3 = 0 ºC</td><td>ºC</td><td>Displays the temperature, in Celsius, of IGBT R3.</td></tr>
<tr><td>SV9.21-Temperature S1 = 0 ºC</td><td>ºC</td><td>Displays the temperature, in Celsius, of IGBT S1.</td></tr>
<tr><td>SV9.22-Temperature S2 = 0 ºC</td><td>ºC</td><td>Displays the temperature, in Celsius, of IGBT S2.</td></tr>
<tr><td>SV9.23-Temperature S3 = 0 ºC</td><td>ºC</td><td>Displays the temperature, in Celsius, of IGBT S3.</td></tr>
<tr><td>SV9.24-Temperature T1 = 0 ºC</td><td>ºC</td><td>Displays the temperature, in Celsius, of IGBT T1.</td></tr>
<tr><td>SV9.25-Temperature T2 = 0 ºC</td><td>ºC</td><td>Displays the temperature, in Celsius, of IGBT T2.</td></tr>
<tr><td>SV9.26-Temperature T3 = 0 ºC</td><td>ºC</td><td>Displays the temperature, in Celsius, of IGBT T3.</td></tr>
<tr><td>SV9.26a-Humidity= 50 %</td><td>%</td><td>Shows the percentage of humidity.</td></tr>
</table>

# Group SV10: DU Measures

This group shows the measured current and voltage for all DU channels (one at a time, depending on the selected channel in [SV10.1]), as well as the number of contactors closed and isolated channels. Please notice it will not be shown in STATCOM.

**Note:** Due to space limitations, it is not possible to visualize multiple channels information in the display. To view the data from several channels at a time, Modbus communication must be used. See section 7.

| Screen | Units | Function |
|---|---|---|
| **SV10.1-Selected channel = 1** | NONE | Shows which DU channel is selected. |
| **SV10.2-Measured current = 0.0 A** | A | Shows the measured current for the selected channel. |
| **SV10.3-Measured voltage = 0.0 V** | V | Shows the measured voltage for the selected channel.<br>**Note:** This parameter will only be shown if G10.2b Enabled NEC = NEC 2014. |
| **SV10.4-Contactors closed = 0** | NONE | Shows the number of contactors closed for the selected channel.<br>**Note**: This parameter will only be shown if the DU type = "Modular". |
| **SV10.5-Isolated channels = 0** | NONE | Shows the number of isolated channels.<br>**Note**: This parameter will only be shown if the DU type = "Modular". |
| **SV10.4-Vdc 1 = 0.0 V** | V | Shows the busbar one DC voltage measure.<br>**Note**: This parameter will only be shown if DU type = "HEM" and for NEC 2017 inverter. |
| **SV10.5-Vdc 2 = 0.0 V** | V | Shows the busbar two DC voltage measure.<br>**Note**: This parameter will only be shown if DU type = "HEM" and for NEC 2017 inverter. |
| **SV10.6-Summation Current = 0.0 A** | A | Allows the summation of currents to be displayed. |
| **SV10.6a-Current DU breaker1 = 0 A** | A | Sum of currents through the Q8 DC switch.<br>Current measurement read by the electronic boards E201 and E203 |
| **SV10.6b-Current DU breaker2 = 0 A** | A | The sum of the currents through the Q9 DC switch.<br>Current measurement read by the the electronic boards E202 and E204 |

EN

# Group SV11: Algorithms

This group shows information about the algorithms run by the inverter.

## Subgroup SV11.1: MPPt

This algorithm searches for the maximum power point of the photovoltaic arrays connected to the inverter.

| Screen | Units | Description |
|---|---|---|
| **SV11.1.1-Status = Off** | NONE | Shows the status of MPPt algorithm:<br>0: Off (deactivated)<br>1: Init (initializing)<br>2: On (activated) |
| **SV11.1.2-Irradiation = Still** | NONE | MPPt irradiation<br>0: Still<br>1: Rise<br>2: Fall |
| **SV11.1.3-Delta irradiation = 0.0 %** | % | MPPt delta irradiation. |
| **SV11.1.4-Current action = 0.0 V** | V | MPPt current action. |
| **SV11.1.5-Current reference = 0.0 V** | V | MPPt current reference. |

## Subgroup SV11.2: Trackers Mode

This alternative supply mode allows a module to act as an uninterrupted supply system to provide power for photovoltaic trackers and auxiliary services circuits.

| Screen | Units | Description |
|---|---|---|
| **SV11.2.1-Status = 0** | NONE | Shows the status of trackers mode:<br>0: Off<br>1: Activation sequence/Activated<br>2: Deactivation sequence |
| **SV11.2.2-Activation signal = 0** | NONE | Shows whether activation conditions are met:<br>0: No<br>1: Yes |
| **SV11.2.3-K1 contactor fb = 0** | NONE | Shows K1 contactor feedback:<br>0: Deactivated<br>1: Activated |
| **SV11.2.4-Voltage UV = 0 V** | V | Shows UV voltage measure, in volts, of the auxiliary services power supply circuit. |
| **SV11.2.5-Voltage VW = 0 V** | V | Shows VW voltage measure, in volts, of the auxiliary services power supply circuit. |
| **SV11.2.6-Voltage WU = 0 V** | V | Shows WU voltage measure, in volts, of the auxiliary services power supply circuit. |

## Subgroup SV11.3: Heating

These parameters show the status of the algorithm for **Heating**. For further information about heating mode, see the descriptions of parameter subgroup G4.6.

| Screen | Units | Description |
|---|---|---|
| **SV11.3.1-Status = 0** | NONE | Shows the status of the Heating algorithm. |
| **SV11.3.2-Hygrostat signal = No** | NONE | Shows whether the hygrostat signal is detected:<br>0: No<br>1: Yes |
| **SV11.3.3-Temperature cond = No** | NONE | Shows whether temperature conditions to run Heating algorithm are met:<br>0: No<br>1: Yes |
| **SV11.3.4-Activation cond = No** | NONE | Shows whether the conditions to activate the Heating algorithm are met:<br>0: No<br>1: Yes |

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## Subgroup SV11.4: Night mode

These parameters show the status of the algorithm for **Night mode**. For further information about night mode operation, see the descriptions of parameter subgroup G4.7.

<table>
<tr><th>Screen</th><th>Units</th><th>Description</th></tr>
<tr><td>SV11.4.1-Status = 0</td><td>NONE</td><td>Shows the status of the night mode algorithm.</td></tr>
<tr><td>SV11.4.2-Activation signal = No</td><td>NONE</td><td>Shows whether night mode activation signal has been received:<br>0: No<br>1: Yes</td></tr>
<tr><td>SV11.4.3-Activation cond = No</td><td>NONE</td><td>Shows whether night mode activation conditions are met:<br>0: No<br>1: Yes</td></tr>
<tr><td>SV11.4.4-Autonight status = A_NIGHT_OFF</td><td>NONE</td><td>Shows the status of the auto-night mode algorithm. This algorithm transitions from night mode to day mode without stopping the inverter. During the transition, the inverter shows status ANGH and performs the transitions of the state machine in both directions (night to day and day to night).<br><br>The possible states are<br><br>
<table>
<tr><th>VALUE</th><th>STATE</th><th>DESCRIPTION</th></tr>
<tr><td>1</td><td>A_NIGHT_OF</td><td>Submachine stopped.</td></tr>
<tr><td>2</td><td>A_NIGHT_WAIT</td><td>Delay after reaching references.</td></tr>
<tr><td>3</td><td>A_NIGHT_DONE</td><td>Transition done (message to the main machine)</td></tr>
<tr><td>4</td><td>A_NIGHT_ISOLATE_DU</td><td>Disconnection from DC.</td></tr>
<tr><td>5</td><td>A_NIGHT_KRAMPS</td><td>Adaptation of references of the new mode.</td></tr>
<tr><td>6</td><td>A_NIGHT_RAMPS</td><td>Adaptation of gains between modes.</td></tr>
<tr><td>7</td><td>A_NIGHT_STOP</td><td>Not used.</td></tr>
<tr><td>8</td><td>A_NIGHT_CLOSE_DU</td><td>Connection to DC.</td></tr>
<tr><td>9</td><td>A_NIGHT_OUT</td><td>Transition done (message to the main machine)</td></tr>
<tr><td>10</td><td>A_NIGHT_KO</td><td>Transition state machine fault state. Fault [F196: Timeout autonight trans] will be reported.</td></tr>
</table>
<br>Note: this parameter will be shown when [G4.7.1-Enable] is configured as Autonight.</td></tr>
</table>

## Subgroup SV11.5: F/P

These parameters show the status of the algorithm **frequency/power**. For further information about power variation depending on the frequency, see the descriptions of parameter subgroup G4.5.

| Screen | Units | Description |
|---|---|---|
| **SV11.5.1-Status = 0** | NONE | Shows the status of the f/P algorithm. |
| **SV11.5.2-Frozen P ref = 0** | % | Shows the frozen power reference. |
| **SV11.5.3-Calc P limit = 0** | % | Shows the calculated P limit. |

## Subgroup SV11.6: V/Q

These parameters show the status of the algorithm **V/Q**.

| Screen | Units | Description |
|---|---|---|
| **SV11.6.1-Status = 0** | NONE | Shows the status of the V/Q algorithm. |
| **SV11.6.2-Calc Q ref = 0** | % | Shows the calculated Q reference. |
| **SV11.6.3-Meas avg Vac = 0** | % | Shows the measured average Vac. |

## Subgroup SV11.7: VRT

These parameters show information about VRT algorithms (Low Voltage Ride Through, LVRT, and Over Voltage Ride Through, OVRT).

| Screen | Units | Description |
|---|---|---|
| **SV11.7.1-LVRT status = Off** | NONE | Shows the status of the LVRT algorithm:<br>0: Off: Algorithm is not running.<br>1: On: Algorithm is running in advanced mode (injecting current).<br>2: Fault avoid: Algorithm was able to avoid the dip.<br>3: Basic [1]: Algorithm is running in basic mode (without injecting current).<br>4: Momentary cessation: Algorithm is not running momentarily.<br>5: Out: Algorithm is coming out of the dip.<br>6: Recovery: Inverter is recovering from the dip event (going back to stable conditions).<br>7: End: Algorithm has finished. |
| **SV11.7.2-LVRT number = 0** | NONE | Number of dips detected since power supply is detected by the control board. If power supply is removed, the counter will restart. |
| **SV11.7.3-Last LVRT depth = 0 %** | % | Voltage value of the last dip. |
| **SV11.7.4-Last LVRT duration = 0 s** | s | Duration, in seconds of the last dip. |
| **SV11.7.5-OVRT status = Off** | NONE | Shows the status of the OVRT algorithm:<br>0: Off: Algorithm is not running.<br>1: On: Algorithm is running in advanced mode (injecting current).<br>2: Fault avoid: Algorithm was able to avoid the overvoltage.<br>3: Basic [1]: Algorithm is running in basic mode (without injecting current).<br>4: Momentary cessation: Algorithm is not running momentarily.<br>5: Out: Algorithm is coming out of the overvoltage.<br>6: Recovery: Inverter is recovering from the overvoltage event (going back to stable conditions).<br>7: End: Algorithm has finished. |
| **SV11.7.6-OVRT number = 0** | NONE | Number of overvoltage conditions detected. |
| **SV11.7.7-Last OVRT depth = 0 %** | % | Voltage value during the last overvoltage ride through. |

[1] Basic mode is not yet implemented. Thus, this status will never appear.

| Screen | Units | Description |
|---|---|---|
| **SV11.7.8-Last OVRT duration = 0 s** | s | Duration of the last overvoltage condition. |
| **SV11.7.9-Vdrop number = 0** | NONE | Number of times a voltage drop has occurred.<br><br>**Note:** Please notice that not all voltage drops are considered dips. Every time a drop occurs, this counter will increase. If, according to the adjusted conditions, a voltage drop is considered a dip, the counter "LVRT number" will increase. |
| **SV11.7.10-Fast Exit Number = 0** | NONE | Number of fast exit retries from LVRT status that is set to G4.2.4 Exit detection. Once the fast exit thresholds are reached, the status is switched to PREEXIT (Check G4.2.4.2) This parameter is always reset when the inverter switches from LVRT to ON status. |

## Subgroup SV11.8: V/P

These parameters show information about V/P algorithm.

| Screen | Units | Description |
|---|---|---|
| **SV11.8.1-Status = Disabled** | NONE | Shows the status of the V/P algorithm:<br>0: Disabled<br>1: Off. Algorithm not running<br>2: On. Algorithm running, limiting P as a function of V.<br>3: On (recovery). Algorith in power recovery phase. Exiting algorithm.<br>4: Out: Finishing algorithm execution. |
| **SV11.8.2-Lim P calculated = 0** | NONE | Value of the P limit calculated by the algorithm with respect to the inverter's nominal power. |
| **SV11.8.3-Avg Vac meas = 0** | NONE | Average value of the three RMS values. |

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## Subgroup SV11.9: Anti-island

These parameters display information related to the anti-island algorithm. For further information, please refer to group G5.6.

| Screen | Units | Description |
|---|---|---|
| **SV11.9.1-Dev. F passive (Hz/s) = 0.0** | Hz/s | Frequency deviation, in Hz/s, for passive anti-island algorithm. |
| **SV11.9.2-Dev. F active = 0.0 Hz** | Hz | Frequency deviation, in Hz, for active anti-island algorithm. |

## Subgroup SV11.10: SVR (Suppression of Voltage Rise)

These parameters display information related to the suppression of voltage rise. For further information, please refer to group G4.11.

| Screen | Units | Description |
|---|---|---|
| **SV11.10.1-Status = Off** | NONE | Status of the SVR algorithm:<br><br>0: Off<br>1: Q injection<br>2: P reduction<br>3: P recovery<br>4: Q recovery |
| **SV11.10.2-Frozen P ref = 0** | NONE | Value of P power before activation of the suppression of voltage rise protection. |
| **SV11.10.3-Frozen Q ref = 0** | NONE | Value of Q power before activation of the SVR protection. |

| Screen | Units | Description |
|---|---|---|
| **SV11.10.4-Calculated P limit = 0** | NONE | Limit value of P calculated by the algorithm. |
| **SV11.10.5-Calculated Q limit = 0** | NONE | Limit value of Q calculated by the algorithm. |
| **SV11.10.6-Meas Vac average = 0** | NONE | Average value of the three RMS values. |

## Subgroup SV11.11: Harmonic Distorsion

These parameters show information about harmonic distortion.

| Screen | Units | Description |
|---|---|---|
| **SV11.11.1-THDi = 0.00 %** | % | Total harmonic distorsión in the current |
| **SV11.11.2-I1 = 0.00 %** | % | Amplitude of harmonic current 1 with respect to the fundamental frequency (50Hz or 60Hz) |
| **SV11.11.3-I2 = 0.00 %** | % | Amplitude of harmonic current 2 with respect to fundamental current |
| **SV11.11.4-I3 = 0.00 %** | % | Amplitude of harmonic current 3 with respect to fundamental current |
| **SV11.11.5-I4 = 0.00 %** | % | Amplitude of harmonic current 4 with respect to fundamental current |
| **SV11.11.6-I5 = 0.00 %** | % | Amplitude of harmonic current 5 with respect to fundamental current |
| **SV11.11.7-I7 = 0.00 %** | % | Amplitude of harmonic current 7 with respect to fundamental current |
| **SV11.11.8-I9 = 0.00 %** | % | Amplitude of harmonic current 9 with respect to fundamental current |
| **SV11.11.9-I11 = 0.00 %** | % | Amplitude of the harmonic current 11 with respect to the fundamental current |
| **SV11.11.10-I13 = 0.00 %** | % | Amplitude of the harmonic current 13 with respect to the fundamental current |

## Subgroup SV11.12: Econmode

These parameters show information about the functionality of Econmode.

<table>
<tr><th>Screen</th><th>Units</th><th>Description</th></tr>
<tr><td>SV11.12.1 Status = Off</td><td>NONE</td><td>Displays the status of the econmode algorithm.
<table>
<tr><th>OPCIÓN</th><th>DESCRIPCIÓN</th></tr>
<tr><td>Off</td><td>The algorithm is deactivated.</td></tr>
<tr><td>Wait</td><td>Appears during the opening mode of the MV switchgear.</td></tr>
<tr><td>Open</td><td>When the MV switchgear is open.</td></tr>
<tr><td>Premagnetize</td><td>Pre-magnetization operation of the MV transformer.</td></tr>
<tr><td>Close</td><td>When the MV switchgear is closed.</td></tr>
</table>
</td></tr>
</table>

# Group SV12: Comms Status

This group shows information about the communications status in the inverter components.

## Subgroup SV12.1: FPGA modules

| Screen | Units | Description |
|---|---|---|
| SV12.1.1-Mod 1 Rx OK frames = 0.0 % | % | Correct reception of CRC frames between the central P2G (central electronics) and the FPGA of module 1 (module electronics) |
| SV12.1.2-Mod 1 Tx OK frames = 0.0 % | % | Correct CRC frame transmission between the central P2G (central electronics) and the module 1 FPGA (module electronics). |
| SV12.1.3-Mod 2 Rx OK frames = 0.0 % | % | Correct reception of CRC frames between the central P2G (central electronics) and the FPGA of module 2 (module electronics) |
| SV12.1.4-Mod 2 Tx OK frames = 0.0 % | % | Correct CRC frame transmission between the central P2G (central electronics) and the module 2 FPGA (module electronics). |
| SV12.1.5-Mod 3 Rx OK frames = 0.0 % | % | Correct reception of CRC frames between the central P2G (central electronics) and the FPGA of module 3 (module electronics) |
| SV12.1.6-Mod 3 Tx OK frames = 0.0 % | % | Correct CRC frame transmission between the central P2G (central electronics) and the module 3 FPGA (module electronics). |
| SV12.1.7-Mod 4 Rx OK frames = 0.0 % | % | Correct reception of CRC frames between the central P2G (central electronics) and the FPGA of module 4 (module electronics) |
| SV12.1.8-Mod 4 Tx OK frames = 0.0 % | % | Correct CRC frame transmission between the central P2G (central electronics) and the module 4 FPGA (module electronics). |
| SV12.1.9-Mod 5 Rx OK frames = 0.0 % | % | Correct reception of CRC frames between the central P2G (central electronics) and the FPGA of module 5 (module electronics) |
| SV12.1.10-Mod 5 Tx OK frames = 0.0 % | % | Correct transmission of CRC frames between the central P2G (central electronics) and the FPGA of module 5 (module electronics) |
| SV12.1.11-Mod 6 Rx OK frames = 0.0 % | % | Correct reception of CRC frames between the central P2G (central electronics) and the module 6 FPGA (module electronics) |
| SV12.1.12-Mod 6 Tx OK frames = 0.0 % | % | Correct CRC frame transmission between the central P2G (central electronics) and the module 6 FPGA (module electronics). |
| SV12.1.13-Mod 7 Rx OK frames = 0.0 % | % | Correct reception of CRC frames between the central P2G (central electronics) and the module 7 FPGA (module electronics) |
| SV12.1.14-Mod 7 Tx OK frames = 0.0 % | % | Correct CRC frame transmission between the central P2G (central electronics) and the module 7 FPGA (module electronics). |
| SV12.1.15-Mod 8 Rx OK frames = 0.0 % | % | Correct reception of CRC frames between the central P2G (central electronics) and the module 8 FPGA (module electronics). |
| SV12.1.16-Mod 8 Tx OK frames = 0.0 % | % | Correct CRC frame transmission between the central P2G (central electronics) and the module 8 FPGA (module electronics). |
| SV12.1.17-Mod 9 Rx OK frames = 0.0 % | % | Correct reception of CRC frames between the central P2G (central electronics) and the module 9 FPGA (module electronics) |
| SV12.1.18-Mod 9 Tx OK frames = 0.0 % | % | Correct CRC frame transmission between the central P2G (central electronics) and the module 9 FPGA (module electronics). |
| SV12.1.19-Mod 10 Rx OK frames = 0.0 % | % | Correct reception of CRC frames between the central P2G (central electronics) and the FPGA of module 10 (module electronics) |
| SV12.1.20-Mod 10 Tx OK frames = 0.0 % | % | Correct CRC frame transmission between the central P2G (central electronics) and the module 10 FPGA (module electronics). |

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## Subgroup SV12.2: DSP uC

| Screen | Units | Description |
|---|---|---|
| **SV12.2.1-Total OK frames = 0.0 %** | % | Total number of correct CRC frames sent internally in the P2G central electronics, between the DSP and the microprocessor |

## Subgroup SV12.3: DU

| Screen | Units | Description |
|---|---|---|
| **SV12.3.1-Total OK frames = 0.0 %** | % | Total number of correct CRC frames sent from the P2G central electronics to the electronics in the DU via Modbus RTU |

## Subgroup SV12.4: MV

| Screen | Units | Description |
|---|---|---|
| **SV12.4.1-MV Rx OK frames = 0.0 %** | % | Total of received correct CRC frames sent from the Mv board to the P2G central electronics |

# Group SV13: Digital I/O status

This group shows information about the status of the digital inputs and outputs.

## Subgroup SV13.1: DSP-uC central

<table>
<tr><th>Screen</th><th>Units</th><th>Description</th></tr>
<tr><td>SV13.1.1-Digital inputs = 0000000000000000</td><td>NONE</td><td>Each bit (from right to left) represents a digital input, as described in the table below:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Trafo Thermal contact</td><td>Shows the status of the trafo thermal contact (J26B-1) (0: Open, 1: Closed).</td></tr>
<tr><td>1</td><td>LC contactor feedback</td><td>Displays status of LC contactor feedback (J26B-3) (0: Open, 1: Closed).</td></tr>
<tr><td>2</td><td>Inductance Thermal contact</td><td>Shows the state of the inductance thermal contact (0: Open, 1: Closed).</td></tr>
<tr><td>3</td><td>Aux contactor feedback</td><td>Shows the status of the aux contactor's feedback. (J26B-5) (0: Open, 1: Closed).</td></tr>
<tr><td>4</td><td>AC overvoltage</td><td>Displays the AC overvoltage status (J26B-7) (0: Open, 1: Closed).</td></tr>
<tr><td>5</td><td>RUN input</td><td>Displays the status of Input Run (J26B-9) (0: Open, 1: Closed).</td></tr>
<tr><td>6</td><td>Isopro Ok</td><td>Shows the status of Isolation monitoring device Ok (0: Open, 1: Closed).</td></tr>
<tr><td>7</td><td>GFDI fuse ok</td><td>Displays GFDI fuse status Ok (J26C-2) (0: open, 1: closed)</td></tr>
<tr><td>8</td><td>External fault input</td><td>Displays the status of the external fault input (J26C-6) (0: open, 1: closed).</td></tr>
<tr><td>9</td><td>Hygrostat feedback</td><td>Shows the status of the hygrostat feedback (J26C-8) (0: open, 1: closed).</td></tr>
<tr><td>10</td><td>Emergency stop 1</td><td>Shows the status of the emergency stop 1 (J26D-1) (0: open, 1: closed).</td></tr>
<tr><td>11</td><td>Emergency stop 2</td><td>Shows the state of the emergency stop 2 (J26D-2) (0: open, 1: closed).</td></tr>
<tr><td>12</td><td>Fan OK Feedback</td><td>Displays the status of the fan's feedback (J26A-2) (0: open, 1: closed).</td></tr>
<tr><td>13</td><td>DC/DC converter OK</td><td>Shows the status of the DC/DC converter (J26A-4) (0: open, 1: closed).</td></tr>
<tr><td>14</td><td>Softcharge feedback</td><td>Displays the status of the soft load backwash (J26A-6) (0: open, 1: closed).</td></tr>
<tr><td>15</td><td>Trackers feedback</td><td>Displays the status of the trackers' backlighting (J26A-8) (0: open, 1: closed).</td></tr>
</table>
</td></tr>
</table>

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<table>
<tr><th>Screen</th><th>Units</th><th>Description</th></tr>
<tr><td>SV13.1.2-Digital outputs = 00000000</td><td>NONE</td><td>Displays information on the following digital outputs:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Softcharge</td><td>Shows the status of the soft load output<br>(J31D-1and2)<br>(0: Open, 1: Closed).</td></tr>
<tr><td>1</td><td>LC contactor</td><td>Shows the status of the LC contactor output<br>(J32A-7 and 8)<br>(0: Open, 1: Closed).</td></tr>
<tr><td>2</td><td>RUN LED</td><td>Shows the status of the RUN LED output<br>(J32A-9 and J32B-1)<br>(0: Open, 1: Closed).</td></tr>
<tr><td>3</td><td>BT extraction FANs</td><td>Shows the output status of the BT extraction fans<br>(J32B-2 and J32C1)<br>(0: Open, 1: Closed).</td></tr>
<tr><td>4</td><td>Trackers enabled</td><td>Shows the output status of the Trackers mode activation<br>(J32A-5and6)<br>(0: open, 1: closed)</td></tr>
<tr><td>5</td><td>General Fault FPGA central</td><td>Displays the status of the general fault output of the central FPGA<br>(J32C2and3)<br>(0: open, 1: closed).</td></tr>
<tr><td>6</td><td>General fault uP</td><td>Shows the status of the general fault output of Up<br>(J32A-3and4)<br>(0: Open, 1: Closed).</td></tr>
<tr><td>7</td><td>LED Warning/Fault</td><td>Displays the status of the Alarm / Fault LED output<br>(J32A-1and2)<br>(0: Open, 1: Closed).</td></tr>
</table></td></tr>
<tr><td>SV13.1.3-Status error = 000</td><td>NONE</td><td>Displays information on the following errors:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Softcharge contactor fault</td><td>Displays the error status of the soft load contactor failure<br>(0:OK, 1: NOT OK).</td></tr>
<tr><td>1</td><td>Trackers contactor fault</td><td>Displays the error status of the tracker mode contactor failure<br>(0:OK, 1: NOT OK).</td></tr>
<tr><td>3</td><td>LC Contactor fault</td><td>Shows the failure status of the LC contactor<br>(0:OK, 1: NOT OK).</td></tr>
</table></td></tr>
</table>

## Subgroup SV13.2: FPGA module interface

<table>
<tr><th>Screen</th><th>Units</th><th>Description</th></tr>
<tr><td>SV13.2.1.1-Mod 1 digital I/O = 00000</td><td>NONE</td><td>Each bit shows information about the following digital inputs and outputs: from the FPGA of module 1:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Close AC cont</td><td>Shows the status of the digital inputs and outputs of the AC contactor closure.<br>(0: Open, 1: Closed).</td></tr>
<tr><td>1</td><td>Fb AC cont</td><td>Shows the status of the digital inputs and outputs of the AC contactors' feedback<br>(0: Open, 1: Closed).</td></tr>
<tr><td>2</td><td>Fans activation</td><td>Shows the status of the digital inputs and outputs of the ventilation activation<br>(0: Open, 1: Closed).</td></tr>
<tr><td>3</td><td>Fb fans ok</td><td>Displays the status of the digital inputs and outputs of the ventilation feedback<br>(0: Open, 1: Closed).</td></tr>
<tr><td>4</td><td>Fb thermal induct</td><td>Shows the status of the digital inputs and outputs of the thermal inductance feedback<br>(0: Open, 1: Closed).</td></tr>
</table></td></tr>
<tr><td>SV13.2.1.2-Status error (1) = 00</td><td>NONE</td><td>Each bit displays information about the following FPGA errors in module 1:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>AC Contactor Fault</td><td>Displays the error status of the AC contactor failure.<br>(0:OK, 1: NOT OK).</td></tr>
<tr><td>1</td><td>FAN contactor fault</td><td>Shows the error status of the fan contactor failure<br>(0:OK, 1: NOT OK).</td></tr>
</table></td></tr>
<tr><td>SV13.2.1.2-Mod 2 digital I/O = 00000</td><td>NONE</td><td>Each bit shows information about the following digital inputs and outputs: from the FPGA of module 2:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Close AC cont</td><td>Shows the status of the digital inputs and outputs of the AC contactor closure.<br>(0: Open, 1: Closed).</td></tr>
<tr><td>1</td><td>Fb AC cont</td><td>Shows the status of the digital inputs and outputs of the AC contactors' feedback<br>(0: Open, 1: Closed).</td></tr>
<tr><td>2</td><td>Fans activation</td><td>Shows the status of the digital inputs and outputs of the ventilation activation<br>(0: Open, 1: Closed).</td></tr>
<tr><td>3</td><td>Fb fans ok</td><td>Displays the status of the digital inputs and outputs of the ventilation feedback<br>(0: Open, 1: Closed).</td></tr>
<tr><td>4</td><td>Fb thermal induct</td><td>Shows the status of the digital inputs and outputs of the thermal inductance feedback<br>(0: Open, 1: Closed).</td></tr>
</table></td></tr>
<tr><td>SV13.2.2.2-Status error (2) = 00</td><td>NONE</td><td>Each bit displays information about the following FPGA errors in module 2:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>AC Contactor Fault</td><td>Displays the error status of the AC contactor failure.<br>(0:OK, 1: NOT OK).</td></tr>
<tr><td>1</td><td>FAN contactor fault</td><td>Shows the error status of the fan contactor failure<br>(0:OK, 1: NOT OK).</td></tr>
</table></td></tr>
</table>

EN

<table>
<tr><th>Screen</th><th>Units</th><th>Description</th></tr>
<tr><td>SV13.2.1.3-Mod 3 digital I/O = 00000</td><td>NONE</td><td>Each bit shows information about the following digital inputs and outputs: from the FPGA of module 3:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Close AC cont</td><td>Shows the status of the digital inputs and outputs of the AC contactor closure.<br>(0: Open, 1: Closed).</td></tr>
<tr><td>1</td><td>Fb AC cont</td><td>Shows the status of the digital inputs and outputs of the AC contactors' feedback<br>(0: Open, 1: Closed).</td></tr>
<tr><td>2</td><td>Fans activation</td><td>Shows the status of the digital inputs and outputs of the ventilation activation<br>(0: Open, 1: Closed).</td></tr>
<tr><td>3</td><td>Fb fans ok</td><td>Displays the status of the digital inputs and outputs of the ventilation feedback<br>(0: Open, 1: Closed).</td></tr>
<tr><td>4</td><td>Fb thermal induct</td><td>Shows the status of the digital inputs and outputs of the thermal inductance feedback<br>(0: Open, 1: Closed).</td></tr>
</table></td></tr>
<tr><td>SV13.2.3.2-Status error (3) = 00</td><td>NONE</td><td>Each bit displays information about the following FPGA errors in module 3:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>AC Contactor Fault</td><td>Displays the error status of the AC contactor failure.<br>(0:OK, 1: NOT OK).</td></tr>
<tr><td>1</td><td>FAN contactor fault</td><td>Shows the error status of the fan contactor failure<br>(0:OK, 1: NOT OK).</td></tr>
</table></td></tr>
<tr><td>SV13.2.1.4-Mod 4 digital I/O = 00000</td><td>NONE</td><td>Each bit shows information about the following digital inputs and outputs: from the FPGA of module 4:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Close AC cont</td><td>Shows the status of the digital inputs and outputs of the AC contactor closure.<br>(0: Open, 1: Closed).</td></tr>
<tr><td>1</td><td>Fb AC cont</td><td>Shows the status of the digital inputs and outputs of the AC contactors' feedback<br>(0: Open, 1: Closed).</td></tr>
<tr><td>2</td><td>Fans activation</td><td>Shows the status of the digital inputs and outputs of the ventilation activation<br>(0: Open, 1: Closed).</td></tr>
<tr><td>3</td><td>Fb fans ok</td><td>Displays the status of the digital inputs and outputs of the ventilation feedback<br>(0: Open, 1: Closed).</td></tr>
<tr><td>4</td><td>Fb thermal induct</td><td>Shows the status of the digital inputs and outputs of the thermal inductance feedback<br>(0: Open, 1: Closed).</td></tr>
</table></td></tr>
<tr><td>SV13.2.4.2-Status error (4) = 00</td><td>NONE</td><td>Each bit displays information about the following FPGA errors in module 4:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>AC Contactor Fault</td><td>Displays the error status of the AC contactor failure.<br>(0:OK, 1: NOT OK).</td></tr>
<tr><td>1</td><td>FAN contactor fault</td><td>Shows the error status of the fan contactor failure<br>(0:OK, 1: NOT OK).</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Screen</th><th>Units</th><th>Description</th></tr>
<tr><td>SV13.2.1.5-Mod 5 digital I/O = 00000</td><td>NONE</td><td>Each bit shows information about the following digital inputs and outputs: from the FPGA of module 5:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Close AC cont</td><td>Shows the status of the digital inputs and outputs of the AC contactor closure.<br>(0: Open, 1: Closed).</td></tr>
<tr><td>1</td><td>Fb AC cont</td><td>Shows the status of the digital inputs and outputs of the AC contactors' feedback<br>(0: Open, 1: Closed).</td></tr>
<tr><td>2</td><td>Fans activation</td><td>Shows the status of the digital inputs and outputs of the ventilation activation<br>(0: Open, 1: Closed).</td></tr>
<tr><td>3</td><td>Fb fans ok</td><td>Displays the status of the digital inputs and outputs of the ventilation feedback<br>(0: Open, 1: Closed).</td></tr>
<tr><td>4</td><td>Fb thermal induct</td><td>Shows the status of the digital inputs and outputs of the thermal inductance feedback<br>(0: Open, 1: Closed).</td></tr>
</table></td></tr>
<tr><td>SV13.2.5.2-Status error(5) = 00</td><td>NONE</td><td>Each bit displays information about the following FPGA errors in module 5:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>AC Contactor Fault</td><td>Displays the error status of the AC contactor failure.<br>(0:OK, 1: NOT OK).</td></tr>
<tr><td>1</td><td>FAN contactor fault</td><td>Shows the error status of the fan contactor failure<br>(0:OK, 1: NOT OK).</td></tr>
</table></td></tr>
<tr><td>SV13.2.1.6-Mod 6 digital I/O = 00000</td><td>NONE</td><td>Each bit shows information about the following digital inputs and outputs: from the FPGA of module 6:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Close AC cont</td><td>Shows the status of the digital inputs and outputs of the AC contactor closure.<br>(0: Open, 1: Closed).</td></tr>
<tr><td>1</td><td>Fb AC cont</td><td>Shows the status of the digital inputs and outputs of the AC contactors' feedback<br>(0: Open, 1: Closed).</td></tr>
<tr><td>2</td><td>Fans activation</td><td>Shows the status of the digital inputs and outputs of the ventilation activation<br>(0: Open, 1: Closed).</td></tr>
<tr><td>3</td><td>Fb fans ok</td><td>Displays the status of the digital inputs and outputs of the ventilation feedback<br>(0: Open, 1: Closed).</td></tr>
<tr><td>4</td><td>Fb thermal induct</td><td>Shows the status of the digital inputs and outputs of the thermal inductance feedback<br>(0: Open, 1: Closed).</td></tr>
</table></td></tr>
<tr><td>SV13.2.6.2-Status error (6) = 00</td><td>NONE</td><td>Each bit displays information about the following FPGA errors in module 6:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>AC Contactor Fault</td><td>Displays the error status of the AC contactor failure.<br>(0:OK, 1: NOT OK).</td></tr>
<tr><td>1</td><td>FAN contactor fault</td><td>Shows the error status of the fan contactor failure<br>(0:OK, 1: NOT OK).</td></tr>
</table></td></tr>
</table>

EN

## Subgroup SV13.3: DU

<table>
<tr><th>Screen</th><th>Units</th><th>Description</th></tr>
<tr><td>SV13.3.1-Digital Inputs = 00000000</td><td>NONE</td><td>State of the digital inputs of the central DU.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Aux</td><td>Shows the status of the auxiliary digital inputs<br>(0: Open, 1: Closed).</td></tr>
<tr><td>1</td><td>DC overvoltage</td><td>Displays the status of the central DU's digital DC overvoltage inputs<br>(0: Open, 1: Closed).</td></tr>
<tr><td>2</td><td>Breaker 2 energized spring</td><td>Displays the status of the breaker 2-spring digital inputs of the central DU<br>(0: Open, 1: Closed).</td></tr>
<tr><td>3</td><td>Breaker 2 close</td><td>Shows the status of the digital inputs of breaker 2 closed of the central DU<br>(0: Open, 1: Closed).</td></tr>
<tr><td>4</td><td>Breaker 2 open</td><td>Shows the status of the digital inputs of the open breaker 2 of the central DU<br>(0: Open, 1: Closed).</td></tr>
<tr><td>5</td><td>Breaker 1 energized spring</td><td>Displays the status of the digital inputs of the breaker 1-spring switch of the central DU<br>(0: Open, 1: Closed).</td></tr>
<tr><td>6</td><td>Breaker 1 close</td><td>Shows the status of the digital inputs of breaker 1 closed of the central DU<br>(0: Open, 1: Closed).</td></tr>
<tr><td>7</td><td>Breaker 1 open</td><td>Displays the status of the digital inputs of breaker 1 open of the central DU<br>(0: Open, 1: Closed).</td></tr>
</table></td></tr>
<tr><td>SV13.3.2-Digital Outputs = 00000</td><td>NONE</td><td>Status of the digital outputs of the central DU:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>GFDI</td><td>Displays the status of the central DU's GFDI digital outputs<br>(0: Open, 1: Closed).</td></tr>
<tr><td>1</td><td>Breaker 1 close</td><td>Displays the status of the digital outputs of breaker 1 closed of the central DU<br>(0: Open, 1: Closed).</td></tr>
<tr><td>2</td><td>Breaker 1 open</td><td>Displays the status of the digital outputs of breaker 1 open of the central DU<br>(0: Open, 1: Closed).</td></tr>
<tr><td>3</td><td>Breaker 2 close</td><td>Displays the status of the digital outputs of breaker 2 closed of the central DU<br>(0: Open, 1: Closed).</td></tr>
<tr><td>4</td><td>Breaker 2 open</td><td>Shows the status of the digital outputs of the open breaker 2 of the central DU<br>(0: Open, 1: Closed).</td></tr>
</table></td></tr>
<tr><td>SV13.3.3-Status error = 00</td><td>NONE</td><td>Central DU error status.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DIS1</td><td>Displays W149 for DC cabinet switch disconnector Q8.<br>(0:OK, 1: NOT OK).</td></tr>
<tr><td>1</td><td>DIS2</td><td>Displays W149 warning for DC cabinet switch disconnector Q9<br>(0:OK, 1: NOT OK).</td></tr>
</table></td></tr>
</table>

## Subgroup SV13.4: MV control interface

This group of visualization parameters only applies to the HEM inverter.

<table>
<tr><th>Pantalla</th><th>Unidades</th><th>Descripción</th></tr>
<tr><td>SV13.4.1-Digital Inputs = 00000000</td><td>NONE</td><td>Status of the difital inputs of the MV switchgear.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>3</td><td>MV switchgear protection open</td><td>Displays the status of the digital inputs of the MV cell protection open<br>(0: Open, 1: Closed).</td></tr>
<tr><td>4</td><td>MV switchgear protection close</td><td>Displays the status of the digital inputs of the MV cell protection close<br>(0: Open, 1: Closed).</td></tr>
<tr><td>6</td><td>MV switchgear open</td><td>Displays the status of the digital inputs of the MV switchgear open<br>(0: Open, 1: Closed).</td></tr>
<tr><td>7</td><td>MV switchgear close</td><td>Displays the status of the digital inputs of the MV switchgear close<br>(0: Open, 1: Closed).</td></tr>
<tr><td>8</td><td>MV earth disconnector close</td><td>Displays the status of the digital inputs of the MV earth disconnector close<br>(0: Open, 1: Closed).</td></tr>
<tr><td>9</td><td>MV earth disconnector open</td><td>Displays the status of the digital inputs of the MV earth disconnector open<br>(0: Open, 1: Closed).</td></tr>
<tr><td>11</td><td>MV fans feedback</td><td>It shows the status of the digital inputs of the MV fans feedback<br>(0: Open, 1: Closed).</td></tr>
</table></td></tr>
<tr><td>SV13.4.2-Digital Outputs = 00000</td><td>NONE</td><td>Status of the digital outputs of the MV switchgear.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>MV switchgear close</td><td>Displays the status of the digital outputs of the MV switchgear close<br>(0: Open, 1: Closed).</td></tr>
<tr><td>1</td><td>MV switchgear protection trip coil</td><td>Displays the status of the digital outputs MV switchgear protection trip coil<br>(0: Open, 1: Closed).</td></tr>
<tr><td>2</td><td>MV transf. fans contactor</td><td>Displays the status of the digital outputs of the MV transformer fans contactor.<br>(0: Open, 1: Closed).</td></tr>
</table></td></tr>
<tr><td>SV13.4.3-Status error = 000</td><td>NONE</td><td>MT cell error status
<table>
<tr><th>BIT</th><th>ESTADO</th><th>DESCRIPCIÓN</th></tr>
<tr><td>0</td><td>MV circuit breaker failure</td><td>Shows the fault status of the circuit breaker in the MV circuit breaker failure<br>(0:OK, 1: NOT OK).</td></tr>
<tr><td>1</td><td>MV tripping coil</td><td>Shows the fault status of the MV cell tripping coil<br>(0:OK, 1: NOT OK).</td></tr>
<tr><td>2</td><td>Contactor Failure Window. Transf</td><td>Displays the failure status of the t Contactor Failure Window. Transformer.<br>(0:OK, 1: NOT OK).</td></tr>
</table></td></tr>
</table>

EN

# PROGRAMMING PARAMETERS

5

Programming parameters enable the user to change the inverter settings.

Once a user has entered his password in G1.4-User password, the user mode will be set G1.3-User and the list of visible and configurable parameters varies depending on the user mode (password protected):

**Note:**

To check the permissions for your user role, please refer to section 2.

Parameters highlighted in coral are hidden for roles “customer” and “qualified customer”.

## Group G1: General

In this group, user can enter the password to change the user mode and adjust date, time and UTC offset.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G1.1-Init parameters = No** | No<br>Yes | Initialize parameters. Refer to section “Parameters Initialization” for further information. | No |
| **G1.2-Access code = 0** | 0 to 65535 | Shows the user mode access code. | Yes |
| **G1.3-User = Customer** | Customer<br>Set up<br>Qualified Customer<br>Power Electronics<br>Developer | Shows the current user mode. It will change depending on the password entered in G1.4-User password.<br><br>According to the active mode, displayed parameters and writing permissions change (access level is password protected and managed by Power Electronics. If you require assistance, please contact our technical department. | Yes |
| **G1.4-User password = 0** | 0 to 65535 | Allows entering the password to change the user mode.<br><br>This parameter allows to select the user mode when the technician is registered in PowerCommsV server as Developer or Power Electronics. | Yes |
| **G1.5-Date/time = 01/01/2018 0:00:00** | 0 to 59<br>0 to 59<br>0 to 23<br>1 to 31<br>1 to 12<br>2000 to 2159 | Set current date and time.<br>Format DD/MM/YYYY H:M:S | Yes |
| **G1.6-UTC offset = 0** | -720 to 840 min | Set the difference, in minutes, from UTC (Universal Time Coordinated) for a particular place, if needed. | Yes |

# Group G2: Running conditions

## Subgroup G2.1: Start

Parameters to start the inverter can be set in this group. Parameters in this group must be adjusted during commissioning to comply with local regulations, they are not set automatically.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G2.1.1-Vdc low = Vbus max (drive select) * 0.7** | Vac nominal (drive select) * 1.4142 * 0.95 to 1500 V | Allows setting the minimum DC voltage in order to start.<br>**Note:**<br>• This parameter is not shown in STATCOM.<br>• The softcharge has a non-configurable start level that is calculated with sqrt (2)Vin*1.03. | Yes |
| **G2.1.2-Vac low = 95%** | 80.0 to 100.0 % | Allows setting the minimum permissible AC voltage in order to start. | Yes |
| **G2.1.3-Vac high = 105.0%** | 100.0 to 120.0 % | Allows setting the maximum permissible AC voltage in order to start. | Yes |
| **G2.1.4-F low = 59.3Hz** | 45.0 to 60.0 Hz | Allows setting the minimum permissible frequency in order to start. | Yes |
| **G2.1.5-F high = 60.5Hz** | 50.0 to 65.0 Hz | Allows setting the maximum permissible frequency in order to start. | Yes |
| **G2.1.6-Start delay = 300s** | 0 to 600 s | Allows setting the delay applied since the AC grid reconnection conditions of voltage and frequency are met. | Yes |
| **G2.1.7-Constant Vdc (V/ºC)= 0.000** | 0.000 to 25.000 | Allows setting the $V_{DC}$ constant, used in the equation to calculate the starting voltage depending on the ambient temperature. | Yes |
| **G2.1.8-Calculated starting voltage = 0 V** | 0 to 2000 V | Shows the calculated starting voltage.<br><br>The new starting voltage will be calculated as a function of the ambient temperature, a counter and the original starting voltage. This allows starting the product at different voltages depending on the time of the year and the ambient conditions.<br><br>The parameters that will be considered to calculate this new voltage are:<br>• Low start Vdc → Voltage below which the unit does not start.<br>• Intake temp. → Temperature measurement of air inlet NTC.<br>• Constant V/ºC → Calculation constant.<br><br>And this voltage will be calculated using the following equation:<br>$Vdc\ start = [\text{G2.1.1-Vdc low}] + [\text{G2.1.7-Constant Vdc}] * (25 - [\text{SV4.1.9-Admission tº}])$<br><br>This new minimum starting voltage replaces the one set in parameter [G2.1.1-Vdc low].<br><br>By default, this functionality is disabled since parameter [G2.1.7-Vdc constant] has a default value of 0. | Yes |

**Note:** Whenever limit power (G3.1.1-P limit) is lower than the minimum stop power (G2.2.2-Min P), start will be prevented to avoid start-stop sequences.

EN

## Subgroup G2.2: Stop

Parameters to stop the inverter can be set in this group.

**Note:** These parameters will not be shown for STATCOM inverters, since no photovoltaic panels are connected to the equipment.

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>G2.2.1-Min P enable = AC</td><td>No<br>AC<br>DC</td><td>Allows to enable or disable the automatic stop of the inverter when the AC or DC power falls below the level configured in G2.2.2-Min P.<br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>No</td><td>Disables the automatic stop of the inverter.</td></tr><tr><td>AC</td><td>Enables the automatic stop of the inverter due to an AC power drop.</td></tr><tr><td>DC</td><td>Enables the automatic stop of the inverter due to a DC power drop.</td></tr></table>Note: In case the Bus Plus option (SG10.1.4a) is activated, it is recommended to configure this parameter as DC.</td><td>Yes</td></tr>
<tr><td>G2.2.2-Min P = 1.2%</td><td>0.1 to 10.0%</td><td rowspan="2">These two parameters allow configuring the automatic inverter shutdown depending on the minimum power (AC or DC, depending on the configuration of G2.2.1-Min P enable).<br><br>If measured power drops below the value of G2.2.2-Min P, and the timeout set in G2.2.3-P min delay, the inverter will stop.</td><td>Yes</td></tr>
<tr><td>G2.2.3-P min delay = 60s</td><td>1 to 300 s</td><td>Yes</td></tr>
<tr><td>G2.2.4-Min Vdc enable = Yes</td><td>No<br>Yes</td><td>Allows enabling/disabling the disconnection due to low DC voltage.<br><br>Note: the default value of this parameter varies depending on the configuration of [SG10.1.4a-Bus Plus]:<br>- If [SG10.1.4a-Bus Plus] = No, the default value of G2.2.4 will also be No.<br>- If [SG10.1.4a-Bus Plus] = any other value, the default value of G2.2.4 will be Yes.</td><td>Yes</td></tr>
<tr><td>G2.2.5-Min Vdc = (Vac nominal (drive select) * 1.4142) - 20V</td><td>(Vac nominal (drive select) * 1.4142) -20V to (Vac nominal (drive select) * 1.4142) -20V</td><td rowspan="2">Allow configuring the automatic inverter shutdown depending on the minimum DC voltage.<br><br>If voltage drops below the value of G2.2.5-Min Vdc, and the timeout set in G2.2.6-Min Vdc delay, the inverter will stop.</td><td>Yes</td></tr>
<tr><td>G2.2.6-Min Vdc delay = 5s</td><td>1 to 300 s</td><td>Yes</td></tr>
<tr><td>G2.2.7-Re start delay = 180.0 s</td><td>0.0 to 600.0 s</td><td>Defines the minimum time that must elapse for being able to restart the inverter once the stop conditions (parameters G2.2.1 to G2.2.6) have been met.</td><td>Yes</td></tr>
<tr><td>G2.2.8-Plim STOP enable = Yes</td><td>No<br>Yes</td><td>Enables or disables the automatic shutdown of the inverter due to minimum power when this minimum P shutdown condition is fulfilled, and the unit power is being limited with the global limit of the inverter (G3.1.1 and G3.1.1b).<br><br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>No</td><td>Only allow stop by Pmin if:<br>[Pmeas, inverter < Pmin, stop(G2.2.2)]<br>and<br>[Plim, inverter (G3.1.1) > Pmin, stop(G2.2.2)]</td></tr><tr><td>Yes</td><td>Allows stop by Pmin if:<br>[Pmeas,inverter < Pmin,stop(G2.2.2)].<br>Regardless of the value of the inverter power limit (G3.1.1) --> will stop if the unit is limited Plim< Pmin.</td></tr></table>Note: the main utility of this functionality is that, during the day, a power curtailment (power reduction Plim<Pmin) does not stop the plant and allows them to continue injecting Q.</td><td>Yes</td></tr>
</table>

## Subgroup G2.3: Reconnection

This group allows configuring reconnection conditions (minimum and maximum AC voltage and frequency, delay, soft start function). Parameters in this group must be adjusted during commissioning to comply with local regulations, they are not set automatically.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G2.3.1-Vac low = 95%** | 80.0 to 100.0% | Defines the minimum permissible AC voltage in order to reconnect. | Yes |
| **G2.3.2-Vac high = 105%** | 100.0 to 120.0% | Defines the maximum permissible AC voltage in order to reconnect. | Yes |
| **G2.3.3-F low = 59.3Hz** | 46.0 to 61.0 Hz | Defines the minimum permissible frequency in order to reconnect. | Yes |
| **G2.3.4-F high = 60.5Hz** | 49.0 to 63.0 Hz | Defines the maximum permissible frequency in order to reconnect. | Yes |
| **G2.3.5-Delay grid fault = 300s** | 0 to 600 s | Defines the delay applied since the AC grid reconnection conditions of voltage and frequency are met. | Yes |
| **G2.3.6-Soft start mode = Disabled** | Disabled<br>Modular<br>Central | **Note**: **Do not modify this parameter unless express authorization has been received from the R&D department.**<br><br>This parameter allows to enable / disable the soft start mode of the equipment. Soft start is a mechanism to avoid peaks during contactors conmutation as each module enters during start.<br><br>**OPTION** / **DESCRIPTION**<br>Disabled: All modules start following the same procedure as in modular mode, but during reconnections the procedure is the one described in central mode.<br>Modular: Enabled modules start sequentially at the conditions established on [G2.4. Modules connection] following the gradient limitation parameters adjusted in [G4.8.1 Start].<br>Central: The equipment starts setting all enabled modules to zero power and increases power following the reconnect gradient adjusted in [G4.8.1 Start]. This mode is used to comply with ramp specifications for some nomatives. | Yes |

## Subgroup G2.4: Modules connection

In this group, conditions for connecting modules can be adjusted.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G2.4.1-Activation level = 30%** [1] | 0 to 100 % | Allows setting the module activation power level. | Yes |
| **G2.4.2-Deactivation level = 30%** [1] | 0 to 100 % | Allows setting the module deactivation power level. | Yes |
| **G2.4.3-Activation time = 1s** | 1 to 300 s | Allows setting the module activation time after detecting the activation power level set on G2.4.1. | Yes |
| **G2.4.4-Deactivation time = 10s** [1] | 1 to 300 s | Allows setting the module deactivation time after detecting the deactivation power level set on G2.4.1. | Yes |
| **G2.4.5-Reset retries counter = No** | No<br>Yes | Whenever a module stops, general auto reset takes no effect (the equipment is not in fault state). The number of retries that a module attempts in case of a failure is defined in SG1.2.4.<br><br>If the number of retries has not been reached, the module will start when the configured delays have elapsed. If it is reached, the module will remain stopped until the next time the inverter stops.<br><br>This parameter allows resetting the number of retry attempts and force a stopped module to restart. | Yes |
| **G2.4.6-Available Q backup = 0%** [1] | 0 to 100 % | This parameter indicates the value of reactive that the inverter should be able to provide with a quick response. Depending on this threshold, the number of modules that need to be started, in order to satisfy the reactive demand, is calculated | Yes |

[1] These parameters will not be shown for M-PCS inverters.

## Subgroup G2.5: Night detection

This group allows setting the parameters to enable and disable the night mode. Please notice that they will not be displayed if the inverter type is STATCOM.

The night functions can only be executed once the night has been detected. Similarly, these functions forcefully cease to operate when the day is detected.

Night is detected when the voltage measured in panels is lower than the inverter disconnection voltage minus a configurable threshold (G2.5.1).

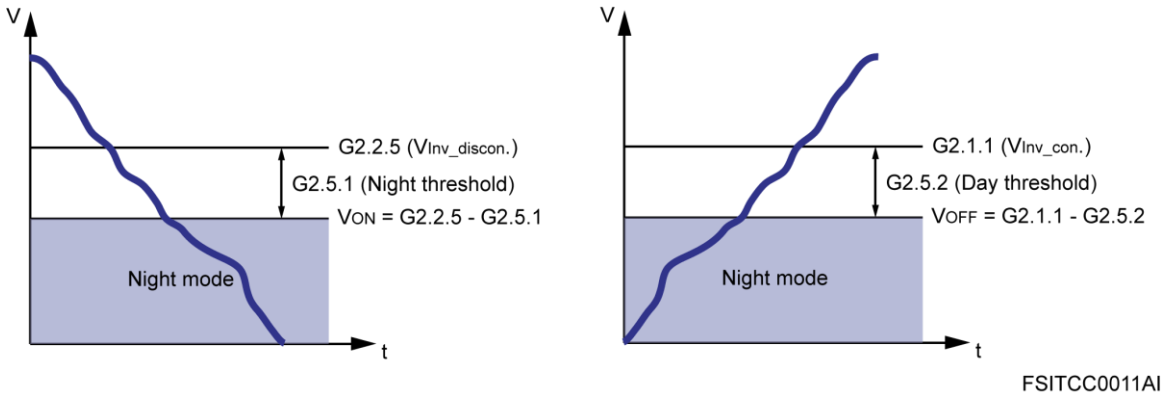
A day is detected when the voltage measured in panels is greater than the starting voltage configured in the inverter (G2.1.1) minus a configurable threshold (G2.5.2).

Night detection is indicated on the display by the "NIGHT" status.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G2.5.1-Night threshold = 50V** | 5 to 400 V | Allows setting the Night/Heating mode input voltage as a hysteresis regarding the DC disconnection voltage set on G2.2.5. | Yes |
| **G2.5.2-Day threshold = 50V** | 5 to 400 V | Allows setting the Night/Heating mode output voltage as a hysteresis regarding the DC connection voltage set on G2.1.1. | Yes |
| **G2.5.3-Day delay = 900 s** | 0 to 3600 s | To switch to NIGHT MODE, the inverter stops the power and opens DU. The PV voltage rises to the open circuit level. The PV voltage in open circuit could be higher than G2.5.2 (daytime voltage threshold) and therefore the logic could lead us back to the RUN state). This parameter allows the inverter to wait for this delay time before returning to the RUN state. | Yes |

When the "NIGHT" status is activated, it is possible to enable the heating or the night mode. If both modes are enabled, the night mode will always be prioritized.

The following figure illustrates the night mode configuration:



## Subgroup G2.6: Softcharge mode

There are several ways to perform the soft charge. Use this group to configure the mode that will be employed.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G2.6-Softcharge mode = CENTRAL AC** | Central AC<br>External DC<br>Internal DC central<br>DU | Selects the soft charge mode, according to the following table:<br><br>**OPTION** / **DESCRIPTION**<br>Central AC: A central soft charge is made from the AC side.<br>External DC: An external soft charge is made from the DC side.<br>Internal DC central: A soft charge using the test off-grid box is made from the DC side. **Note:** This option is only shown if SV4.4.1.1-Inverter type is not configured as Statcom.<br>DU: A soft charge is made from the DC side in the DU. **Note:** This option is only shown if SV4.4.1.1-Inverter type is not configured as PCS. | Yes |

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## Subgroup G2.7: Alternative softcharge

If G2.6 = Central AC, and the soft charge fails, this parameter allows defining an alternative process by using the MODULE AC.

**Note**: This group will only be shown in V1500 inverters, as long as they are not STATCOM.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G2.7-Alternative softcharge = No** | No<br>Yes | This parameter will only be shown if SV4.4.11, "DC mod contactors", is set to "Yes".<br><br>When the soft charge process through the CENTRAL AC fails, it is possible to make the soft charge through the MODULE AC, depending on the value of this parameter:<br><br>**OPT.** / **DESCRIPTION**<br>No: If CENTRAL AC soft charge fails, the inverter will not start.<br>Yes: The inverter will automatically attempt to make the soft charge through the AC module. | Yes |

## Subgroup G2.8: Softcharge offset

**Note:** this parameter is only shown in inverters with DU NEC 2014.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G2.8-Softcharge offset = 50 V** | Off = 0;<br>1 to 150 V | Since not all strings will be connected to the Disconnecting Unit, it is possible that the highest voltage from the PV field is higher than the read values.<br>This parameter defines an offset that will be applied to the voltage measured in the DU, and this will be the target value to reach with the softcharge.<br>In HEM the field voltage is read only in some strings. The highest measured value will be the one taken by the DSP as the maximum value of the field.<br>However, as there may be some strings (among those that have not been measured) with a higher voltage value, a short circuit may occur when closing the DU once the soft load is over.<br>Of all the values read, the DSP stays with the maximum (which is not necessarily the maximum of the field, because not all the strings have been measured).<br>With respect to this maximum, G2.8 is added and it will be the target value to reach with the soft load. | Yes |

# Group G3: Limits

## Subgroup G3.1: P/Q

This group allows setting the active and reactive power limits, as well as defining which one takes priority.

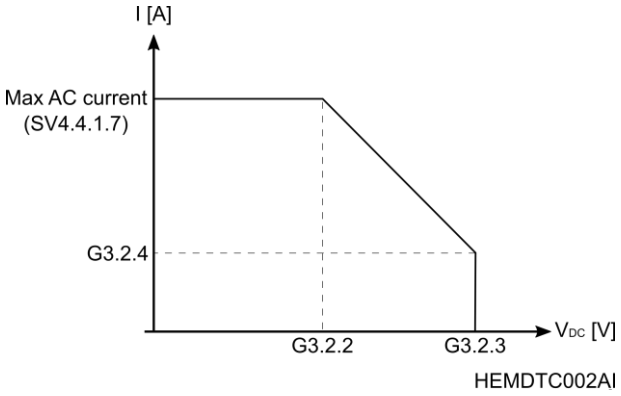
| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G3.1.1-P limit = 100.00%** | 0.00 to 100.00% | Allows setting the active power limit.<br><br>**Note**:<br>• This parameter not shown in STATCOM.<br>• The default value of G3.1.1 'P limit' cannot exceed the value set in SG10.3.3. | Yes |
| **G3.1.1b-P limit charge = -100.00%** | -100.00 to 0.00 % | Allows setting the active power limit during charge.<br><br>**Note:** this parameter is shown when the inverter is Bus Plus ([SG10.1.4a-Bus Plus] = Yes) and it is not a Statcom ([SG10.1.4-Inverter type] ≠ Statcom) | Yes |
| **G3.1.2-Q limit = 100.00%** | 0.00 to 150.00% | Allows setting the reactive power limit.<br><br>**Note:** The default value of G3.1.2 'Q limit' cannot exceed the value set in SG10.3.4. | Yes |
| **G3.1.3-P/Q priority = P** | P<br>Q | This parameter allows deciding which magnitude P or Q will take priority to keep S constant at a maximum value. P selects active power priority and Q selects reactive power priority. | Yes |
| **G3.1.4-S limit =100.00%** | 0.00 to 200.00% | Allows defining the S limit in percentage.<br><br>**Note:**<br>• This parameter not shown in STATCOM.<br>• The default value of G3.1.4 'S limit' cannot exceed the value set in SG10.3.5. | Yes |
| **G3.1.5-P algorithm priority = Minimum** | Minimum<br>P(f) priority<br>P(V) priority | If algorithms P(V) and P(f) are enabled, this parameter allows choosing which takes priority.<br>If this parameter is set to "Minimum", the applied power will be the lowest between the two algorithms. | Yes |

## Subgroup G3.2: Derating V

Parameters to configure the derating by voltage function.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G3.2.1-Enable = Yes** | No<br>Yes | OPTION / DESCRIPTION<br>No – The derating V is disabled.<br>Yes – The derating V is enabled.<br><br>If this parameter is set to Yes, when the voltage reaches the value set in G3.2.2, derating will start.<br><br>At G3.2.3, current will be equal to the value set in G3.2.4.<br><br>If voltage keeps rising and overcomes 1500V, fault F2 (HW Vbus) will be triggered and the inverter will stop. | Yes |
| **G3.2.2-Start Vdc = 1310V** | 0V to G3.2.3 | Allows setting the DC voltage derating start expressed in volts. | Yes |
| **G3.2.3-Max Vdc = 1460V** | G3.2.2 to 1500V | Allows setting the DC maximum voltage expressed in volts. | Yes |
| **G3.2.4-Imin (Vdc max) = 0A** | 0 to 150A | Allows setting the current at the maximum DC voltage defined on G3.2.3. | Yes |

The following figure illustrates the voltage derating function:



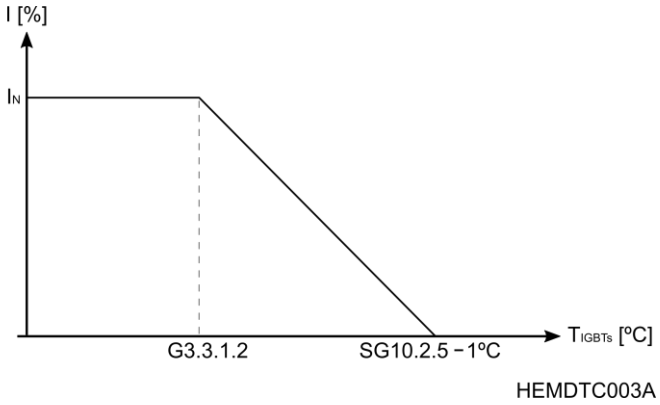

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## Subgroup G3.3: Derating T

Parameters to configure the module maximum current derating by temperature function.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G3.3.1.1-Enable = Yes** | No<br>Yes | OPTION / DESCRIPTION<br>No – The temperature derating is disabled.<br>Yes – The temperature derating is enabled.<br>If temperature derating is enabled, when the IGBT temperature reaches the threshold value (G3.3.2), derating will start.<br>Stop threshold is defined as SG10.2.5 – 1°. If the temperature keeps rising and reaches the protection threshold value (SG10.2.5), fault F40 (AC mod overtemperature) will be triggered. | Yes |
| **G3.3.1.2-Initial Tª IGBTs = 0 ºC** | 0 to 65535 ºC | It allows setting initial point of derating by IGBTs overtemperature. Expressed in Celsius | Yes |
| **G3.3.1.3-TªIGBTsDeviationMax = 15 ºC** | 5 to 15 ºC | Allows setting the maximum admissible difference between the highest IGBT temperature and the average of the highest IGBT temperatures per module, so if that difference is not exceeded the derating starts according to the highest temperature of all operating modules, and if exceeded, begins according to the highest temperature that is not out of range. | Yes |
| **G3.3.2.1-Admission enable = Yes** | No<br>Yes | Enable / disable derating by admission temperature. If this parameter is enabled, use parameters G3.3.7 to G3.3.12 to configure three derating points. | Yes |
| **G3.3.2.2-Admission T1 = 25 ºC** | -35 to 40 ºC | Allows setting the admission temperature derating in degrees for the first point | Yes |
| **G3.3.2.3-Admission I1 = 100.00 %** | 0.02 to 100.00 % | Allows setting the admission temperature derating percentage for the first point. | Yes |
| **G3.3.2.4-Admission T2 = 50 ºC** | -34 to 50 ºC | Allows setting the admission temperature derating for the second point. | Yes |
| **G3.3.2.5-Admission I2 = 90.90 %** | 0.01 to 100.00 % | Allows setting the admission temperature derating percentage for the second point | Yes |
| **G3.3.2.6-Admission T3 = 55 ºC** | -33 to 55 ºC | Allows setting the admission temperature derating for the third point. | Yes |
| **G3.3.2.7-Admission I3 = 86.36 %** | 0.00 to 86.36 % | Allows setting the admission temperature derating percentage for the third point | Yes |

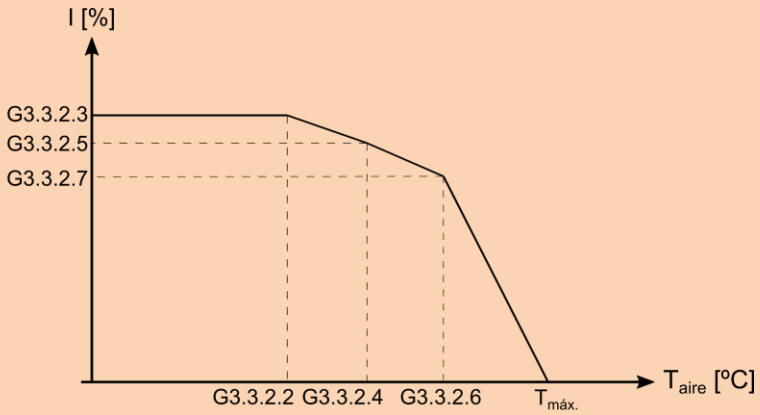
The following figure illustrates the derating function depending on IGTB temperature:




HEMDTC003A

The definitions of parameters SG10.2.5 and SG10.1.3a and SG10.1.3b are defined in the service group.

On the other hand, the following figure illustrates the derating function depending on air admission temperature:




FSDTC0058BE

Similarly, the standard configuration in case of derating by admission temperature depends on the inverter configuration. The usual values are shown in the following table:

| | SG10.1.3a= Normal | | SG10.1.3a= Plus | | SG10.1.3a= 1500 | |
|---|---|---|---|---|---|---|
| | SG10.1.3b = Full | SG10.1.3b = Hybrid | SG10.1.3b = Full | SG10.1.3b = Hybrid | SG10.1.3b = Full | SG10.1.3b = Hybrid |
| **Parameter** | **Value** | **Value** | **Value** | **Value** | **Value** | **Value** |
| **G3.3.1.1-Enable** | YES | YES | YES | YES | YES | YES |
| **G3.3.1.2-Initial Tª IGBTs** | 99ºC | 106ºC | 102ºC | 121ºC | 102ºC | 121ºC |
| **G3.3.1.3-TªIGBTs DeviationMax** | 15ºC | 15ºC | 15ºC | 15ºC | 15ºC | 15ºC |
| **G3.3.2.1-Admission enable** | YES | YES | YES | YES | YES | YES |
| **G3.3.2.2-Admission T1** | 25ºC | 25ºC | 40ºC | 40ºC | 45ºC | 45ºC |
| **G3.3.2.3-Admission I1** | 100% | 100% | 100% | 100% | 100% | 100% |
| **G3.3.2.4-Admission T2** | 50ºC | 50ºC | 50ºC | 50ºC | 50ºC | 50ºC |
| **G3.3.2.5-Admission I2** | 90.9% | 90.9% | 96.75% | 96.75% | 96.75% | 96.75% |
| **G3.3.2.6-Admission T3** | 53ºC | 53ºC | 53ºC | 53ºC | 53ºC | 53ºC |
| **G3.3.2.7-Admission I3** | 86.36% | 86.36% | 86.36% | 86.36% | 86.36% | 86.36% |

### Subgroup G3.3.4: Cold derating

Parameters to configure the cold derating are set within this subgroup.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| G3.3.4.1-Enable = No | No<br>Yes | Enables (Yes) or disables (No) the cold derating. | Yes |
| G3.3.4.2-Tª IGBT initial = 25 ºC | 20 to 50ºC | Defines the initial temperature. | Yes |
| G3.3.4.3-Tª IGBT final = 0 ºC | 0 to 20ºC | Defines the initial temperature. | Yes |
| G3.3.4.4-I1(T final) = 550 A | 300 to 550 A | Sets the current I1. | Yes |
| G3.3.4.5-I2 (Tfinal) = 200 A | 0 to 200 A | Sets the current I2. | Yes |

## Subgroup G3.4: Current

In this group, the limits for current can be configured.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| G3.4.1-Idc Max = 535A[2] | 0 a 65535 A | Defines the base value to configure parameters G3.4.2 and G3.4.3. | Yes |
| G3.4.2-DC discharge = 100.00%[2] | 0.00 to 100.00% | DC current limit for batteries discharge. | Yes |
| G3.4.3-DC charge = 100.00%[2] | 0.00 to 100.00% | DC current limit for batteries charge. | Yes |
| G3.4.4-Autonight extra Id = 0.0% | 0.0 to 100.0% | This parameter and [SV11.4.4-Autonight status] improve the day-night transitions in terms of times, limits and ramps.<br><br>**Note:** this parameter will only be displayed if [G4.7.1-Enable] is set to Autonight . | Yes |
| G3.4.5-Is temporal VRT = 1.00% | 0.00 to 1.00% | Global limit of the P/Q power in VRT. | Yes |
| G3.4.6-Is DIR VRT = 1.00% | 0.00 to 1.00% | Limit of the direct sequence of P/Q in VRT. | Yes |
| G3.4.7-Is INV VRT = 1.00% | 0.00 to 1.00% | Limit of the inverse sequence of P/Q in VRT. | Yes |

## Subgroup G3.5: Voltage

In this group, the maximum and minimum voltages for batteries charge can be configured. Please notice these parameters are only used in PCS inverters.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| G3.5.1-Min Vdc = 750V[2] | 0 to 1050V | Batteries charge minimum voltage. | Yes |
| G3.5.2-Max Vdc = 1050V[2] | 750 to 1500V | Batteries charge maximum voltage. | Yes |

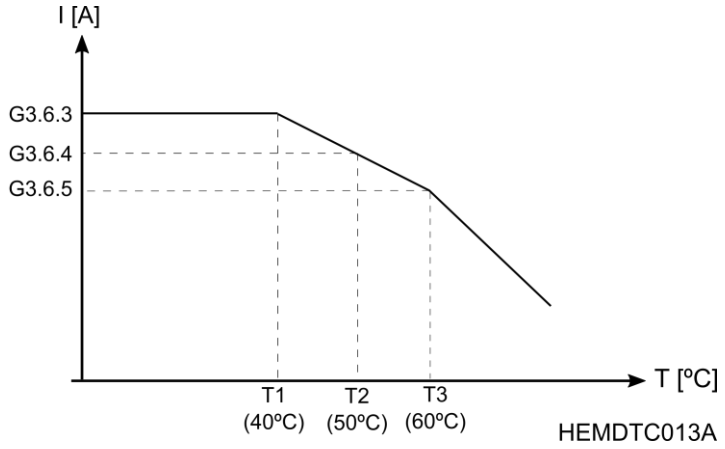
[2] This parameter is only used in PCS inverters

## Subgroup G3.6: DU limits

Projects with Bus Plus functionality can lead to the installation of a large photovoltaic power. This can cause overheating on the DC side where the inverter cannot control the power of the DCDC converter which is managed by the SCADA. Basically, the inverter will limit its own power (not the power of the DCDC converter) by reading the current of the DU and the inlet temperature in order not to exceed the above limits.

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>G3.6.1-DU Curr limit enable = No</td><td>No<br>Yes</td><td>Enables the DC current limitation.<br>To enable this parameter, G10.2b = NEC 2017, SG10.1.13 = Yes and SG10.1.4a Bus = Enabled.</td><td>Yes</td></tr>
<tr><td>G3.6.2-DU total Cur limit = 3970 A</td><td>0 to 65535 A</td><td>It allows to set the limit of the total DC current.<br>This current depends on the type set in parameter SG10.1.12.<br><table><tr><th>SG10.1.12</th><th>G3.6.2</th></tr><tr><td>Aluminium</td><td>3970A</td></tr><tr><td>Copper</td><td>6000A</td></tr></table></td><td>Yes</td></tr>
<tr><td>G3.6.3-DC Switch Cur lim 40C = 2400 A</td><td>0 to 65535 A</td><td>Allows to set the current limit of the DC switch to 40ºC. T1<br>This current depends on the type set in parameter SG10.1.12.<br><table><tr><th>SG10.1.12</th><th>G3.6.3</th></tr><tr><td>Aluminium</td><td>2400A</td></tr><tr><td>Copper</td><td>3200A</td></tr></table></td><td>Yes</td></tr>
<tr><td>G3.6.4-DC Switch Cur lim 50C = 2400 A</td><td>0 to 65535 A</td><td>Allows the current limit of the DC switch to be set at 50°C. T2<br>This current depends on the type set in parameter SG10.1.12.<br><table><tr><th>SG10.1.12</th><th>G3.6.4</th></tr><tr><td>Aluminium</td><td>2400A</td></tr><tr><td>Copper</td><td>3050A</td></tr></table></td><td>Yes</td></tr>
<tr><td>G3.6.5-DC Switch Cor lim 60C = 2400 A</td><td>0 a 65535 A</td><td>Allows to set the current limit of the DC switch to 60ºC. T3<br>This current depends on the type set in parameter SG10.1.12..<br><table><tr><th>SG10.1.12</th><th>G3.6.5</th></tr><tr><td>Aluminium</td><td>2400A</td></tr><tr><td>Copper</td><td>2920A</td></tr></table></td><td>Yes</td></tr>
</table>

The following graph shows the power limitation as a function of parameters G3.6.3, G3.6.4 and G3.6.5.



# Group G4: Algorithms

Parameters for running the different algorithms to ensure optimum performance.

## Subgroup G4.1: MPPt

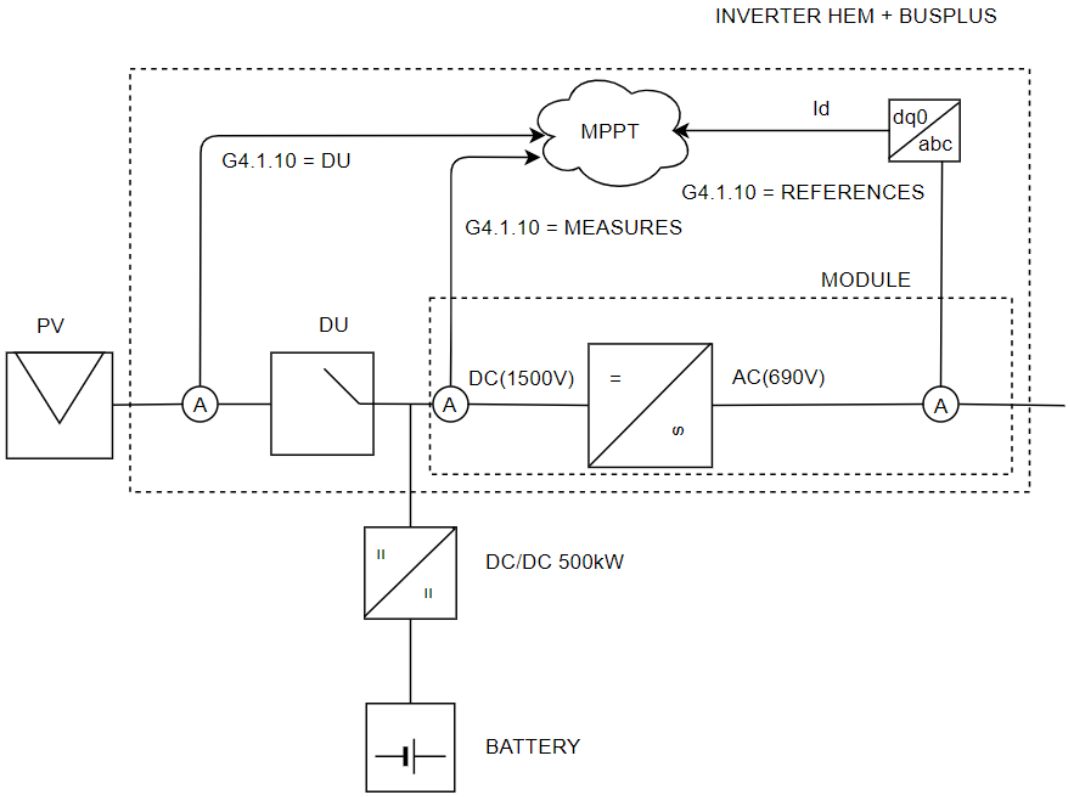
The MPPt (Maximum Power Point Tracking) algorithm searches constantly for the point of maximum power, which helps maximize the power generated by the panels and contributes to battery maintenance.

**Note**: These parameters are not shown in STATCOM.

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>G4.1.1-Min Vdc = Vac nominal * 2^0.5V</td><td>Vac nominal * 2^0.5 to 1250 V</td><td>Minimum voltage for the MPP algorithm.</td><td>Yes</td></tr>
<tr><td>G4.1.2-Max Vdc = 1350V</td><td>975 to 1500 V</td><td>Maximum voltage for the MPP algorithm</td><td>Yes</td></tr>
<tr><td>G4.1.3-Start voltage = 85.0%</td><td>65.0 to 95.0 %</td><td>Multiplying factor applied to the no-load voltage to start searching for MPP.</td><td>Yes</td></tr>
<tr><td>G4.1.4-Sample time = 3.0s</td><td>0.1 to 10.0 s</td><td>Time interval between executions of the MPP algorithm.<br>Note: the default value of this parameter varies depending on whether the inverter is Bus Plus (10 seconds) or not (3 seconds).</td><td>Yes</td></tr>
<tr><td>G4.1.5-Max delta Vdc = 10V</td><td>1 to 10 V</td><td>Maximum voltage variation (increase or decrease) resulting from the MPP algorithm.</td><td>Yes</td></tr>
<tr><td>G4.1.6-Delta Vdc = 2.5V</td><td>0.1 to 10.0 V</td><td>Increase or decrease the panel's reference voltage based on the MPP control.</td><td>Yes</td></tr>
<tr><td>G4.1.7-Delta P = 10.00%</td><td>0.01 to 10.00 %</td><td>Power MPP constant threshold.</td><td>Yes</td></tr>
<tr><td>G4.1.8-Delta Irr = 2.50 %</td><td>0.01 to 20.00 %</td><td>Irradiation threshold to detect irradiation changes.</td><td>Yes</td></tr>
<tr><td>G4.1.9-Hyst Vac = 2%</td><td>0 to 25%</td><td>Minimum AC voltage value that the MPPt algorithm can set. This parameter enables controlling the minimum MPPt voltage as a percentage of the real AC voltage.</td><td>Yes</td></tr>
<tr><td>G4.1.10 Mode = References</td><td>Measures<br>References<br>DU</td><td>This parameter allows to stablish the reference modes:
<table>
<tr><th>Note: the default value of this parameter varies depending on whether the inverter is Bus Plus or not (SG10.1.4a). If it is Bus Plus, the default value will be DU mode. Otherwise, the default value will be References mode.OPTION</th><th>DESCRIPTION</th></tr>
<tr><td>Measures</td><td>The MPPt is calculated with the Vdc measurements and the Idc sum of the modules</td></tr>
<tr><td>References</td><td>The MPPt is calculated with the MPPt reference and the Id reference (transformed to Iac)</td></tr>
<tr><td>DU</td><td>DU mode requires the uP to send to the DSP the sum of the DC current measurements of all the DU channels. If there is no current measurement the DU mode cannot be selected.</td></tr>
</table></td><td>Yes</td></tr>
</table>

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<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>G4.1.11- Negative Power MPPT = 0.0%</td><td>0.0 to 100.0%</td><td>Parameter to allow charging batteries from the grid depending on the inverter type (SG10.1.4) and following MPPT algorithm.<br><br>Usually, the inverters regulates DC Voltage BUS according to the MPPt algorithm.<br><br>There are different options depending on the inverter type and permission to charge grid batteries:<br><br>
<table>
<tr><th>OPTION</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Standard inverter. A solar inverter is usually not able to consume power from grid.<br>If this is the case, G4.1.11 must be set to 0%.</td></tr>
<tr><td>0</td><td>Bus Plus inverter type with no permissions to charge batteries. In the case of a Bus Plus inverter type with no permissions to get power from grid to charge the batteries, G4.1.11 must be set to 0%.</td></tr>
<tr><td>Nx550 / Sbase</td><td>Bus Plus inverter type with permissions to charge batteries.<br>In the case of a Bus Plus inverter type, the DCDC external converter could be requested to charge batteries when there is not enough PV power.<br>In that, case, this parameter allows to get power from grid to charge the batteries.<br><br>If this is accepted by the electric grid operator (or grid rules), this parameter G4.1.11 can be configured different than 0%.<br>The parameter setting must be in relation to the power of the number of DCDCs (N) installed and what the network operator allows for charging the batteries.<br><br>In order to set this parameter note that base power (Sbase) of the inverter is calculated by SG10.1.7 Nº modules x SG10.1.6 Nominal S module (kVA).</td></tr>
</table>
<br>Contact to Power Electronics if there are doubts about this parameter.</td><td>Yes</td></tr>
<tr><td>G4.1.12-Dynamic = No</td><td>No<br>Yes</td><td>Allows to sensitize the dynamic MPPT threshold according to the current irradiance level measured by the inverter.</td><td>Yes</td></tr>
</table>




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## Subgroup G4.2: LVRT

The LVRT (Low Voltage Ride Through) algorithm helps controlling the behavior of the inverter in case of an AC voltage dip, to prevent it from stopping due to that voltage drop.

### G4.2.1 Main

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G4.2.1.1-Enable = Mode 1** | Disabled<br>Mode 1<br>Mode 2<br>Mode 3 | This parameter allows enabling the algorithm to define the behavior of the inverter during a voltage dip event.<br><br>**OPTION** / **DESCRIPTION**<br>Disabled: LVRT algorithm is disabled.<br>Mode 1: Enable low voltage ride-through. The reference voltage corresponds with the direct sequence voltage.<br>Mode 2: Enable low voltage ride-through. The reference voltage corresponds with the lowest value of phase-to-phase voltages.<br>Mode 3: Enable the algorithm. The LVRT algorithm takes into account the estimated voltage in the POI to access or not to LVRT. | Yes |
| **G4.2.1.2.1-Threshold = 85.0%** [3] | 0.0 to 95.0% | Sets the threshold for the LVRT algorithm. If the voltage goes under this value, the algorithm will be executed. | Yes |
| **G4.2.1.2.2-Moment. cessation = Off** [3] | 0.0 to 100.0%<br>Off = 1001 | This parameter allows setting a delay before starting to execute the LVRT algorithm.<br><br>If the value defined in this parameter is overcome, the algorithm will wait for a period (set in SG1.2.15) before starting to run.<br><br>If the value of this parameter is not overcome, the algorithm will run without delay. | Yes |

[3] Parameters G4.2.1.2.1 to G4.2.1.2.6 will only be displayed if the algorithm is enabled (G4.2.1.1 = Mode 1 or 2).

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>G4.2.1.3-Configuration = S prev + kdV [3]</td><td>kdV<br>Q prev + kdV<br>S prev + kdV<br>P max + Q prev + kdV<br>Constant<br>Advanced</td><td>Selects the behavior in case of a dip event:<br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>kdV</td><td>Response is proportional to voltage variation.</td></tr><tr><td>Q prev + kdV</td><td>Response is proportional to voltage variation + previous reactive power.</td></tr><tr><td>S prev + kdV</td><td>Response is proportional to voltage variation + previous apparent power.</td></tr><tr><td>P max + Q prev + kdV</td><td>Response is proportional to voltage variation + maximum active power and previous reactive power.</td></tr><tr><td>Constant</td><td>Injected current is a constant value that is set internally. It can only be modified by Power Electronics personnel.</td></tr><tr><td>Advanced</td><td>Allows configuring a customized behavior.<br>Once this option is selected, parameters [G4.2.2.1-Injection mode Id] and [G4.2.2.2-Injection mode Iq] will be shown.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G4.2.1.4-Control mode P = Current [3][</td><td>Current<br>Power</td><td>Allow selecting the P control mode:<br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Current</td><td>Control by current.</td></tr><tr><td>Power</td><td>Control by power.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G4.2.1.5-Control mode Q = Current [3][</td><td>Current<br>Power</td><td>Allow selecting the Q control mode:<br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Current</td><td>Control by current.</td></tr><tr><td>Power</td><td>Control by power.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G4.2.1.6-P/Q priority = Q priority [3]</td><td>P<br>Q</td><td>This parameter allows deciding which magnitude P or Q will take priority to keep S constant at a maximum value. P selects active power priority and Q selects reactive power priority.<br>Note: this parameter has been reorganized in version 3.0.0. As of that version it is [G4.2.2.2.1-P/Q].</td><td>Yes</td></tr>
<tr><td>G4.2.1.7-Sequence priority = Direct</td><td>Direct<br>Inverse</td><td>Defines the LVRT sequence priority:<br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Direct</td><td>Direct sequence in current injection takes priority.</td></tr><tr><td>Inverse</td><td>Inverse sequence in current injection takes priority.</td></tr></table><br>Note: This parameter will only be shown if G4.2.1.3 = Advanced.<br>Note: this parameter has been reorganized in version 3.0.0. As of that version it is [G4.2.2.2.2-Sequence].</td><td>Yes</td></tr>
<tr><td>G4.2.1.8-Enable P balancing = No</td><td>No<br>Yes</td><td>Enables the activation of power balancing between phases during LVRT.</td><td>Yes</td></tr>
</table>

### G4.2.2 Control

**Note:** this group will be displayed if G4.2.1.1 = Mode 1 or Mode 2.

**Note:** parameters G4.2.2.4 to G4.2.2.6 define the proportional constant (K) to multiply the voltage (dV) in both direct and reverse sequences (KdV).

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>G4.2.2.1-Injection mode Id = Constant</td><td>x prev<br>Constant<br>MPPt</td><td>Allows selecting the Id injection mode:<br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>x prev</td><td>Injects the previous value, where x = P or I depending on the P control set previously in G4.2.1.4.</td></tr><tr><td>Constant</td><td>Injects a constant value.</td></tr><tr><td>MPPt</td><td>Uses MPPt algorithm for injection.</td></tr></table>Note: This parameter will only be shown if G4.2.1.3 = Advanced.<br>Note: this parameter has been reorganized in version 3.0.0. As of that version it is [G4.2.2.3.1-Injection mode Id].</td><td>Yes</td></tr>
<tr><td>G4.2.2.2-Injection mode Iq = x prev + kdV</td><td>kdV<br>x prev + kdV<br>Constant</td><td>Allows selecting the Iq injection mode:<br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>kdV</td><td>Injects a value proportional to the dip depth.</td></tr><tr><td>x prev + kdV</td><td>Injects the previous value, where x depends on the configuration set in G4.2.1.3.</td></tr><tr><td>Constant</td><td>Injects a constant value.</td></tr></table>Note: This parameter will only be shown if G4.2.1.3 = Advanced.<br>Note: this parameter has been reorganized in version 3.0.0. As of that version it is [G4.2.2.3.2-Injection mode Iq].</td><td>Yes</td></tr>
<tr><td>G4.2.2.3-Id ref dir = 0.0 %</td><td>0.0 to 100.0 %</td><td>Sets the reference for Id current in direct sequence during operation in LVRT.<br>Note: This parameter will only be shown if G4.2.2.1 = Constant.<br>Note: this parameter has been reorganized in version 3.0.0. As of that version it is [G4.2.2.3.3-Id ref dir].</td><td>Yes</td></tr>
<tr><td>G4.2.2.4-K dir = 2.0</td><td>0.0 to 10.0</td><td>Adjusts the K multiplying factor for direct sequence during operation in LVRT.<br>Note: This parameter will only be shown if G4.2.2.2 = kdV or x prev + kdV.<br>Note: this parameter has been reorganized in version 3.0.0. As of that version it is [G4.2.2.3.6-K dir].</td><td>Yes</td></tr>
<tr><td>G4.2.2.5-Iq ref dir = 100.0 %</td><td>0.0 to 100.0 %</td><td>Sets the reference for Iq current in direct sequence during operation in LVRT.<br>Note: This parameter will only be shown if G4.2.2.2 = Constant.<br>Note: this parameter has been reorganized in version 3.0.0. As of that version it is [G4.2.2.3.4-Iq ref dir].</td><td>Yes</td></tr>
<tr><td>G4.2.2.6-K inv = 0.0</td><td>0.0 to 10.0</td><td>Adjusts the K multiplying factor, proportional to the voltage (dV), for reverse sequence during operation in LVRT.<br>Note: This parameter will only be shown if G4.2.2.2 = kdV or x prev + kdV.<br>Note: this parameter has been reorganized in version 3.0.0. As of that version it is [G4.2.2.3.7-K inv].</td><td>Yes</td></tr>
<tr><td>G4.2.2.7-Iq ref inv = 0.0 %</td><td>0.0 to 100.0 %</td><td>Sets the reference for Iq current in reverse sequence during operation in LVRT.<br>Note: This parameter will only be shown if G4.2.2.2 = Constant.<br>Note: this parameter has been reorganized in version 3.0.0. As of that version it is [G4.2.2.3.5-Iq ref inv].</td><td>Yes</td></tr>
<tr><td>G4.2.2.8- Vset = Off</td><td>G4.2.1.2.1 to 120.0%</td><td>Allows to set the power injection quantity based on Vset in LVRT algorithm. Parameter G4.2.2.8 LVRT SET is a p.u value that will be used to calculate deltaV used in the G4.2 LVRT algorithm.</td><td>Yes</td></tr>
<tr><td>G4.2.2.9- K Balancing P = 100.0 %</td><td>0.0 to 100.0 %</td><td>Constant to control the power balance, it allows to apply more or less balancing.</td><td>Yes</td></tr>
</table>

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| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G4.2.2.10-Fast dV filter = 0.02 s** | 0.00 to 1.00 s | Parameters G4.2.2.10 to G4.2.2.13 were created in version 4.3.0 to simulate that special voltage filters can be configured in case of a drop. They do not have any logic associated, nor do they affect the functionality of the algorithms.<br><br>Frequency cut-off slow filter.<br><br>**Note**: this parameter will only be shown if the LVRT algorithm is enabled. | Yes |
| **G4.2.2.11-Slow dV filter = 0.02 s** | 0.00 to 1.00 s | Frequency cut-off fast filter.<br>**Note**: this parameter will only be shown if the LVRT algorithm is enabled. | Yes |
| **G4.2.2.12-Hist filt f-s (pu) = 0.05** | 0.00 to 1.00 | Threshold for switching from fast to slow filter.<br>**Note**: this parameter will only be shown if the LVRT algorithm is enabled. | Yes |
| **G4.2.2.13-Hist filt l-r (pu) = 0.05** | 0.00 to 1.00 | Threshold for switching from slow to fast filter.<br>**Note**: this parameter will only be shown if the LVRT algorithm is enabled. | Yes |

### G4.2.2.2: Priority

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G4.2.2.2.1-P/Q = Q** | P<br>Q | Allows setting the LVRT priority (P or Q).<br>**Note**:this parameter has been reorganized in version 3.0.0. Prior to that version it was G4.2.1.6. | Yes |
| **G4.2.2.2.2-Sequence = Direct** | Direct<br>Inverse<br>Proportional | Allows setting the LVRT sequence priority.<br>**Note**:this parameter has been reorganized in version 3.0.0. Prior to that version it was G4.2.1.7. | Yes |

### G4.2.2.3. References

The references for the LVRT algorithm for the different configurations are defined in this subgroup.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G4.2.2.3.1-Injection mode Id = Constant** | x prev = 0<br>Constant = 1<br>MPPt = 2 | Sets the LVRT Injection mode Id.<br><br>**Note:** this parameter will be hidden if [G4.2.1.3-Configuration] is lower than 5 (Advanced").<br>**Note**:this parameter has been reorganized in version 3.0.0. Prior to that version it was G4.2.2.1. | Yes |
| **G4.2.2.3.2-Injection mode Iq = x prev + kdV** | kdV = 0<br>x prev + kdV = 1<br>Constant = 2 | Sets the LVRT Injection mode Iq.<br><br>**Note:** this parameter will be hidden if [G4.2.1.3-Configuration] is lower than 5 (Advanced).<br>**Note**:this parameter has been reorganized in version 3.0.0. Prior to that version it was G4.2.2.2. | Yes |
| **G4.2.2.3.3-Id ref dir = 0.0 %** | 0.0 to 100.0 % | Configures the direct Id reference.<br><br>**Note:** this parameter will only be shown if [G4.2.2.3.1-Injection mode Id] is set to "Constant".<br>**Note**:this parameter has been reorganized in version 3.0.0. Prior to that version it was G4.2.2.3. | Yes |
| **G4.2.2.3.4-Iq ref dir = 100.0 %** | 0.0 to 100.0 % | Configures the direct Iq reference.<br><br>**Note:** this parameter will only be shown if [G4.2.2.3.2-Injection mode Iq] is set to "Constant".<br>**Note**:this parameter has been reorganized in version 3.0.0. Prior to that version it was G4.2.2.5. | Yes |
| **G4.2.2.3.5-Iq ref inv = 0.0 %** | 0.0 to 100.0 % | Configures the inverse Iq reference.<br><br>**Note:** this parameter will only be shown if [G4.2.2.3.2-Injection mode Iq] is set to "Constant".<br>**Note**:this parameter has been reorganized in version 3.0.0. Prior to that version it was G4.2.2.7. | Yes |

| Screen | Range | Function | Set on run |
|---|---|---|---|
| G4.2.2.3.6-K dir = 2.0 | 0.0 to 10.0 | Sets the direct K factor.<br><br>**Note:** this parameter is shown if [G4.2.1.3-Configuration] = Constant or ([G4.2.1.3-Configuration] = Advanced and [G4.2.2.3.2- G4.2.2.3.2-Injection mode Iq] = Constant).<br><br>**Note**:this parameter has been reorganized in version 3.0.0. Prior to that version it was G4.2.2.4. | Yes |
| G4.2.2.3.7-K inv = 0.0 | 0.0 to 10.0 | Sets the inverse K factor.<br><br>**Note:** this parameter is shown if [G4.2.1.3-Configuration] = Constant or ([G4.2.1.3-Configuration] = Advanced and [G4.2.2.3.2- G4.2.2.3.2-Injection mode Iq] = Constant).<br><br>**Note**:this parameter has been reorganized in version 3.0.0. Prior to that version it was G4.2.2.6. | Yes |

### G4.2.3 Exit

**Note:** This group will be displayed if G4.2.1.1 = Mode 1 or Mode 2.

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| Screen | Range | Function | Set on run |
|---|---|---|---|
| G4.2.3.1-Delay threshold = Infinite | 0.005 to 60.000 s<br>Infinite | Sets a delay, in seconds, to the LVRT algorithm output.<br>When set to 60001, time will be "infinite". | Yes |
| G4.2.3.2-Delay hysteresis = 0.005 s | 0.005 to 60.000 s | Sets a delay, in seconds, to the LVRT algorithm output if the hysteresis condition (G4.2.3.3) is met. | Yes |
| G4.2.3.3-Hysteresis = 5 % | 1 to 15 % | Defines the hysteresis to obtain the output voltage value (voltage at which there is no dip).<br>If the voltage is below the threshold (G4.2.1.2.1), a dip event will be considered, and it will be over once voltage is G4.2.1.2.1 + G4.2.3.3. | Yes |
| G4.2.3.4a-Recover ramp P = Off | 20 to 3000 %/s<br>Off = 19 | Recovery ramp to the LVRT output.<br>If this parameter is set to Off, ramp will be disabled. Otherwise, the value set will determine the slope (percentage per second) of P. | Yes |
| G4.2.3.4b-Recover ramp Q = Off %/s | 20 to 3000 %/s<br>Off = 19 | Recovery ramp to the LVRT output.<br>If this parameter is set to Off, ramp will be disabled. Otherwise, the value set will determine the slope (percentage per second) of Q. | Yes |
| G4.2.3.5-Recover P = 100.0 % | 0.0 to 100% | Sets the active power output of LVRT recovery. This is, the P power value that the algorithm will apply on dip recovery before normal control starts acting again. | Yes |
| G4.2.3.6-Recover Q = 100.0 % | 0.0 to 100% | Sets the reactive power output of LVRT recovery. This is, the Q power value that the algorithm will apply on dip recovery before normal control starts acting again, | Yes |
| G4.2.3.7-Recover time = 0.1 s | Off = 0<br>0.1 to 60.0 s | Recovery time of the LVRT output. During the time lapse set in this parameter, parameters G4.2.3.5 and G4.2.3.6 will be applied. | Yes |

### G4.2.4 Exit Detection

**Note:** This group refers to an algorithm to exit from LVRT in a fast way.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| G4.2.4.1-Delay = 60 ms | 30 to 100 ms | Once inverter enters to LVRT mode, a timer is started and this timer has to exceed G4.2.4.1 in order to start the algorithm to exit from LVRT in a fast way. | Yes |
| G4.2.4.2-Confirmation time = 10 ms | 10 to 100 ms | Defines the confirmation time the inverter must go through before switching from LVRT to RUN. | Yes |
| G4.2.4.3-Number of Retries = 2 | 1 to 10 | During "PREEXIT" substate any of the exiting conditions are still met, then it will go to LVRT and G4.2.4.3 counter will increase. | Yes |
| G4.2.4.4-Threshold exit 1 (pu/s) = 120 | 10 to 200 | Once the fast exit algorithm is started, then it is checking the conditions G4.2.4.4-Threshold exit 1 (dv/dt) and G4.2.4.5-Threshold exit 2 (Voltage threshold) in order to exit. When one of these conditions has been met, then the algorithm goes to a substate called "PREEXIT", where it has to be a period greater than the defined in G4.2.4.2-Confirmation time to bring the inverter from LVRT to RUN. | Yes |
| G4.2.4.5-Threshold exit 2 = 120 % | 100 to 140 % | | Yes |

## Subgroup G4.3: OVRT

The OVRT (Over Voltage Ride Through) algorithm is similar to the LVRT algorithm but used in case there is a voltage peak. It helps maintaining inverter operation stable.

### G4.3.1: Main

**Note:** G4.3.1.2.1 to G4.3.1.6 will only be displayed if the algorithm is enabled (G4.3.1.1 = Mode 1 or 2).

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>G4.3.1.1-Enable = Disabled</td><td>Disabled<br>Mode 1<br>Mode 2</td><td>This parameter allows enabling a special behavior during OVRT.<br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>OVRT algorithm is disabled.</td></tr><tr><td>Mode 1</td><td>Enable high voltage ride-through. The reference voltage corresponds with the direct sequence voltage.</td></tr><tr><td>Mode 2</td><td>Enable high voltage ride-through. The reference voltage corresponds with the highest value of phase-to-phase voltages.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G4.3.1.2.1-Threshold = 110.0%</td><td>105.0 to 120.0 %</td><td>Sets the input voltage threshold to enter in OVRT mode.</td><td>Yes</td></tr>
<tr><td>G4.3.1.2.2-OVRT Moment Cessation = Off</td><td>100.0 to 120.0%<br>Off = 1201</td><td>This parameter allows setting a delay before starting to execute the OVRT algorithm.<br><br>If the value defined in this parameter is overcome, the algorithm will wait for a period (set in SG1.2.15) before starting to run.<br>If the value of this parameter is not overcome, the algorithm will run without delay.</td><td>Yes</td></tr>
<tr><td>G4.3.1.3-Config = Q prev + kdV</td><td>kdV<br>Q prev + kdV<br>S prev + kdV<br>P max + Q prev + kdV<br>Constant<br>Advanced</td><td>Selects the behavior in case of an overvoltage.:<br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>kdV</td><td>Response is proportional to voltage variation.</td></tr><tr><td>Q prev + kdV</td><td>Response is proportional to voltage variation + previous reactive power.</td></tr><tr><td>S prev + kdV</td><td>Response is proportional to voltage variation + previous apparent power.</td></tr><tr><td>P max + Q prev + kdV</td><td>Response is proportional to voltage variation + maximum active power and previous reactive power.</td></tr><tr><td>Constant</td><td>Injected current is a constant value that is set internally. It can only be modified by Power Electronics personnel.</td></tr><tr><td>Advanced</td><td>Allows configuring a customized behavior.<br>Once this option is selected, parameters [G4.3.2.1-Injection mode Id] and [G4.3.2.2-Injection mode Iq] will be shown.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G4.3.1.4-Control mode P = Current</td><td>Current<br>Power</td><td>Allow selecting the P control mode:<br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Current</td><td>Control by current.</td></tr><tr><td>Power</td><td>Control by power.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G4.3.1.5-Control mode Q = Current</td><td>Current<br>Power</td><td>Allow selecting the Q control mode:<br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Current</td><td>Control by current.</td></tr><tr><td>Power</td><td>Control by power.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G4.3.1.6-P/Q priority = Q</td><td>P<br>Q</td><td>This parameter allows deciding which magnitude P or Q will take priority to keep S constant at a maximum value. P selects active power priority and Q selects reactive power priority.<br>Note: this parameter has been reorganized in version 3.0.0. As of that version it is [G4.3.2.9.1-P/Q].</td><td>Yes</td></tr>
</table>

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G4.3.1.7-Seq priority = Direct** | Direct<br>Inverse | Defines the OVRT sequence priority:<br><br>**OPTION** / **DESCRIPTION**<br>Direct: Direct sequence in current injection takes priority.<br>Inverse: Inverse sequence in current injection takes priority.<br><br>**Note**: This parameter will only be displayed if G4.3.1.3 = Advanced.<br>**Note:** this parameter has been reorganized in version 3.0.0. As of that version it is [G4.3.2.9.2-Sequence]. | Yes |

### G4.3.2: Control

**Note**: This group will only be displayed if G4.3.1.1 = Mode 1 or Mode 2.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G4.3.2.1 Injection mode Id = Constant** | x prev<br>Constant<br>MPPt | Allows selecting the Id injection mode:<br><br>**OPTION** / **DESCRIPTION**<br>x prev: Injects the previous value, where x = P or I depending on the configuration of G4.3.1.4.<br>Constant: Injects a constant value.<br>MPPt: Uses MPPt algorithm for injection.<br><br>**Note**: This parameter will only be displayed if G4.3.1.3 = Advanced. | Yes |
| **G4.3.2.2-Injection mode Iq = x prev + kdV** | kdV<br>x prev + kdV<br>Constant | Allows selecting the Iq injection mode:<br><br>**OPTION** / **DESCRIPTION**<br>kdV: Injects a value proportional to overvoltage.<br>x prev + kdV: Injects the previous value, where x depends on the configuration set G4.3.1.3.<br>Constant: Injects a constant value.<br><br>**Note**: This parameter will only be displayed if G4.3.1.3 = Advanced. | Yes |
| **G4.3.2.3-Id ref dir = 0.0 %** | 0.0 to 100.0 % | Sets the reference for Id current in direct sequence during operation in OVRT.<br>**Note:** This parameter will only be shown if G4.3.2.1 = Constant**.** | Yes |
| **G4.3.2.4-K dir = 2.0** | 0.0 to 10.0 | Adjusts the K multiplying factor for direct sequence during operation in OVRT.<br>**Note:** This parameter will only be shown if G4.3.2.2 = kdV or x prev + kdV**.** | Yes |
| **G4.3.2.5-Iq ref dir = 100.0 %** | 0.0 to 100.0 % | Sets the reference for Iq current in direct sequence during operation in OVRT.<br>**Note:** This parameter will only be shown if G4.3.2.2 = Constant. | Yes |
| **G4.3.2.6-K inv = 0.0** | 0.0 to 10.0 | Adjusts the K multiplying factor for reverse sequence during operation in OVRT.<br>**Note:** This parameter will be hidden if G4.3.2.2 = Constant. | Yes |
| **G4.3.2.7-Iq ref inv = 0.0 %** | 0.0 to 100.0 % | Sets the reference for Iq current in reverse sequence during operation in OVRT.<br>**Note:** This parameter will only be shown if G4.3.2.2 = Constant. | Yes |
| **G4.3.2.8-Vset = Off** | Off = 899;<br>90.0% a G4.3.1.2.1 | Allows to set the power injection quantity based on Vset in OVRT alorithm. Parameter G4.3.2.8 OVRT SET is a p.u value that will be used to calculate deltaV used in the G4.3 OVRT algorithm | Yes |
| **G4.3.2.10-Fast dV filter = 0.02 s** | 0.00 to 1.00 s | Parameters G4.3.2.10 to G4.3.2.13 Have been created in version 4.3.0 to simulate the creation of special filters in case of a voltage drop. They do not have any logic associated, nor do they affect the functionality of algorithms.<br><br>Allows to set the value, in seconds, for the fast dV filter.<br><br>**Note:** this parameter will only be shown if the OVRT algorithm is enabled. | Yes |
| **G4.3.2.11-Slow dV filter = 0.02 s** | 0.00 to 1.00 s | Allows to set the value, in seconds, for the slow dV filter.<br>**Note:** this parameter will only be shown if the OVRT algorithm is enabled. | Yes |

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| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G4.3.2.12-Hist filt f-s (pu) = 0.05** | 0.00 to 1.00 | Threshold to change from fast to slow filter.<br>**Note:** this parameter will only be shown if the OVRT algorithm is enabled. | Yes |
| **G4.3.2.13-Hist filt l-r (pu) = 0.05** | 0.00 to 1.00 | Threshold to change from slow to fast filter.<br>**Note:** this parameter will only be shown if the OVRT algorithm is enabled. | Yes |

### G4.3.2.9-Priority

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G4.3.2.9.1-P/Q = Q** | P<br>Q | This parameter allows deciding which magnitude P or Q will take priority to keep S constant at a maximum value. P selects active power priority and Q selects reactive power priority.<br>**Note**:this parameter has been reorganized in version 3.0.0. Prior to that version it was G4.3.1.6. | Yes |
| **G4.3.2.9.2-Sequence = Direct** | Direct<br>Inverse<br>Proportional | Defines the OVRT sequence priority.<br>**Note**:this parameter has been reorganized in version 3.0.0. Prior to that version it was G4.3.1.7. | Yes |

### G4.3.3: Exit

**Note:** This group will be displayed if G4.2.1.1 = Mode 1 or Mode 2.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G4.3.3.1-Delay threshold = Infinite** | 0.005 to 60.000 s<br>Infinite | Sets a delay, in seconds, to the OVRT algorithm output.<br>When set to 60001, time will be "infinite". | Yes |
| **G4.3.3.2-Delay hysteresis = 0.005 s** | 0.005 to 60.000 s | Sets a delay, in seconds, to the OVRT algorithm output if the hysteresis condition (G4.3.3.3) is met. | Yes |
| **G4.3.3.3-Hysteresis = 2 %** | 1 to 15 % | OVRT algorithm output hysteresis. | Yes |
| **G4.3.3.4a-Recover ramp P= Off %/s** | 20 to 3000 %/s<br>Off = 19 | Recovery ramp to the OVRT output.<br>If this parameter is set to 19, ramp will be disabled. Otherwise, the value set will determine the slope (percentage per second)of P. | Yes |
| **G4.3.3.4b-Recover ramp Q = Off %/s** | 20 to 3000 %/s<br>Off = 19 | Recovery ramp to the OVRT output.<br>If this parameter is set to Off, ramp will be disabled. Otherwise, the value set will determine the slope (percentage per second)of Q. | Yes |
| **G4.3.3.5-Recover P = 100.0 %** | 0.0 to 100% | Sets the active power output of OVRT recovery. This is, the P power value that the algorithm will apply on dip recovery before normal control starts acting again. | Yes |
| **G4.3.3.6-Recover Q = 100.0 %** | 0.0 to 100% | Sets the reactive power output of OVRT recovery. This is, the Q power value that the algorithm will apply on dip recovery before normal control starts acting again, | Yes |
| **G4.3.3.7-Recover time = Off s** | Off = 0<br>0.1 to 60.0 s | Recovery time of the OVRT output. During the time lapse set in this parameter, parameters G4.3.3.5 and G4.3.3.6 will be applied. | Yes |

## Subgroup G4.5: F/P

Parameters within this group are used to configure the power variation depending on the frequency.

**Note**: This group is not shown in STATCOM inverters.

### G4.5.1: Overfrequencies

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G4.5.1.1-Enable = No** | No<br>Yes | Enable / disable frequency regulation; active power is reduced depending on the frequency. | Yes |
| **G4.5.1.2-Activation freq = 60.30 Hz** | 50.00 to 65.00 Hz | Set function activation frequency ($f_{on}$). This is, the frequency in which the power begins to be reduced according to the droop set in G4.5.1.4. For more information, see figure below.<br>**Note**: The default value is set internally. | Yes |
| **G4.5.1.3-Hysteresis = 0.00 Hz** | 0.00 to 655.35 Hz | Hysteresis to recover active power with frequency reduction. Frequency to stop regulation is defined as:<br>$f_{off} = f_{on} - G4.5.1.3$<br>**Notes:**<br>- This parameter is only displayed if G4.5.1.1-Enable = Yes.<br>- The maximum value that can take this parameter depends on an internal variable whose default value is 500. | Yes |
| **G4.5.1.4-Droop = 4.0 %** | 2.0 to 20.0 % | Defines the frequency variation per each 100% load variation:<br>$Droop[\%] = 100/\ (f_n[Hz] * slope[pu/Hz]\ )$<br>For example, a 2% droop corresponds to a ramp of 1pu/Hz at 50Hz. | Yes |

### G4.5.2: Underfrequencies

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G4.5.2.1-Enable = No** | No<br>Yes | Enable / disable frequency regulation; active power is reduced depending on the frequency. | Yes |
| **G4.5.2.2-Activation freq = 59.70 Hz** | 45.00 to 65.00 Hz | Set function activation frequency ($f_{on}$). This is, the frequency in which the power begins to be reduced according to the droop set in G4.5.2.4.<br>**Note**: The default value is set internally. | Yes |
| **G4.5.2.3-Hysteresis = 0.00 Hz** | 0.00 to 655.35 Hz | Hysteresis to recover active power with frequency reduction. Frequency to stop regulation is defined as:<br>$f_{off} = f_{on} + G4.5.2.3$<br>**Notes:**<br>- This parameter is only displayed if G4.5.2.1-Enable = Yes.<br>- The maximum value that can take this parameter depends on an internal variable which default value is 500. | Yes |
| **G4.5.2.4-Droop = 4.0 %** | 2.0 to 20.0 % | Defines the frequency variation per each 100% load variation:<br>$Droop[\%] = 100/\ (f_n[Hz] * slope[pu/Hz]\ )$ | Yes |
| **G4.5.2.5-Power backup = 0.0 %** | 0.0 to 10.0 % | Defines the power reserve the inverter makes with respect to the nominal. For example, if this value is set to 10%, the inverter will only produce 90% of the nominal, and the remaining 10% will be reserved for operation in case of under frequencies. | Yes |

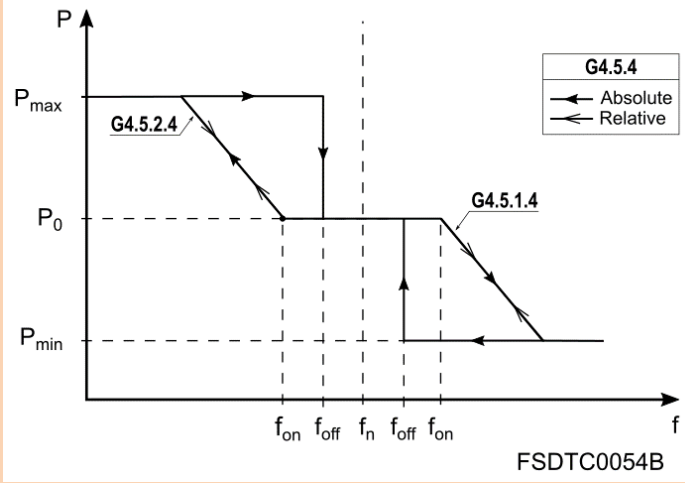
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**Common parameters:**

Parameters shown in the table below are common for both over frequencies and under frequencies:

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G4.5.3-P reference = Nominal** | Relative<br>Nominal | Sets the percentage of reduction / increase per Hertz that will be applied on the power at the instant when activation frequency is reached (G4.5.3 = Relative) or on the rated power (G4.5.3 = Nominal). | Yes |
| **G4.5.4-Recover mode = Relative** | Absolute<br>Relative | Allows choosing the recovery mode:<br>**OPTION** / **DESCRIPTION**<br>Absolute: Power limitation obtained when frequency recovery begins, it is maintained until deactivation frequency is reached.<br>Relative: An adjustable hysteresis is applied. Power limitation is maintained during hysteresis; then, power limitation will be proportional to the frequency. | Yes |
| **G4.5.5-Gradient = Off** | Off<br>0.01 to 655.00 %/s | Adjusts the gradient that limits power variation when F/P control is used. | Yes |
| **G4.5.6-Response time = 5.00 s** | 0.00 s to 5.00 s | Adjusts the response time in seconds of the P(f) algorithm. Response time is defined as the time that takes reaching 90% of the reference since the moment when the change occurs. | Yes |

The following figure shows the power variation depending on the frequency:




| Over frequency | Under frequency |
|---|---|
| $f_{on} = G4.5.1.2$ | $f_{on} = G4.5.2.2$ |
| $f_{off} = f_{on} - G4.5.1.3$ | $f_{off} = f_{on} + G4.5.2.3$ |

## Subgroup G4.6: Heating

Heating Mode is configured using parameters G4.6.1 to G4.6.5. These parameters can only be configured by Power Electronics' staff and are not shown in STATCOM.

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>G4.6.1-Enable = Internal</td><td>Disabled<br>Internal<br>External</td><td>This parameter allows enabling or disabling the heating mode.<br>The heating will be activated, as long the inverter is in night status.<br><br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Disables the heating option.</td></tr><tr><td>Internal</td><td>Enables internal heating option. The heating is activated through the reading of the temperature and humidity of the central control board. When one of these variables reach the configured value, the heating is activated. The temperature and humidity thresholds must be defined through G4.6.7-Temperature threshold and G4.6.8-Humidity threshold.</td></tr><tr><td>External</td><td>Enable external heating option. The heating is activated through the digital output of a commercial hygrostat with a specific temperature and humidity.</td></tr></table><br>Note: If Night Mode (G4.7) is activated, it will have priority over this mode.</td><td>Yes</td></tr>
<tr><td>G4.6.1b-Modules Enabled= 0000XX0000</td><td>0000000000 a XXXXXXXXXX</td><td>Indicates which modules switch their IGBTs to produce the heating. Each bit represents one of the modules of the equipment (generally they are six but depending on the configuration of the drive select they could be up to ten.). The first bit from left to right corresponds to module 1.<br><br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>X</td><td>Switches its IGBTs.</td></tr><tr><td>0</td><td>Switches its IGBTs.</td></tr></table><br>Note: Physically, module 1 is the closest to the main control cabinet. See figure below.</td><td>Yes</td></tr>
<tr><td>G4.6.2-Q ref = 0.0%</td><td>0.0 to 30.0%</td><td>Allows setting a reactive power reference for the Heating mode.</td><td>Yes</td></tr>
<tr><td>G4.6.3-Vbus ref = 1120V</td><td>500 to 1500V</td><td>Allows fixing the bus voltage reference for the Heating mode.</td><td>Yes</td></tr>
<tr><td>G4.6.4-Desactivation = 90ºC</td><td>Minimum [G4.6.5] to 120 ºC</td><td>Allows setting the deactivation heating temperature. The minimum value matches the minimum value of [G.4.6.5].</td><td>Yes</td></tr>
<tr><td>G4.6.5-Reactivation = 60ºC</td><td>0 to Maximum [G4.6.4] ºC</td><td>Allows setting the reactivation heating temperature. The maximum value matches the maximum value of [G4.6.4].</td><td>Yes</td></tr>
<tr><td>G4.6.7-Temperature threshold = 5 ºC</td><td>-30 to 60 ºC</td><td>Allows to define the temperature threshold used for the active heating functionality, when G4.6.1-Enable = Internal.<br><br>Note: This parameter is only displayed if G4.6.1-Enable = Internal.</td><td>Yes</td></tr>
<tr><td>G4.6.8-Humidity threshold = 65 %</td><td>0 to 100 %</td><td>Allows to define the humidity threshold used for the active heating functionality when G4.6.1-Enable = Internal.<br><br>Note: This parameter is only displayed if G4.6.1-Enable = Internal.</td><td>Yes</td></tr>
</table>

The activation/deactivation of the Heating Mode does not depend on parameter configuration, but on the humidity and temperature readings received by the uP.

Operation in Heating Mode will start if the conditions are fulfilled and the inverter is at night.

This mode requires the DC cabinet to isolate the inverter from the solar panels. Fans can be activated depending on the configured operation mode.

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Once conditions for Heating Mode are fulfilled, the procedure is the following:

1. Open DC cabinet.
2. DC bus soft charge at voltage reference level set in G4.6.3 through central AC soft charge without allowing DC cabinet closing once the process is finished.
3. Start all available modules (network connection and IGBTs activation).
4. DC bus regulation at voltage set in G4.6.3.
5. Reactive current regulation at the value set in G4.6.2.

Operation in Heating Mode is also determined by the maximum IGBTs temperature, as it will stop if this temperature overcomes a configurable threshold (G4.6.4) and will start when it is below a threshold that can also be configured (G4.6.5).

Heating Mode is shown in the display as warning W12:Heating.

If during Heating Mode there are no available modules (fault, disabled…) the inverter stops and triggers warning W6:No modules.

Particular cases according to the reactive reference set in G4.6.2.

**Basic Heating**:

If the reference is 0, there is no reactive current regulation; thus, all modules work regulating the configured Vbus. This operation mode is called *Basic Heating.*

**Extended Heating**:

If the reference is other than 0, each module regulates a different Q reference, calculated as a function of the reference set in G4.6.2 and the number of active modules. The reference is distributed by producing current recirculation between modules so that injected power balance is 0. This operation mode is called *Extended Heating.*

Q reference will be the same for all modules in magnitude, but half the modules will have a positive sign (capacitive) and the other half will have negative sign (inductive), in order to guarantee the power balance.

Example:

| MODULE | 6 | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|---|
| **Status** | RUN | RUN | OFF | RUN | OFF | RUN |
| **Sign** | + | - | | + | | - |
| **Magnitude** | Q | Q | | Q | | Q |
| **Coefficient** | 1 | 1 | | 1 | | 1 |
| **Reference** | -Q | +Q | | +Q | | -Q |

$Balance: 2(-Q) + 2Q = 0$

**Note**:

Active heating is enabled to avoid condensation. The thermostat + hygrostat or the main control board readings trigger the activation command, but On/Off is made based on the IGBTs temperature; thus, it is possible to change the parameters to reduce consumption and avoid power losses.

Depending on the external temperature control On/Off can be performed with On = 18ºC (G4.6.5) and Off = 30ºC (G4.6.4). Q power can also be adjusted (G4.6.2) between 0 and 100%.

By changing these three parameters it is possible to optimize active heating to reduce consumption and minimize losses.

## Subgroup G4.7: Night mode

Night Mode is configured using parameters G4.7.1 to G4.7.4. Please notice that these parameters will not be shown in STATCOM.

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<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>G4.7.1-Enable = Disabled</td><td>Disabled<br>Standard<br>Autonight</td><td>This parameter allows enabling or disabling the Night mode.<br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Disables the Night mode option.</td></tr><tr><td>Standard</td><td>Standard type transition means that the inverter will stop before going to NIGHT MODE (IGBTs to OFF and open DU, aftewards the reactive power will start to be regulated).<br>The inverter stops for G2.2 stop conditions, and stays in READY state.<br>When night conditions are detected (G2.5), and if night mode is active(G4.7.4) , then it will go to night mode.</td></tr><tr><td>Autonight</td><td>Enables Autonight mode option. Means that a smooth transition(Without stopping the inverter) to NIGHT MODE will be done when conditions are fulfilled.<br>When Ppv is lower than G2.2.2-Min P and time is greater than G2.2.3-Pmin delay, then the inverter goes to “AUTO NIGHT ON” state.</td></tr></table>Once Night Mode has been enabled, parameters G4.7.2 to G4.7.4 will be shown.<br>Note: If the Heating Mode (G4.6) is activated, night mode G4.7) has priority.</td><td>Yes</td></tr>
<tr><td>G4.7.2-Q ref = 0.0%</td><td>-100.0% to +100.0%</td><td>Allows setting a reactive power reference for the Night mode.</td><td>Yes</td></tr>
<tr><td>G4.7.3-Vbus ref = 1120V</td><td>500 to 1500 V</td><td>Allows fixing the bus voltage reference for the Night mode.</td><td>Yes</td></tr>
<tr><td>G4.7.4-Night mode activate = No</td><td>No<br>Yes</td><td>Allows activating/deactivating the Night mode if it is enabled on G4.7.1.</td><td>Yes</td></tr>
</table>

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>G4.7.5-Force Night mode = No</td><td>No<br>Yes</td><td>Allows to force the activation of the night mode if parameter G4.7.1 is enabled.<br>This parameter forces a night mode with standard transition (Stop, open DU and regular reactive). There are two options depending on the option selected in G4.7.1:<br><table><tr><th>G4.7.1</th><th>DESCRIPTION</th></tr><tr><td>Autonight</td><td>If the type of transition selected is auto-night the options for its use will depend on the value defined in parameter G2.2.1.<br><table><tr><th>G2.2.1</th><th>DESCRIPTION</th></tr><tr><td>AC</td><td>SCADA must reduce the power limit G3.1.1 Active Power to 0% (O below G2.2.3)<br>G4.7.5 Force Night mode (Omit night conditions G2.5)</td></tr><tr><td>DC</td><td>SCADA must reduce the power of the DCDC converter to 0%.<br>Reduce the power limit G3.1.1 Active Power to 0% (Or at least below G2.2.3)<br>G4.7.5 Force Night Mode (Omit night conditions G2.5)</td></tr><tr><td>NO</td><td>With this option it is not possible to force the night mode</td></tr></table></td></tr><tr><td>Standard</td><td>If the type of transition is standard, then this parameter is used as follows:<br>The inverter has to be in READY status.<br>Apply G4.7.5=Yes, the inverter will go into night mode (No check of G2.5).</td></tr></table><br>Note: this parameter is shown if [G4.7.6-Enable float Night mod] = No and [SV4.4.1.1-Inverter type] ≠ Statcom.</td><td>Yes</td></tr>
<tr><td>G4.7.6-Enable float Night mod = No</td><td>No<br>Yes</td><td>Enables (Yes) or disables (No) the floating night mode.<br>Note: this parameter is shown if [G4.7.1-Enable] >= Standard and [SV4.4.1.1-Inverter type] ≠ Statcom.</td><td>Yes</td></tr>
<tr><td>G4.7.7-Retry Time N to D = 3600 s</td><td>0 to 43200 s</td><td>Night mode retry time night to day.<br>Note: this parameter is shown if [G4.7.6-Enable float Night mod] = Yes and [SV4.4.1.1-Inverter type] ≠ Statcom.</td><td>Yes</td></tr>
<tr><td>G4.7.8-Countdown Retry N to D = Not started</td><td>0 = Not started<br>1 to 65535 s</td><td>Night mode retry countdown night to day.<br>Note: this parameter is shown if [G4.7.6-Enable float Night mod] = Yes and [SV4.4.1.1-Inverter type] ≠ Statcom.</td><td>Yes</td></tr>
<tr><td>G4.7.9-Autonight Min Vdc = 1250 V</td><td>850 to 1400 V</td><td>Autonight minimum DC voltage.<br>Note: this parameter will be shown when [G4.7.1-Enable] is configured as Autonight.</td><td>Yes</td></tr>
<tr><td>G4.7.10-Autonight gfdi = No</td><td>No<br>Mode 1<br>Mode 2</td><td>GFDI mode in autonight.</td><td>Yes</td></tr>
</table>

**Note:** Once the device is in NIGHT MODE, only two modules are executed instead of all the modules. And depending on the Q_ref in night mode, more modules will be started. In case G2.4.6-Available Q backup is set to 100%, then when entering night mode all modules will be started.

The inverter will operate in night mode if the conditions are fulfilled and it is in NIGHT mode.
This mode requires the DC cabinet (or DU) to isolate the inverter from the solar panels. Fans can be activated depending on the configured operation mode.

Once conditions for Night Mode are fulfilled, the procedure is the following:

1. Open DU.
2. DC bus soft charge at voltage reference level set in G4.7.3-Vbus ref through central AC soft charge without allowing DU closing once the process is finished.
3. Start all available modules (network connection and IGBTs activation).
4. DC bus regulation at voltage set in G4.7.3-Vbus ref.
5. Reactive current regulation at the value set in G4.7.2-Q ref. All modules regulate at the same Q reference, so a power injection into the network is produced.

When the inverter is operating in Night Mode, the corresponding warning (W13:Night mode) will be hown in the display.

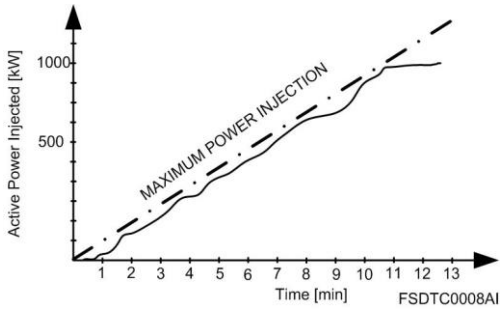
If there are no available modules (fault, disabled…) the inverter stops and triggers warning W6:No modules.

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## Subgroup G4.8: Gradients

Parameters to configure the power gradients (start, rise, drop and stop) can be found in this group.

### G4.8.1: Start

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G4.8.1.1-Enable P = Yes** [4] | No<br>Yes | Enable or disable the active power start gradient.<br><br>When this option is enabled, power is injected gradually. The converter will increase its output power per minute depending on the set value.<br>  <br><br>**Note:** The value set in this parameter is retained after restarting the converter.<br>If gradient is disabled, the inverter defaults to 50%/sec. | Yes |
| **G4.8.1.2-P gradient = 10.0%/s** [4] | 0.1 to 3000.0 %/s | Set the active power start gradient value. | Yes |
| **G4.8.1.3-Enable Q = Yes** | No<br>Yes | Enable or disable the reactive power start gradient. | Yes |
| **G4.8.1.4-Q gradient = 10.0%/s** | 0.1 to 3000.0 %/s | Set the reactive power start gradient value.<br>**Note:** This parameter is only shown when the reactive power start gradient is enabled. | Yes |

### G4.8.2: Rise

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G4.8.2.1-Enable P = Yes** [5] | No<br>Yes | Enable or disable the active power rise gradient while the inverter is running. The converter will increase its output power per minute depending on the set value. | Yes |
| **G4.8.2.2-P gradient = 10.0%/s** [5] | 0.1 to 3000.0 %/s | Set the active power rise gradient value. | Yes |
| **G4.8.2.3-Enable Q = Yes** [6] | No<br>Yes | Enable or disable the reactive power rise gradient while the inverter is running. The converter will increase its output power per minute depending on the set value. | Yes |
| **G4.8.2.4-Q gradient = 10.0%/s** [6] | 0.1 to 3000.0 %/s | Set the reactive power rise gradient value.<br>**Note:** This parameter is only shown when the reactive power rise gradient is enabled. | Yes |

[4] Parameter not shown in STATCOM.
[5] These parameters will only be shown if [G4.8.1.1-Enable P] is set to YES and the inverter is not STATCOM.
[6] These parameters will only be shown if [G4.8.1.3-Enable Q] is set to YES and the inverter is not STATCOM.

### G4.8.3: Drop

| Screen | Range | Function | Set on run |
|---|---|---|---|
| G4.8.3.1-Enable P = No [5] | No<br>Yes | Enable or disable the active power drop gradient while the inverter is running. The converter will decrease its output power per minute depending on the set value. | Yes |
| G4.8.3.2-P gradient = 10.0%/s [5] | 0.1 to 3000.0 %/s | Set the active power drop gradient value. | Yes |
| G4.8.3.3-Enable Q = No [6] | No<br>Yes | Enable or disable the reactive power drop gradient while the inverter is running. The converter will decrease its output power per minute depending on the set value. | Yes |
| G4.8.3.4-Q gradient = 10.0%/s [6] | 0.1 to 3000.0 %/s | Set the reactive power drop gradient value. | Yes |

### G4.8.4: Stop

| Screen | Range | Function | Set on run |
|---|---|---|---|
| G4.8.4.1-Enable P = No [5] | No<br>Yes | Enable or disable the active power stop gradient. The converter will gradually decrease its output power until reaching zero. | Yes |
| G4.8.4.2-P gradient = 10.0%/s [5] | 0.1 to 3000.0 %/s | Set the active power stop gradient value. | Yes |
| G4.8.4.3-Enable Q = No [6] | No<br>Yes | Enable or disable the reactive power stop gradient. The converter will gradually decrease its output power until reaching zero. | Yes |
| G4.8.4.4-Q gradient = 10.0%/s [6] | 0.1 to 3000.0 %/s | Set the reactive power stop gradient value. | Yes |

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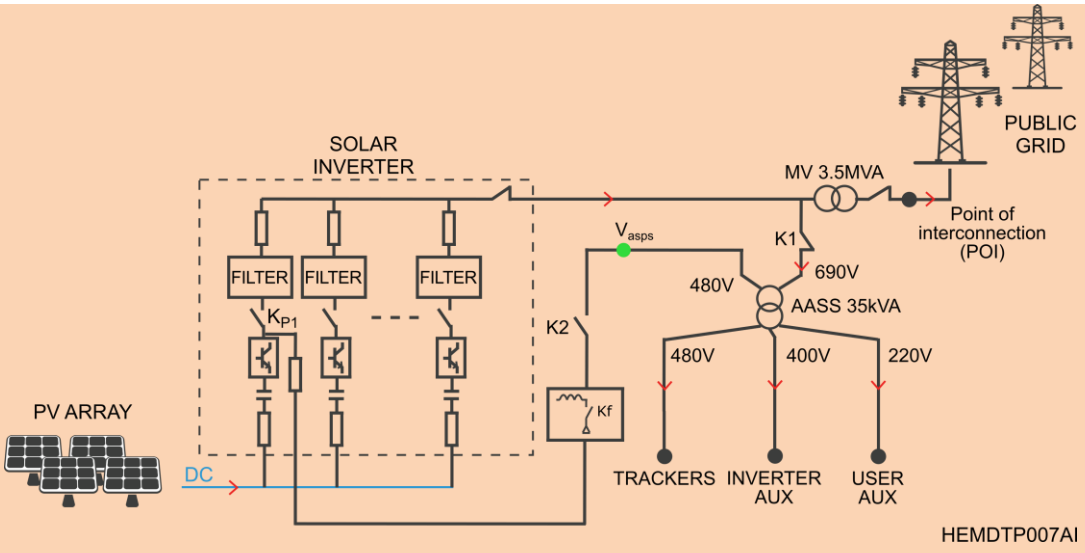
### Others

The following parameters allow configuring the behavior of the product in case of communications loss between the PPC and the inverter.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| G4.8.5-P rec ramp = 10.0 %/s | 0.1 to 50.0 %s | Defines the value of the P recovery ramp that will be applied when communications are restored..<br><br>**Note:** this parameter will be shown if [G6.3.9-Ramp selector] = Recovery ramps. | Yes |
| G4.8.6-Q rec ramp = 10.0 %/s | 0.1 to 50.0 %s | Defines the value of the Q recovery ramp.<br><br>**Note:** this parameter will be shown if [G6.3.9-Ramp selector] = Recovery ramps. | Yes |
| G4.8.7-AEMO Gradient rise P = 10.0 %/s | 0.3 to 3000.0 %s | Parameters G4.8.7 to G4.8.10 define the gradients for the rise and drop references whenever communications with the PPC are lost.<br><br>[G4.8.7-AEMO Gradient rise P] defines the P rise gradient.<br>[G4.8.8-AEMO Gradient rise Q] defines the Q rise gradient<br>[G4.8.9-AEMO Gradient drop P] defines the P drop gradient.<br>[G4.8.10-AEMO Gradient drop Q] defines the Q drop gradient.<br><br>**Note:** these parameters are shown if [G6.3.9-Ramp selector] = AEMO. | Yes |
| G4.8.8-AEMO Gradient rise Q = 10.0 %/s | 0.3 to 3000.0 %s | | Yes |
| G4.8.9-AEMO Gradient drop P = 10.0 %/s | 0.3 to 3000.0 %s | | Yes |
| G4.8.10-AEMO Gradient drop Q = 10.0 %/s | 0.3 to 3000.0 %s | | Yes |

## Subgroup G4.9: Trackers mode

Trackers mode helps guarantee an uninterrupted power supply for the inverter and the auxiliary services circuit in case there is a power loss due to a voltage dip or drop. Please notice that this function is not available in STATCOM.




To perform the corresponding maneuver, the inverter implements a hardware solution that maintains electronics' power supply long enough so that control can activate this operation mode. Once activated, the own voltage generated by the auxiliary services circuit (Vasps in the figure above) will feed the equipment's electronic. Only the module electrically prepared to inject power to the auxiliary services circuit will be able to collaborate.

The following table describes all parameters necessary to operate in Trackers Mode:

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G4.9.1-Enable = No** | No<br>Yes | Enable or disable trackers mode. | Yes |
| **G4.9.2-Vac out = 480V** | 0 to [SV4.4.1.2] V | Set the output voltage. Possible values for this parameter vary between 0 and the nominal AC voltage (SV4.4.1.2-Nominal AC voltage). | Yes |
| **G4.9.3-Vac min = 80.0%** | 0.0 to 100.0% | Set the minimum grid voltage, as a percentage, to activate trackers mode. | Yes |
| **G4.9.4-Time start = 1.0s** | 0.1 to 1.5 s | Set trackers mode activation time. | Yes |

Trackers Mode operation will be activated after a network fault if the following conditions are fulfilled:

- The fault produced was a Low Vac Fault (F6) or Islanding Fault (F11). Previously, a voltage dip may or may not have occurred.
- Network voltage Vac is below a configurable threshold (G4.9.3).
- The algorithm is enabled (G4.9.1).
- Inverter selector is On.

Trackers Mode operation will stop if any of the following conditions is fulfilled:

- A Trackers Mode specific warning is produced.
- Network voltage Vac is restored, and a configurable threshold is overcome (G4.9.3).
- The algorithm is disabled (G4.9.1).
- The inverter selector is Off.

Central control handles a single relay output to activate / deactivate the contactors involved in this operation mode: Kf, K2 and K1. All of them are electrically interlocked so that when Kf and K2 are closed, K1 is open. Likewise, when Kf and K2 are open, K1 is closed.

Once Trackers Mode conditions have been detected, the activation sequence begins:

- Verify that the module prepared to work in Trackers Mode is available.
- Open contactor K1 and close contactors Kf and K2.
- Module is activated without closing Kp1 contactor (AC contactor).
- Module generates the Vac set in G4.9.2.

Trackers Mode deactivation sequence will be:

- Stop the active module.
- Open contactors K2 and Kf, close contactor K1.
- The equipment enters in status READY and waits for start conditions to be fulfilled.

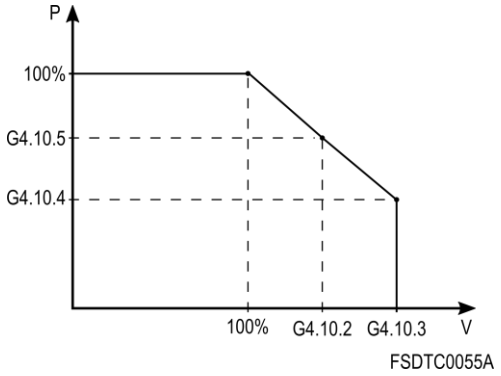
In both sequences (activation and deactivation) and during operation in Trackers Mode, a series of messages (W21, W22, W23, W24, W25) may be triggered to indicate if any problem is detected. Most of them stop operation, which will cause a power loss in the equipment electronics. Please refer to section 3 for further information about these messages.

Additional information is provided in visualization group SV11.2.

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## Subgroup G4.10: V/P

These parameters allow configuring the algorithm for power control depending on the voltage. Please notice that they will not be shown in STATCOM inverters.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G4.10.1-V/P Enable = No** | No<br>Yes | Allows enabling (Yes) or disabling (No) V/P control. Power will depend on the voltage. | Yes |
| **G4.10.2-V/P V1 = 110.00 %** | 100.00 to [G4.10.3] % | To define the voltage/power curve, the algorithm uses four points: two voltage values (G4.10.2 and G4.10.3) and two power values (G4.10.4 and G4.10.5).<br>  <br> | Yes |
| **G4.10.3-V/P V2 = 120.00 %** | [G4.10.2] to 140.00 % | | Yes |
| **G4.10.4-V/P P1 = 100.00 %** | [G4.10.5] to 100.00 % | | Yes |
| **G4.10.5-V/P P2 = 0.00 %** | 0.00 to [G4.10.4] % | | Yes |
| **G4.10.6-V/P Hysteresis = 0.0%** | 0.0 to 10.0 % | Hysteresis to recover active power with voltage reduction. | Yes |
| **G4.10.8-V/P Delay hysteresis = 60.0 sec** | 0.0 to 90.0 s | Allows defining a delay, in seconds, to the P(V) curve. If the value set in G4.10.6 is overcome, then this delay will be applied before starting the algorithm. | Yes |
| **G4.10.9-V/P Response time = 10.0 s** | 0.0 s to 90.0 s | Adjusts the time response in seconds of the P(V) algorithm. Response time is defined as the time that takes reaching 90% of the reference since the moment when the change occurs. | Yes |
| **G4.10.10-V/P Reference power = Relative** | Relative<br>Nominal | Defines the reference power mode for the P(V) event.<br>• Relative mode: before the perturbation, P(V) event, the active power output which is modulated is the value of power before the event.<br>• Nominal mode: the active power output which is modulated is the value of rated power. | Yes |
| **G4.10.11-V/P Recover mode = Absolute** | Absolute<br>Relative | Defines the recovery mode after the P(V) event. Recovery mode defines how power will be recovered after reaching the activation voltage (V1). There are two modes:<br>• Absolute: keeps the power limitation existent at the beginning of recovery until reaching deactivation voltage.<br>• Relative: power limitation is kept during hysteresis, then, power is recovered with the voltage. | Yes |

## Subgroup G4.11: Sup. Voltage Rise

These parameters allow configuring the algorithm for suppression of voltage rise (SVR).

This function is a requirement of the Japanese normative JEAC 9701-2012 to limit an uncontrolled rise in grid voltage.

Two voltage thresholds are defined, in G4.11.2 and G4.11.8. Whenever voltage overcomes the first threshold (G4.11.2), the inverter will inject reactive inductive power with the gradient set in G4.11.4. the injection of reactive power is limited by G4.11.6.

If voltage rising cannot be suppressed, and the second threshold (G4.11.8) is overcome, the injection of active power is reduced to zero with the gradient set in G4.11.9. Power factor limitation (G4.11.6) must be kept at all times.

Active power that was present before voltage disturbance is recovered with a gradient adjustable in G4.11.10 once voltage has stabilized and goes back to being lower than G4.11.7 – G4.11.8. Power factor limitation is kept until voltage has stabilized and goes back to being lower than G4.11.2 – G4.11.3. In that instant, injection of reactive inductive power is reduced with the gradient G4.11.5 until reaching reactive power that was present before disturbance. Q priority must be enabled.

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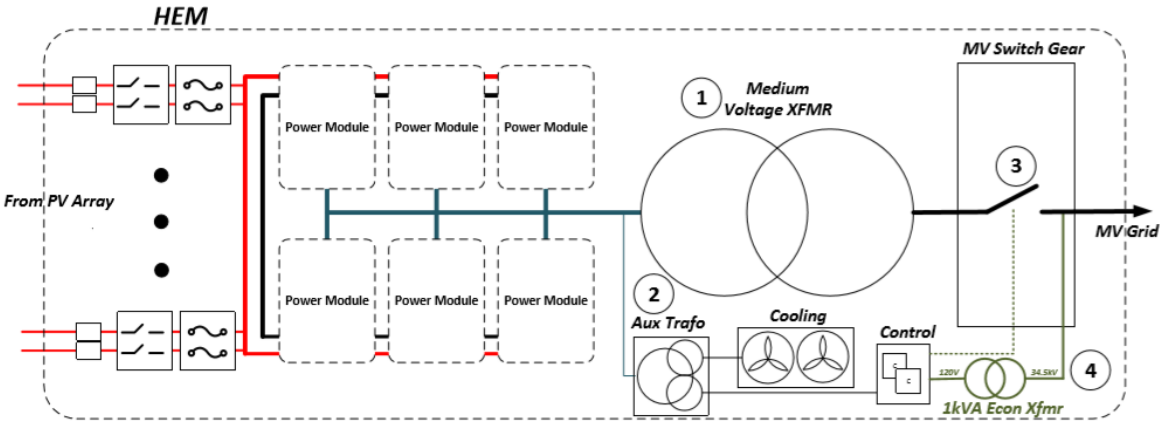
| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G4.11.1-Enable = No** | No<br>Yes | Allows enabling or disabling the algorithm for suppression of voltage rise.<br>**Note:** Parameters G4.11.2 to G4.11.10 will only be shown once the SVR algorithm has been enabled. | Yes |
| **G4.11.2-V1 Threshold = 105.0 %** | 100.0% to [G4.11.7] | Lower threshold to activate the SVR algorithm. If grid voltage overcomes this value, the inverter will start injecting reactive power with the gradient set in G4.11.4 and limited by G4.11.6. | Yes |
| **G4.11.3-Hysteresis V1 = 2.0 %** | 0.5 to 10.0 % | When grid voltage is lower than V1 Threshold minus this hysteresis (G4.11.2 – G4.11.3), the injection of reactive inductive power will be reduced. | Yes |
| **G4.11.4-Grad. Inj. Q = 0.3 %/s** | 0.1 to 200.0 %/s | Sets the gradient for Q injection when V1 threshold (G4.11.2) is overcome. | Yes |
| **G4.11.5-Grad. Recover Q = 1.7 %/s** | 0.1 to 200.0 %/s | Sets the gradient for Q recovery when voltage is below *V1 Threshold – Hysteresis V1*. | Yes |
| **G4.11.6-CosFi limit = 0.85** | 0.80 to 1.00 | Sets the power factor limit for Q injection. | Yes |
| **G4.11.7-V2 Threshold = 107.0 %** | [G4.11.2] to 120.0 % | Sets the second threshold from which P will be reduced to zero with the gradient set in G4.11.9 and limited by G4.11.6. | Yes |
| **G4.11.8-Hysteresis V2 = 2.0 %** | 0.5 to 10.0 % | When grid voltage is lower than V2 Threshold minus this hysteresis (G4.11.7 – G4.11.8), active power injection will be reduced. | Yes |
| **G4.11.9-Grad. Reduc. P = 0.3 %/s** | 0.1 to 200.0 %/s | Sets the gradient for P reduction when V2 threshold (G4.11.7) is overcome. | Yes |
| **G4.11.10-Grad. Recover P = 1.7 %/s** | 0.1 to 200.0 %/s | Sets the gradient for P recovery when voltage is below *V2 Threshold – Hysteresis V2*. G4.11.6 must be kept. | Yes |

**Note**: These parameters will not be shown in STATCOM inverters.

## Subgroup G4.12: Econmode

The Econmode functionality allows the medium voltage transformer to be disconnected from the grid during the night in order to remove the transformers losses.

During the morning start, it also pre-magnetizes the transformer to avoid inrush current (Re-closure of the MV switchgear).




Disconnection from the grid during the night:

When night conditions are detected (meaning the array power reaches a value below the minimum DC power of the inverter), the stop sequence of the inverter will bring it stand-by mode and stop producing active power. The ventilation system will continue operating during a time (configurable). Once this time is elapsed, a delay in the MV switchgear relay opening can also be included. With these times (ventilation + MV switchgear relay opening delay), it is possible to consider the time needed by the trackers to move to security position before opening the MV switchgear relay.

In this moment, the Econmode transformer will start to supply power to the control system and user auxiliaries connected to it.

Econmode is not compatible with “night mode”, neither active heating or DC switch disconectors opening during that period and will be ignored until the next day. Thus, if some of the specified modes are activated, Econ mode would be disabled, and MV protection switchgear will not be open. There will exist a configuration for “ECON-Mode Priority” where the relevance of each mode will be defined.

Opening and closure of the protection MV switchgear will be automatized and do not allow the user to do it manually.

Re-closure of the MV switch (avoiding inrush current):

In the morning when solar irradiance starts to increase, the DC voltage will rise until it reaches the minimum DC voltage configured for day conditions. Once this is condition is reached, pre-magnetization of the MV transformer starts. Firstly, voltage is measured on the medium voltage side which is connected to the grid. Meanwhile, with little DC power needed, one module will start-up as a voltage source and start injecting a very small current towards the transformer. This current delivered by the module will increase steadily, until the voltage on the inverter side is equal to the voltage measured on the MV line side. When the same value is measured, the switch-disconnector of the MV protection switchgear is closed.

This pre-magnetization will avoid two things:

1. Inrush current at the transformer
2. Closing switch-disconnector under load conditions, thus not affecting electrical contacts of the device itself. (Note: mechanical wear and a tare still occurs from the opening and closing of the switch).

The only case where the pre-magnetization cannot be achieved is when the DC switch disconnectors open as a result of voltage event in the grid at night, during Econmode. In this case, the MV switch disconnector must be hard-closed to energize the inverter automatically (without pre-magnetization). This switching maneuver has a limit on the number of operations due to the in-rush stress suffered on those situations.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G4.12.1-Enable = No** | No<br>Yes | Parameter to enable the Econmode functionality in the inverter.<br>**Note**: This function depends on (SG10.6.1 Econmode Ready = Yes and SG10.1.3 = HEM ). | Yes |
| **G4.12.2-Activation = No** | No<br>Yes | This parameter allows the SCADA to decide to activate the Econmode algorithm during the night.<br>**Note**: this function depends on (G4.12.1= Yes and SG10.1.3 = HEM). | Yes |
| **G4.12.3-Priority = No** | No<br>Yes | Sets the priority of the Econmode over night modes (night mode, active heating and DC disconnection)<br>There are two situations depending on the priority selected:<br><br>**YES:** It disables night modes activation,a active heating (internal or external) and DU night opening (SG1.2.3).<br><br>N**O:** The activation of night modes have priority over Econmode. This parameter can only be changed with the MV switchgear closed in order to avoid several openings at night.<br>It is not allowed to change the Econmode priority with the MV switchgear open, a warning will appear.<br>**Note**: this function depends on (G4.12.1 = Yes and SG10.1.3 = HEM). | Yes |
| **G4.12.4-Open Delay = 30 s** | 0 to 60 s | Configurable delay in the opening of the MV switchgear by Econmode activation.<br>**Note**: This function depends on (G4.12.1 = Yes and SG10.1.3 = HEM). | Yes |
| **G4.12.5-Time Fans Off = 1800 s** | 60 to 3600 s | Time to open the MV Switchgear since the fans of impulsion are extinguished. This is to avoid thermal rebounds of the MV transformer.<br>**Note**: This function depends on (G4.12.1 = YES and SG10.1.3 = HEM). | Yes |
| **G4.12.6-Clos Witho Prm Allow = No** | No<br>Yes | Parameter that allows to close the MV Switchgear disconnector without pre-magnetizing.<br>According to the explanations in F192, if the DU is open and cannot be closed, this parameter allows the Econmode sequence to close the MV switchgear automatically without pre-magnetizing.<br><br>**Note**: This function depends on (G4.12.1 Enable = No and SG10.1.3-Product = HEM). | Yes |

## Subgroup G4.13: AntiPID

PID - Potential induces degradation. It is the degradation of power suffered by the solar panels when the inverters are in isolated poles due to the parasite currents that appear. If the effect of the degradation has already started it is possible to control and even reverse it. The AntiPID device is fed by an auxiliary voltage (230V) so it is incompatible with the Econmode.

This functionality is compatible with night and heating mode because both open the DU.

The parameters of this group will be visible if the DU is configured as isolated poles (G10.2= Standard) and the equipment is not PCS.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G4.13.1-Enable = No** | No<br>Yes | Enables the anti-PID function. | Yes |
| **G4.13.2-Activate = No** | No<br>Yes | The activation closes a digital output of the Central DU. This is used to close the GFDI counter. Yes G4.13.1= No this parameter will be disabled To activate the anti-PID function the equipment must be in night mode and the DU can be open or closed. | Yes |
| **G4.13.3-Time Aft Night Cond = 30 s** | 0 to 60 s | Time after night conditions. | Yes |

## Subgroup G4.14: POD

Power oscillation damping functionality allows to attenuate the voltage oscillations from 5 to 55 Hz.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G4.14.1-POD Enable = No** | No<br>Yes | Enables POD (Power Oscillation Damping) function. | Yes |
| **G4.14.2-POD Damping = 1** | 1 to 100 | Allows to configure the damping that must be applied by the algorithm.<br><br>**Note:** This parameter is only displayed if G4.14.1-POD Enable = Yes | Yes |
| **G4.14.3-POD AC Voltage Filter = 1.00 Hz** | 0.01 to 100.00 Hz | Allows to define the cut frequency for AC voltage. This frequency is needed for the LPF (Low Pass Filter).<br><br>**Note:** This parameter is only displayed if G4.14.1-POD Enable = Yes | Yes |

# Subgroup G4.15: Simple – Signal

## G4.15.1-Enable

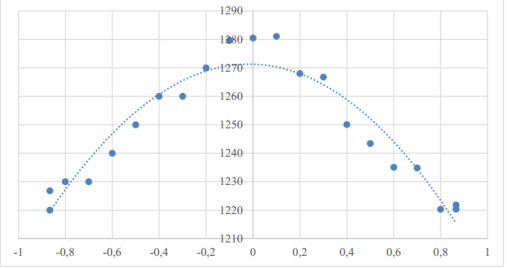
| Screen | Range | Function | Set on run |
|---|---|---|---|
| G4.15.1.1-Modulation type = Simple-Double | Double<br>Simple-Double | Sets the type of modulation to apply.<br><br>OPTION / DESCRIPTION<br>Double: Keeps better balance between two DC-Link semi-buses.<br>Simple-Double: Dynamic changes between single and double modulation Double modulation keeps two DC-Link semibuses the same, so IGBTs support the same voltage. But single modulation has better results in THDi emission. This option checks the voltage of the DC-Link so that when a Vsemibus is higher than 655 V when Q=0 then the double modulation will be followed.<br><br>**Note**: this parameter, in versions prior to 4.3.0, used to be [SG1.3.a1-Modulation type]. | Yes |

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## G4.15.2-Double Changes

**Note:** parameters within this subgroup will only be displayed if [G4.15.1.1-Modulation type] = Simple-Double.

### G4.15.2.1-Operation region

| Screen | Range | Function | Set on run |
|---|---|---|---|
| G4.15.2.1.1-Ofset curv Vdc-Q = 0.00 V | -50.00 to 150.00 V | Modulation applies an offset to the threshold define by the curve V(Q).<br>The following graph represents the V(Q) curve according to the value of this parameter.<br>  <br><br>**Note**: this parameter, in versions prior to 4.3.0, used to be [SG1.3.a2-Modulation offset]. | Yes |
| G4.15.2.1.2-Max Q = 50.00 % | 0.00 to 100.00 % | Maximum reactive power that allows working in single modulation. From this value, it is necessary to change to double modulation.<br><br>**Note**: this parameter, in versions prior to 4.3.0, used to be [SG1.3.a4-Max Q]. | Yes |

### G4.15.2.2-Gradients

| Screen | Range | Function | Set on run |
|---|---|---|---|
| G4.15.2.2.1-Max P/s = 20.00 %/s | 0.10 to 100.00 %/s | Maximum increase of active power in a time range (defined in [G4.15.2.4-Single delay]). When this increase is reached in the given time interval, the modulation will switch to double.<br><br>**Note:** this parameter, in versions prior to 4.3.0, used to be [SG1.3.a5-Max P/s]. | Yes |
| G4.15.2.2.2-Max Q/s = 3.00 %/s | 0.10 to 100.00 %/s | Maximum increase of reactive power in a time range (defined in [G4.15.2.4-Single delay]). When this increase is reached in the given time interval, the modulation will switch to double.<br><br>**Note:** this parameter, in versions prior to 4.3.0, used to be [SG1.3.a6-Max Q/s]. | Yes |

### G4.15.2.3-Bus unbalance

| Screen | Range | Function | Set on run |
|---|---|---|---|
| G4.15.2.3.1-Instant Semibus = 120 V | 50 to 200 V | Allows to set the allowed bus unbalance, from which the modulation change to double will be established (the FPGA measurement is taken as a reference). Once the bus unbalance is restored, the modulation will return to single.<br><br>**Note:** this parameter, in versions prior to 4.3.0, used to be [SG1.3.a8-FPGA change threshol]. | Yes |
| G4.15.2.3.2-Filtered semibus = 80 V | 50 to 200 V | Voltage deviation threshold. When this value is reached, the module modulation will change to double (according to DSP measurement).<br><br>**Note:** this parameter, in versions prior to 4.3.0, used to be [SG1.3.a7-DSP Change Thresh]. | Yes |
| G4.15.2.3.3-Anti-Glitch delay = 250 µs | 10 to 2000 µs | Allows to set the delay between the detection of an unbalance and the change to double modulation.<br><br>**Note:** this parameter, in versions prior to 4.3.0, used to be [SG1.3.a14-Modulat. delay FPGA]. | Yes |
| G4.15.2.4-Single delay = 5.00 s | 0.10 to 100.00 s | Defines the delay time for the module to switch to single modulation once the requirements to switch back to single modulation are met.<br><br>**Note:** this parameter, in versions prior to 4.3.0, used to be [SG1.3.a3-Single delay]. | Yes |

### G4.15.3-Protection

**Note:** parameters within this subgroup will only be shown if [G4.15.1.1-Modulation type] = Single-Double. Additionally, parameters G4.15.3.2 to G4.15.3.5 will only be visible when [G4.15.3.1-Enable] ≠ Disabled.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| G4.15.3.1-Enable = Warning | Warning<br>Fault<br>Disabled | It allows configuring the reaction of the inverter when the number of changes of modulation reaches the value set in parameter [G4.15.3.2-Consec changes].<br><br>OPTION / DESCRIPTION<br>Warning: When the number of changes in modulation exceeds the value set in [G4.15.3.2-Consec changes], [W35:Force Double] will be triggered.<br>Fault: When the number of changes in modulation exceeds the value set in [G4.15.3.2-Consec changes], [F200:Simple Modulation] will be triggered.<br>Disabled: No protection is enabled when the number of changes in modulation exceeds the value in [G4.15.3.2-Consec changes].<br><br>**Note:** this parameter, in versions prior to 4.3.0, used to be [SG1.3.a9-Excessive changes]. | Yes |

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G4.15.3.2-Consec changes = 10** | 0 to 30 | Maximum number of changes of modulation allowed per minute. When this value is overcome, the protection configured in [G4.15.3.1-Enable] will be triggered.<br><br>**Note:** this parameter, in versions prior to 4.3.0, used to be [SG1.3.a11-Max N changes/min]. | Yes |
| **G4.15.3.3-Additional time = 130.0 %** | 100.0 to 150.0 % | Defines the extra hysteresis time.<br><br>**Note:** this parameter, in versions prior to 4.3.0, used to be [SG1.3.a12-Extra t hysteresis]. | Yes |
| **G4.15.3.4-Pmin threshold = 10.0 %** | 0.0 to 100.0 % | Allows to define the minimum power value under which the changes to double signal are not counted. Under this level of power, the changes to double signal are not considered as dangerous from a thermal perspective.<br><br>**Note:** this parameter, in versions prior to 4.3.0, used to be [SG1.3.a10-Pmin count changes]. | Yes |
| **G4.15.3.5-Allowed Double time = 10.0 %** | 0.0 to 100.0 % | It allows to configure the maximum period in which the inverter is allowed to work in double modulation. Once this period has ended, [W35:Force Double] and [F200:Simple Modulation], are triggered.<br><br>**Note:** this parameter, in versions prior to 4.3.0, used to be [SG1.3.a13-Allowed Double time] | Yes |

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# Group G5: Protections

This group allows configuring (enable / disable, set limits) the different available protections.

The high and low voltage protections (G5.X) save the values of the low measured AC voltage or the one estimated in the POI according to the value set in parameter SG1.2.26.

## Subgroup G5.1: Low AC voltage

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.1.1-Enable = YYNNN** | Slow: Y / N<br>Fast: Y / N<br>Fast 2: Y / N<br>Very fast: Y / N<br>Very fast 2: Y / N | There are five low voltage protection levels:<br><br>- Slow protection<br>- Fast protection<br>- Fast 2 protection<br>- Very fast protection<br>- Very fast 2 protection<br><br>To enable each protection, set the corresponding bit to Yes and adjust the voltage values and delay as desired (parameters G5.1.2 to G5.1.11). | Yes |
| **G5.1.1b-Interpolation = No** | No<br>Yes | **Note: This parameter only applies for Egypt grid code.**<br><br>Whenever **interpolation of low voltage protections** is enabled (**G5.1.1b = YES)**, the inverter will calculate, in real time, the time to trigger protections depending on the current voltage and the curves/points configured in parameters G5.1.2 to G5.1.11.<br><br>To enable interpolation of high voltage protections, go to parameter G5.2.1b. | Yes |
| **G5.1.2-Slow protection =** [7] | 0.0 to 100.0 % | Protection levels can be set from 0.0% up to 100.0%. 100.0% corresponds with the rated voltage.<br><br>The slow protection delay can be set from 0.0s up to 6550.0s. Fast protections delay can be set from 0.00s up to 655.00s.<br>Very fast protections delay range varies between 0.000 and 65.500s<br><br>It is possible to enable/disable each protection level individually in G5.1.1. Default values, on the other hand, depend on the selected normative. | Yes |
| **G5.1.3-Delay slow protection =** [7] | 0.0 to 6550.0 s | | Yes |
| **G5.1.4-Fast protection =** [7] | 0.0 to 100.0% | | Yes |
| **G5.1.5-Delay fast protection =** [7] | 0.0 to 655.00 s | | Yes |
| **G5.1.6-Fast 2 protection =** [7] | 0.0 to 100.0 % | | Yes |
| **G5.1.7-Delay fast 2 protection =** [7] | 0.00 to 655.00 s | | Yes |
| **G5.1.8-Very fast protection =** [7] | 0.0 to 100.0% | | Yes |
| **G5.1.9-Delay very fast prot =** [7] | 0.000 to 65.500 s | | Yes |
| **G5.1.10-Very fast 2 protection =** [7] | 0.0 to 100.0% | | Yes |
| **G5.1.11-Delay very fast 2 prot =** [7] | 0.000 to 65.500 s | | Yes |

[7] These values depend on the normative selected. See table below.

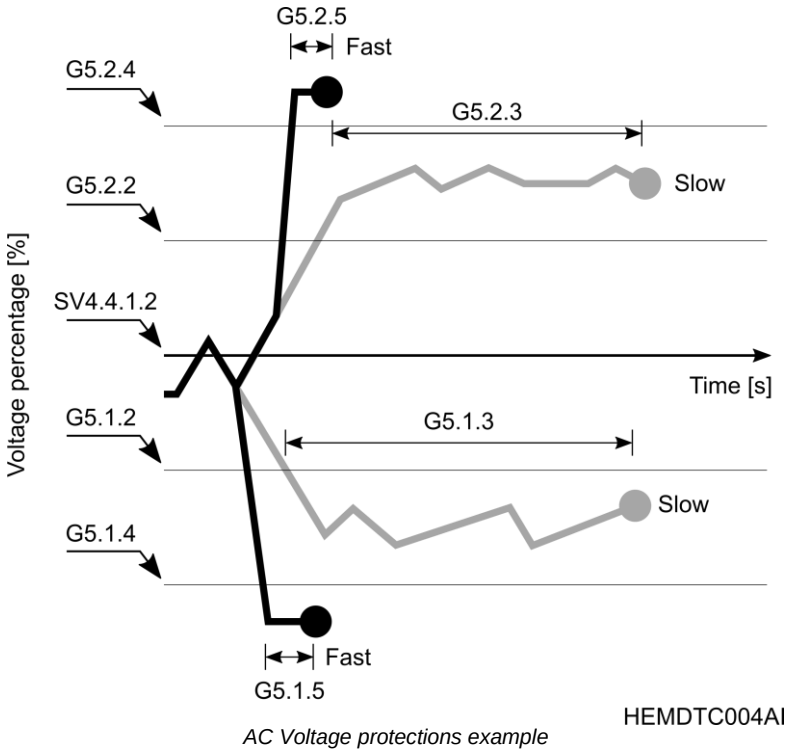
## Subgroup G5.2: High AC voltage

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.2.1-Enable = YYNNN** | Slow: Y / N<br>Fast: Y / N<br>Fast 2: Y / N<br>Very fast: Y / N<br>Very fast 2: Y / N | There are five protection levels:<br><br>- Slow protection<br>- Fast protection<br>- Fast 2 protection<br>- Very fast protection<br>- Very fast 2 protection<br><br>To enable each protection, set the corresponding bit to Yes and adjust the voltage values and delay as desired (parameters G5.1.2 to G5.2.11). | Yes |
| **G5.2.1b-Interpolation = No** | No<br>Yes | **Note: This parameter only applies for Egypt grid code.**<br><br>Whenever **interpolation of high voltage protections** is enabled (**G5.2.1b = YES**), the inverter will calculate, in real time, the time to trigger protections depending on the current voltage and the curves/points configured in parameters G5.2.2 to G5.2.11.<br><br>To enable interpolation of high voltage protections, go to parameter G5.1.1b. | Yes |
| **G5.2.2-Slow protection = [7]** | 100.0 to 150.0 % | Protection levels can be set from 0.0% up to 150.0%. 100.0% corresponds with the rated voltage.<br><br>The slow protection delay can be set from 0.0s up to 6550.0s. Fast protections delay can be set from 0.00s up to 655.00s.<br>Very fast protections delay range varies between 0.000 and 65.500s<br><br>It is possible to enable/disable each protection level individually in G5.2.1. Default values, on the other hand, depend on the selected normative. | Yes |
| **G5.2.3-Delay slow prot = [7]** | 0.0 to 6550.0 s | | Yes |
| **G5.2.4-Fast protection = [7]** | 100.0 to 150.0 % | | Yes |
| **G5.2.5-Delay fast protection = [7]** | 0.00 to 655.00 s | | Yes |
| **G5.2.6-Fast 2 protection = [7]** | 100.0 to 150.0 % | | Yes |
| **G5.2.7-Delay fast 2 protection = [7]** | 0.00 to 655.00 s | | Yes |
| **G5.2.8-Very fast protection = [7]** | 100.0 to 150.0% | | Yes |
| **G5.2.9-Delay very fast prot = [7]** | 0.000 to 65.500 s | | Yes |
| **G5.2.10-Very fast 2 protection = [7]** | 100.0 to 150.0% | | Yes |
| **G5.2.11-Delay very fast 2 prot = [7]** | 0.000 to 65.500 s | | Yes |

[7] These values depend on the normative selected. See table below.

The figure below shows an operation example with four voltage protections enabled (two low and two high).

EN




*AC Voltage protections example*

## Subgroup G5.3: Low AC freq

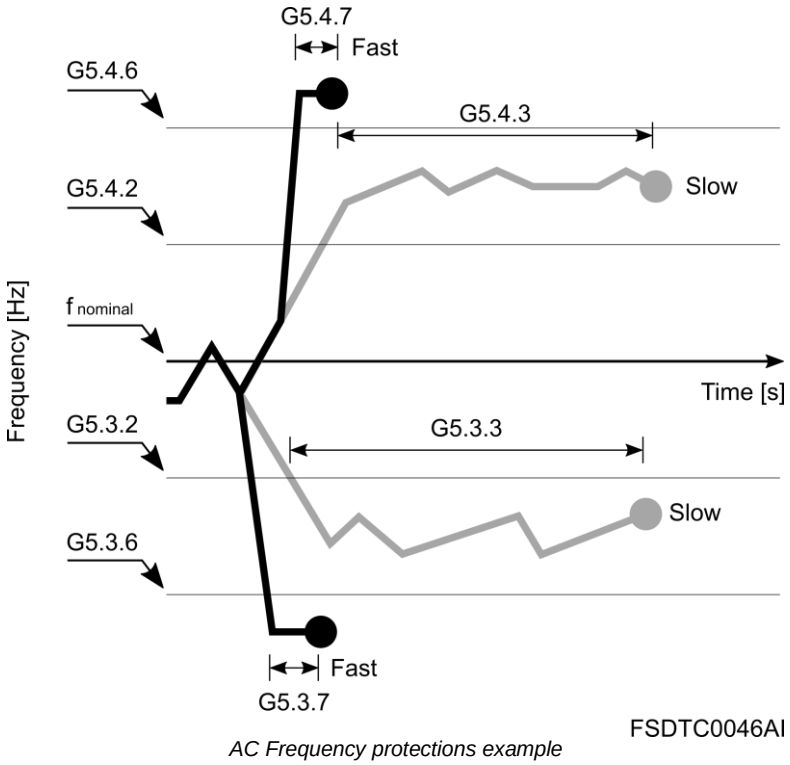
| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.3.1-Enable = YNYNN** | Slow: Y / N<br>Slow 2: Y / N<br>Fast: Y / N<br>Very fast: Y / N<br>Very fast 2: Y / N | There are five low frequency protection levels:<br><br>- Slow protection<br>- Slow 2 protection<br>- Fast protection<br>- Very fast protection<br>- Very fast 2 protection<br><br>To enable each protection, set the corresponding bit to Yes.<br><br>Protection levels can be set from 45.0Hz up to 60.0Hz.<br><br>The slow protection delay can be set from 0.0s up to 6550.0s. Fast protections delay can be set from 0.00s up to 655.00s.<br><br>Very fast protections delay range varies between 0.000 and 65.500s<br><br>It is possible to enable/disable each protection level individually in G5.3.1. Default values, on the other hand, depend on the selected normative. | Yes |
| **G5.3.2-Slow protection =** [7] | 45.0 to 60.0 Hz | | Yes |
| **G5.3.3-Delay slow protection =** [7] | 0.0 to 6550.0 s | | Yes |
| **G5.3.4-Slow 2 protection =** [7] | 45.0 to 60.0 Hz | | Yes |
| **G5.3.5-Delay slow 2 protection =** [7] | 0. to 6550.0 s | | Yes |
| **G5.3.6-Fast protection =** [7] | 45.0 to 60.0 Hz | | Yes |
| **G5.3.7-Delay fast protection =** [7] | 0.00 to 655.00 s | | Yes |
| **G5.3.8-Very fast protection =** [7] | 45.0 to 60.0 Hz | | Yes |
| **G5.3.9-Delay very fast prot =** [7] | 0.000 to 65.500 s | | Yes |
| **G5.3.10-Very fast 2 protection =** [7] | 45.0 to 60.0 Hz | | Yes |
| **G5.3.11-Delay very fast 2 prot =** [7] | 0.000 to 65.500 s | | Yes |

[7] These values depend on the normative selected. See table below.

## Subgroup G5.4: High AC freq

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.4.1-Enable = NNYNN** | Slow: Y / N<br>Slow 2: Y / N<br>Fast: Y / N<br>Very fast: Y / N<br>Very fast 2: Y / N | There are five low frequency protection levels:<br><br>- Slow protection<br>- Slow 2 protection<br>- Fast protection<br>- Very fast protection<br>- Very fast 2 protection<br><br>To enable each protection, set the corresponding bit to Yes.<br><br>Protection levels can be set from 50.0Hz up to 65.0Hz.<br><br>Slow protection delays can be set from 0. 0s up to 6550.0s. The fast protection delay can be set from 0.00s up to 655.00s.<br>Very fast protections delay range varies between 0.000 and 65.500s<br><br>It is possible to enable/disable each protection level individually in G5.4.1. Default values, on the other hand, depend on the selected normative. | Yes |
| **G5.4.2-Slow protection =** [7] | 50.0 to 65.0 Hz | | Yes |
| **G5.4.3-Delay slow protection =** [7] | 0.0 to 6550.0 s | | Yes |
| **G5.4.4-Slow protection =** [7] | 50.0 to 65.0 Hz | | Yes |
| **G5.4.5-Delay slow protection =** [7] | 0.0 to 6550.0 s | | Yes |
| **G5.4.6-Fast protection =** [7] | 50.0 to 65.0 Hz | | Yes |
| **G5.4.7-Delay fast protection =** [7] | 0.00 to 655.00 s | | Yes |
| **G5.4.8-Very fast protection =** [7] | 50.0 to 65.0 Hz | | Yes |
| **G5.4.9-Delay very fast prot =** [7] | 0.000 to 65.500 s | | Yes |
| **G5.4.10-Very fast 2 protection =** [7] | 50.0 to 65.0 Hz | | Yes |
| **G5.4.11-Delay very fast 2 prot =** [7] | 0.000 to 65.500 s | | Yes |

The figure below shows an operation example with four frequency protections enabled (two low and two high).




*AC Frequency protections example*

[7] These values depend on the normative selected. See table below.

[7] The default values summarized below are established internally by the inverter to comply with the selected national normative.

| NORMATIVE / PARAMETER | UL 1741 (IEEE 1547) | BRAZIL | ERCOT | HECO 1 | HECO 2 | RULE 21 | JORDAN |
|---|---|---|---|---|---|---|---|
| G5.1.2-Slow protection | 88.0 | 90.0 | Disabled | 88.0 | 88.0 | 88.0 | 90.0 |
| G5.1.3-Delay slow protection | 2.0 | 5.0 | Disabled | 20.2 | 20.2 | 20.2 | 180.0 |
| G5.1.4-Fast protection | 50.0 | 85.0 | 90.0 | 50.0 | 50.0 | 70.0 | 80.0 |
| G5.1.5-Delay fast protection | 0.16 | 1.00 | 1.75 | 0.40 | 0.40 | 10.20 | 2.50 |
| G5.1.6-Fast 2 protection | Disabled | 20.0 | 45.0 | Disabled | Disabled | 50.0 | 0.0 |
| G5.1.7-Delay fast 2 protection | Disabled | 0.50 | 0.95 | Disabled | Disabled | 1.20 | 0.25 |
| G5.1.8-Very fast protection | Disabled | Disabled | 0.0 | Disabled | Disabled | Disabled | Disabled |
| G5.1.9-Delay very fast protection | Disabled | Disabled | 0.150 | Disabled | Disabled | Disabled | Disabled |
| G5.1.10-Very fast 2 protection | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled |
| G5.1.11-Delay very fast 2 prot | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled |
| G5.2.2-Slow protection | 110.0 | 110.0 | 110.0 | 110.0 | 110.0 | 110.0 | 110.0 |
| G5.2.3-Delay slow protection | 1.0 | 2.5 | 1.0 | 0.95 | 0.95 | 12.2 | 60.0 |
| G5.2.4-Fast protection | 120.0 | Disabled | 115.0 | 120.0 | 120.0 | 120.0 | 115.0 |
| G5.2.5-Delay fast protection | 0.16 | Disabled | 0.50 | 0.01 | 0.01 | 0.01 | 0.20 |
| G5.2.6-Fast 2 protection | Disabled | Disabled | 117.5 | Disabled | Disabled | Disabled | Disabled |
| G5.2.7-Delay fast 2 protection | Disabled | Disabled | 0.20 | Disabled | Disabled | Disabled | Disabled |
| G5.2.8-Very fast protection | Disabled | 120.0 | 120.0 | Disabled | Disabled | Disabled | 120.0 |
| G5.2.9-Delay very fast protection | Disabled | 0.001 | 0.001 | Disabled | Disabled | Disabled | 0.001 |
| G5.2.10-Very fast 2 protection | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled |
| G5.2.11-Delay very fast 2 prot | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled |
| G5.3.2-Slow protection | 59.3 | 58.5 | 59.4 | 57.0 | 57.0 | 58.5 | 47.5 |
| G5.3.3-Delay slow protection | 300.0 | 20.0 | 540.0 | 20.2 | 20.2 | 299.2 | 20.0 |
| G5.3.4-Slow 2 protection | Disabled | Disabled | 58.4 | Disabled | Disabled | Disabled | 47.0 |
| G5.3.5-Delay slow 2 protection | Disabled | Disabled | 30.0 | Disabled | Disabled | Disabled | 0.5 |
| G5.3.6-Fast protection | 57.0 | Disabled | 58.0 | 56.0 | 50.0 | 57.0 | Disabled |
| G5.3.7-Delay fast protection | 0.16 | Disabled | 2.00 | 0.01 | 0.01 | 0.01 | Disabled |
| G5.3.8-Very fast protection | Disabled | 56.0 | 57.5 | Disabled | Disabled | Disabled | Disabled |
| G5.3.9-Delay very fast protection | Disabled | 0.001 | 0.001 | Disabled | Disabled | Disabled | Disabled |
| G5.3.10-Very fast 2 protection | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled |
| G5.3.11-Delay very fast 2 prot | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled |
| G5.4.2-Slow protection | Disabled | 62.5 | 60.6 | 63.0 | 63.0 | 60.5 | 51.5 |
| G5.4.3-Delay slow protection | Disabled | 10.0 | 540.0 | 20.2 | 20.2 | 299.2 | 0.5 |
| G5.4.4-Slow 2 protection | Disabled | Disabled | 61.6 | Disabled | Disabled | Disabled | Disabled |
| G5.4.5-Delay slow 2 protection | Disabled | Disabled | 30.0 | Disabled | Disabled | Disabled | Disabled |
| G5.4.6-Fast protection | 60.5 | Disabled | Disabled | 64.0 | 65.0 | 62.0 | Disabled |
| G5.4.7-Delay fast protection | 0.16 | Disabled | Disabled | 0.01 | 0.01 | 0.01 | Disabled |
| G5.4.8-Very fast protection | Disabled | 63.0 | 61.8 | Disabled | Disabled | Disabled | Disabled |
| G5.4.9-Delay very fast protection | Disabled | 0.001 | 0.001 | Disabled | Disabled | Disabled | Disabled |
| G5.4.10-Very fast 2 protection | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled |
| G5.4.11-Delay very fast 2 prot | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled |
| G5.6.1-Anti islanding mode | Active | Active | Active | Active | Active | Active | Active |

**Note:** In the Std IEEE 1547, there is an intermediate low frequency protection which shall be set between 59.8Hz and 57Hz, and 0.16s and 300s. A delay with hundredths of accuracy can be set on the parameter G5.3.7-Delay fast prot. A delay with tenths of accuracy can be set on the parameters G5.3.3-Delay slow prot and G5.3.5-Delay slow 2 prot.

| NORMATIVE / PARAMETER | MEXICO | NERC PRC-024 EASTERN | NERC PRC-024 WESTERN | NERC PRC-024 QUEBEC | NERC PRC-024 ERCOT | JAPAN 50Hz (JEAC 9701-12HV) | JAPAN 60Hz (JEAC 9701-12HV) |
|---|---|---|---|---|---|---|---|
| G5.1.2-Slow protection | 90.0 | 90.0 | 90.0 | 90.0 | 90.0 | 80.0 | 80.0 |
| G5.1.3-Delay slow protection | 1.50 | 3.0 | 3.0 | 3.0 | 3.0 | 1.0 | 1.0 |
| G5.1.4-Fast protection | 45.0 | 75.0 | 75.0 | 75.0 | 75.0 | 30.0 | 30.0 |
| G5.1.5-Delay fast protection | 1.05 | 2.00 | 2.00 | 2.00 | 2.00 | 0.30 | 0.30 |
| G5.1.6-Fast 2 protection | 0.0 | 65.0 | 65.0 | 65.0 | 65.0 | Disabled | Disabled |
| G5.1.7-Delay fast 2 protection | 0.45 | 0.30 | 0.30 | 0.30 | 0.30 | Disabled | Disabled |
| G5.1.8-Very fast protection | Disabled | 45.0 | 45.0 | 45.0 | 45.0 | Disabled | Disabled |
| G5.1.9-Delay very fast protection | Disabled | 0.150 | 0.150 | 0.150 | 0.150 | Disabled | Disabled |
| G5.1.10-Very fast 2 protection | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled |
| G5.1.11-Delay very fast 2 prot | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled |
| G5.2.2-Slow protection | 120 | 110.0 | 110.0 | 110.0 | 110.0 | 115.0 | 115.0 |
| G5.2.3-Delay slow protection | 0.20 | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 |
| G5.2.4-Fast protection | Disabled | 115.0 | 115.0 | 115.0 | 115.0 | Disabled | Disabled |
| G5.2.5-Delay fast protection | Disabled | 0.50 | 0.50 | 0.50 | 0.50 | Disabled | Disabled |
| G5.2.6-Fast 2 protection | Disabled | 117.5 | 117.5 | 117.5 | 117.5 | Disabled | Disabled |
| G5.2.7-Delay fast 2 protection | Disabled | 0.20 | 0.20 | 0.20 | 0.20 | Disabled | Disabled |
| G5.2.8-Very fast protection | Disabled | 120.0 | 120.0 | 120.0 | 120.0 | Disabled | Disabled |
| G5.2.9-Delay very fast protection | Disabled | 0.001 | 0.001 | 0.001 | 0.001 | Disabled | Disabled |
| G5.2.10-Very fast 2 protection | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled |
| G5.2.11-Delay very fast 2 prot | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled |
| G5.3.2-Slow protection | 58.8 | Disabled | 59.4 | 59.4 | 59.4 | Disabled | Disabled |
| G5.3.3-Delay slow protection | 1800.0 | Disabled | 180.0 | 660.0 | 540.0 | Disabled | Disabled |
| G5.3.4-Slow 2 protection | 58.2 | Disabled | 58.4 | 58.5 | 58.4 | Disabled | Disabled |
| G5.3.5-Delay slow 2 protection | 900.0 | Disabled | 30.0 | 90.0 | 30.0 | Disabled | Disabled |
| G5.3.6-Fast protection | Disabled | Disabled | 57.8 | 57.5 | 58.0 | 47.5 | 57.0 |
| G5.3.7-Delay fast protection | Disabled | Disabled | 7.50 | 10.00 | 2.00 | 1.00 | 1.00 |
| G5.3.8-Very fast protection | 57 | 57.8 | 57.3 | 56.5 | 57.5 | Disabled | Disabled |
| G5.3.9-Delay very fast protection | 0.001 | 0.001 | 0.750 | 0.350 | 0.001 | Disabled | Disabled |
| G5.3.10-Very fast 2 protection | Disabled | Disabled | 57.0 | 55.5 | Disabled | Disabled | Disabled |
| G5.3.11-Delay very fast 2 prot | Disabled | Disabled | 0.001 | 0.001 | Disabled | Disabled | Disabled |
| G5.4.2-Slow protection | 61.2 | Disabled | 60.6 | 60.6 | 60.6 | Disabled | Disabled |
| G5.4.3-Delay slow protection | 1800.0 | Disabled | 180.0 | 660.0 | 540.0 | Disabled | Disabled |
| G5.4.4-Slow 2 protection | 61.8 | Disabled | 61.6 | 61.5 | 61.6 | Disabled | Disabled |
| G5.4.5-Delay slow 2 protection | 900.0 | Disabled | 30.0 | 90.0 | 30.0 | Disabled | Disabled |
| G5.4.6-Fast protection | Disabled | Disabled | Disabled | 63.0 | Disabled | 51.5 | 61.8 |
| G5.4.7-Delay fast protection | Disabled | Disabled | Disabled | 5.00 | Disabled | 1.00 | 1.00 |
| G5.4.8-Very fast protection | 63.0 | 61.8 | 61.7 | Disabled | 61.8 | Disabled | Disabled |
| G5.4.9-Delay very fast protection | 0.001 | 0.001 | 0.001 | Disabled | 0.001 | Disabled | Disabled |
| G5.4.10-Very fast 2 protection | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled |
| G5.4.11-Delay very fast 2 prot | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled | Disabled |
| G5.6.1-Anti islanding mode | Passive | Active | Active | Active | Active | Off | Off |

EN

## Subgroup G5.4b: Volt/freq Combined

These parameters are related to the grid code of Egypt, which requires a switchgear opening protection under certain voltage and frequency conditions.

### G5.4b.1: Combined [Hf_Hv]

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.4b.1.1-Enable = No** | No<br>Yes | Enables / disables the combined protection for high frequency and high voltage.<br><br>If voltage overcomes the value of V1 and frequency is higher than f1 during the time t1, fault **F190: Comb prot Volt Frq** will be triggered. | Yes |
| **G5.4b.1.2-V1 = 120.0 %** | 100.0 to 140.0 % | | Yes |
| **G5.4b.1.3-f1 = 62.00 Hz** | 50.00 to 67.00 Hz | | Yes |
| **G5.4b.1.4-t1 = 1.0 s** | 0.0 to 1800.0 s | | Yes |

### G5.4b.2: Combined [Hf_Lv]

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.4b.2.1-Enable = No** | No<br>Yes | Enables / disables the combined protection for high frequency and low voltage.<br><br>If voltage drops below the value of V2 and frequency is higher than f2 during the time t2, fault **F190: Comb prot Volt Frq** will be triggered. | Yes |
| **G5.4b.2.2-V2 = 80.0 %** | 0.0 to 100.0 % | | Yes |
| **G5.4b.2.3-f2 = 62.00 Hz** | 50.00 to 67.00 Hz | | Yes |
| **G5.4b.2.4-t2 = 1.0 s** | 0.0 to 1800.0 s | | Yes |

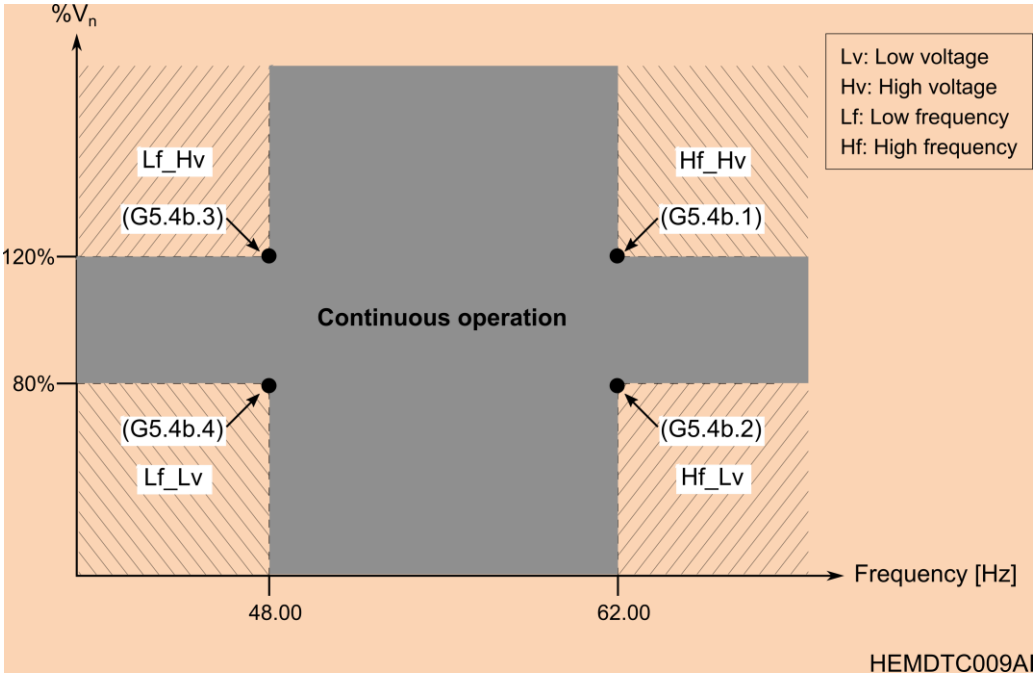
### G5.4b.3: Combined [Lf_Hv]

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.4b.3.1-Enable = No** | No<br>Yes | Enables / disables the combined protection for low frequency and high voltage.<br><br>If voltage overcomes the value of V3 and frequency is lower than f3 during the time t3, fault **F190: Comb prot Volt Frq** will be triggered. | Yes |
| **G5.4b.3.2-V3 = 120.0 %** | 100.0 to 140.0 % | | Yes |
| **G5.4b.3.3-f3 = 58.00 Hz** | 45.00 to 60.00 Hz | | Yes |
| **G5.4b.3.4-t3 = 1.0 s** | 0.0 to 1800.0 s | | Yes |

### G5.4b.4: Combined [Lf_Lv]

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.4b.4.1-Enable = No** | No<br>Yes | Enables / disables the combined protection for low frequency and low voltage.<br><br>If voltage drops below the value of V4 and frequency is lower than f4 during the time t4, fault **F190: Comb prot Volt Frq** will be triggered. | Yes |
| **G5.4b.4.2-V4 = 80.0 %** | 0.0 to 100.0 % | | Yes |
| **G5.4b.4.3-f4 = 58.00 Hz** | 45.00 to 60.00 Hz | | Yes |
| **G5.4b.4.4-t4 = 1.0 s** | 0.0 to 1800.0 s | | Yes |

The following figure shows an operation example of the combined voltage / frequency protections, usign the default values of parameters G5.4b.x.x.




*Voltage / frequency combined protections example*

## Subgroup G5.5: Unbalanced

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.5.1-Enable = YY** | I unbalance: Y / N<br>V unbalance: Y / N | Enable/disable current and voltage unbalance protections individually. | Yes |
| **G5.5.2-Unbalance I = 7%** | 1 to 100 % | Set the current unbalance protection threshold in percentage. | Yes |
| **G5.5.2b-Unbalance I min = 30%** | 0 to 100% | Sets the minimum current, in percentage, to start applying the unbalance algorithm. | Yes |
| **G5.5.3-Delay unbalance I = 10s** | 1 to 60 s | Set the delay introduced since the current unbalance protection threshold is exceeded until the fault is reported. | Yes |
| **G5.5.4-Unbalance V = 25%** | 1 to 100 % | Set the voltage unbalance protection threshold in percentage. | Yes |
| **G5.5.5-Delay unbalance V = 5s** | 1 to 60 s | Set the delay introduced since the voltage unbalance protection threshold is exceeded until the fault is reported. | Yes |
| **G5.5.6-DU I unbal threshold = 25%** | 4 to 100 % | Set the DU current unbalance protection threshold in percentage.<br>**Note:** This parameter is only shown if the inverter type is PCS. | Yes |
| **G5.5.7-DU I unbal time = 15s** | 0 to 120 s | Set the time that can elapse before triggering a DU current unbalance fault.<br>**Note:** This parameter is only shown if the inverter type is PCS. | Yes |
| **G5.5.8-Mod-DSP I balance = Disabled %** | 0 to 100%<br>Disabled = 101 | Defines the maximum unbalance allowed between the sum of currents measured by the modules and the current measured in the low voltage side of the MV transformer (this measurement is made by central control). | Yes |
| **G5.5.8b-Mod-DSP I balan tim = 10.00 s** | 0.00 to 10.00 s | It is the sum of the RMS values of the 3 phases, MAX_RMS - AVG_RMS or MIN_RM - AVG_RMS.<br>If this threshold is overcome during the time set in **G5.5.8b-Mod-DSP I balan tim,** fault **F187:Cur Bal DSPMD** will be triggered.<br><br>**Note:** These parameters only apply to HEM inverters. | Yes |
| **G5.5.9 Module Homopolar I = Disabled A** | Disabled = 0;<br>1 to 500 A | Parameter that defines a homopolar protection for the FPGA with the sum of currents in the AC part of the modules during a period of time. The sum of the three currents will be of low value.<br>When the threshold is exceeded, the module fails and triggers warning W125: Module Homopolar I. | Yes |

## Subgroup G5.6: Anti islanding

The use of this parameter group is reserved to Power Electronics staff. Be careful if you make any changes; otherwise, the inverter may malfunction.

An anti-island condition occurs when the inverter keeps injecting on the grid although it does not have voltage because it comes into resonance with some load. This is dangerous because all the grid should be without current but there is a section (island) where current is flowing without being controlled.

Currently, there are two modes: active and passive.

In active mode, the inverter causes a disturbance in the grid (slightly distorts the current waveform). If the inverter is connected to the grid frequency cannot vary, but, if there is an island, the frequency will vary due to the distortion. Parameters used in this mode are G5.6.2 to G5.6.4.

In passive mode, the inverter only checks frequency variation during a time lapse. If the frequency variation ($\Delta freq$) overcomes a threshold (G5.6.5) during the time lapse set in G5.6.6, the anti-island protection will be triggered.

$$\Delta freq = current\ frequency - previous\ frequency$$

**Simultaneous** mode is a combination of active and passive modes.

The following table shows parameters related to anti islanding function:

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>G5.6.1-Mode = Off [7]</td><td>Off<br>Active<br>Passive<br>Simultaneous</td><td>This command allows selecting the anti-islanding mode:<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Off</td><td>Disable anti-islanding mode.</td></tr><tr><td>Active</td><td>Enable anti-islanding mode.</td></tr><tr><td>Passive</td><td>Enable ROCOF anti-islanding.</td></tr><tr><td>Simultaneous</td><td>Active and passive modes operate at the same time.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G5.6.2-Drift = 0.0% [8]</td><td>0.00 to 5.00 %</td><td>Constant drift added in the frequency.<br>Note: Parameter available if anti-islanding mode is Active or Simultaneous.</td><td>Yes</td></tr>
<tr><td>G5.6.3-K error = 0.0% [8]</td><td>0.0 to 15.0 %</td><td>Drift added depending on the frequency error differential.<br>Note: Parameter available if anti-islanding mode is Active or Simultaneous.</td><td>Yes</td></tr>
<tr><td>G5.6.4-Fault level (º) = 15 [8]</td><td>1 to 180º</td><td>Threshold for triggering the anti-islanding protection<br>Note: Parameter available if anti-islanding mode is Active or Simultaneous.</td><td>Yes</td></tr>
<tr><td>G5.6.5-ROCOF (Hz/s) = 4.0 [9]</td><td>1.0 to 50.0</td><td>If the value set in this parameter is overcome, anti-island will be activated.<br>Note: Parameter available if anti-islanding mode is Passive or Simultaneous.</td><td>Yes</td></tr>
<tr><td>G5.6.6-Fault delay = 0.25 s [9]</td><td>0.01 to 10.00 s</td><td>Configure a delay before triggering an anti-islanding fault when ROCOF mode is active.<br>Note: Parameter available if anti-islanding mode is Passive or Simultaneous.</td><td>Yes</td></tr>
</table>

[8] Parameters used in active mode.
[9] Parameters used in passive mode (ROCOF).

## Subgroup G5.7: Temperature

Parameters to configure the temperature protections for IGBTs, AC module and the medium voltage transformer. Please notice protections for the transformer (G5.7.3.1-MVTransformrThreshld to G5.7.3.3-MVTransTcriticThld) only apply to HEM inverters.

### G5.7.1 IGTBs

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.7.1.1-IGBTs temp thres flt = 125 ºC** | 20 to 127 ºC | Displays the threshold for IGBT temperature protection.<br>**Note:** This parameter cannot be modified, it is set in the *drive select*. | Yes |
| **G5.7.1.2-IGBTsTempHysteresis = 10 ºC** | 0 to 50 ºC | This command allows defining the hysteresis to clear the IGBTs overtemperature fault.<br>When the temperature decreases to the recorded value of the difference between the temperature recorded in G5.7.1 minus G5.7.2 it will allow resetting the fault shown when the temperature protection threshold is exceeded. | Yes |
| **G5.7.1.3-EstDeltaTªIGBT Warn = 5 ºC** | Disabled = 51;<br>0 to Disabled ºC | This parameter triggers the W20 warning in the case that ΔT_IGBT is greater than the value defined in G5.7.1.3 | Yes |
| **G5.7.1.4-EstDeltaTªIGBTfault = Disabled ºC** | Disabled = 51;<br>0 to Disabled ºC | This parameter triggers the warning W127 in case ΔT_IGBT is greater than the value defined in G5.7.1.4 | Yes |

### G5.7.2 Módulos

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.7.2.1-ModuleTempThresFlt = 75 ºC** | 20 to 87 ºC | Sets the internal overtemperature protection threshold.<br>When the threshold is exceeded, Fault 104 is displayed. | Yes |
| **G5.7.2.2-ModuleTempHysteresis = 10 ºC** | 0 to 50 ºC | This command allows defining the hysteresis to clear the module internal overtemperature fault. | Yes |
| **G5.7.2.3-Mean delta T thresh = 15** | 0 to 100<br>Disabled = 101 | Sets the internal AC module overtemperature protection threshold. | Yes |

### G5.7.3 Transformer MV

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.7.3.1-MVTransformrThreshld = 125 ºC** | 120 to 150 ºC | Sets the overtemperature protection threshold for the MV transformer. **Only apply to HEM inverter.** | Yes |
| **G5.7.3.2-MV transformr delay = 120 s** | 0 to 300 s | Sets the delay (in seconds) for the MV transformer overtemperature protection. **Only apply to HEM inverter.** | Yes |
| **G5.7.3.3-MVTransTcriticThld = 150 ºC** | 130 to 160 ºC | Sets the temperature threshold to trigger **F148: MV Critical Tº Trans** and open the circuit breaker. **Only apply to HEM inverter.** | Yes |

### G5.7.4 Inverter

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.7.4.1-Main control board = 70 ºC** | 20 to 85ºC<br>Disabled = 86 | Temperature protection threshold for the P2G control board.<br>When the threshold is exceeded, Fault 40 is displayed. | Yes |
| **G5.7.4.2-DU central board = Disabled ºC** | 20 to 85ºC<br>Disabled = 86 | Temperature protection threshold for the central DU.<br>When the threshold is exceeded, Fault 149 is displayed. | Yes |

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## Subgroup G5.8: GFDI current

Group G5.8 is only shown if group G10.2 is configured for ground (GFDI + IMI).

**Note:** These parameters are not shown in STATCOM.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.8.1-Enable = No** | No<br>Yes | Allows enabling or disabling the earth leak protection GFDI. | Yes |
| **G5.8.2-Current threshold = 4.00A** | 0.30 to 10.00 A | Define the maximum current allowed through the GFDI. If this threshold is overcome, fault F53:GFDI current leak will be triggered once G5.8.3-Trip time has elapsed. | Yes |
| **G5.8.3-Trip time = 5.0s** | 1.0 to 10.0 s | Define the maximum time that can elapse before triggering fault F53:GFDI current leak. | Yes |
| **G5.8.4-Measured current=0.00A** | 0.00 to 50.00A | Shows the present current measured trough the GFDI. | Yes |
| **G5.8.5 Contactor GFDI Open = Yes** | No<br>Yes | Enables GFDI contactor opening functionality in the event of fault **F53:GFDI current leak** | Yes |

## Subgroup G5.9: HV Phase lost

In this group user can configure the protections against a phase loss in high voltage. Please notice parameters G5.9.1, G5.9.2 and G5.9.6 are common to HEM and HEMK inverters, while the others only apply to HEMK inverters.

**Note:** Parameters within this group will only be shown if the protection has been enabled (G5.9.1 = Yes).

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.9.1-Enable = No** | No<br>Yes | Allows enabling or disabling the HV phase lost.<br>**Note:** Once this protection has been enabled, parameters G5.9.2 to G5.9.8 will be shown. | Yes |
| **G5.9.2-Activation level = 5%** | 0 to 100 % | Defines the minimum power threshold to trigger the fault. Below this level, the fault will not be triggered. | Yes |
| **G5.9.3-Scale = 1000** [10] | 10<br>100<br>1000<br>10000 | Scale of the measured fault level. | Yes |
| **G5.9.4-Fault level = 0** [10] | 0 to 65535 | Defines the maximum level allowed for the inverse voltage sequence. If this level is overcome, fault F22 will be triggered after the time defined in [G5.9.6] has elapsed. | Yes |
| **G5.9.5-Filter = 0.100Hz** [10] | 0.001 to 30.000 Hz | Filters the measured fault level. | Yes |
| **G5.9.6-Fault delay = 2.0s** | 0.1 to 30.0 s | Define the maximum elapsed time before triggering the fault F22. | Yes |
| **G5.9.7-Start test = No** [10] | No<br>Yes | Allows enabling or disabling the start test.<br>The start test is carried out before starting the inverter with reactive power, at the level defined in [G5.9.8]. | Yes |
| **G5.9.8-Test level = -20%** [10] | -100 to 100 % | Defines the level for the start test. | Yes |

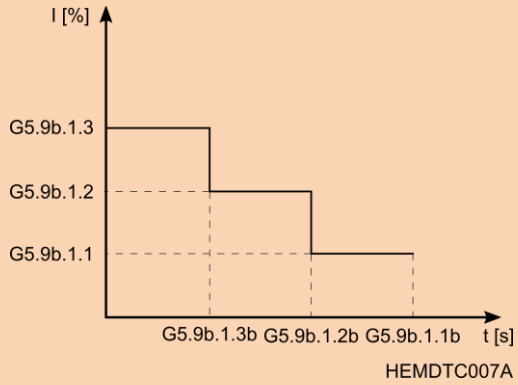
[10] These parameters only apply to HEMK inverters.

## Subgroup G5.9b Transformer

**Note:** These protections only apply to HEM inverters. In future versions, it is foreseen that protections for the low voltage side (subgroup G5.9b.1) will also apply to HEMK.

**Subgroup G5.9b.1: Low voltage**

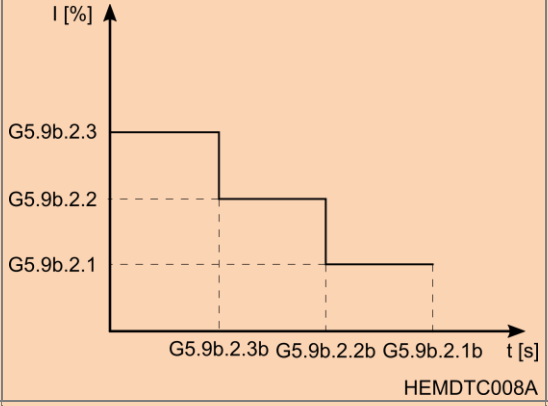
Protections for the low voltage side of the MV transformer are configured in this subgroup.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.9b.1.1-Overcurr threshold = Disabled %** | 100 to 400%<br>Disabled = 401 | These parameters allow defining three overcurrent thresholds in the low voltage side of the transformer, as a percentage of the current set in the drive select, and the timeout for each.<br><br>If **any of the three** thresholds is overcome during its corresponding timeout, fault **F36:Ovc Inrush LV** will be triggered**:**<br><br>- If **G5.9b.1.1** is overcome during **G5.9b.1.1b,** F36 will be triggered.<br>- If **G5.9b.1.2** is overcome during **G5.9b.1.2b,** F36 will be triggered.<br>- If **G5.9b.1.3** is overcome during **G5.9b.1.3b,** F36 will be triggered. | Yes |
| **G5.9b.1.1b-Overcurr timeout = 65.00 s** | 0.00 s to 65.00 s | | Yes |
| **G5.9b.1.2-Overload threshold = Disabled** | 100 to 400%<br>Disabled = 401 | These thresholds can be disabled and/or configured independently. The only restriction that must be taken into account is:<br><br>$$G5.9b.1.3 > G5.9b.1.2 > G5.9b.1.1$$<br><br>The figure below illustrates this functionality:<br><br>  <br><br>HEMDTC007A | Yes |
| **G5.9b.1.2b-Overload timeout = 65.00 s** | G5.9b.1.1b-Overload timeout to 65.00 s | | Yes |
| **G5.9b.1.3-Inrush threshold = Disabled %** | 100 to 400%<br>Disabled = 401 | | Yes |
| **G5.9b.1.3b-Inrush timeout = 65.00 s** | G5.9b.1.2b-Overload timeout to 65.00 s | | Yes |
| **G5.9b.1.4-I off thresh = Disabled %** | 0 to 100%<br>Disabled = 101 | When the inverter is not running, no currents are expected.<br><br>These two parameters allow configuring the protection to detect if there is any current in the low voltage side of the transformer while the equipment is stopped, due to an internal short circuit.<br><br>If the threshold set in G5.9b.1.4 is overcome during the timeout adjusted in G5.9b.1.4b, fault **F38:CurrentOff LV** will be triggered. | Yes |
| **G5.9b.1.4b-I off timeout = 65.00 s** | 0.00 s to 65.00 s | | Yes |
| **G5.9b.1.5-Differential protec = Disabled %** | 0 to 100%<br>Disabled = 101 | These two parameters allow configuring the protection against zero-sequence currents in the low voltage side of the transformer.<br><br>If current overcomes the threshold set in **G5.9b.1.5-Differential protec** during the time adjusted in **G5.9b.1.5b-Differential time**, fault **F34:Hom Diff LV** will be triggered. | Yes |
| **G5.9b.1.5b-Differential time = 1000 ms** | 0 ms to 1000 ms | | Yes |
| **G5.9b.1.6-I unbalance = Disabled %** | 0 to 100%<br>Disabled = 101 | These two parameters define the maximum unbalance threshold allowed between phases in the low voltage side of the transformer and the delay to trigger the corresponding fault.<br><br>If the threshold (**G5.9b.1.6**) is overcome and the time set in **G5.9b.1.6b** elapses, fault **F185:Cor Des LV** will be triggered**.** | Yes |
| **G5.9b.1.6b-I unbalance tim = 1000 ms** | 0 ms to 1000 ms | | Yes |
| **G5.9b.1.7-Negative P = Disabled %** | 0 to 100%<br>Disabled = 101 | These two parameters allow configuring the protection for negative power in the low voltage side of the transformer.<br><br>Active power must always be positive. If a negative (or zero) power value is detected, it overcomes the threshold set in **G5.9b.1.7** and the timeout adjusted in **G5.9b.1.7b** elapses, fault **F183:Neg Pow LV** will be triggered. | Yes |
| **G5.9b.1.7b-Neg P timeout = 1000 ms** | 0 ms to 1000 ms | | Yes |

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## G5.9b.2: Medium voltage

Protections for the medium voltage side of the MV transformer are configured in this subgroup.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.9b.2.1-Overcurr threshold = Disabled %** | 100 to 400%<br>Disabled = 401 | These parameters allow defining three overcurrent thresholds in the medium voltage side of the transformer, as a percentaje of the current set in the drive select, and the timeout for each.<br><br>If **any of the three** thresholds is overcome during its corresponding timeout, fault **F37:Ovc Inrush MV** will be triggered**:**<br><br>- If **G5.9b.2.1** is overcome during **G5.9b.2.1b,** F37 will be triggered.<br>- If **G5.9b.2.2** is overcome during **G5.9b.2.2b,** F37 will be triggered.<br>- If **G5.9b.2.3** is overcome during **G5.9b.2.3b,** F37 will be triggered.<br><br>These thresholds can be disabled and/or configured independently. The only restriction that must be taken into account is:<br><br>$G5.9b.2.3 > G5.9b.2.2 > G5.9b.2.1$<br><br>The figure below illustrates this functionality:<br><br>  <br> | Yes |
| **G5.9b.2.1b-Overcurr timeout = 65.00 s** | 0.00 s to 65.00 s | | Yes |
| **G5.9b.2.2-Overload threshold = Disabled %** | 100 to 400%<br>Disabled = 401 | | Yes |
| **G5.9b.2.2b-Overload timeout = 65.00 s** | G5.9b.2.1b-Overcurr timeout to 65.00 s | | Yes |
| **G5.9b.2.3-Inrush threshold = Disabled %** | 100 to 400%<br>Disabled = 401 | | Yes |
| **G5.9b.2.3b-Inrush timeout = 65.00 s** | G5.9b.2.2b-Overload timeout to 65.00 s | | Yes |
| **G5.9b.2.4-I off thresh = Disabled %** | 0 to 100%<br>Disabled = 101 | When the inverter is not running, no currents are expected.<br><br>These two parameters allow configuring the protection to detect if there is any current in the medium voltage side of the transformer while the equipment is stopped, due to an internal short circuit.<br><br>If the threshold set in G5.9b.2.4 is overcome during the timeout adjusted in, fault **F39:CurrentOff MV** will be triggered. | Yes |
| **G5.9b.2.4b-I off timeout = 65.00 s** | 0.00 s to 65.00 s | | Yes |
| **G5.9b.2.5-Differential protec = Disabled %** | 0 to 100%<br>Disabled = 101 | These two parameters allow configuring the protection against zero-sequence currents in the medium voltage side of the transformer.<br><br>If current overcomes the threshold set in **G5.9b.2.5** during the time adjusted in **G5.9b.2.5b**, fault **F181:Hom Diff MV** will be triggered. | Yes |
| **G5.9b.2.5b-Differential time = 1000 ms** | 0 ms to 1000 ms | | Yes |
| **G5.9b.2.6-I unbalance = Disabled %** | 0 to 100%<br>Disabled = 101 | These two parameters define the maximum unbalance threshold allowed between phases in the medium voltage side of the transformer and the delay to trigger the corresponding fault.<br><br>If the threshold (**G5.9b.2.6**) is overcome and the time set in **G5.9b.2.6b** elapses, fault **F186:Cur Unbl MV** will be triggered. | Yes |
| **G5.9b.2.6b-I unbalance tim = 1000 ms** | 0 ms to 1000 ms | | Yes |
| **G5.9b.2.7-Negative P = Disabled %** | 0 to 100%<br>Disabled = 101 | **These two parameters will be implemented in future versions.** They will allow configuring the protection for negative power in the medium voltage side of the transformer<br><br>Active power must always be positive. If a negative (or zero) power value is detected, it overcomes the threshold set in **G5.9b.2.7** and the timeout adjusted in **G5.9b.2.7b** elapses, fault **F184:Neg Pow MV** will be triggered. | Yes |
| **G5.9b.2.7b-Neg P timeout = 1000 ms** | 0 ms to 1000 ms | | Yes |

**Others**

Parameters to configure additional protections for the medium voltage transformer.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.9b.3-LV-MV I balance = Disabled %** | 0 to 100%<br>Disabled = 101 | Compares the current measures of the tranformer in the low and medium voltage sides to determine whether there is any unbalance. If threshold G5.9b.3 is overcome during the timeout set in G5.9b.3b, fault **F182:Curr Bal LVMV** will be triggered. | Yes |
| **G5.9b.3b-LV-MV I balance T = 10.00 s** | 0.00 s to 10.00 s | | Yes |
| **G5.9b.4-LV-MV P balance = Disabled %** | 0 to 100%<br>Disabled = 101 | **Estos dos parámetros se implementarán en futuras versiones.** They will allow configuring the protection for power unbalance in the medium voltage side. | Yes |
| **G5.9b.4b-LV-MV P bal time = 1000 ms** | 0 ms to 1000 ms | | Yes |
| **G5.9b.5-MV THDi protection = Disabled %** | 0 to 100%<br>Disabled = 101 | Defines the threshold for harmonic distortion (THDi). If this threshold is overcome during the time adjusted in G5.9b.5b, fault **F189:THDi MV** will be triggered. | Yes |
| **G5.9b.5b-MV THDi prot. Time = 6.0 s** | 0.0 s to 100.0 s | | Yes |

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# Subgroup G5.10: Special protections

## G5.10.1: Special faults

This group allows configuring up to thirty faults which will have a special fault behavior.

When any of the following faults occurs, actions will be carried out on the inverter according to the parametrization of group G.10.X.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| G5.10.1.1-Special fault 1 = F25:Low Vdc | No faults to F299 | As it has been mentioned before, some faults have additional actions associated. This is, faults with a special behavior.<br><br>Use parameters G5.10.1.x to configure up to thirty **inverter faults** that will have a special behavior. Simply set as parameter value the corresponding fault code.<br><br>To configure module warnings/faults to have a special behavior, use group G5.10b.1.<br><br>Please refer to section 3 for the list of fault codes. | Yes |
| G5.10.1.2-Special fault 2 = F109:Low module Vdc | No faults to F299 | | Yes |
| G5.10.1.3-Special fault 3 = F43:Emergency stop | No faults to F299 | | Yes |
| G5.10.1.4-Special fault 4 = No faults | No faults to F299 | | Yes |
| G5.10.1.5-Special fault 5 = No faults | No faults to F299 | | Yes |
| G5.10.1.6-Special fault 6 = No faults | No faults to F299 | | Yes |
| G5.10.1.7-Special fault 7 = No faults | No faults to F299 | | Yes |
| G5.10.1.8-Special fault 8 = No faults | No faults to F299 | | Yes |
| G5.10.1.9-Special fault 9 = No faults | No faults to F299 | | Yes |
| G5.10.1.10-Special fault 10 = No faults | No faults to F299 | | Yes |
| G5.10.1.11-Special fault 11 = No faults | No faults to F299 | | Yes |
| G5.10.1.12-Special fault 12 = No faults | No faults to F299 | | Yes |
| G5.10.1.13-Special fault 13 = No faults | No faults to F299 | | Yes |
| G5.10.1.14-Special fault 14 = No faults | No faults to F299 | | Yes |
| G5.10.1.15-Special fault 15 = No faults | No faults to F299 | | Yes |
| G5.10.1.16-Special fault 16 = No faults | No faults to F299 | | Yes |
| G5.10.1.17-Special fault 17 = No faults | No faults to F299 | | Yes |
| G5.10.1.18-Special fault 18 = No faults | No faults to F299 | | Yes |
| G5.10.1.19-Special fault 19 = No faults | No faults to F299 | | Yes |
| G5.10.1.20-Special fault 20 = No faults | No faults to F299 | | Yes |
| G5.10.1.21-Special fault 21 = No faults | No faults to F299 | | Yes |
| G5.10.1.22-Special fault 22 = No faults | No faults to F299 | | Yes |
| G5.10.1.23-Special fault 23 = No faults | No faults to F299 | | Yes |
| G5.10.1.24-Special fault 24 = No faults | No faults to F299 | | Yes |
| G5.10.1.25-Special fault 25 = No faults | No faults to F299 | | Yes |
| G5.10.1.26-Special fault 26 = No faults | No faults to F299 | | Yes |
| G5.10.1.27-Special fault 27 = No faults | No faults to F299 | | Yes |

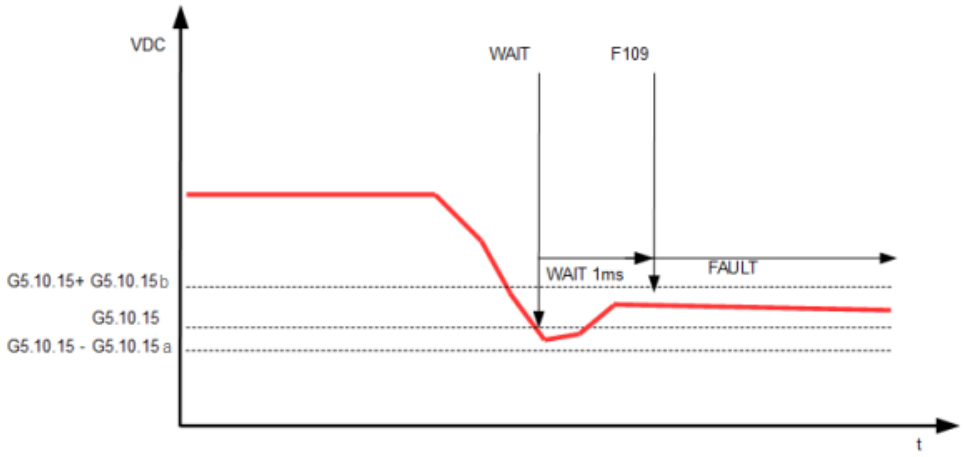
| Screen | Range | Function | Set on run |
|---|---|---|---|
| G5.10.1.28-Special fault 28 = No faults | No faults to F299 | | Yes |
| G5.10.1.29-Special fault 29 = No faults | No faults to F299 | | Yes |
| G5.10.1.30-Special fault 30 = No faults | No faults to F299 | | Yes |

## G5.10.X: Others

| Screen | Range | Function | Set on run |
|---|---|---|---|
| G5.10.2-Special protections = Circuit breaker | Fault<br>Emergency stop<br>Circuit breaker | Defines how should a fault be treated:<br>OPTION: Fault — DESCRIPTION: Treat as fault.<br>Emergency stop — Treat as if the stop pushbutton was pushed (opens DC).<br>Circuit breaker — Open the circuit breaker and isolate the inverter (opens DC and medium voltage AC). | Yes |
| G5.10.3-Unbal bus enable = YY | Bus unbalance: Y / N<br>V max half bus Y / N | Enables or disables the faults for unbalanced bus (F106) and maximum half bus voltage (F108). These faults have a fixed threshold:<br>• F106: Vbus unbalanced: triggered at 200V.<br>• F108: Half bus Vmax: triggered at 800V. | Yes |
| G5.10.4-Unbal bus threshold = 100 V | 50 to 150V | If "Bus unbalance" is enabled in G5.10.3, these two parameters define the voltage threshold and delay time to trigger fault F66. | Yes |
| G5.10.5-Delay unbalance bus = 3s | 1 to 10s | | Yes |
| G5.10.6-V max half bus = 800V | 300 to 900V | If "V max half bus" is enabled in G5.10.3, these two parameters define the voltage threshold and delay time to trigger fault F108. | Yes |
| G5.10.7-Delay V max half bus = 3s | 1 to 10s | | Yes |
| G5.10.8-Unb Pac Pdc threshold = 30% | 0 to 100%<br>Off | Sets the threshold at which fault F105 will be triggered.<br>To disable this protection, set this value to 101 = Off. | Yes |
| G5.10.9-Deriv. Idc threshold = 100A | 0 to 250A | If current is detected in a module while it is stopped, a fault will be triggered. These parameters allow defining the threshold to trigger an $I_{DC}$ or $I_{AC}$ fault (F98 or F99, respectively). | Yes |
| G5.10.10-Deri. Iac threshold = 50A | 0 to 250A | | Yes |
| G5.10.11-Smoke detector DU = No | No<br>Yes | If this parameter is set to Yes, when the DU smoke detector reports an anomaly, fault F61 will be triggered.<br>**Note**: This option is not available in STATCOM.<br><br>**Note:** this parameter will be shown if [SG1.1.23-Enable smoke detector ] = Yes and [SV4.4.1.1-Inverter type] ≠ Statcom. | Yes |
| G5.10.12-Smoke detector FPGA = No | No<br>Yes | If this parameter is set to Yes, when the FPGA smoke detector reports an anomaly, fault F97 will be triggered.<br>**Note:** this parameter has been hidden in version 4.3.0 | Yes |
| G5.10.13-Thr Activ Flt Idref = 100 A | 50 to 400 A | This parameter sets the threshold to activate fault / warning F107 / W107:Iac not reached.<br>When the output current does not reach, as minimum, 50% of the reference current, having overcome the threshold set in G5.10.13, system will display a fault or warning 107. | Yes |
| G5.10.14-Idc overcurrent = Off | Off to 1200 | Sets the threshold for DC overcurrent detection. If this parameter is enabled, and its value overcome, fault **F67: Mod. DC overcurrent** will be triggered. | Yes |
| G5.10.15-Low module Vdc = 650V | Off = 49<br>50 to 950V | The default value of G5.10.15 is used for the protection from voltage F109 will change from fixed to variable depending on the following equation.<br><br>DEF(G5.10.15)= Sqrt(2)*Vac_configured-300V<br><br>When the voltage configured in SG10.1.5 is lower than 250V, the Vdc undervoltage protection will be disabled. | Yes |

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| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.10.15a-Low mod Vdc High = 50 V** | 0 to 200 V | To avoid fake failures, this parameter allows the device to be less sensitive to fault F109 which can mean short circuit.<br>If the value of the voltage is lower than the value configured in this parameter, it will stop switching.<br>If the voltage value is between the thresholds configured during 1ms, F109 will appear.<br>If the voltage rises again above the value set in G5.10.15 in less than 1 ms, it will switch on.<br><br>**Note**: See graph at the end of the section | Yes |
| **G5.10.15b-Low mod Vdc Low = 100 V** | 0 to 200 V | To avoid fake failures this parameter allows the equipment to be less sensitive to fault F109 which can mean short circuit.<br>If the voltage value is lower than the value configured in this parameter, the F109 fault will occur.<br>If the voltage value is between the thresholds configured during 1ms, F109 will appear.<br>If the voltage rises again above the value set in G5.10.15 in less than 1 ms, it will switch on.<br><br>**Note**: See graph at the end of the section | Yes |
| **G5.10.16-Low DSP Vdc = 300V** | 20 to 500 V | Defines the margin with respect to the rectified current, which is used to calculate the threshold to trigger a low voltage fault:<br><br>$< V_{ac}\sqrt{2} - G5.10.16$<br><br>Whenever bus voltage measured by the DSP is below this threshold, fault F25 will be triggered. | Yes |
| **G5.10.17-Delay Low DSP Vdc = 3.0ms** | 1.0 to 100.0 ms<br>Off = 1001 | Defines the delay that can exist between the low voltage detection and the moment in which the fault will be triggered.<br>When set to 1001, the delay will be deactivated. | Yes |
| **G5.10.20-Overvoltage DC DSP = 100 V** | 0 to 100 V | Allows to configure the voltage threshold that is applied to increase the DC voltage of the inverter. | Yes |



## G5.10b.1: List of interlockable module faults

Usually, when a module fault is triggered, the module is stopped (IGBTs OFF and AC contactors open). Then, after a certain time, a restart order will be sent, and the module will attempt to restart as many times as the value of **SG1.2.4 Modules retries.**

However, in some cases, it is convenient to ensure the module is checked before sending the restart command (e.g. if there is a thermal failure, but it is not fixed, the module would restart and, since temperature will be still too high, it will stop again. A loop of restart/stop sequences will take place).

Thus, a list of parameters can be created using group G5.10b.1., to include those module faults that will disable the module (instead of stopping + restarting it) until a technician checks the module, fixes whatever is necessary and sends the restart command.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.10b.1.1-Lock mod fault 1 = F83:Mod. Multiple desat** | No faults to F129 | This parameter group is similar to group G5.10.1, but these are used to define a special behavior to **module faults**, not inverter faults.<br><br>It is worth recalling that a module warning becomes a fault when it is added to the list of special faults.<br><br>Up to fifteen faults can be configured to have a special behavior. Simply set as parameter value the corresponding fault code.<br><br>When a module fault included in this list is triggered, the module will be disabled and the autoreset logic of SG1.2.4 will no longer apply. The module will be out of service until a technician solves the condition that triggers the fault and sends the restart command.<br><br>Please refer to section 3 for the list of fault codes. | Yes |
| **G5.10b.1.2-Lock mod fault 2 = F90:Mod. IGBTs tº HW** | No faults to F129 | | Yes |
| **G5.10b.1.3-Lock mod fault 3 = F93:Mod. Thermal LCL** | No faults to F129 | | Yes |
| **G5.10b.1.4-Lock mod fault 4 = F117:Mod. Common thermal** | No faults to F129 | | Yes |
| **G5.10b.1.5-Lock mod fault 5 = F125: Module Homopolar I** | No faults to F129 | | Yes |
| **G5.10b.1.6-Lock mod fault 6 = F127:EstDeltaTªIGBTmod fault** | No faults to F129 | | Yes |
| **G5.10b.1.7-Lock mod fault 7 = F126:IGBTs temp deviation** | No faults to F129 | | Yes |
| **G5.10b.1.8-Lock mod fault 8 = F103:IGBTs temperature** | No faults to F129 | | Yes |
| **G5.10b.1.9-Lock mod fault 9 = F96:Mod. Amb tº HW** | No faults to F129 | | Yes |
| **G5.10b.1.10-Lock mod fault 10 = F104:Ambient temperature** | No faults to F129 | | Yes |
| **G5.10b.1.11-Lock mod fault 11 = F107:Iac not reached** | No faults to F129 | | Yes |
| **G5.10b.1.12-Lock mod fault 12 = F215:LC fuse - dV/dt** | No faults to F129 | | Yes |
| **G5.10b.1.13-Lock mod fault 13 = F216:LC fuse - dI/dt** | No faults to F129 | | Yes |
| **G5.10b.1.14-Lock mod fault 14 = No faults** | No faults to F129 | | Yes |
| **G5.10b.1.15-Lock mod fault 15 = No faults** | No faults to F129 | | Yes |

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## Subgroup G5.11: More protections

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.11.1-Earth current = 30.0%** | 0.0 to 100.0 % | Allows setting the earth current. | Yes |
| **G5.11.2-Delay high Vdc = 30 s** | 0 to 60 s | Allows setting a trip delay in high voltage conditions.<br>**Note**: This parameter is not shown in STATCOM. | Yes |
| **G5.11.3-DU overcurrent = 400.0 A** | 0.0 to 600.0 A | Allows setting a current value for the DU channels. If this value is overcome, a fault will be triggered.<br>**Note**: This parameter is not shown in STATCOM. | Yes |
| **G5.11.3a-Inv current prot = -350.0 A** | -3501 to -190 A | In a photovoltaic field sometimes inverse or negative currents can appear that we are going to read through the current transformers (in each channel) that are in the DC cabinet (DU).<br>This parameter defines the equipment protections based on these currents.<br>When the negative current of a channel in the DC cabinet (DU) exceeds G5.11.3a, fault F156 occurs and the inverter stops (but does not open the DU).<br><br>The protection is activated if there are current sensors DU (SG10.1.13 DU Current measurements) and G10.2b NEC enabled=NEC 2014.<br><br>**Note**: In the case of PCS or DU inverters with NEC 2017, the fault will be disabled.<br><br>When the parameter has the value -3501 the protection is disabled.<br>In order to be able to implement this functionality, the current measurement board must be:<br>• E996BA - Current measurement board. This version can measure negative currents (Note: This board is connected to the E995CB voltage measurement board). The first versions of HEM go with these boards for the NEC 2014.<br>• E990xx - Current measurement board. It is the board used in the final design of HEM for both NEC 2014 and NEC2017. | Yes |
| **G5.11.3c-DU pos I detect = 350.0 A** | 20.0 to 350.0 A | Allows user to configure positive and negative current of the DU, to activate the opening logic of the NEC 2014 contactors.<br><br>When the DU is to be opened (either by selector, or special fault), the DU will only be opened if the current is between parameters G5.11.3c and G5.11.3d). | Yes |
| **G5.11.3d-DU neg I detect = -100.0 A** | -350.0 to -20.0 A | | Yes |
| **G5.11.3e1-DUtotal neg cur = Disabled A** | Disabled = 1001;<br>0 to Disabled A | These parameters have been created for the protection "PV negative power", operating with an inverter and DC/DC system for battery connection.<br>This protection is designed to ensure that the DCDC's connected to the Bus+ do not inject current into the PV panels, when there's an unexpected stop of the inverter.<br><br>F194 will be trigger upon these conditions:<br><br>When SG10.16(DU Current) is below - G5.11.3d1 DU total negative current protection for more than G5.11.3d2.delay. F194 will be trigger upon these conditions<br><br>This parameter will be enabled depending on the following settings:<br>Therefore, the current is monitored at the PV interface to ensure that there is no outgoing current to the PV panels.<br>The DU type set in G10.1 = HEM and the maximum power at which the Bus Plus works must be less than 1x 500 KW | Yes |
| **G5.11.3e2-DUtotal neg cur del = 2 s** | 0 to 10 s | This parameter follows the functionality of parameter G5.11.3d1.<br>Allows to set a delay time in seconds for negative current protection.<br><br>This parameter will be enabled depending on the following settings:<br><br>This parameter will be activated according to the following settings:<br>It must be ensured that the DU type set in G10.1 = HEM and that the maximum power at which the Bus Plus works is less than 1x 500 KW | Yes |
| **G5.11.4-MV manometer fault = Off** | Off<br>1 to 4095 | These two parameters allow to configure when fault **F143: Air flow MV** should be triggered.<br><br>G5.11.4 defines the ventilation pressure protection value for the manometer at the medium voltage (MV) side. If this threshold is overcome during the time set in G5.11.5, F143 will be triggered.<br><br>**Note:** These parameters only apply to HEM inverters. | Yes |
| **G5.11.5-MV manometer delay = 10s** | 0 to 300 s | | Yes |

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.11.6-LV manometer fault = Off** | 0 to 4095 | These two parameters allow to configure when fault **F142: Air flow LV** should be triggered.<br><br>G5.11.6 defines the ventilation pressure protection value for the manometer at the low voltage (LV) side. If this threshold is overcome during the time set in G5.11.7, F142 will be triggered. | Yes |
| **G5.11.7-LV manometer delay = 10s** | 0 to 300 s | | Yes |
| **G5.11.8-Fb DC breaker fault = Fault** | Disabled<br>Warning<br>Fault | Allows a fault or warning to be displayed depending on the status of the circuit breakers<br>If warning is selected, in case of discrepancy between the command (Close and open) and the state of the circuit breaker (Closed or open), then the warning W148 (Circuit breaker case 1) or W149 (circuit breaker case 2) will be displayed.<br>If fault is selected, in case of discrepancy between the command (Close and open) and the state of the circuit breaker (Closed or open), then fault F153 (Circuit breaker case 1) or F154 (circuit breaker case 2) will be output. | Yes |
| **G5.11.9-Vpv-Vdc mismat. thres = 30 V** | 10 to 500 V | Allows to define the maximum voltage threshold of $V_{PV} – V_{BUS}$. If this threshold is exceeded during a period of time greater than the defined in G5.11.10-Vpv-Vdc mismat. dlay, the fault F12:Vdc mismatch will be triggered. | Yes |
| **G5.11.10-Vpv-Vdc mismat. dlay = 5.0 s** | 0.5 to 50.0 s | Allows to define the maximum voltage threshold of $V_{PV} – V_{BUS}$. If this threshold is exceeded during a period of time greater than the defined in G5.11.9-Vpv-Vdc mismat. thres, the fault F12:Vdc mismatch will be triggered. | Yes |
| **G5.11.11-Transf Aux Thermal = Fault** | Warning<br>Fault | When the thermal circuit breaker that protects the auxiliary transformer is triggered, it sends a digital signal to the main control board.<br>Depending on the configuration of this parameter, the system will display a fault or a warning.<br><br>**OPTION** / **DESCRIPTION**<br>Warning: W161:Trans Aux Therm will be triggered.<br>Fault: F152:Transf Aux Thermal will be triggered and the inverter will stop working. | Yes |

## Subgroup G5.12: Open DU Protections

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G5.12.1-Open DU fault 1 = F41:GFDI fuse** | No faults to F299 | In contrast to the special fault list that opens the DU side and also the AC side, the DU fault list allows PE staff to set the faults that are only needed to open the DU and then the inverter will go to the OFF state.<br><br>In G5.12.x parameters, it is possible to configure up to 20 faults for the DC cabinet opening protection. | Yes |
| **G5.12.2-Open DU fault 2 = F45:Insulation** | No faults to F299 | | Yes |
| **G5.12.3-Open DU fault 3 = F49:IMI autotest** | No faults to F299 | | Yes |
| **G5.12.4-Open DU fault 4 = F53:GFDI current leak** | No faults to F299 | | Yes |
| **G5.12.5-Open DU fault 5 = F17:DU feedback** | No faults to F299 | | Yes |
| **G5.12.6-Open DU fault 6 = F62:DU comms** | No faults to F299 | | Yes |
| **G5.12.7-Open DU fault 7 = F153:Fb DC Breaker 1** | No faults to F299 | | Yes |
| **G5.12.8-Open DU fault 8 = F154:Fb DC Breaker 2** | No faults to F299 | | Yes |
| **G5.12.9-Open DU fault 9 = F51:PPC comms** | No faults to F299 | | Yes |
| **G5.12.10-Open DU fault 10 = F191:Critical Low Battery V** | No faults to F299 | | Yes |
| **G5.12.11-Open DU fault 11 = F12:Vdc mismatch** | No faults to F299 | | Yes |
| **G5.12.12-Open DU fault 12 = No faults** | No faults to F299 | | Yes |
| **G5.12.13-Open DU fault 13 = No faults** | No faults to F299 | | Yes |
| **G5.12.14-Open DU fault 14 = No faults** | No faults to F299 | | Yes |
| **G5.12.15-Open DU fault 15 = No faults** | No faults to F299 | | Yes |

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| Screen | Range | Function | Set on run |
|---|---|---|---|
| G5.12.16-Open DU fault 16 = No faults | No faults to F299 | | Yes |
| G5.12.17-Open DU fault 17 = No faults | No faults to F299 | | Yes |
| G5.12.18-Open DU fault 18 = No faults | No faults to F299 | | Yes |
| G5.12.19-Open DU fault 19 = No faults | No faults to F299 | | Yes |
| G5.12.20-Open DU fault 20 = No faults | No faults to F299 | | Yes |

## Subgroup G5.13: High DC voltage Night

| Screen | Range | Function | Set on run |
|---|---|---|---|
| G5.13.1-Overvolt DC level = 1350 V | 800 to 1550 V | Defines the overvoltage DC level. | Yes |
| G5.13.2-Overvolt DC restart = 1340 V | 800 to 1550 V | Defines the DC to restart due to overvoltage. | Yes |
| G5.13.3-Overvolt time restart = 0.3 s | 0.0 to 10.0 s | Defines the overvoltage time to restart. | Yes |
| G5.13.4-Overvolt DSP = 355 V | 0 to 600 V | Defines the DSP overvoltage. | Yes |
| G5.13.5-Overvolt FPGA = 385 V | 0 to 600 V | Defines the FPGA overvoltage. | Yes |

## Subgroup G5.14: Resonance LLCL prot

Parameters to prevent the LCL filter from being damaged when the modules go into resonance.

When the protection is configured in fault mode, the product will only be stopped when "n" modules are in dVI/dt fault.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| G5.14.1-Configuration = Disabled | Disabled<br>Warning<br>Fault | Enables or disables the protection.<br>If this parameter is set to fault, [F215:LC fuse - dV/dt] or [F216:LC fuse - dI/dt] will be triggered.<br>If this parameter is set to warning, W215 or W216 will be triggered. | Yes |
| G5.14.1a.-N mod resonanc = 3 | 1 to 6 | Defines the number of modules “n” that must go into fault for the protection to be activated.<br><br>**Note:** this parameter will be shown when [G5.14.1-Configuration] ≠ Fault. | Yes |
| G5.14.2-Protection type = Dynamic [11] | Dynamic<br>Fixed | Configures the protection as dynamic or fixed type.<br><br>**Dynamic**: raw offset adjustment on the defined voltage or current curve (G5.14.2a or G5.14.2b).<br><br>**Fixed**: fixed threshold (G5.14.2c or G5.14.2d). | Yes |
| G5.14.2a-Offset dV/dt = 80 %/s [11] | Off = 0;<br>1 to 400 %/s | Sets the offset dV/dt. | Yes |
| G5.14.2b-Offset dI/dt = 60 %/s [11] | Off = 0;<br>1 to 400 %/s | Sets the offset dI/dt. | Yes |
| G5.14.2c-Threshold dV/dt = 200 %/s [11] | Off = 0;<br>1 to 400 %/s | Configures the dV/dt threshold. | Yes |
| G5.14.2d-Threshold dI/dt = 200 %/s [11] | Off = 0;<br>1 to 400 %/s | Configures the dI/dt threshold. | Yes |

[11] These parameters will be shown if the protection is enabled.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| G5.14.3-dx/dt Mismatch detec = Disabled [11] | Disabled<br>Enabled | Enables or disables dx/dt mismatch detection. This is an additional protection which detects a possible mismatch between calculation in each phase (RS, ST, TR).<br><br>If enabled, the maximum difference between dxdt values of all modules will be chosen.<br><br>The inverter will trip if these two conditions are met:<br><br>1. [G5.14.1-Configuration] is enabled and G5.14.3a or b are set.<br>2. The maximum dxdt value of all modules is greater than the configured threshold. | Yes |
| G5.14.3a-dV/dt mismatch TH = 50 %/s [11] | Off = 0;<br>1 to 400 %/s | Configures the dV/dt TH mismatch. | Yes |
| G5.14.3b-dI/dt mismatch TH = 50 %/s [11] | Off = 0;<br>1 to 400 %/s | Configures the dI/dt TH mismatch. | Yes |

## Subgroup G5.15: Module homopolar I

Anomalies in the operation of the modules should be detected from the measurement of excessive homopolar current.

The fault is implemented in a stepped process because when the module stops due to excessive homopolar current (faulty module, usually due to problems in the hardware), the product can continue to operate correctly, since the current of the other modules will be balanced and the homopolar level will be in the correct range because the faulty module has been stopped.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| G5.15.1-Enable = No | No<br>Yes | Enable (Yes) or disable (No) module homopolar current protection.<br><br>**Note**: if this parameter is disabled, the rest of parameters within this subgroup will not be shown. | Yes |
| G5.15.2-Initial Ih = 50 A | 10 to 250 A | Minimum homopolar current above which the protection timeout would be enabled. | Yes |
| G5.15.3-Initial timeout = 60 s | 0 to 100 s | Failure time corresponding to the initial homopolar current (minimum). | Yes |
| G5.15.4-Maximum Ih = 60 A | 15 to 400 A | Maximum allowed homopolar current. | Yes |
| G5.15.5-Time out (Maximum Ih) = 1 s | 0 to 100 s | Fault time for maximum allowed homopolar current. | Yes |

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# Group G6: Communications

Parameters for Modbus and Ethernet communication protocols can be set in this group.

## Subgroup G6.1: Modbus

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G6.1.1-Display baudrate = 115200 bps** | 9600<br>19200<br>57600<br>115200<br>230400<br>460800 | Set the baud rate (bits per second) of the display. | Yes |
| **G6.1.2-Modbus address = 150** | 1 to 247 | Set the unique unit address (slave address) for Modbus communication.<br>In a standard Modbus network, there is one Master and up to 247 Slaves, each with a unique Slave Address from 1 to 247 | Yes |
| **G6.1.3-Modbus baudrate = 19200 bps** | 9600<br>19200<br>57600<br>115200<br>230400<br>460800 | Set the baud rate (bits per second) of the serial connection. | Yes |
| **G6.1.4-Modbus parity = None** | None<br>Odd<br>Even | Modbus parity setting. It is used for data validation. If you do not want to validate the data set this parameter to 'NONE'. Parity selection must match the master communication bus of the inverter. | Yes |
| **G6.1.5-Modbus stop bits = 1bit** | 1 bit<br>2 bits | Set the number of stop bits for each byte sent. | Yes |
| **G6.1.7-Display error CRC = 0** | 0 to 65535 | Keeps count of the lost frames in communication between the display and the inverter. | Yes |
| **G6.1.8-Modbus error CRC = 0** | 0 to 65535 | Keeps count of the frames lost in Modbus communication. | Yes |
| **G6.1.9-Modbus OK frames = 0** | 0 to 65535 | Keeps count of the frames that have been successfully delivered in Modbus communication. | Yes |
| **G6.1.10-Modbus state = 0** | 0 to 65535 | Only used by developers. | Yes |
| **G6.1.11-Modbus recover = No** | No<br>Yes | Only used by developers. | Yes |
| **G6.1.12-ADP Mode = No** | No<br>Yes | **Parameter added in HEM_1.2.1 for the customer Nextera.**<br>It allows enabling a special mode in which they can write multiple registers with Modbus RTU-RS485 without having exceptions in the following cases:<br>• Parameters which do not exist.<br>• Password protected parameters.<br>• Parameters which are disabled due to the selected operation mode.<br>• Read-only parameters.<br>• Parameters out of range. | Yes |
| **G6.1.13-Poll Latency = 50 ms** | 50 to 250 ms | Configures how often should Modbus requests be sent.<br><br>**Note:** this parameter is only available for Developers. | Yes |
| **G6.1.14-Comms DU faster = No** | No<br>Yes | Allows enabling (Yes) or disabling (No) the functionality to interleave every two RS485 frames a reading of the current registers of the DU.<br><br>When the functionality is enabled, after querying a device other than the central DU, the RS485 state machine will be forced to make two consecutive requests (states and measurements) to the central DU. Asynchronous requests cannot be executed in these requests. Once the queries are finished, the RS485 queue will continue where it left off, querying the next device. If the device is not available, it will pass to the next available device (it will not force a query to the DU, since it has not made a request to another device).<br><br>**Note:** this parameter is only available for Developers. | Yes |

## Subgroup G6.2: Ethernet

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G6.2.1-Automatic IP = No** | No<br>Yes | Allows selecting the IP assignment method of the module.<br><br>**OPT.** / **DESCRIPTION**<br>No: The inverter will take the IP address, subnet mask and gateway set by the user from parameters [G6.2.1.1] to [G6.2.1.3].<br>Yes: DHCP protocol is enabled and the network server will automatically set the IP address, subnet mask and gateway. | Yes |
| **G6.2.1.1-Assigned IP = 0.0.0.0.0** | 0 to 255 | Shows the IP address assigned to the module when DHCP is active. | Yes |
| **G6.2.1.2-Assigned subnet = 0.0.0.0.0** | 0 to 255 | Shows the subnet mask address assigned to the module. | Yes |
| **G6.2.1.3-Assigned gateway = 0.0.0.0** | 0 to 255 | Shows the gateway address assigned to the module. | Yes |
| **G6.2.2-IP address = 192.168.1.143** | 1 to 255 | Set the IP address in the user's local network, according to the guidelines given by the network administrator.<br><br>IP address format is: A.B.C.D.<br><br>To adjust this address, introduce a value in each of the four parameters that compose it. | Yes |
| **G6.2.3-Subnet Mask = 255.255.255.0** | 1 to 255 | Set the subnet mask address in the user's local network, according to the guidelines given by the network administrator.<br><br>Subnet mask address format is: A.B.C.D.<br><br>To adjust this address, introduce a value in each of the four parameters that compose it. | Yes |
| **G6.2.4-Gateway = 0.0.0.1** | 0 to 255 | Set the gateway address in the user's local network so that the inverter is able to communicate to an external network. This address must follow the guidelines given by the network administrator.<br><br>Gateway's address format is: A.B.C.D.<br><br>To adjust this address, introduce a value in each of the four parameters that compose it. | Yes |
| **G6.2.5-MAC address = 00-1B-77-81-EE-42** | 0 to 255 | Set the MAC address. This address is unique and exclusive, and it is related with the LAN / inverter board. It must be given by Power Electronics.<br><br>MAC address format is: A.B.C.D.E.F.<br><br>To adjust this address, introduce a value in each of the six parameters that compose it | Yes |

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## Subgroup G6.3: PPC

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G6.3.1-Reset refs time = 4s** [12] | Off<br>1 to 60 s | If no communication is detected between the Power Plant Controller and the inverter for a longer time than the timeout set in [G6.3.1], all references will be set to G6.3.3 and parameter G6.3.4.<br>If [G8.1.1-P control mode] = P.<br>- P Ref = [G6.3.3 – Home P Ref]<br>- Q Ref = [G6.3.4 – Home Q Ref].<br>If [G8.1.1-P control mode] = MPPT.<br>- P Ref = [G6.3.3 - Home P Ref]<br>- Q Ref = [G6.3.4 - Home Q Ref].<br>- If [G6.3.3 - Home P Ref] > P limit in SG10.3.3, this value will be configured to P value of SG10.3.3.<br>- If [G6.3.3 - Home P Ref] <= P limit in SG10.3.3, this value will be configured to P value of [G6.3.3]. | Yes |
| **G6.3.2-Comms fault time = 60s** [12] | Off<br>1 to 600 s | [G6.3.2], on the other hand, defines the maximum time communications can be down before triggering a PPC communications fault (F51:PPC comms). | Yes |
| **G6.3.3-Home P ref = 0.00 %** [12] | 0 to 100.00% | Active power setpoint value when the time set in G6.3.1 has elapsed.<br>**Note:** This parameter is not shown for M-PCS inverters. | Yes |
| **G6.3.4-Home Q ref = 0.00 %** [12] | -100.00 to 100.00 % | Reactive power setpoint when the time set in G6.3.1 has elapsed. | Yes |
| **G6.3.5-PPC IP filter = No** | No<br>Yes | If there are several PPCBox within the same communications network, it is possible to configure the inverter so it only listens to one PPCBox.<br>User must set this parameter to YES and configure the IP address which corresponds to the PPCBox in G6.3.6. | Yes |
| **G6.3.6-PPC IP Address = 0.0.0.0** | 0 to 255 | If parameter G6.3.5 has been set to YES, this parameter allows defining the IP address of the PPCBox with which the inverter will communicate.<br><br>IP address format is: A.B.C.D.<br><br>To adjust this address, introduce a value in each of the four parameters that compose it. | Yes |
| **G6.3.7-PPC Heartbeat Reg = 0** | 0 to 65535 | Allows to check if the communication with the PPC is working properly. If the communication is lost, the value of this parameter will be frozen. | Yes |
| **G6.3.8-Ramp act time = 2.0 s** | 0.1 to 5.0 s | Configures the ramp activation time.<br>**Note**: this parameter will be shown if [G6.3.9-Ramp selector] = Recovery ramps. | Yes |
| **G6.3.9-Ramp selector = Disabled** | Disabled<br>Recovery ramps<br>AEMO | Allows selecting which ramp will be applied if communications are lost. | No |
| **G6.3.10-AEMO Max losses = 2** [13] | 1 to 10 | If the AEMO behavior has been enabled, this parameter defines the maximum number of consecutive losses to send the stop command to the product. | Yes |
| **G6.3.11-AEMO Max time = 60 min** [13] | 1 to 600 min | If the AEMO behavior has been enabled, this parameter defines the maximum time between losses to consider them consecutive losses. | Yes |
| **G6.3.12-AEMO Timeout 1st lost = 30 s** [13] | 5 to 60 s | If the AEMO behavior has been enabled, this parameter sets the maximum duration of the loss of communications. If this value is exceeded, the shutdown manoeuvre will be initiated sending the inverter to home references (parameters [G6.3.3-Ref P home] and [G6.3.4-Ref Q home]), at which time fault [F157:Communications AEMO] will be triggered. | Yes |
| **G6.3.13-AEMO Detect lost = 0.5 s** [13] | 0.1 to 5.0 s | If the AEMO behavior has been enabled, this parameter configures the maximum time to detect a communication loss with the PPC. | Yes |
| **G6.3.14-Delta time PPC = 200 ms** [13] | 0 to 500 ms | Defines the delta time of the PPC to start checking the intersection between PPC and inverter reference. | Yes |

**Notes**:

- The SCADA controller must start the watchdog timer when the program first starts.
- To start the watchdog timer, a 16-bit number between 1 and 65355 must be written into Modbus register 44252.

[12] Parameters G6.3.1 to G6.3.4 will be displayed if [G6.3.9-Ramp selector] is not “AEMO”.
[13] Parameters G6.3.10 to G6.3.14 will only be shown if [G6.3.9-Ramp selector] = AEMO.

- Modbus register 44252 must be re-written (same value is ok) in less than 4 seconds (G6.3.1 default value).
- If 44252 is not updated, the inverter power output will automatically be set to the values in G6.3.3 and G6.3.4 (PPC Home Reference) (Default: P% = 0 and Q% = 0).
- If 44252 is not updated in 60 seconds (G6.3.2 default value), then the inverter will trip showing F51 PPC comms.
- If the PPC/SCADA writes a ZERO at 44252, the system is disabled; this has the same effect than to set G6.3.1 and G6.3.2 to Off.

## Subgroup G6.5: Cerbero Direct

Parameters to set up the connection to Cerbero Direct server where the information is saved for 7 days corresponding to the data of the SD cards. Allows to change/read the configuration parameters.



| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G6.5.1-Enable connection = No** | No<br>Yes | Enables connection with cerbero direct. | Yes |
| **G6.5.2-IP address = 0.0.0.0** | 0.0.0.0 to 255.255.255.254 | Allows to set Ip address. This Ip address shall be configurated with. IP: 173.249.7.94 | Yes |
| **G6.5.3-Port = 8888** | 1 to 65535 | Allows to set port in Cerbero Direct settings. SFTP server port22 | Yes |
| **G6.5.4-Plant name = Plant** | 0 to 255 | Allows to set plant name in Cerbero Direct settings | Yes |
| **G6.5.5-Inverter name = inverter** | 0 to 255 | Allows to set inverter name in Cerbero Direct settings. | Yes |
| **G6.5.10-Connection status = Not connected** | Not connected<br>Connected | Shows the connection status in Cerbero Direct settings | Yes |
| **G6.5.11-Status config vbm = Sent LOGs** | Not sent LOGs<br>Sent LOGs | Enables the option to send or not Logs in Cerbero Direct settings | Yes |

# Group G7: Auto reset

Parameters to configure the auto reset function are described in the table below.

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>G7.1-Enable = Yes</td><td>No<br>Yes</td><td>Allows to enable (Yes) or disable (No) Auto reset function. This function resets the faults of the equipment automatically.<br><br><table><tr><th>OPT.</th><th>DESCRIPTION</th></tr><tr><td>No</td><td>Disables the Auto Reset function.</td></tr><tr><td>Yes</td><td>Enables the Auto Reset function. Once this function is enabled, all the faults will be reset automatically as many times as set in G7.2-Number of retries and with the delay between retries set in G7.3-Delay between retries. Also, it is possible to modify the operation of the algorithm using the G7.5-All faults enabled, G7.6-Faults allowed number and G7.7-Exception number parameters.</td></tr></table><br>⚠ Caution: The Auto Reset may cause unexpected automatic starts. Before activating this function, make sure the facility meets the requirements to be configured this way without creating risks of any personal injury or damage.</td><td>Yes</td></tr>
<tr><td>G7.2-Number of retries = 5</td><td>1 to 30</td><td>This parameter allows setting the maximum number of reset attempts that the inverter will execute.<br>To carry out a secure automatic reset of the faults, it is important to configure this parameter and G7.4-Reset retries timeout correctly.</td><td>Yes</td></tr>
<tr><td>G7.3-Delay between retries = 30s</td><td>5 to 300 s</td><td>This parameter allows to set the time lapse between one reset attempt and the next.</td><td>Yes</td></tr>
<tr><td>G7.4-Reset retries timeout = 30min</td><td>1 to 60 min</td><td>This parameter allows to set the time before the Auto Reset attempt counter will reset to zero.<br><br>There are two possible situations:<br><br>a) The reset time of the counter elapses before the inverter exhausts all the Auto Reset attempts defined in parameter G7.2-Number of retries. In this case, the counter will reset to 0.<br><br>b) The inverter exhausts the maximum number of Auto reset attempts, but between these attempts, the defined time to reset the counter has not been elapsed. In this case, the fault remains active and G7.4 keeps the value of time in which the last failed attempt to reset happened.</td><td>Yes</td></tr>
<tr><td>G7.5-All faults enabled = Yes</td><td>No<br>Yes</td><td>If the Auto reset function is enabled (G7.1-Enable), this parameter allows to define if it must be applied to all faults of the equipment or not.<br><br><table><tr><th>OPT.</th><th>DESCRIPTION</th></tr><tr><td>No</td><td>The Auto reset function will not be applied to all equipment faults.<br>Through the G7.6-Faults allowed number parameter it is possible to define which faults must be reset using the Auto reset function.</td></tr><tr><td>Yes</td><td>The Auto reset function will be applied to all equipment faults.<br>Some faults can be defined as exceptions (this is, each fault will do its predefined reset) through the G7.7-Exception number parameter.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G7.6-Faults allowed number = 0</td><td>0 to 10</td><td>Sets the number of faults allowed to be configured for auto reset when not all faults have been enabled for it (G7.5-All faults enabled =No).<br>Note: This parameter is only displayed when G7.5-All faults enabled =No.</td><td>Yes</td></tr>
<tr><td>G7.6.x-Fault allowed x = 0</td><td>0 to 199</td><td>The user can configure up to 10 faults allowed, according to section 3. The faults not allowed will do its predefined reset.<br><br>Note: x varies between 1 and 10, depending on the value of G7.6-Faults allowed number.</td><td>Yes</td></tr>
</table>

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G7.7-Exception number = 3** | 0 to 10 | If all faults are enabled for auto reset (G7.5-All faults enabled = Yes), this parameter allows to define the number of faults that must do their predefined reset, this is, they will be exceptions for the auto reset function.<br><br>These exceptions can be configured in the following parameters ([G7.7.x]).<br><br>**Note:** *x* varies between 1 and 10, depending on the value of G7.7-Exception number. | Yes |
| **G7.7.x-Exception x = No faults** | 0 to 199 | The user can configure up to 8 faults allowed, according to section 3.<br><br>*x* varies between 1 and 10, depending on the value of G7.7-Exception number.<br><br>**Note:** Parameters G7.7.1-Exception 1 = F41:GFDI fuse and G7.7.2-Exception 2 = F53:GFDI current leak are configured by default. User can configure other exceptions through the parameters G7.7.3-Exception 3 to G7.7.10-Exception 10. | Yes |
| **G7.7.1-Exception 1 = F41:GFDI fuse** | F41:GFDI fuse | Default exception for F41:GFDI fuse. | Yes |
| **G7.7.2-Exception 2 = F53:GFDI current leak** | F53:GFDI current leak | Default exception for F53:GFDI current leakage. | Yes |
| **G7.7.3-Exception 3 = F157:Communications AEMO** | F157:Communications AEMO | Default exception for F157:Communications AEMO. | Yes |

**Notes:**

- If it is not a special fault and it is not in the autoreset exception list, the manual reset will be unlimited.
- If it is a special fault, the maximum number of manual reset retries will be one.
- If it is a fault that is in the autoreset exceptions, the maximum manual reset number will be two.
- If the user is personal to Power Electronics staff, the maximum number of manual reset retries will be unlimited.

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# Group G8: Control

## Subgroup G8.1: References

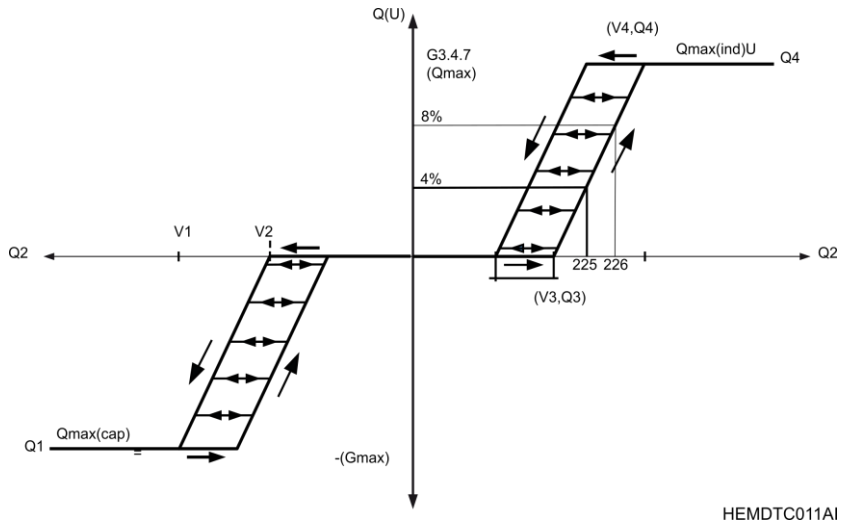
<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>G8.1.1-P control mode = MPPt</td><td>MPPt<br>I<br>P<br>Vdc</td><td>Allows setting the control mode for the active power. There are four control modes, as shown in the table below:<br><br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>MPPt</td><td>The inverter is always looking for maximum power point. This option only applies to SOLAR type inverters.</td></tr><tr><td>I</td><td>The inverter maintains constant the output active current set on [G8.1.3 Id reference].</td></tr><tr><td>P</td><td>The inverter maintains constant the output active power set on [G8.1.3 P reference]. This option only applies to PCS type inverters.</td></tr><tr><td>Vdc</td><td>The inverter holds the DC voltage set on [G8.1.3 Vdc reference].</td></tr></table><br>Note: Options I, P and Vdc are used for testing and may not be visible to the user. Contact Power Electronics for assistance.</td><td>No</td></tr>
<tr><td>G8.1.2-Q control mode = Q</td><td>Cos phi<br>Q<br>I<br>V/Q</td><td>Allows setting the control mode for the reactive power. Options are shown in the table below:<br><br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Cos phi</td><td>The inverter maintains constant the output cos(phi) set on G8.1.4 / Cos phi reference absolute.</td></tr><tr><td>Q</td><td>The inverter maintains constant the output reactive power set on G8.1.4 / Q reference.</td></tr><tr><td>I</td><td>The inverter maintains constant the output reactive current set on G8.1.4 / Iq reference.</td></tr><tr><td>V/Q</td><td>Reactive power depending on voltage curve. The reference values are set in [G8.1.4 Vref], [G8.1.5 V1], [G8.1.6 V2], [G8.1.7 V3 voltage] and G8.1.8 to G8.1.15.</td></tr></table></td><td>No</td></tr>
<tr><td>G8.1.3-P reference = 100.00% [14]</td><td>0.00 to 100.00%</td><td rowspan="3">The reference for the active power depends on which control mode has been chosen in [G8.1.1 P control mode].<br><br>Thus, parameter G8.1.3 will vary as shown in the following table:<br><br><table><tr><th>G8.1.1</th><th>G8.1.3</th><th>DESCRIPTION</th></tr><tr><td>MPPt</td><td>Not shown</td><td>-</td></tr><tr><td>I</td><td>Id reference</td><td>Set an active current reference in percentage.</td></tr><tr><td>P</td><td>P reference</td><td>Set an active power reference in percentage.</td></tr><tr><td>Vdc</td><td>Vdc reference</td><td>Set a DC voltage reference in volts.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G8.1.3-Id reference = 100.00% [14]</td><td>0.00 to 100.00%</td><td>Yes</td></tr>
<tr><td>G8.1.3-Vdc reference = 1100V [14]</td><td>975 to 1500 V</td><td>Yes</td></tr>
<tr><td>G8.1.3.x-Px reference = 100.00 % [15]</td><td>-100.00 to 100.00 %</td><td rowspan="2">Parameters for M-PCS, where x is the module number and varies between 1 and 7.<br><br>If G8.1.3 = P reference, a set of parameters to set the P reference for each module will be shown (P1 reference, P2 reference…).<br><br>If G8.1.3 = Id reference, a set of parameters to set the Id reference for each module will be shown (Id1 reference, Id2 reference...).</td><td>Yes</td></tr>
<tr><td>G8.1.3.x-Idx reference = 100.00 % [15]</td><td>-100.00 to 100.00 %</td><td>Yes</td></tr>
</table>

[14] Parameter G8.1.3 depends on the P control mode set in G8.1.1.
[15] These parameters are only shown in M-PCS inverters and depend on the P reference set in G8.1.3.

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>G8.1.4-Cos phi reference = 1.000 [16]</td><td>0.001 to 1.000</td><td rowspan="2">The reference for the reactive power depends on which control mode has been chosen in [G8.1.2 P control mode].<br><br>Thus, parameter G8.1.4 will vary as shown in the following table:<br><table><tr><th>G8.1.2</th><th>G8.1.4</th><th>DESCRIPTION</th></tr><tr><td>Cos phi</td><td>Cos phi reference</td><td>Allows setting a cos(phi) absolute reference.</td></tr><tr><td>Q</td><td>Q reference</td><td>Allows setting a reactive power reference in percentage.</td></tr><tr><td>I</td><td>Iq reference</td><td>Allows setting a reactive current reference in percentage.</td></tr><tr><td>V/Q</td><td>Vref</td><td>Allows setting the voltage reference.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G8.1.4-Iq reference = 0 [16]</td><td>-100.00 to 100.00%</td><td>Yes</td></tr>
<tr><td>G8.1.4-Q reference = 0.00% [16]</td><td>-100.00 to 100.00%</td><td>Yes</td></tr>
<tr><td>G8.1.4-Vref = 100.00% [16]</td><td>90.00 to 110.00 %</td><td>No</td></tr>
<tr><td>G8.1.5-Excitation = Capacitive [16]</td><td>Capacitive<br>Inductive</td><td>This parameter depends on the Q control mode set in [G8.1.2].<br><table><tr><th>G8.1.2</th><th>G8.1.5</th><th>DESCRIPTION</th></tr><tr><td>Cos phi</td><td>Excitation</td><td>Allows setting a cos(phi) reference sign.</td></tr><tr><td>Q</td><td>Not shown</td><td>-</td></tr><tr><td>I</td><td>Not shown</td><td>-</td></tr><tr><td>V/Q</td><td>V1</td><td>Allows setting the voltage at which the capacitive reactive power injection corresponds with [G8.1.9 Q1].</td></tr></table></td><td>Yes</td></tr>
<tr><td>G8.1.5-V1 = 90.00% [16]</td><td>0.00 to (G8.1.6/100) %</td><td>No</td></tr>
<tr><td>G8.1.6-V2 = 92.00% [16]</td><td>(G8.1.5/100) to (G8.1.4 – G8.1.13)/100 %</td><td>This parameter depends on the Q control mode set in [G8.1.2].<br><table><tr><th>G8.1.2</th><th>G8.1.5</th><th>DESCRIPTION</th></tr><tr><td>Cos phi</td><td>Not shown</td><td>-</td></tr><tr><td>Q</td><td>Not shown</td><td>-</td></tr><tr><td>I</td><td>Not shown</td><td>-</td></tr><tr><td>V/Q</td><td>V2</td><td>Allows setting the voltage at which the inductive reactive power corresponds with [G8.1.12 Q2].</td></tr></table></td><td>No</td></tr>
<tr><td>G8.1.7-V3 = 108.00% [16]</td><td>(G8.1.4 + G8.1.13)/100 to (G8.1.8/100)</td><td>This parameter depends on the Q control mode set in [G8.1.2].<br><table><tr><th>G8.1.2</th><th>G8.1.5</th><th>DESCRIPTION</th></tr><tr><td>Cos phi</td><td>Not shown</td><td>-</td></tr><tr><td>Q</td><td>Not shown</td><td>-</td></tr><tr><td>I</td><td>Not shown</td><td>-</td></tr><tr><td>V/Q</td><td>V3</td><td>Allows setting the voltage at which the inductive reactive power injection corresponds with [G8.1.12 Q3].</td></tr></table></td><td>No</td></tr>
<tr><td>G8.1.8-V4 = 110.00% [17]</td><td>(G8.1.7 V3 /100) to 140.00 %</td><td>Allows setting the voltage at which the inductive reactive power corresponds with [G8.1.12 Q4].</td><td>No</td></tr>
<tr><td>G8.1.9-Q1 = 48.43% [17]</td><td>0.00 to 100.00 %</td><td>Allows setting the reactive power when the voltage is equal to [G8.1.5 V1].</td><td>No</td></tr>
<tr><td>G8.1.10-Q2 = 0.00% [17]</td><td>0.00 to 100.00 %</td><td>Allows setting the reactive power when the voltage is equal to [G8.1.6 V2].</td><td>No</td></tr>
<tr><td>G8.1.11-Q3 = 0.00% [17]</td><td>0.00 to 100.00 %</td><td>Allows setting the reactive power when the voltage is equal to [G8.1.7 V3].</td><td>No</td></tr>
<tr><td>G8.1.12-Q4 = 48.43% [17]</td><td>0.00 to 100.00 %</td><td>Allows setting the reactive power when the voltage is equal to [G8.1.8 V4].</td><td>No</td></tr>
<tr><td>G8.1.13-V hysteresis = 1.0% [17]</td><td>0 to 10.0 %</td><td>Sets the voltage hysteresis for the V/Q control mode.</td><td>No</td></tr>
<tr><td>G8.1.14-P activation = 20.0% [17]</td><td>0.00 to 100.00 %</td><td>Sets the activation active power for the V/Q control mode.</td><td>No</td></tr>
<tr><td>G8.1.15-P deactivation = 5.0% [17]</td><td>0.00 to 100.00 %</td><td>Sets the deactivation active power for the V/Q control mode.</td><td>No</td></tr>
<tr><td>G8.1.16-Response delay = 0.3s [17]</td><td>0.0 to 10.0 s</td><td>Defines how long to wait before activating the V/Q control whenever activation conditions are met.</td><td>No</td></tr>
<tr><td>G8.1.17-Response time = 5.0 s [17]</td><td>0.0 s to 90.0 s</td><td>Adjusts the time response in seconds of the Q/V control. Response time is defined as the time that takes reaching 90% of the reference since the moment when the change occurs.</td><td>No</td></tr>
</table>

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[16] Parameters G8.1.4 to G8.1.5 depend on the Q control mode set in G8.1.2.
[17] Parameters G8.1.6 to G8.1.17 are only shown when the Q control mode in (G8.1.2) is set to V/Q.




## Subgroup G8.2: Advanced configuration

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G8.2.1-Kp pll = 10.0%** | 0.0 to 100% | PLL proportional constant. | Yes |
| **G8.2.2-Ki pll = 10.0%** | 0.0 to 100% | PLL integral constant | Yes |
| **G8.2.3-Kp Vdc = 5.0%** | 0.0 to 100% | DC voltage loop proportional constant. | Yes |
| **G8.2.4-Ki Vdc = 10.0%** | 0.0 to 100% | DC voltage loop integral constant. | Yes |
| **G8.2.5-Kp I = 10.0%** | 0.0 to 100% | Current loop proportional constant. | Yes |
| **G8.2.6-Ki I = 10.0%** | 0.0 to 100% | Current loop integral constant | Yes |
| **G8.2.7-Kp freq = 10.0%** | 0.0 to 100.0% | Value of Kp for frequency control. | Yes |
| **G8.2.8-Ki freq = 10.0%** | 0.0 to 100.0% | Value of Ki for frequency control | Yes |
| **G8.2.9-Inverse sequence = Yes** | No<br>Yes | Enables or disables current injection with inverse sequence. | Yes |
| **G8.2.10-Kp Vdc noche= 5%** | 0.0 to 100.0 % | Kp value for vdc in night.<br>If this parameter is set to 0, the PID can not react to the night mode Q demand<br>This control keep the DC bus voltage stable in case of reactive variations that may be required from the inverter | Yes |
| **G8.2.11-Ki Vdc noche = 10,0 %** | 0.0 to 100.0 % | Ki value for vdc in night.<br>If this parameter is set to 0, the PID can not react to the night mode Q demand.<br>This control keep the DC bus voltage stable in case of reactive variations that may be required from the inverter | Yes |
| **G8.2.12-Inverse Filter = 1** | 1 to 5000 | Parameter to activate the filter on AC reverse currents.<br>It generates the gain control and discretization of the filter.<br>Contact to Power Electronics to set this parameter. | Yes |
| **G8.2.13-Kp inverse = 10.0 %** | 0.1 to 100.0 % | Proportional constant PI control gains for the inverter current | Yes |
| **G8.2.14-Ki inverse = 10.0 %** | 0.1 to 100.0 % | Integral constant PI control gains for the inverter current | Yes |

**Note:** To avoid an oscillation in the DC bus from appearing, parameters **G8.2.3-Kp Vdc** and **G8.2.4-Ki Vdc** must be configured according to the values set in the operation and sart-up gradients (G4.8.1.2 and G4.8.2.2).

The values to be configured depending on the gradients are shown in the following table:

| Set value | Value to configure | |
|---|---|---|
| G4.8.1.2-P gradient (start) and G4.8.2.2-P gradient (rise) | G8.2.3-Kp Vdc | G8.2.4-Ki Vdc |
| 0.1%/s | 0.5 | 1 |
| 0.5%/s | 1 | 1 |
| 1%/s | 1 | 2 |
| 2%/s | 1 | 4 |
| 3%/s | 2 | 4 |
| 4%/s | 2.5 | 5 |
| 5%/s | 3.5 | 6 |
| >10%/s | 5 | 10 |

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## Subgroup G8.3 Grid

**Egypt's grid code** requires reactive injection depending on the voltage at the POI. This group allows configuring the parameters so the inverter can estimate the voltage present in the grid and calculate how much energy to inject.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G8.3.1-SCR = 20.0** | 0.0 to 1000.0 | **These two parameters will be implemented in future versions.** G8.3.1 will define the short circuit ratio and G8.3.2 the line impedance. | Yes |
| **G8.3.2-X/R = 5.0** | 0.0 to 1000.0 | | Yes |
| **G8.3.3-To POI XI = 0.00 %** | 0.00 to 600.00 % | These are line parameters. They are defined by Power Electronics for each plant and are related to the distance between the POI and the plant. | Yes |
| **G8.3.4-To POI R = 0.00 %** | 0.00 to 600.00 % | **Note:** Do not modify this parameter without authorization from Power Electronics. | Yes |

# Group G9: Leak protection

This group allows enabling/disabling two leak protections (DC and AC) and configuring its parameters.

## Subgroup G9.1: DC

**Note**: parameters G9.1.2 to G9.1.24 will only be shown when the leak protection type [G9.1.1-IMI Type] = Power Electronics.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G9.1.1-IMI Type = Power Electronics** | No IMI<br>Power Electronics<br>External | Selects the protection type:<br>**OPTION** / **DESCRIPTION**<br>No IMI – There is no leak protection.<br>Power Electronics – The IMI used is Power Electronics'.<br>External – The IMI used is external. | No |
| **G9.1.1a-Activation =Yes** | No<br>Yes | Allows the activation or not of the measurement pulses as long as the type of protection selected in parameter G9.1.1 is Power Electronics<br>**OPTION** / **DESCRIPTION**<br>No – The insulation monitor does not send measurement pulses<br>Yes – The insulation monitor sends out measurement pulses | Yes |
| **G9.1.2-Fault threshold = 5.0kOhms** | 0.5 to 300.0 kOhms | Allows setting the fault threshold. | Yes |
| **G9.1.3-Fault recover threshold = 115%** | 101 to 130 % | Allows defining a hysteresis to emerge from fault state. | Yes |
| **G9.1.4-Warning threshold = 50.0kOhms** | 0.5 to 300.0 kOhms | Allows setting the warning threshold. | Yes |
| **G9.1.7-Self test exec period = 1440min** | 5 to 1440 min | Allows setting the self-test execution period.<br>Autotest BIST (self-test that checks if a known internal resistance is correctly measured) does not run at isolation monitoring device power up.<br>**OPTION** / **DESCRIPTION**<br>G9.1.7= 5min – The BIST autotest is not executed periodically.<br>G9.1.7>5min – In case of failure, F49 will be triggered. | Yes |
| **G9.1.8-Self test manual command = No** | No<br>Yes | Allows activating manually the IMI self-test. | Yes |
| **G9.1.10-Night Activation = Yes** | No<br>Yes | Enables (Yes) or disables (No) the protection in night mode. | Yes |
| **G9.1.11-Night Fault threshold = 5.0 kOhms** | 0.5 to 300.0 kOhms | Configures the fault threshold in night mode. | Yes |
| **G9.1.12-Night Flt recov thld = 115 %** | 101 to 130 % | Configures the recovery threshold in night mode. | Yes |
| **G9.1.13-Night Warning thld = 50.0 kOhms** | 0.5 to 300.0 kOhms | Configures the warning threshold in night mode. | Yes |
| **G9.1.16-Self test exec period = 1440 min** | 5 to 1440 min | Defines the self test execution period. | Yes |
| **G9.1.17-Self test manual comm = No** | No<br>Yes | Allows activating manually the IMI self-test. | Yes |
| **G9.1.19-N Consecutive Faults = 20** | 1 to 50 | Allows setting the minimum number of consecutive low insulation measurements before generating fault [F45: Insulation]. | Yes |
| **G9.1.20-N Conse Faults Nigth = 3** | 1 to 10 | Similar to [G9.1.19-N Consecutive Faults] but for night mode. | Yes |
| **G9.1.21-Thld diff measur Vdc = Disabled V** | Disabled = 0;<br>1 to 50 V | Threshold of the difference of the last two measurements of Vdc Bus. | Yes |
| **G9.1.22-Los admissible frames = 20** | Disabled = 0;<br>1 to 30 | Defines the admissible number of lost frames.<br>**Note**: this parameter is only available for Developers. | Yes |
| **G9.1.23-Timeout Wrong Comp = Disabled** | Disabled = 0;<br>1 to 10 | Allows to set the maximum number of insulation measurement attempts before the measurement is considered stable. | Yes |
| **G9.1.24-Timeout Wg Comp Night = Disabled** | Disabled = 0;<br>1 to 10 | Similar to [G9.1.23-Timeout Wrong Comp] but for night mode. | Yes |

## Subgroup G9.2: AC

**Note**: parameters G9.2.2 to G9.2.8 will only be shown when the leak protection type [G9.2.1-IMI Type] = Power Electronics.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G9.2.1-Type = No IMI** | No IMI<br>Power Electronics<br>External | Selects the protection type:<br>**OPTION** / **DESCRIPTION**<br>No IMI – There is no leak protection.<br>Power Electronics – The IMI used is Power Electronics'.<br>External – The IMI used is external. | No |
| **G9.2.1a-Activation = Yes** | No<br>Sí | Allows the activation or not of the measurement pulses as long as the type of protection selected in parameter G9.2.1 is Power Electronics<br>**OPTION** / **DESCRIPTION**<br>No – The insulation monitor does not send measurement pulses<br>Yes – The insulation monitor sends out measurement pulses | Yes |
| **G9.2.2-Fault threshold = 5.0 kOhms** | 0.5 to 100.0 kOhms | Allows setting the fault threshold. | Yes |
| **G9.2.3-Fault recover threshold = 115** | 101 to 130 % | Allows defining a hysteresis to emerge from fault state. | Yes |
| **G9.2.4-Warning threshold = 50.0 kOhms** | 0.5 to 100.0 kOhms | Allows setting the warning threshold. | Yes |
| **G9.2.7-Self test exec period = 60** | 5 to 1440 | Allows setting the self-test execution period. | Yes |
| **G9.2.8-Self test manual command = 0** | 0 to 1 | Allows activating manually the IMI self-test. | Yes |
| **G9.2.10-N Consecutive Faults = 3** | 1 to 10 | Allows setting the minimum number of consecutive low insulation measurements before generating fault [F45: Insulation]. | Yes |

EN

## Subgroup G9.3: Insulation algorithm

The insulation algorithm is executed in the morning, right before starting the inverter. A problem has been reported when bus voltage is not stable: if the algorithm runs when voltage in photovoltaic panels is not stable, a delay on start will be applied and less energy will be generated.

To address this issue, a new mode has been created, “stable conditions” to ensure algorithm will be executed when stable voltage conditions exist. When this mode is enabled, the inverter will look for the ideal conditions to run the algorithm, store the result and apply it later, at the moment of start.

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>G9.3.1-Algorithm execution = Before Run</td><td>Before Run<br>Stable conditions</td><td>Selects insulation measurement mode. The insulation algorithm calculates insulation at night, when the voltage in photovoltaic panels is constant.<br><br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Before Run</td><td>Runs insulation algorithm before start, in the morning.</td></tr><tr><td>Stable conditions</td><td>Runs insulation algorithm with stable $V_{DC}$ conditions, at night. It will look for periods with constant VDC and execute the normal algorithm.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G9.3.2-Check period = 300 s</td><td>30 to 7200 s</td><td>Time lapse during which voltage has to vary less than Delta Vmax (G9.3.3) with respect to the voltage present at the moment of starting the search.<br>Note: This parameter applies only to “Stable conditions” mode of G9.3.1 parameter.</td><td>Yes</td></tr>
<tr><td>G9.3.3-Delta V max = 10 V</td><td>1 to 100 V</td><td>Maximum voltage increment to consider that voltage is stable.<br>Note: This parameter applies only to “Stable conditions” mode of G9.3.1 parameter.</td><td>Yes</td></tr>
<tr><td>G9.3.4-Min V PV = 10 V</td><td>1 to 600 V</td><td>Minimum voltage in photovoltaic panels to execute the search for stable voltage.</td><td>Yes</td></tr>
<tr><td>G9.3.5-Algorithm status = Idle [18]</td><td>Idle<br>Period update<br>Check DU closed<br>Check why DU open<br>Stop night mode<br>Close DU<br>Wait to DU closed<br>Wait algorithm finish<br>Open DU<br>Wait to DU open</td><td>Shows the status of the algorithm, according to the following table:<br><br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Idle</td><td>In idle state</td></tr><tr><td>Period update</td><td>Looks for a period of stable $V_{DC}$.</td></tr><tr><td>Check DU closed</td><td>Verifies if the DU is closed.</td></tr><tr><td>Check why DU open</td><td>Checks the reason why the DU is open, before deciding whether to open it.</td></tr><tr><td>Stop night mode</td><td>Stops the night mode.</td></tr><tr><td>Close DU</td><td>Closes the DU.</td></tr><tr><td>Wait to DU closed</td><td>Waits for the DU to close.</td></tr><tr><td>Wait algorithm finish</td><td>Waits until the algorithm finishes.</td></tr><tr><td>Open DU</td><td>Opens the DU.</td></tr><tr><td>Wait to DU open</td><td>Waits until the DU opens.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G9.3.6-Algorithm result = Unknown</td><td>Unknown<br>OK<br>Fault</td><td>Shows the result of the algorithm, according to the following table:<br><br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Unknown</td><td>The algorithm is unable to determine whether there is an insulation condition or not. This is the initial algorithm state.</td></tr><tr><td>OK</td><td>Insulation OK.</td></tr><tr><td>Fault</td><td>Insulation not OK</td></tr></table></td><td>Yes</td></tr>
<tr><td>G9.3.7-Period initial V = Not available [18]</td><td>0 V to 65534<br>Not available</td><td>Shows the voltage value at the moment of starting the search for stable voltage. Value “not available” means the search is not running.</td><td>Yes</td></tr>
</table>

[18] These parameters cannot be modified (read-only parameters).

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G9.3.8-Period current V = Not available** [18] | 0 V to 65534<br>Not available | Shows the current value during stable voltage search. If "not available" is shown, that indicates that the search is not running. | Yes |
| **G9.3.9-Current period time = Not available** [18] | 0 V to 65534<br>Not available | Shows the elapsed time since stable voltage search started.<br>If its value is "not available", it means that the search is not in progress. | Yes |
| **G9.3.10-Date algorithm exec = 01/01/2000 00:00** [18] | 01/01/2000 00:00 to 31/12/2127 23:59 | Shows the date and time of the last execution of the algorithm. | Yes |

**Note:** To perform the search for stable conditions, the inverter must meet the following criteria:

- Have a grounded DC cabinet (DU).
- Be configured to run the algorithm when stable conditions are detected (G9.3.1-Algorithm execution).
- Result of the algorithm (G9.3.6-Algorithm result) is "unknown".
- Inverter is stopped.
- Heating is off
- BFPS is off.
- Selector is in On position.
- There are no faults.
- Insulation algorithm has not started.

For further information about stable conditions search, please contact Technical Service.

EN

# Group G10: DU configuration

This group contains the parameters to configure the Disconnecting Unit (DU).

**Note**: These group is not shown in STATCOM.

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>G10.1-DU Type = HEM</td><td>Disabled<br>HEM</td><td>Sets the DU type from the following options:<br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Without Disconnecting Unit.</td></tr><tr><td>HEM</td><td>HEM DC cabinet.</td></tr></table></td><td>No</td></tr>
<tr><td>G10.2-Working mode = GFDI + IMI</td><td>Standard<br>GFDI + IMI</td><td>Sets the working mode from the following options:<br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Standard</td><td>Ungrounded system. This option applies to both NEC 2014 and NEC 2017.</td></tr><tr><td>GFDI + IMI</td><td>Grounded system with NEC 2014 or NEC 2017 solution.</td></tr></table>Note: This parameter will be shown if G10.1 ≠ Disabled.</td><td>No</td></tr>
<tr><td>G10.2b-Enabled NEC = NEC 2017</td><td>NEC 2014<br>NEC 2017</td><td>Selects the NEC regulation.<br>The inverter settings will be modified automatically for it to comply with the normative set in this parameter.<br>When G10.2b = 2017, current breakers will be enabled.<br>Likewise, when G10.2b = 2014 contactors will be enabled.<br>Hardware must support this procedure.<br><br>Note:<br>• This parameter depends on the working mode set in G10.2.</td><td>No</td></tr>
<tr><td>G10.2c-V measures = Central DU</td><td>Measures board<br>Central DU</td><td>Allows defining how to measure voltage on the panels side.<br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Measures board</td><td>Voltages are measured with the DC acquisition board (for modular DUs).</td></tr><tr><td>Central DU</td><td>Voltages are measured directly in the central control board. This option only applies to inverters with SW version HEM_1.2.0 or later.</td></tr></table>Note: Consult with Power Electronics before modifying this parameter to verify compatibility if your equipment supports this functionality.</td><td>No</td></tr>
<tr><td>G10.2d-Current offset = No</td><td>No<br>Yes</td><td>Enable or disable the DU current offset.<br>Note: this parameter is shown when [G10.1-DU type] ≠ 0.</td><td>Yes</td></tr>
<tr><td>G10.3-Isolation meas tout = 600 s [19]</td><td>1 to 900 s</td><td>Time with the GFDI contactor opened. It applies with the NEC 2014 solution.</td><td>Yes</td></tr>
<tr><td>G10.4-Number of DU channels = 0</td><td>0 to 36</td><td>Shows the number of DU input channels. This parameter cannot be modified, the number of channels is calculated automatically based on the values of G10.4a to G10.4e.<br><br>Note: The number of channels depends on the number of measure boards and the channels on each board.</td><td>Yes</td></tr>
<tr><td>G10.4a-DU measures boards = 0 [19]</td><td>0 to 4</td><td>User must enter in this parameter the number of measure boards in the DC cabinet.<br>Then, he must configure parameters G10.4b to G10.4e to set the number of channels per measure board.</td><td>Yes</td></tr>
<tr><td>G10.4b-Channels DU meas 1 = 0 [19]</td><td>0 to 12</td><td>Allows setting the number of channels in the measure board 1.</td><td>Yes</td></tr>
<tr><td>G10.4c-Channels DU meas 2 = 0 [19]</td><td>0 to 12</td><td>Allows setting the number of channels in the measure board 2.</td><td>Yes</td></tr>
<tr><td>G10.4d-Channels DU meas 3 = 0 [19]</td><td>0 to 12</td><td>Allows setting the number of channels in the measure board 3.</td><td>Yes</td></tr>
<tr><td>G10.4e-Channels DU meas 4 = 0 [19]</td><td>0 to 12</td><td>Allows setting the number of channels in the measure board 4.</td><td>Yes</td></tr>
</table>

[19] These parameters will only be shown if G10.1 = HEM.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G10.4.x-Channel x ID = Off** [20] | Off<br>1 to 40 | Channel x identifier number.<br>x varies between 1 and 40, depending on the value of [G10.4]. | Yes |
| **G10.5-Select action = Disabled** [20] | Disabled<br>Reset channel status | Enable or disable channel status resetting. | Yes |

User must configure the DU operating mode and channels to be used:

- **G10.1, DU Type:**
  - **Disabled**
  - **HEM**: For HEM and HEMK inverters. It is compatible with NEC 2017, NEC 2014 and IEC.

    In these inverters control is centralized in the central control board, which communicates with the FPGA of each module through optical fiber. Besides, it uses an RS485 bus to communicate with the impulsion fans (cyclones), DC insulation monitor and the DU (also known as DC cabinet). Refer to section "Communications scheme" for further details.

- **G10.2, Working mode,** according to the available options.
- **G10.3, Isolation measure timeout:** Time lapse (in seconds) when, if no new measure from the insulation monitor is received, the current status is considered to be the same as the previous.
- **G10.4, Number of DU channels:** Number of available channels.
- **G10.4.x, Channel x ID:** Identifier (rotary switch) for each channel.

  **Note:** to avoid confusion, the number of channel and its identifier should match (e.g. Channel 1 – ID 1, Channel 7 – ID 7) and, in case an identifier is unavailable, leave the corresponding channel with the ID "disabled". Identifiers range is 1 to 40.
- **G10.4a to G10.4d, Channels DU measure:** For HEM and HEMK inverters, the number of measure boards must be set. Also, the number of channels per measure board must be configured.
- **G10.5, Select action:** Although there is a full diagnosis group, it is only available to users with access to the Service Group password. Thus, this parameter has been created to perform the action "Reset channel status".

Once all parameters have been configured, current (SV10.2-Measured current) and voltage (SV10.3-Measured voltage) measures can be visualized individually from the channel selected in SV10.1-Selected channel.

Due to space limitations, the display can only show information from one channel at a time. For instructions on how to visualize multiple channels, please refer to section "Disconnecting Unit Visualization Parameters".

[20] These parameters will only be shown when G10.1 = Modular.

# Group G11: Fans

Parameters to control inverter ventilation system can be set in this group.

Module fans can be configured in subgroup G11.1[21] and extraction fans + cyclones in subgroup G11.2.

See figures "HEM ventilation layout" and "HEMK ventilation layout" for details about ventilation elements and sides.

## Subgroup G11.1: Module

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G11.1.1-Fans mode = Temperature** | Off<br>Run<br>Fixed<br>Temperature | Selects the operation mode of module fans, according to the following options:<br><br>**OPTION** / **DESCRIPTION**<br>Off – Module fans are disabled.<br>Run – The module fans are connected when the inverter is injecting current and they are stopped three minutes after the end of current injection process.<br>Fixed – Module fans are always connected.<br>Temperature – Module fans are connected when the temperature threshold defined in 'G11.1.2' is overcome. | NO |
| **G11.1.2-Activation Tº = 70ºC** | 20 to 120ºC | Set the temperature threshold for enabling the module fans.<br>**Note**: This parameter is only displayed if 'G11.1.1 Fans Mode' is set to 'Temperature'. | Yes |
| **G11.1.3-Deactiv Temperature = 45 ºC** | 20 ºC to<br>(G11.1.2 – 1) | Set the temperature threshold for disabling the module fans<br>**Note**: This parameter is only displayed if 'G11.1.1 Fans Mode' is set to 'Temperature'. | Yes |

## Subgroup G11.2: Impulsion / Extraction

Parameters from this group allow configuring the cyclone ventilation system, which uses the cyclones (one in HEMK, two in HEM) and extraction fans for air impulsion / extraction.

**G11.2.0 Detection:**

The aim of this group is to identify the type of fan installed within the different existing fan types.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G11.2.0.1-Start Search = No** | No<br>Yes | Starts the process of searching for the type of installed drive fans | Yes |
| **G11.2.0.2- Search State = Disabled** | Disabled<br>Not detected<br>Detected<br>Searching Automatically<br>Searching Manually | Indicates the status of the fan search. | Yes |
| **G11.2.0.3- Detection Mode = Automatic** | Automatic<br>Manual | Allows selecting the fan search mode. | Yes |

[21] Please notice this group requires elevated privileges. Go to G1.4 to introduce the password or contact Power Electronics for assistance.

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>G11.2.0.4-Fan to detect = Emb-Papst</td><td>Emb-Papst<br>Zeihl<br>Tesoer<br>None</td><td>Parameter to select the type of fan to search.<br><br><table><tr><th>VALUE</th><th>TYPE</th></tr><tr><td>0</td><td>Emb-Papst</td></tr><tr><td>1</td><td>Zeihl</td></tr><tr><td>2</td><td>Tesoer</td></tr><tr><td>4</td><td>Ninguno</td></tr></table></td><td>Yes</td></tr>
<tr><td>G11.2.0.5-Fan to search = Fan 1</td><td>Fan 1<br>Fan 2</td><td>Parameter to select which fan will be searched for.<br><br><table><tr><th>VALUE</th><th>FAN</th></tr><tr><td>10</td><td>Fan 1</td></tr><tr><td>11</td><td>Fan 2</td></tr></table><br>Note: this parameter is shown if G11.2.0.4-Fan to detect = Tesoer.</td><td>Yes</td></tr>
<tr><td>G11.2.0.5-Num fans to search = 1</td><td>1 to 2</td><td>Number of fans to search.<br><br>Note: this parameter is shown if G11.2.0.4-Fan to detect = EMB-Papst.</td><td>Yes</td></tr>
<tr><td>G11.2.0.5- Fan 1 SN = 0000000000</td><td>0000000000 a 995900ZZZZ</td><td>Parameter to be used when searching manually. The serial number of the EBM fan must be entered. If only one serial number is entered, the other fan is automatically detected.<br>Note: Only visible when [G11.2.0.3] = “Manual”.</td><td>Yes</td></tr>
</table>

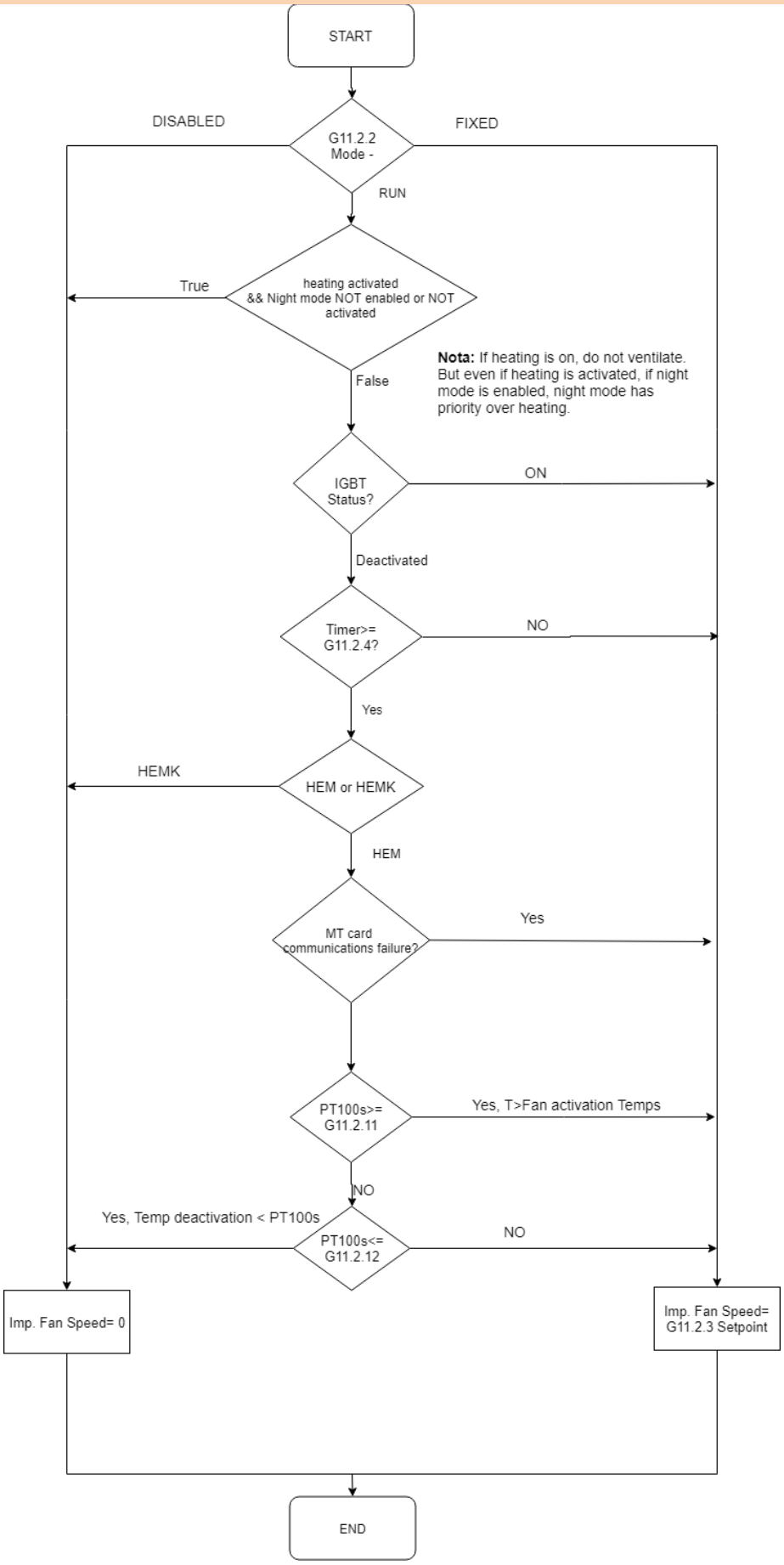
EN

### G11.2 Impulsion / Extraction:

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>G11.2.1-Enable = Yes</td><td>No<br>Yes</td><td>Allows enabling the cyclone ventilation system.</td><td>Yes</td></tr>
<tr><td>G11.2.2-Mode = Run</td><td>Disabled<br>Run<br>Fixed<br>Extraction</td><td>Allows adjusting ventilation mode:<br><br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Deactivates the impulsion ventilation.</td></tr><tr><td>Run</td><td>Cyclones and extraction fans operate at maximum capacity while the inverter is running. When the equipment is stopped, a delay is applied to the ventilators, so they keep operating during a time to cool down the transformer and then temperature hysteresis will take place.</td></tr><tr><td>Fixed</td><td>Activates the impulsion ventilation and sets the fans to the setpoint specified in G11.2.3</td></tr><tr><td>Extraction</td><td>When the product is operating, the impulsion and extraction fans are activated.<br>Whenever the product stops for any reason, the extraction fans will remain on and the ventilation will shut off once the time set in [G11.2.4-Delay off after run] elapses.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G11.2.3-Setpoint = 100 %</td><td>0 to 100 %</td><td>Adjusts the setpoint of impulsion ventilation.</td><td>Yes</td></tr>
<tr><td>G11.2.4-Delay off after run = 3600s</td><td>0 to 28800 s</td><td>Sets a delay to shut off ventilation after an equipment stop.</td><td>Yes</td></tr>
</table>

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>G11.2.5-Error type = Warning</td><td>Fault = 0<br>Warning = 1</td><td>Indicates how to manage ventilation faults, either as a fault or as a warning. Allows defining if a change in fans (either extraction fans or cyclone) feedback should be treated as a warning (W140 to W141) or a fault which will stop the equipment (F144 to F147).<br><br>It is valid for communication failures and for fan failure:<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Fault</td><td>Communication fault: Fault F147 and W140 will appear.<br>Fan reported fault: Fault 144 and W141 will appear.</td></tr><tr><td>Warning</td><td>Communication fault: W140 will appear and a 30% power reduction is applied in P, Q, S in approximately 2 seconds. (See SV11.11).<br>Fan reported fault: W141 will appear and a 30% power reduction is applied in P, Q, S in approximately 2 seconds. (See SV11.11).</td></tr></table>After approximately 15 seconds without communication or fan fault, the faults and warnings will disappear and the system will return to its previous values.</td><td>Yes</td></tr>
<tr><td>G11.2.6-Imp fan 1 status = Disabled</td><td>Disabled = 0<br>Fan Correct = 1<br>Communications error = 2<br>Error = 3</td><td rowspan="2">Shows the status of the impulsion fans (cyclones) 1 respectively.<br>Note: These parameters cannot be modified. Also, please notice cyclone 2 only exists in HEM inverters.</td><td>Yes</td></tr>
<tr><td>G11.2.7-Imp fan 2 status = OK</td><td>Disabled = 0<br>Fan Correct = 1<br>Communications error = 2<br>Error = 3</td><td>Yes</td></tr>
<tr><td>G11.2.8-Imp fan 1 error = 0x00 Hex</td><td>0x00 to 0xFFFF Hex</td><td>Displays the fan error code. To identify and fix it, check the manual of the installed fan.</td><td>Yes</td></tr>
<tr><td>G11.2.9-Imp fan 2 error = 0x00 Hex</td><td>0x00 to 0xFFFF Hex</td><td>Displays the fan error code. To identify and fix it, check the manual of the installed fan.</td><td>Yes</td></tr>
<tr><td>G11.2.10-Request fan 1 = 0x00 Hex</td><td>0x00 to 0xFFFF Hex</td><td>Current request sent to the impulsion fan 1 (G11.2.10) or 2 (G11.2.10b).</td><td>Yes</td></tr>
<tr><td>G11.2.10b-Request fan 2 = 0x00 Hex</td><td>0x00 to 0xFFFF Hex</td><td>Note: these parameters are only available for users with role Developer and [G11.2.16a-Fans Type] <3.</td><td>Yes</td></tr>
<tr><td>G11.2.11- Ext LV fb disabled = No</td><td>No<br>Yes</td><td>Enable the threshold to activate extraction fans and cyclones.<br>Note: This parameter only applies to HEM inverters.</td><td>Yes</td></tr>
<tr><td>G11.2.12- Ext MV fb disabled = No</td><td>No<br>Yes</td><td>Enable the threshold to deactivate extraction fans and cyclones.<br>Note: This parameter only applies to HEM inverters.</td><td>Yes</td></tr>
<tr><td>G11.2.13- T Fans Activation = 60 ºC</td><td>40 to 140ºC</td><td>Allows to set fan activation temperature.</td><td>Yes</td></tr>
<tr><td>G11.2.14- T Fans Deactivation = 30ºC</td><td>20 to 100ºC</td><td>Allows to set fan deactivation temperature</td><td>Yes</td></tr>
<tr><td>G11.2.15-Avaliables Fan Nº = 2</td><td>1 to 2</td><td>Shows the number of fans available</td><td>Yes</td></tr>
<tr><td>G11.2.16a Fans Type = Emb-Papst</td><td>Emb-Papst<br>Zeihl<br>Tesoer<br>None</td><td>Indicates the type of fan detected.<br><br>If there is only one type of fan, G11.2.16a is shown and parameters G11.2.16b and G11.2.16c are hidden.</td><td>Yes</td></tr>
<tr><td>G11.2.16b- Fan 1 Type = Emb-Papst</td><td rowspan="2">Emb-Papst<br>Zeihl<br>Tesoer<br>None</td><td rowspan="2">If there is more than one type of fan, then G11.2.16a is hidden and the other two are shown.<table><tr><th>VALUE</th><th>TYPE</th></tr><tr><td>0</td><td>Emb-Papst</td></tr><tr><td>1</td><td>Zeihl</td></tr><tr><td>2</td><td>Tesoer</td></tr><tr><td>4</td><td>Ninguno</td></tr></table></td><td>Yes</td></tr>
<tr><td>G11.2.16c- Fan 2 Type = Emb-Papst</td><td>Yes</td></tr>
<tr><td>G11.2.17-Fans Power Reduction Min=30.0%</td><td>0.0 to 200.1%<br>(Disabled = 200.1%)</td><td>Sets the final value of power derating when fans are in warning (W140 y W141). The power is increased/decreased to this value. The time of rating is 2s aprox.<br>Note: After modifying this parameter, a SW reset must be performed (Write value 1903 in modbus address 2008).</td><td>Yes</td></tr>
</table>

The following diagram details the features in the sequence of activating or deactivating the inverter's ventilation functionalities.



# Group G12: BFPS

## NOTICE

This group does not apply to HEM nor HEMK inverters.

The Back-Feed Power Supply (BFPS) system helps guarantee an uninterrupted power supply in case no voltage is measured from the panels (e.g. at night, or when the measure is too low).

**Note**: This group is not shown in STATCOM.

It prevents the risk of getting an electric shock in case there is any power feedback from an output other than the main inverter output.

Central control receives the order to activate / deactivate the BFPS contactor from the Station Hub. The inverter only follows the command if “night” has been detected, this means inverter status is “NIGHT”. The inverter will remain in “NIGHT” status while BPFS is active, even if “day” conditions are detected.

BFPS functionality requires a system DU. The DU will open as previous step to activating the BFPS contactor. Once BFPS operation has finished and contactor is deactivated, whenever “day” is detected, the inverter will perform the bus soft charge to close the DU and be ready to start.

Central control facilitates modules operation in Heating/Nigh Modes while the BFPS contactor may be activated. This means, operation in Heating/Night Mode is not compatible with activation / deactivation of the BFPS contactor.

The BFPS provides voltage and current measurements. BFPS current and voltage measurement, information about its state and other configuration parameters are available in group G12. They are described in the two following subsections.

## WARNING

**Follow safety procedures when BFPS system is enabled.**

Voltage might be present in the photovoltaic field at night if this system is enabled and you may get an electric shock.

## Subgroup G12.1: Configuration

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G12.1.1-Max Power = 10kW** | 0 to 15 kW | Maximum power supported by the BFPS. | Yes |
| **G12.1.2-Delay Max Power fault = 10s** | 0 to 60 s | Defines the maximum time that can elapse when the power is higher than [G12.1.1] before triggering a warning. | Yes |
| **G12.1.3-Max I = 100A** | 25 to 125 A | Maximum current supported by the BPFS.<br>If the voltage is considered to be stable, this option is used to apply current control. | Yes |
| **G12.1.4-Delay Max I fault = 1s** | 0 to 60 s | Defines the maximum time that can elapse when the current is higher than [G12.1.3] before triggering a warning. | Yes |
| **G12.1.5-Max Voltage fault = 120%** | 100 to 140% | Sets the highest limit for the BPFS voltage. 100% → 420V. | Yes |
| **G12.1.6-Min Voltage fault = 88%** | 0 to 100 % | Sets the lowest limit for the BPFS voltage. 100% → 420V. | Yes |

## Subgroup G12.2: Others

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>G12.2-Enable = No</td><td>No<br>Yes</td><td>Enable or disable the BFPS</td><td>Yes</td></tr>
<tr><td>G12.3-Status = 00000000000000</td><td>00000000000000 to XXXXXXXXXXXXXX</td><td>Shows the BFPS status.<br>Each bit (from right to left) represents a status.<br><table><tr><th>BIT</th><th>DESCRIPTION</th></tr><tr><td>0</td><td>Online. 0 = Offline, 1 = Online</td></tr><tr><td>1</td><td>Contactor. 0 = Closed 1 = Open</td></tr><tr><td>2</td><td>Feedback contactor. 0 = Open 1 = Closed</td></tr><tr><td>3</td><td>Current present error (*)</td></tr><tr><td>4</td><td>Feedback close error (*)</td></tr><tr><td>5</td><td>Feedback open error (*)</td></tr><tr><td>6</td><td>Comms error (*)</td></tr><tr><td>7</td><td>No conditions error (*)</td></tr><tr><td>8</td><td>Max power fault (*)</td></tr><tr><td>9</td><td>Max current fault (*)</td></tr><tr><td>10</td><td>High voltage fault (*)</td></tr><tr><td>11</td><td>Low voltage fault (*)</td></tr><tr><td>12</td><td>Contactor fault (*)</td></tr><tr><td>13</td><td>Blown fuse fault (*)</td></tr></table>(*) For bits 3 to 13 → If the bit value is 1, there is a fault.</td><td>Yes</td></tr>
<tr><td>G12.4-Event = Disabled</td><td>Disabled<br>Close contactor<br>Open contactor</td><td>Allows forcing a command to the BFPS.</td><td>Yes</td></tr>
<tr><td>G12.5-Current = 0.0A</td><td>0.0 to 6553.5 A</td><td>Shows the measured current.</td><td>Yes</td></tr>
<tr><td>G12.6-Voltage = 0.0V</td><td>0.0 to 6553.5 V</td><td>Shows the measured voltage.</td><td>Yes</td></tr>
</table>

EN

# Group G13: Normative

Based on the CAISO (Canadian Independent System Operator), this group allows to synchronize all the inverters of the field and, using an algorithm, store information when any of these events occur in a specific inverter:

- Frequency variations.
- LVRT/OVRT events.
- Transitional grid overvoltage that causes a cessation of reactive power.
- Faults that cause an inverter stop.

Once one of the previous events has occurred, the algorithm stores the following data:

- Date.
- Time.
- Event.
- Fault.
- Warning.
- Output inverter voltage.
- Output inverter current.
- Frequency.
- DC Bus voltage.
- DC Bus current.
- Active power.
- Reactive power.

## Subgroup G13.1: Fast logger

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **G13.1.1-Enable fast logger = Enabled** | Disabled<br>Enabled | It allows to enable and disable the algorithm hiding the parameters of this group from the client roles. | Yes |
| **G13.1.2-Activate fast logger = Yes** [22] | No<br>Yes | Allows to activate and deactivate the algorithm. | Yes |
| **G13.1.3-Pre-shoot samples = 15** [22] | 0 to 200 | Allows to configure the number of samples that are stored and sent to the SD card before the selected event occurs. | Yes |
| **G13.1.4-Post-shoot samples = 100** [22] | 0 to 200 | Allows to configure the number of samples that are stored and sent to the SD card after the selected event occurs. | Yes |
| **G13.1.5-Shoot over frequency = 0.50 Hz** [22] | 0.00 to 5.00 Hz | Allows to configure the algorithm to be executed when the frequency reaches a level above the defined. | Yes |
| **G13.1.6-Shoot sub frequency = -0.50 Hz** [22] | -5.00 to 0.00 Hz | Allows to configure the algorithm to be executed when the frequency reaches a level below the defined. | Yes |
| **G13.1.7-Shoot LVRT = Enabled** [22] | Disabled<br>Enabled | Allows to configure the algorithm to be executed when a LVRT (Low Voltage Ride Through) event occur. | Yes |
| **G13.1.8-Shoot OVRT = Enabled** [22] | Disabled<br>Enabled | Allows to configure the algorithm to be executed when a OVRT (Over Voltage Ride Through) event occur. | Yes |
| **G13.1.9-Shoot fault = Enabled** [22] | Disabled<br>Enabled | Allows to configure the algorithm to be executed when a fault occur. | Yes |
| **G13.1.10-Event force = No** [22] | No<br>Yes | Allows to force any kind of event to execute the algorithm. | Yes |
| **G13.1.11-Sample time = 10 ms** [22] | 1 to 100 ms | Allows to define the acquisition time of any sample. | Yes |
| **G13.1.12-Logger mode = Normative** | Normative<br>Debug | This parameter has been implemented to carry on tests on the test bench and simulate different situations to check the algorithm. In field it must be configured as G13.1.12-Logger mode = Normative. | Yes |
| **G13.1.13-Counter events/day = 0**[22] | 0 to 65535 | Allows to visualize how many times the algorithm has been executed during the day. At the end of the day this counter returns to zero. | Yes |

[22] These parameters will only be displayed when the fast logger is enabled (G13.1.1-Enable fast logger = Enabled). For further information please contact Power Electronics.

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>G13.1.14-Sync force = No [22]</td><td>No<br>Yes</td><td>Allows to activate and deactivate the forced time synchronization between the microprocessor and the DSP of the inverters of the plant using the GPS.<br><br>Depending on the plant, the number of synchronized inverters could change.<br><br>Note: This parameter will only be shown if G13.1.16-Clock source = GPS.</td><td>Yes</td></tr>
<tr><td>G13.1.15-Sync status = Out of sync [22]</td><td>Out of sync<br>Syncing<br>Synchronized<br>Error host unreachable</td><td>Allows to check the status of the forced synchronization (G13.1.14-Sync force).<br><br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Out of sync</td><td>The inverter is not synchronized.</td></tr><tr><td>Syncing</td><td>The synchronization process has begun.</td></tr><tr><td>Synchronized</td><td>The inverter is synchronized.</td></tr><tr><td>Error host unreachable</td><td>The synchronization process has failed because the host cannot be reached.</td></tr></table><br>Note: This parameter will only be shown if G13.1.16-Clock source = GPS.</td><td>Yes</td></tr>
<tr><td>G13.1.16-Clock source = GPS [22]</td><td>GPS<br>Internal</td><td>Allows to select the clock source (GPS or internal clock) for the reading of the logs generated by the CAISO algorithm.<br><br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>GPS</td><td>This option allows the use of an external GPS for the synchronization of the inverters.</td></tr><tr><td>Internal</td><td>This option allows the equipment without external GPS to use the CAISO functionality.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G13.1.17-Clear SD logger wrn = No [22]</td><td>No<br>Yes</td><td>Allows clearing the SD warnings log.<br>Note: this parameter is shown if [G13.1.1-Enable fast logger] is enabled.</td><td>Yes</td></tr>
</table>

EN

# COMMISSIONING RECOMMENDATIONS

6

## Communications Scheme

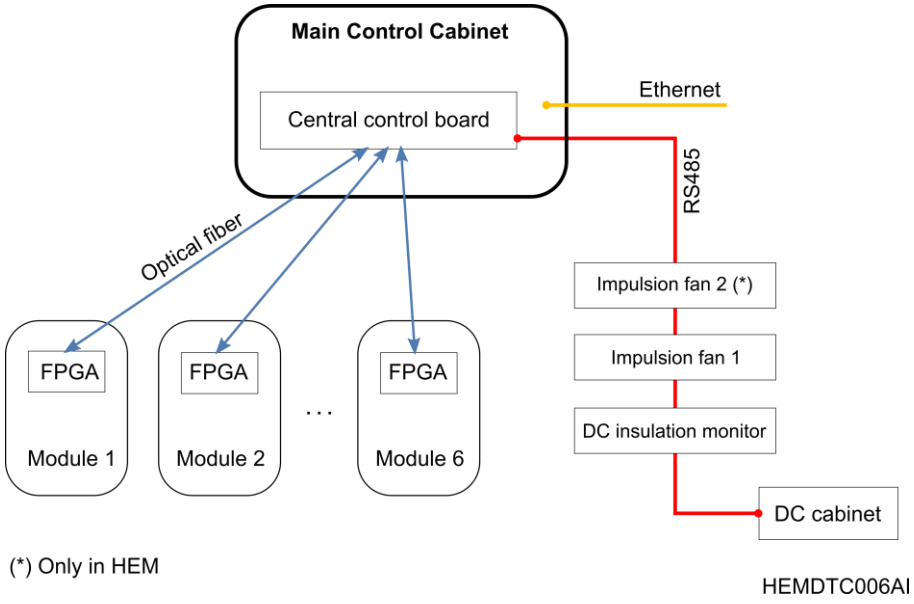
HEM and HEMK inverters have a central control board. It is located in the Main Control Cabinet and contains all the electronics and communication terminals to control the equipment.

The central control board communicates with the FPGA of each module through optical fiber. Besides, it uses an RS485 bus to communicate with the impulsion fans (cyclones), DC insulation monitor and the DC cabinet (or DU).

There is also an Ethernet connection between the central control board and the user local network. It allows stablishing TCP/IP connections between the inverter and a LAN. With a LAN connection we can:

- Communicate any data logger that implements Modbus TCP with the inverter in order to add data for its next computer process.
- Make a system control and monitoring using a Modbus TCP client.
- Receive orders from other Power Electronics peripherals as the Power Plant Controller (PPC).

Inverters' communication scheme is shown below. Please notice that Impulsion fan 2 only exists in HEM inverters, as HEMK inverters have only one impulsion fan.




*Communications scheme*

# Before Giving Auxiliary Power

The system is carefully tested and quality checked before packing and delivering. After receiving and unpacking, please check for possible damages on transportation.

Refer to the *Hardware and Installation Manual* for commissioning instructions. Once all steps have been followed, it is recommended to make a safe start the first time the system is ready to start.

# P2G Card Monitoring Configuration

## SD Card Configuration

Before inserting the micro SD card into the P2G card slot, located in the Main Control Cabinet of the inverter, the P2G card power supply must be disconnected. SD card needs to be formatted before installing it in the inverter's P2G card slot (this step is not needed when the SD card is new).

The following table shows the configuration process for the SD card:

| Parameter | Setting | Description |
|---|---|---|
| SG3.2 / SD card status | OK | Check that SD card is detected and working properly |
| SG3.3 / SD operation | Format SD card | Format SD card whenever it is not brand new |
| SG3.4 / SD apply changes | Yes | Press enter at screen G3.3 |
| SG3.2 / SD card status | Formatted | Shows SD card status |
| SG3.3 / SD operation | Initialize SD card | Choose initialize SD card |
| SG3.4 / SD apply changes | Yes | Press enter at screen G3.3 |
| SG3.2 / SD card status | OK | Shows OK status if SD card is operating normally |
| SG3.1.2 / SD params logger list 1 | 1000 | Grid voltage RS |
| SG3.1.3 / SD params logger list 2 | 1001 | Grid voltage ST |
| SG3.1.4 / SD params logger list 3 | 1002 | Grid voltage TR |
| SG3.1.5 / SD params logger list 4 | 1003 | Grid current U |
| SG3.1.6 / SD params logger list 5 | 1004 | Grid current V |
| SG3.1.7 / SD params logger list 6 | 1005 | Grid current W |
| SG3.1.8 / SD params logger list 7 | 1007 | Grid power P |
| SG3.1.9 / SD params logger list 8 | 1008 | Grid power Q |
| SG3.1.10 / SD params logger list 9 | 1050 | DC bus voltage |
| SG3.1.11 / SD params logger list 10 | 1122 | Current status |
| SG3.1.12 / SD params logger list 11 | - | These parameters can be configured as required. |
| SG3.1.13 / SD params logger list 12 | - | |
| SG3.1.14 / SD params logger list 13 | - | |
| SG3.1.15 / SD params logger list 14 | - | |
| SG3.1.16 / SD params logger list 15 | - | |
| SG3.1.1 / SD params logger enable | Yes | Enables saving Modbus values in the SD card |
| SG3.5 / SD Logger enable | Yes | Enables saving faults in the SD card |

The inverter will store automatically in the SD card the 30 most relevant parameters for monitoring.

The following tables show the structure of the .csv file which can be downloaded from the SD card. The file contains relevant information about the equipment. The order shown in the rows is the same as the one that appears in the columns in the .csv file:

The SD card includes the SW version information of all electronic devices (in the first row of the LOG.csv file)

- uP, DSP, DUs, FPGAs, insulation monitor and MV Board
- The IP address of the inverter

**Log 1:**

<table>
<tr><th>Variable</th><th>Parameter</th><th>Screen</th><th>Address</th></tr>
<tr><td>1 Vdc</td><td>SV2.1</td><td>DC voltage bus</td><td>41051</td></tr>
<tr><td>2 Vpanels</td><td>SV2.1</td><td>DC voltage PV</td><td>41052</td></tr>
<tr><td>3 Idc</td><td>SV2.6</td><td>DC input current</td><td>41062</td></tr>
<tr><td>4 Iac</td><td>SV1.4</td><td>Grid current I1</td><td>41004</td></tr>
<tr><td>5 Iac</td><td>SV1.5</td><td>Grid current I2</td><td>41005</td></tr>
<tr><td>6 Iac</td><td>SV1.6</td><td>Grid current I3</td><td>41006</td></tr>
<tr><td>7 Vac</td><td>SV1.1</td><td>Grid voltage RS</td><td>41001</td></tr>
<tr><td>8 Vac</td><td>SV1.2</td><td>Grid voltage ST</td><td>41002</td></tr>
<tr><td>9 Vac</td><td>SV1.3</td><td>Grid voltage TR</td><td>41003</td></tr>
<tr><td>10 P</td><td>SV1.7</td><td>P</td><td>41008</td></tr>
<tr><td>11 Q</td><td>SV1.8</td><td>Q</td><td>41009</td></tr>
<tr><td>12 Freq</td><td>SV1.11</td><td>Grid frequency</td><td>41012</td></tr>
<tr><td>13 Pref</td><td>G8.1.3</td><td>Depending on the selected control mode, one address or another will be displayed:<br>- MPPt<br>- P<br>- I<br>- Vdc</td><td><table><tr><th>Option</th><th>Address</th></tr><tr><td>MPPt</td><td>N/A</td></tr><tr><td>P</td><td>40553</td></tr><tr><td>I</td><td>40554</td></tr><tr><td>Vdc</td><td>40555</td></tr></table></td></tr>
<tr><td>14 PLim</td><td>SV4.2.1</td><td>P limit inst</td><td>41141</td></tr>
<tr><td>15 Qref</td><td>G8.1.4</td><td>Depending on the selected control mode, one address or another will be displayed:<br>- Cos Fi<br>- I<br>- Q<br>- V/Q</td><td><table><tr><th>Option</th><th>Address</th></tr><tr><td>Cos Fi</td><td>40556</td></tr><tr><td>I</td><td>40557</td></tr><tr><td>Q</td><td>40558</td></tr><tr><td>V/Q</td><td>N/A</td></tr></table></td></tr>
<tr><td>16 QLim</td><td>SV4.2.2</td><td>Q limit inst</td><td>41142</td></tr>
<tr><td>17 Fault</td><td>SV4.1.1</td><td>Current fault</td><td>41121</td></tr>
<tr><td>18 Warning</td><td>SV4.1.2</td><td>Current warning</td><td>41122</td></tr>
<tr><td>19 DSP status</td><td>SV4.1.3</td><td>Current status of the DSP or the inverter.</td><td>41123</td></tr>
<tr><td>20 uP Diagnosis</td><td></td><td>Inverter status diagnostics</td><td><table><tr><th>Nº BIT</th><th>Definition</th></tr><tr><td>0</td><td>Selector</td></tr><tr><td>1</td><td>Start</td></tr><tr><td>2</td><td>Stop</td></tr><tr><td>3</td><td>Dsp_Du_Request</td></tr><tr><td>4</td><td>Up_Du_Request</td></tr><tr><td>5</td><td>Du_ACK</td></tr><tr><td>6</td><td>DU_FB_1_Open_Status</td></tr><tr><td>7</td><td>DU_FB_1_Close_Status</td></tr><tr><td>8</td><td>DU_FB_2_Open_Status</td></tr><tr><td>9</td><td>DU_FB_2_Close_Status</td></tr><tr><td>10</td><td>Reserved</td></tr><tr><td>11</td><td>Reserved</td></tr><tr><td>12</td><td>Reserved</td></tr><tr><td>13</td><td>Reserved</td></tr><tr><td>14</td><td>Reserved</td></tr><tr><td>15</td><td>Reserved</td></tr></table></td></tr>
</table>

**Log 2:**

| Variable | Parameter | Screen | Address |
|---|---|---|---|
| **Modules P** | - | Power of each module.<br>7 addresses are shown, one for each module[1]. | **45008**<br>**45040**<br>**45072**<br>**45104**<br>**45136**<br>**45168**<br>**45200** |
| **Modules Status** | - | Status of each module.<br>7 addresses are shown, one for each module[1]. | **45000**<br>**45032**<br>**45064**<br>**45096**<br>**45128**<br>**45160**<br>**45192** |

**Log 3:**

| Variable | Parameter | Screen | Address |
|---|---|---|---|
| **Temp IGBTs** | - | Maximum IGBT temperature per module.<br>7 addresses are shown, one for each module[1]. | **45012**<br>**45044**<br>**45076**<br>**45108**<br>**45140**<br>**45172**<br>**45204** |
| **Temp PCB** | - | Ambient temperature 6 PCB (printed circuit board) module.<br>7 addresses are shown, one for each module[1]. | **45011**<br>**45043**<br>**45075**<br>**45107**<br>**45139**<br>**45171**<br>**45203** |
| **Internal Tº** | SV4.1.4 | Internal temperature of the central PCB | **41124** |
| **Admisión Tº** | SV4.1.9 | Admittance temperature of the air entering the equipment. | **41129** |
| **Humedad PCB** | | Humidity 6 PCB (printed circuit board) module.<br>7 directions are shown. One for each module[1]. | **45032**<br>**45064**<br>**45096**<br>**45128**<br>**45160**<br>**45192**<br>**45224** |
| **Internal Humidity** | SV4.1.11 | Internal humidity inside the equipment | **41132** |
| **Temperature of MV transformer** | | One record per transformer secondary winding | **41193**<br>**41194**<br>**41195** |

**Custom log:**

This row will only appear if a customized log has been configured in SG3.1.

| Variable | Parameter | Screen | Address |
|---|---|---|---|
| **Var1 register** | SG3.1.2 | Register entered by user (if exists) | **42112** |
| **Var1 content** | - | Content of the register (if the register exists) | **-** |
| **Var2 register** | SG3.1.3 | Register entered by user (if exists) | **42113** |
| **Var2 content** | - | Content of the register (if the register exists) | **-** |
| ... | | | |
| **Var15 register** | SG3.1.16 | Register entered by user (if exists) | **42126** |
| **Var15 content** | - | Content of the register (if the register exists) | **-** |

[1] HEM/K inverters have 6 modules maximum, so the last address will remain empty.

Please notice that if a register does not exist (is not used), columns will be rearranged. See SG3.1 for further information.

It is recommended to store in the custom log some parameters related to temperature. In order:

- 1192, PT100 (1)
- 1193, PT100 (2)
- 1194, PT100 (3)
- 1123, Internal temperature
- 1124, Maximum IGBTs temperature
- 1128, Admission temperature
- 1125, Maximum modules temperature

From HEM_1.4.0 the Blackbox registry is also saved in the SD (according to the configuration adjusted in **SG3.7-Download Blackbox**).

**Blackbox Registry:**

The Blackbox registry is an independent file in which identifies the module, the date and time and when the file was downloaded with the following structure. The real value of each variable is obtained after multiplying the value shown in the file by a factor. This calculation is performed automatically by PowerCommsV.

| IR | IS | IT | VRS | VST | VTR | VBUS_POS | STATUS | VBUS_NEG | DESAT | ADC | ADC | IDC | S/D |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1234 | 1234 | 1234 | 3210 | 3210 | 3210 | 1212 | 7 | 1212 | 16773120 | 0 | 0 | | 1 |
| 1234 | 1234 | 1234 | 3210 | 3210 | 3210 | 1212 | 7 | 1212 | 16773120 | 0 | 0 | | 1 |

The file will structure the information in 7 columns:

- Go: Current in phase R.
- Is: Current in phase S.
- It: Current in phase T.
- Vrs: Voltage between phases R and S.
- Vst: Voltage between phases S and T.
- Vbus_Pos: Voltage at the positive bus.
- Status: FPGA status.
- Vbus_Neg: Negative bus voltage.
- Desat: Registry that stores a saturation protection of the IGTB.
- ADC: 15V supply voltage of the module board.
- ADC: 24V supply voltage of the module board.
- Idc. Module DC input current.
- S/D: Indicates if the modulation of the FPGA is single (S) or double (D). 0 indicates it is single and 1 indicates it is double.

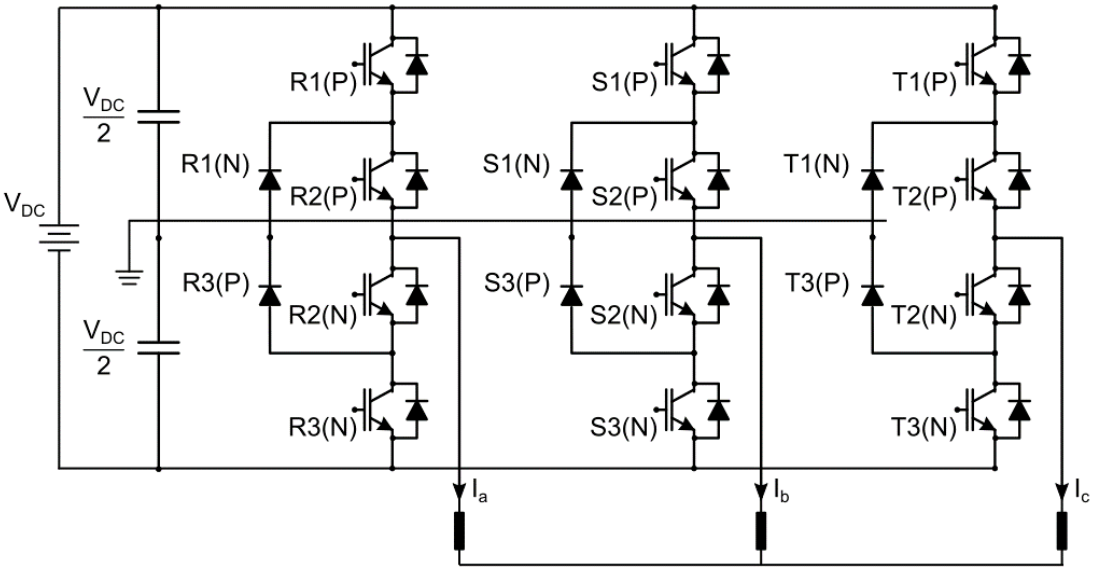
In detail, the DESAT parameter (18 bits) is saved as an integer, and must be set to binary, each bit means: {DESAT1H_R, DESAT2H_R, DESAT2L_R, DESAT3L_R, DESAT1H_S, DESAT2H_S, DESAT2L_S, DESAT3L_S, DESAT1H_T, DESAT2H_T, DESAT2L_T, DESAT3L_T, DESAT1L_R, DESAT3H_R, DESAT1L_S, DESAT3H_S, DESAT1L_T, DESAT3H_T};

**Note**: Example for the Branch phase R

DSAT1H_R=R1(P)-F71/W71, DESAT1L_R=R1(N) F119/W119

DSAT2H_R=R2(P)-F72/W72, DESAT2L_R=R2(N) F73/W73

DSAT3H_R=R3(P)-F74/W74, DESAT3L_R=R3(N) F120/W




HEMDTP002A

**Fast logger:**

This register will only be stored if the parameter G13.1.1-Enable fast logger is configured as Enabled. The information is stored as shown below:

| Variable | Parameter | Screen | Address |
|---|---|---|---|
| **1 Freq** | SV1.11 | Grid frequency | **41012** |
| **2 Fault** | SV4.1.1 | Current fault | **41121** |
| **3 Warning** | SV4.1.2 | Current warning | **41122** |
| **4 P** | SV1.7 | P | **41008** |
| **5 Q** | SV1.8 | Q | **41009** |
| **6 Ir** | SV1.4 | Grid current I1 | **41004** |
| **7 Is** | SV1.5 | Grid current I2 | **41005** |
| **8 It** | SV1.6 | Grid current I3 | **41006** |
| **9 Vrs** | SV1.1 | Grid voltage RS | **41001** |
| **10 Vst** | SV1.2 | Grid voltage ST | **41002** |
| **11 Vrt** | SV1.3 | Grid voltage TR | **41003** |
| **12 Vdc** | SV2.1 | DC voltage bus | **41051** |
| **13 Idc** | SV2.6 | DC input current | **41062** |

## FileZilla Client Installation

After installing and executing FileZilla client in the computer, the following fields need to be filled to stablish a connection:

- Server: (Inverter IP address)
- Username: sat
- Password: sat
- Port: 21

We can then download the .csv file, which contains the information registered from the P2G memory (approx. 147 lines per minute). It is possible to register all faults, warnings and updates of the inverter, since the day the mini SD card was installed up to 333 days approximately (4000Mb / 12Mb per day = 333 days) for a 4Gb SD card.

The .csv file will show inverter faults highlighted in red, warnings in yellow and log parameters in green.

For each one of them, the file shows, from left to right:

- MAC address of the P2G.
- Date of the data saving.
- Hour of the data saving.
- MSG whenever the log shows a message or log parameter.
- EEP whenever a value is modified in EEPROM. It will show the modified index and value.

**Note:** When the connection is made by Filezilla to download the SD logs, the uP stops saving data in the logs.

# Out of Service Module

If a module is damaged and it is not possible to repair it immediately, we recommend leaving it out of service. The system will disregard it. This can be done by disconnecting the auxiliary circuit breaker. (This can be done remotely by acting on the parameter SG1.5)

Ask Power Electronics' technical department for more information.

## NOTICE

Whenever a module of the inverter must be disabled due to a fault or malfunctioning, consult Power Electronics for the safety sequence for replacement. The power module must always be isolated from the other modules.

# MODBUS COMMUNICATION

7

EN

## Supported Modbus Function Codes

Serial communications protocol provided by FREESUN inverters adheres to Modicon's industrial standard communications protocol, Modbus. The inverter uses the following two reading and writing functions from the Modbus protocol:

| Function | Description | Registers Number |
|---|---|---|
| 3 | Registers Reading | 120 |
| 16 | Registers Writing | 120 |

The implementation of this function code in the inverter allows reading up to 120 registers into a Parameters Group in a frame. If you want to access to a consecutive memory register, but belonging to different groups, you should access in so many frames as groups are involved.

## Modbus Function Code Nº 3: Registers Reading

This function code allows the Modbus controller (master) reading the content of the data registers indicated in the inverter (slave). This function code only admits unicast addressing. Broadcast or groupcast addressing are not possible with this function code.

The implementation of this function code in the inverter allows reading up to 120 registers with consecutive addresses of the inverter in a single frame.

In the next example, a frame is shown where the master tries to read the minimum DC voltage via communications. The information that should be attached in the request frame is the following one:

- Data address of the inverter.
- Modbus function code (3 Registers reading).
- Starting Data address.
- Number of registers to read.
- CRC-16 code.

The answer of the inverter (slave) should contain the following fields:

- Data address of the slave.
- Modbus function code (3 Registers reading).
- Bytes number for reading.
- Bytes number / 2 registers.
- CRC-16 code.

Each register consists of 2 bytes (2x8bits=16 bits). This is the default registry length of the inverter.

### Operation Example of Modbus Function Code Nº3 (Registers Reading)

Suppose we want to read via communications the minimum DC voltage to start the inverter. This data corresponds to parameter [G2.1.1 Vdc low]. The frame that should be transmitted is:

| Modbus Address | Modbus Function Code | Starting Data Address (40282) | Registers Number | CRC-16 |
|---|---|---|---|---|
| 0x0A | 0x03 | 0x00A0 | 0x0001 | 0x8553 |

Suppose that minimum DC Voltage of the equipment is 1050V (Modbus value 1050 in decimal is equivalent to 0x41A in hexadecimal). The answer of the slave would be:

| Modbus Address | Modbus Function Code | Bytes Number | Data (address 20) | CRC-16 |
|---|---|---|---|---|
| 0x0A | 0x03 | 0x02 | 0x41A | 0x9D84 |

## Modbus Function Code Nº 16: Registers Writing

This function code allows the Modbus controller (master) to write the content of the data registers indicated in the inverter (slave), as long as those registers are not “Read only”. Registers writing by the master does not impede the later modification of those registers by the slave.

The implementation of this function code in the inverter allows writing up to 5 registers of the inverter in a single frame.

Up next, a frame is shown where the master tries to write the content of a register. The information that should be attached in the request frame is:

- Data address of the slave.
- Modbus function code (16 Registers writing).
- Starting data address.
- Number of registers to write.
- Bytes number for writing.
- Content of registers for writing.
- CRC-16 code.

The answer of the slave includes:

- Data address of the slave.
- Modbus function code (16 Registers writing).
- Starting data address.
- Written registers number.
- CRC-16 code.

# Addressing Modes

## Broadcast Addressing Mode

Broadcast addressing mode allows the master to access at the same time to all the slaves connected to the Modbus network. The Modbus function code that admits this global addressing mode is:

| Function | Description |
|---|---|
| 16 | Registers Writing |

To access to all the equipment connected in a Modbus network, use the address 0.

When this address is used, all the slaves in the Modbus network execute the required task but they do not prepare any answer.

Broadcast addressing mode can be used during standard inverter operation, as well as if night mode is active.

EN

# Remote Control Functions

| **START** | |
|---|---|
| Screen | - |
| Range | 0 – 1 |
| Modbus address | **43010** |
| Modbus range | 0 to 1 |
| Read / Write | Yes |
| Description | Manual start: Send the start command to the inverter through the communications network. |

| **STOP** | |
|---|---|
| Screen | - |
| Range | 0 – 1 |
| Modbus address | **43011** |
| Modbus range | 0 to 1 |
| Read / Write | Yes |
| Description | Send the stop command to the inverter through the communications network. |

| **RESET** | |
|---|---|
| Screen | - |
| Range | 0 – 1 |
| Modbus address | **43012** |
| Modbus range | 0 to 1 |
| Read / Write | Yes |
| Description | Send a reset command to the inverter through the communications network. |

| **EMERGENCY STOP** | |
|---|---|
| Screen | - |
| Range | 0 – 1 |
| Modbus address | **43200** |
| Modbus range | 0 to 1 |
| Read / Write | Yes |
| Description | Send an emergency stop command to the inverter through the communications network. |

# Disconnecting Unit Visualization Parameters

The display can only show information about one channel at a time, as mentioned before. To visualize information from multiple channels, please refer to the following table:

<table>
<tr><th>DU channel</th><th>Description</th><th>Addresses</th></tr>
<tr><td>Channel 1</td><td>Channel 1 parameters<br>Each channel has six parameters [1]:</td><td>45801 to 45806</td></tr>
<tr><td></td><td colspan="2">
<table>
<tr><th>Parameter</th><th>Units</th><th>Address</th><th>Function</th><th>Modbus conversion</th></tr>
<tr><td>Channel version (MSB)</td><td>NONE</td><td>45801</td><td rowspan="2">The value of these parameters is shown as a 32-bits DWord, two 16-bits words (MSB and LSB).<br><br>The high byte (MSB) corresponds with the lowest Modbus address (45801) and the low byte (LSB) corresponds with the highest Modbus address (45802). Thus, to correctly read the value, perform the calculation:<br>$V = MSB * 2^{16} + LSB$</td><td>Real value = Modbus value</td></tr>
<tr><td>Channel version (LSB)</td><td>NONE</td><td>45802</td><td>Real value = Modbus value</td></tr>
<tr><td>Channel calibration status</td><td>NONE</td><td>45803</td><td>8-bits register for current (bits 0 to 3) and voltage (bits 4 to 7) minimum and maximum calibration values.
<table>
<tr><th>Bit</th><th>Status</th><th>Bit</th><th>Status</th><th>If bit = 1</th></tr>
<tr><td>0</td><td>I:No</td><td>4</td><td>V:No</td><td>No limits have been set</td></tr>
<tr><td>1</td><td>I:Min</td><td>5</td><td>V:Min</td><td rowspan="2">Only min or max is set</td></tr>
<tr><td>2</td><td>I:Max</td><td>6</td><td>V:Max</td></tr>
<tr><td>3</td><td>I:Ok</td><td>7</td><td>V:Ok</td><td>Min and max values are set</td></tr>
</table></td><td>Real value = Modbus value</td></tr>
<tr><td>Status</td><td>NONE</td><td>45804</td><td>
<table>
<tr><th>Bit</th><th>Status</th><th>Bit</th><th>Status</th></tr>
<tr><td>0</td><td>Online</td><td>6</td><td>Comms error</td></tr>
<tr><td>1</td><td>Contactor</td><td>7</td><td>Negative pole</td></tr>
<tr><td>2</td><td>Contactor feedback</td><td>8</td><td>Negative pole online</td></tr>
<tr><td>3</td><td>Current present</td><td>9</td><td>Channel isolated</td></tr>
<tr><td>4</td><td>Contactor locked opening</td><td>10</td><td>Channel miswired</td></tr>
<tr><td>5</td><td>Contactor locked closing</td><td></td><td></td></tr>
</table></td><td>Real value = Modbus value</td></tr>
<tr><td>Current measure</td><td>dA</td><td>45805</td><td>Channel 1 current expressed in deciamperes (dA).</td><td>Real value = Modbus value</td></tr>
<tr><td>Voltage measure</td><td>dV</td><td>45806</td><td>Channel 1 voltage expressed in decivolts (dV).</td><td>Real value = Modbus value</td></tr>
</table>
</td></tr>
</table>

[1] Channels 2 to 40 follow the same structure; the user simply must enter the Modbus address that corresponds with each parameter. The address range for channels 2 to 40 is:

| DU channel | Parameter addresses | DU channel | Parameter addresses |
|---|---|---|---|
| Channel 2 | 45807 to 45812 | Channel 22 | 45927 to 45932 |
| Channel 3 | 45813 to 45818 | Channel 23 | 45933 to 45938 |
| Channel 4 | 45819 to 45824 | Channel 24 | 45939 to 45944 |
| Channel 5 | 45825 to 45830 | Channel 25 | 45945 to 45950 |
| Channel 6 | 45831 to 45836 | Channel 26 | 45951 to 45956 |
| Channel 7 | 45837 to 45842 | Channel 27 | 45957 to 45962 |
| Channel 8 | 45843 to 45848 | Channel 28 | 45963 to 45968 |
| Channel 9 | 45849 to 45854 | Channel 29 | 45969 to 45974 |
| Channel 10 | 45855 to 45860 | Channel 30 | 45975 to 45980 |
| Channel 11 | 45861 to 45866 | Channel 31 | 45981 to 45986 |
| Channel 12 | 45867 to 45872 | Channel 32 | 45987 to 45992 |
| Channel 13 | 45873 to 45878 | Channel 33 | 45993 to 45998 |
| Channel 14 | 45879 to 45884 | Channel 34 | 45999 to 46004 |
| Channel 15 | 45885 to 45890 | Channel 35 | 46005 to 46010 |
| Channel 16 | 45891 to 45896 | Channel 36 | 46011 to 46016 |
| Channel 17 | 45897 to 45902 | Channel 37 | 46017 to 46022 |
| Channel 18 | 45903 to 45908 | Channel 38 | 46023 to 46028 |
| Channel 19 | 45909 to 45914 | Channel 39 | 46029 to 46034 |
| Channel 20 | 45915 to 45920 | Channel 40 | 46035 to 46040 |
| Channel 21 | 45921 to 45926 | | |

# Module Visualization Parameters

The display can only show information about one module at a time, as mentioned before. To visualize information from multiple modules, please refer to the following table:

<table>
<tr><th>Module number</th><th>Description</th><th>Addresses</th></tr>
<tr><td>Module 1</td><td>Module 1 parameters<br>Each module has 24 parameters (see SV9.2 for further details) [2]:<br><br>
<table>
<tr><th>Parameter</th><th>Units</th><th>Address</th><th>Modbus conversion</th></tr>
<tr><td>Module status</td><td>NONE</td><td>45001</td><td>Real value = Modbus value</td></tr>
<tr><td>Current R</td><td>A</td><td>45002</td><td>Real value = (Modbus value / 10)</td></tr>
<tr><td>Current S</td><td>A</td><td>45003</td><td>Real value = (Modbus value / 10)</td></tr>
<tr><td>Current T</td><td>A</td><td>45004</td><td>Real value = (Modbus value / 10)</td></tr>
<tr><td>DC current</td><td>A</td><td>45005</td><td>Real value = (Modbus value / 10)</td></tr>
<tr><td>DC voltage (P)</td><td>V</td><td>45006</td><td>Real value = Modbus value</td></tr>
<tr><td>DC voltage (N)</td><td>V</td><td>45007</td><td>Real value = Modbus value</td></tr>
<tr><td>DC voltage</td><td>V</td><td>45008</td><td>Real value = Modbus value</td></tr>
<tr><td>P</td><td>kW</td><td>45009</td><td>Real value = (Modbus value / 10)</td></tr>
<tr><td>Q</td><td>kVAr</td><td>45010</td><td>Real value = (Modbus value / 10)</td></tr>
<tr><td>Ambient temperature<br>Module PCB Temperature</td><td>ºC</td><td>45011</td><td>Real value = Modbus value</td></tr>
<tr><td>Max temperature<br>Maximum Module IGBT Temperature</td><td>ºC</td><td>45012</td><td>Real value = Modbus value</td></tr>
<tr><td>I/O status</td><td>NONE</td><td>45013</td><td>Real value = Modbus value</td></tr>
<tr><td>Hw version</td><td>NONE</td><td>45014</td><td>Real value = Modbus value</td></tr>
<tr><td>Sw version</td><td>NONE</td><td>45015 – MSB<br>45061 – LSB</td><td>Real value = Modbus value</td></tr>
<tr><td>Temperature R1</td><td>ºC</td><td>45023</td><td>Real value = Modbus value</td></tr>
<tr><td>Temperature R2</td><td>ºC</td><td>45024</td><td>Real value = Modbus value</td></tr>
<tr><td>Temperature R3</td><td>ºC</td><td>45025</td><td>Real value = Modbus value</td></tr>
<tr><td>Temperature S1</td><td>ºC</td><td>45026</td><td>Real value = Modbus value</td></tr>
<tr><td>Temperature S2</td><td>ºC</td><td>45027</td><td>Real value = Modbus value</td></tr>
<tr><td>Temperature S3</td><td>ºC</td><td>45028</td><td>Real value = Modbus value</td></tr>
<tr><td>Temperature T1</td><td>ºC</td><td>45029</td><td>Real value = Modbus value</td></tr>
<tr><td>Temperature T2</td><td>ºC</td><td>45030</td><td>Real value = Modbus value</td></tr>
<tr><td>Temperature T3</td><td>ºC</td><td>45031</td><td>Real value = Modbus value</td></tr>
</table>
All possible module status messages are described in section 3.</td><td>45001 to 45032</td></tr>
</table>

[2] Modules 2 to 10 follow the same structure; the user simply must enter the Modbus address that corresponds with each parameter. The address range for modules 2 to 10 is:

| DU Module | Parameter addresses |
|---|---|
| Module 2 | 45033 to 45064 |
| Module 3 | 45065 to 45096 |
| Module 4 | 45097 to 45128 |
| Module 5 | 45129 to 45160 |
| Module 6 | 45161 to 45192 |

**Note:** To read the value of those parameters which are represented by more than one register (e.g. Sw version), please notice that the high byte (MSB) corresponds with the lowest Modbus address and the low byte (LSB) corresponds with the highest Modbus address.

Thus, to correctly read the value, perform the calculation:

$$V = MSB * 2^{16} + LSB$$

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# Energy Visualization Parameters

The display can only show information about one module at a time, as mentioned before. To visualize information from multiple modules, please refer to the following table:

<table>
<tr><th>Module number</th><th>Description</th><th>Addresses</th></tr>
<tr><td>Module 1</td><td>Module 1 active and reactive energy parameters.<br><br>Each module has 10 parameters, five related to active energy and five to reactive energy. See SV5.1 and SV5.2 for further details [3]:<br><br>
<table>
<tr><th>Parameter</th><th>Units</th><th>Address</th><th>Modbus conversion</th></tr>
<tr><td>Daily energy [*]</td><td rowspan="10">kWh</td><td>43521 – MSB<br>43522 – LSB</td><td>Real value = Modbus value</td></tr>
<tr><td>Yesterday energy [*]</td><td>43523 – MSB<br>43524 – LSB</td><td>Real value = Modbus value</td></tr>
<tr><td>Current month energy [*]</td><td>43525 – MSB<br>43526 – LSB</td><td>Real value = Modbus value</td></tr>
<tr><td>Last month energy [*]</td><td>43527 – MSB<br>43528 – LSB</td><td>Real value = Modbus value</td></tr>
<tr><td>Total energy [*]</td><td>43529 – MSB<br>43530 – LSB</td><td>Real value = Modbus value</td></tr>
<tr><td>Daily reactive energy [*]</td><td>43531 – MSB<br>43532 – LSB</td><td>Real value = Modbus value</td></tr>
<tr><td>Yesterday reactive energy [*]</td><td>43533 – MSB<br>43534 – LSB</td><td>Real value = Modbus value</td></tr>
<tr><td>Current month reactive energy [*]</td><td>43535 – MSB<br>43536 – LSB</td><td>Real value = Modbus value</td></tr>
<tr><td>Last month reactive energy [*]</td><td>43537 – MSB<br>43538 – LSB</td><td>Real value = Modbus value</td></tr>
<tr><td>Total reactive energy [*]</td><td>43539 – MSB<br>43540 – LSB</td><td>Real value = Modbus value</td></tr>
</table>
[*] The value of these parameters is shown as a 32-bits DWord, two 16-bits words (MSB and LSB).<br><br>The high byte (MSB) corresponds with the lowest Modbus address and the low byte (LSB) corresponds with the highest Modbus address.<br><br>Thus, to correctly read the value, perform the calculation:<br><br>$V = MSB * 2^{16} + LSB$<br><br>Example: Daily energy = [43521 Modbus value] * $2^{16}$ + [43522 Modbus value]</td><td>43521 to 43540</td></tr>
</table>

[3] Modules 2 to 10 follow the same structure; the user simply must enter the Modbus address that corresponds with each parameter. The address range for modules 2 to 10 is:

| DU Module | Parameter addresses |
|---|---|
| Module 2 | 43541 to 43560 |
| Module 3 | 43561 to 43580 |
| Module 4 | 43581 to 43600 |
| Module 5 | 43601 to 43620 |
| Module 6 | 43621 to 43640 |
| Module 7 | 43641 to 43660 |
| Module 8 | 43661 to 43680 |
| Module 9 | 43681 to 43700 |
| Module 10 | 43701 to 43720 |

It is also possible to visualize the total active and reactive energy produced by all modules. For this, follow the same parameter structure as for individual modules, but using the addresses 43501 to 43520.

## PCS Visualization Parameters

**Note:** This group is only shown for inverters for batteries (PCS).

To visualize information from the battery, please refer to the following table:

| Parameter | Units | Address | Function | Modbus Conversion |
|---|---|---|---|---|
| P command | % | **44202** | Power command in the PECS. When it is positive, the battery is discharging and when it is negative, the battery is charging. | Real value = (Modbus value / 10000) |
| Q command | % | **44203** | Reactive power command in the PECS. When it is positive, the battery is discharging and when it is negative, the battery is charging. | Real value = (Modbus value / 10000) |
| Max discharge current continuous | % | **44227** | DC current limit for the battery discharge. The converter cannot request a higher DC current. Positive value. | Real value = (Modbus value / 10000) |
| Max charge current continuous | % | **44228** | DC current limit for the battery charge. The converter cannot provide a higher DC current to the battery. Negative value. | Real value = (Modbus value / 10000) |
| PPC heartbeat | - | **44252** | This parameter is a 0 to 65535 counter that allows verifying communication between the converter and the Scada system. PCS reverts commands to 0 if the heartbeat doesn't change within an adjustable time in seconds set in the following parameter. | Real value = Modbus value |
| PPC heartbeat time clear refs | s | **44253** | It is the time that the heartbeat count needs to be frozen before the converter sets the P and Q references to 0. | Real value = Modbus value |
| PPC heartbeat time comms lost | s | **44254** | The PCS will wait a second timeout to trip the inverter on "communications lost", if it is set to 0 seconds, it will set P and Q to 0 and trip immediately. | Real value = Modbus value |

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## Customizable Visualization Parameters

Registers from 44501 to 44620 (4500 to 4619) are available in Modbus for Scada or Modscan (not in display), to be configured to show specific information according to the user needs and for each particular application. These registers can be easily accessed through the Scada system.

To be able to read and write these registers, first the password (1710) needs to be entered in SG1.1.12 / Custom Modbus key (Modbus address 42015).

If password is set, registers from 4500 to 4619 will show the Modbus address of the registers pointed. The addresses can be overwritten and will be saved in EEPROM.

If password is not set, registers from 4500 to 4619 will show the current value of the variable that is actually being pointed. If a write request is performed this will apply to the variable pointed as if its real Modbus address was being written directly.

**Note:** Variables included in these registers, will keep their usual restricted access.

# SERVICE GROUP PARAMETERS

8

## Group SG1: Service

### Subgroup SG1.1: uP service

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG1.1.1-New modbus map = No** | No<br>Yes | Set a new Modbus map. | Yes |
| **SG1.1.3-Init factory def. = 0** | 0 to 65535 | Set all parameters to factory values.<br><br>Init Factory: This should be avoided unless exceptional circumstances occur.<br><br>This option deletes all data, including calibration values of currents, voltages.<br><br>It sets all parameters to their default value, except the MAC address.<br><br>If the new SW version changes the measurement system, it will be necessary to restore the calibrations and calibrate afterwards.<br><br>If it does not change, avoid this step if the equipment is not to be calibrated. | No |
| **SG1.1.4-Init service group = No** | No<br>Yes | Initialize service groups. | No |
| **SG1.1.5-NTP IP address = 98.175.203.200** | 0 to 255 | NTP, Network Time Protocol is a protocol for clock synchronization between different equipment. This parameter allows defining the IP address of the NTP server that will be used for time synchronization.<br><br>IP address format is: A.B.C.D.<br><br>To adjust this address, introduce a value in each of the four parameters that compose it. | Yes |
| **SG1.1.5b-NTP time request = 5 min** | 1 to 30 min | Allows to configure a delay for a periodic synchronization through GPS. | Yes |
| **SG1.1.6-System reboot = 0** | 0 to 65535 | Reboot the system once a password is entered. Similar to turning off and on the control. | No |
| **SG1.1.7-Enable IO board = Yes** | No<br>Yes | Enable or disable the I/O board. | Yes |
| **SG1.1.8-Qualif. customer key = 12127** | 0 to 65535 | Set the password to enable “Qualified customer” mode. | Yes |
| **SG1.1.9-GFDI fuse input = Yes** | No<br>Yes | Use Digital Input 4 from the control board for the GFDI fuse.<br>**Note**: This parameter is not shown in STATCOM. | Yes |
| **SG1.1.10-TCP logger = No** | No<br>Yes | This parameter is reserved for future use. | Yes |

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>SG1.1.11-Selector config = Key</td><td>Key<br>Remote<br>Key + remote</td><td>Allows configuring when the bus charge should start (connection to Disconnection Unit), depending on the selector status.<br><br>There are two inputs that can be used as a selector: The physical key (in the AC cabinet door) or a virtual selector that is configured in SG1.1.11 according to the following options.<br><br><table><tr><th>Function</th><th>Description</th></tr><tr><td>Key</td><td>It will work according to the position of the physical key. The selector key must be placed and in ON position. The bus will charge as soon as charge conditions are detected.</td></tr><tr><td>Remote</td><td>Regardless the position of the key, bus will charge if will instead function according to the setting of SG1.4.2. [SG1.4.2-Selector status]= On.</td></tr><tr><td>Key + remote</td><td>It is a combination of the previous options.<br>The selector key must be placed and in ON position. Besides, [SG1.4.2-Selector status] must be “On”. Then, the bus charge Will start.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG1.1.12-Custom modbus key = 0</td><td>0 to 65535</td><td>Password to edit the customized Modbus map.</td><td>Yes</td></tr>
<tr><td>SG1.1.13-Ext fault/OVGR in. = Off</td><td>Off<br>Normally Closed<br>Normally Open<br>Positive Pulse<br>Negative Pulse</td><td>Digital Input 6 of the control board can be used either as external fault input or to connect an OVGR relay.It must be activated in relation to the OVGR used in the project.<br><br>Select the option that applies to the device connected at J26C:6, this will determine when F55 “External fault” should be triggered.</td><td>Yes</td></tr>
<tr><td>SG1.1.14-IMI input = Central</td><td>DU<br>Central</td><td>This parameter is used to select where the digital signal of the insulation monitor is connected.<br><br><table><tr><th>Function</th><th>Description</th></tr><tr><td>DU</td><td>The insulation monitor digital output is connected to the DU microcontroller.</td></tr><tr><td>Central</td><td>The insulation monitor digital output is connected to the central microcontroller.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG1.1.15-Keep alive status = Normal</td><td>Disabled<br>Normal<br>Special</td><td>V1500 inverters have a redundant control system; two control boards are installed and, if one fails, the controller will enable the other.<br><br>The control board sends a “keep alive” status to the controller to show it is operating correctly. If the controller stops receiving the “keep alive” status, it will cut power supply and turn on the redundant control board.<br><br>This parameter defines the time lapse that the controller can wait before enabling the redundant device:<br><br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Keep alive status is not sent. This option should only be used in inverters without control redundany.</td></tr><tr><td>Normal</td><td>The control board sends the “keep alive” status every 15 ms.</td></tr><tr><td>Special</td><td>A longer time lapse is defined (adjusted internally). This option is useful when a longer task is going to take place (e.g. remote upgrade), the control board lets the controller know it must maintain the power supply on during this time.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG1.1.17-Humidity/Temp sensor = Yes</td><td>No<br>Yes</td><td>This parameter allows enabling / disabling humidity and temperature protections for the electronic board and the active heating.<br><br>If SG1.1.17 = Yes, protections will be enabled.<br>To disable protections, set SG1.1.17 = No.</td><td>Yes</td></tr>
<tr><td>SG1.1.19-OverT LCL box error = Warning</td><td>Warning<br>Fault<br>Disabled</td><td>Allows to configure the response of the equipment when an overtemperature is detected inside the LCL filter cabinet.<br><br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Warning</td><td>W163:Exc temp LCL box is triggered.</td></tr><tr><td>Fault</td><td>F159:Exc temp LCL box is triggered.</td></tr><tr><td>Disabled</td><td>Does not trigger a failure nor a warning.</td></tr></table></td><td>Yes</td></tr>
</table>

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| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG1.1.20-P lim fans fault = No** | No<br>Yes | This parameter allows to reflect or not in [SV4.2.1-P limit inst] the power limitation that is applied to the product whenever any of the ventilation fans fails.<br><br>**OPTION** / **DESCRIPTION**<br>No: SV4.2.1 will not take into account the power limitation no tendrá en cuenta la limitación de potencia imposed by the fans in case any of them fails.<br>Yes: SV4.2.1 will show the most restrictive available power between the one imposed by the fans (set in [G11.2.17-Fans Power Reduction Min]) and the one from other deratings or limits. | Yes |
| **SG1.1.21-F/W bit display = No** | No<br>Yes | Enables or disables the visualization of the faults and warnings bits. | Yes |
| **SG1.1.22-Cause derat. monitor = No** | No<br>Yes | This parameter enables or disables the possibility of triggering [W37:Derating frequency]. | Yes |
| **SG1.1.23-Smoke detector = No** | No<br>Yes | Allows to enable (and show) or disable (and hide) the smoke detector [G5.10.11-Smoke detector DU] in those products which include it. | Yes |

## Subgroup SG1.2: DSP service

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG1.2.1-Bus discharge = Yes** | No<br>Yes | Enable or disable the manual bus discharge. | Yes |
| **SG1.2.3-Open night dc cont = No** | No<br>Yes | Open DC contactor at night and close back in the morning. | Yes |
| **SG1.2.4-Modules retries = 10** | 0 to 20 | Sets the number of attempts to restart the module. Then the DSP launches W116 for 30 min, then restarts autoreset sequences indefinitely.<br>Set the number of attempts to restart the module after any fault.<br>If the module fails to restart SG1.2.4 times, the DSP will not attempt to restart it until the next day.<br>This parameter also sets the number of retries for the modules to activate the AC contactors before the fault F88 is triggered. | Yes |
| **SG1.2.5-DU open Vmax = 10 %** | 0 to 60 | Set the maximum voltage to open the Disconnecting Unit.<br>**Note:** Do not modify this parameter without authorization from the R&D department. | Yes |
| **SG1.2.6-Eficiency = 98.0 %** | 90.0 to 100.0 % | Defines the accuracy of the DC limit per module. | Yes |
| **SG1.2.8-Mod. Start fault = 6** | 1 to 7 | Number of modules that must fail on start before triggering a fault F27 and stopping the inverter.<br><br>**Note:** Default and maximum values of this parameter are set in Modbus addresses 43386 and 43387, respectively. By default, its default value is 6 and maximum value 7. | Yes |
| **SG1.2.11-ROCOF fine (Hz/s) = 0.5** | 0.0 to 2.0 | Fine adjustment of the anti-island algorithm.<br><br>**Note**: This parameter is only visible for users with role “Developer” and for anti-islanding mode “Passive”. | Yes |
| **SG1.2.12-ROCOF delay fine = 0.000 s** | 0.000 to 1.000 s | Delay between ROCOF anti-island detection and fault trigger.<br><br>**Note**: This parameter is only visible for users with role “Developer” and for anti-islanding mode “Passive”. | Yes |
| **SG1.2.15-IGBTs blocking time = 50** [1] | 0 to 1000 | This parameter is used to define a delay, in miliseconds, between restarts in case of a voltage event. The number of restarts is defined in [SG1.3.2-VRT Retries]. | Yes |
| **SG1.2.16-Vac feed = No** [1] | No<br>Yes | Allows defining who determines the module output voltage.<br><br>If this parameter is set to YES, the DSP tells the module the voltage that must be injected. Otherwise, if set to NO the module regulator will determine the voltage to be injected. | Yes |

[1] Parameters related to LVRT and OVRT algorithms.

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>SG1.2.21-Filter Vdc = 25.0 Hz</td><td>1.0 to 150.0 Hz</td><td>To visualize the DC measure, a filter is used. This parameter sets the cut frequency of the filter.</td><td>Yes</td></tr>
<tr><td>SG1.2.22-MV board enable = Yes</td><td>No<br>Yes</td><td>Enables triggering F30: MV comms when there is no medium voltage board.<br><br>Note: This parameter is used for test benches without MV board and we want to avoid being bothered by the fault. Do not modify this parameter without express approval from R&D department.</td><td>Yes</td></tr>
<tr><td>SG1.2.23-Night power = 10.0 %</td><td>10.0 to 100.0 %</td><td>This parameter allows modifying the power limit at night. Whenever the inverter is used as DC source or during testing at the laboratory, it can be modified to provide full equipment power.<br><br>Note: Do not modify this parameter without express approval from R&D department.</td><td>Yes</td></tr>
<tr><td>SG1.2.24-En. Undue LVRT = Yes</td><td>No<br>Yes</td><td>Enables triggering F33:Unable Reconnect when the algorithm for fast reconnections is running.<br><br>Note: This parameter is used when the algorithm does not work properly on field. Do not modify this parameter without express approval from R&D department.</td><td>Yes</td></tr>
<tr><td>SG1.2.25-Protect V trigger = Worst</td><td>Worst<br>All</td><td>Allows defining when voltage protections must be enabled. It applies both to the preexisting protections and the interpolation protections.<br><br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Worst</td><td>Whenever voltage in any phase (RS, ST or TR) overcomes the threshold, the protection will be triggered:<br>- High voltage faults are checked against the maximum voltage.<br>- Low voltage faults are checked against the minimum voltage.</td></tr><tr><td>All</td><td>The protection will be triggered when voltage in all phases (RS, ST and TR) has overcome the threshold:<br>- High voltage faults are checked against the minimum voltage.<br>- Low voltage faults are checked against the maximum voltage.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG1.2.26-Protect V location = Inverter</td><td>Inverter<br>POI<br>Both</td><td>Allows determining where the measure for voltage protections (overvoltages and low voltages) will be taken in the medium voltage side.<br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Inverter</td><td>Voltages are measured at the inverter terminals.</td></tr><tr><td>POI</td><td>Voltages are estimated/calculated from inverter voltage, current and line impedance.</td></tr><tr><td>Both</td><td>Any voltage that overcomes the limits will trigger the protections</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG1.2.27-Autonight closure setting DU = 0.60 %</td><td>0.00 to 10.00 %</td><td>This parameter allows making a fine adjustment of the DC voltage level at which the DC cabinet closes DURING AUTONIGHT for transition to day, this means, the DC cabinet will close with a voltage level: Vbus = (Vpv - SG1.2.27- Autonight closure setting DU).<br><br>This way, fault [F194:DU Neg current exceeded] is avoided, sinde the PV consumption is forced when Vpv is adjusted.<br><br>Note: this parameter is only shown if [G4.7.1-Enable] is set to “Autonight”</td><td>Yes</td></tr>
<tr><td>SG1.2.28-Feedforward Idc = No</td><td>No<br>Yes</td><td>Allows enabling the feed forward I_DC algorithm. This algorithm helps controlling the bus voltage in case of disturbances in the DC voltage.</td><td>Yes</td></tr>
<tr><td>SG1.2.29-Idc filter = 10 Hz</td><td>1 to 5000 Hz</td><td>Sets the cut-off frequency of the Idc measurement filter.<br><br>Note: this parameter is only shown if [SG1.2.28-Feedforward Idc] is set to “Yes”.</td><td>Yes</td></tr>
</table>

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## Subgroup SG1.2a: VRT Fault Avoid

Group to autoreset modules and avoid failure in a voltage drop event.

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>SG1.2.a1 VRT Mode = Mode 1</td><td>Mode 1<br>Mode 2</td><td>This parameter allows to stablish a mode to reset modules<br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Mode 1</td><td>Automatically resets the FPGA of the inverter module(s) when there are 3 or more modules (SG1.2.a 4-Sudden n drop) in overcurrent warning (W65:AC overcurrent).</td></tr><tr><td>Mode 2</td><td>Automatically resets module (FPGA) when there are sudden variations in voltage (8%) or frequency (10%) in a period shorter than 100 ms.</td></tr></table>In any of these modes, after module reset the above reference is applied a factor of 0.8 (SG1.2.a5-Recovery).</td><td>Yes</td></tr>
<tr><td>SG1.2.a2-Sudden V drop = 0.079</td><td>0.00 to 1.00</td><td>Sets the sudden voltage drop with respect to the voltage measured by the central board when a voltage dip occurs.<br><br>In case a sudden drop is detected, but conditions to enable LVRT algorithm are not met, the overcurrent fault will be avoided and inverter will restart immediately once the time set in SG1.2.15 has elapsed.<br><br>Note: This parameter is displayed if SG1.2.a1-VRT Mode = Mode 2.</td><td>Yes</td></tr>
<tr><td>SG1.2.a3-Sudden f drop = 0.100</td><td>0.00 to 1.00</td><td>Sets the sudden frequency drop, with respect to the frequency measurement of the central board when a voltage dip occurs.<br><br>Similar to [SG1.2.a2-Sudden V drop], but for frequency variations.<br><br>Note: This parameter is displayed if SG1.2.a1-VRT Mode = Mode 2.</td><td>Yes</td></tr>
<tr><td>SG1.2.a4-Sudden n drop = 3</td><td>1 to 10</td><td>Automatically resets inverter module(s) (FPGA) when there are 3 or more (SG1.2.a 4-Sudden n drop) modules in overcurrent warning (W65:AC overcurrent).<br><br>Note: This parameter is displayed if SG1.2.a1-VRT Mode = Mode 1.</td><td>Yes</td></tr>
<tr><td>SG1.2.a5-Recovery = 80 %</td><td>0 to 150 %</td><td>Factor that will be applied to the reference after the module reset.</td><td>Yes</td></tr>
<tr><td>SG1.2.a6-VRT Entrance Priority = Disabled</td><td>Disabled<br>LVRT<br>OVRT</td><td>Defines the priority for LVRT/OVRT algorithm activation.<br><br>Once it has been activated and the DSP has entered LVRT logic, if the grid disturbance continues and there are OVRT algorithm activation conditions, it would only be possible to activate OVRT and inject inductive power if the priority is set to OVRT, and vice versa (so far OVRT always has had priority).<br><br>Disabled means that the complementary algorithm cannot be activated until the current algorithm logic has finished completely and the product logic returns to RUN.</td><td>Yes</td></tr>
</table>

## Subgroup SG1.2b: Balancing P

Group to set parameters of power balance algorithm. This algorithm allows avoiding oscillations in the busbar when the grid operates in inverse sequence.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG1.2.b1 Balancing mode = Off** | 0 to 2 | This parameter allows to stablish the working mode of the power balance algorithm.<br>**Option** / **Description**<br>Off: Balancing mode is disabled.<br>Mode 1: Balancing mode is activated when **voltage increase** due to inverse sequence exceeds a value of increase calculated internally.<br>Mode 2: Balancing mode is activated when **voltage value** due to inverse sequence exceeds a threshold stablished internally. | Yes |
| **SG1.2.b2-Balancing time = 0.10 s** | 0.10 to 1.00 | Time (seconds) during which the power balance will be activated.<br>**Note**: This parameter will remain hidden when SG1.2.b1 is off (0). | Yes |
| **SG1.2.b3-Balancing gain = 1.00** | 0.00 to 1.00 | Constant to control the balancing of P. It allows to set higher or lower balancing (100% would correspond to the maximum possible balancing).<br>**Note**: This parameter will remain hidden when SG1.2.b1 is off (0). | Yes |
| **SG1.2.b4-Balancing threshold = 0.05** | 0.00 to 1.00 | Threshold for activating the balance of P. This same threshold can be configured for both mode 1 and mode 2.<br>**Note**: This parameter will remain hidden when SG1.2.b1 is off (0). | Yes |

## Subgroup SG1.3: FPGA service

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG1.3.1-FPGA Faults En. = XXXXXXXXX** | 000000000 to XXXXXXXXX | Allows enabling or disabling different FPGA faults.<br>Each bit (beginning from the left) represents a fault (0 = Not active; 1 or X = Active).<br>• W118:Thermal drift (not yet implemented)<br>• W115:Module sensor<br>• W114:Module Fans<br>• W109:Low module Vdc<br>• W108:Half bus Vmax<br>• W99:Mod. Iac derivation<br>• W98:Mod. Idc derivation<br>• W88:Mod. AC contactor<br>• W71-83: Desaturation<br>• W67:DC overcurrent<br>**Note**: This parameter should not be modified, please contact Power Electronics before making any changes. | Yes |
| **SG1.3.2-VRT Retries = 5** | 0 to 50 | In case of a voltage event, overcurrent will be detected. The module will attempt to restart as many times as the value set in this parameter and with the delay between retries defined in [SG1.2.15-IGBTs blocking time].<br>If the maximum number of retries has been reached, and the module has not been able to restart, fault F65: Mod. AC overcurrent will be triggered.<br>**Note**: This parameter should not be modified, please contact Power Electronics before making any changes. | Yes |

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| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG1.3.3-Module stop meas = No** | No<br>Yes | When the module is stopped, no current should be measured and group SV9 should display:<br><br>SV9.4-Current R = 0.0 A<br>SV9.5-Current S = 0.0 A<br>SV9.6-Current T = 0.0 A<br><br>If any sensor measures a value other than zero, then there is a problem in the current measurement. This parameter helps identifying where the problem lies.<br><br>If set to NO (default value), the inverter will force to zero the values of SV9.4, SV9.5 and SV9.6 so they always display 0.0 A, even if there is a problem in any of the current measurement elements (sensor, adapter board or even the central board). | Yes |
| **SG1.3.4 Kp balanced Vbus = 2** | 0 to 8 | Kp gain for the control PI. It is configured according to the guidelines of R&D. | Yes |

## Subgroup SG1.3a: Modulation

**Note:** parameters within this subgroup were reorganized, in version 4.3.0, in subgroup G4.15: Simple Signal. Each parameter indicates which one is the new parameter from that version onwards; refer to the descriptions in the new parameters.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG1.3.a1-Modulation type = Simple-Double** | Double<br>Simple-Double | **Note:** this parameter, from version 4.3.0, is [G4.15.1.1-Modulation type]. | Yes |
| **SG1.3.a2-Modulation offset = 0.00 V** | -50.00 a 150.00 V | **Note**: this parameter, from version 4.3.0, is [G4.15.2.1.1-Ofset curv Vdc-Q]. | Yes |
| **SG1.3.a3-Single delay = 5.00 s** | 0.10 a 100.00 s | **Note**: this parameter, from version 4.3.0, is [G4.15.2.4-Single delay]. | Yes |
| **SG1.3.a4-Max Q = 50.00 %** | 0.00 a 100.00 % | **Note**: this parameter, from version 4.3.0, is [G4.15.2.1.2-Max Q]. | Yes |
| **SG1.3.a5-Max P/s = 20.00 %/s** | 0.10 a 100.00 %/s | **Note**: this parameter, from version 4.3.0, is [G4.15.2.2.1-Max P/s]. | Yes |
| **SG1.3.a6-Max Q/s = 3.00 %/s** | 0.10 a 100.00 %/s | **Note**: this parameter, from version 4.3.0, is [G4.15.2.2.2-Max Q/s]. | Yes |
| **SG1.3.a7-DSP Change Thresh = 80 V** | 50 a 200 V | **Note**: this parameter, from version 4.3.0, is [G4.15.2.3.2-Filtered semibus]. | Yes |
| **SG1.3.a8-FPGA change threshol = 120 V** | 50 a 200 V | **Note:** this parameter, from version 4.3.0, is [G4.15.2.3.1-Instant Semibus]. | Yes |
| **SG1.3.a9-Excessive changes = Warning** | Warning<br>Fault<br>Disabled | **Note:** this parameter, from version 4.3.0, is [G4.15.3.1-Enable]. | Yes |
| **SG1.3.a10-Pmin count changes = 10.0 %** | 0.0 a 100.0 % | **Note:** this parameter, from version 4.3.0, is [G4.15.3.4-Pmin threshold]. | Yes |
| **SG1.3.a11-Max N changes/min = 10** | 0 a 30 | **Note:** this parameter, from version 4.3.0, is [G4.15.3.2-Consec changes]. | Yes |
| **SG1.3.a12-Extra t hysteresis = 130.0 %** | 100.0 a 150.0 % | **Note:** this parameter, from version 4.3.0, is [G4.15.3.3-Additional time]. | Yes |
| **SG1.3.a13-Allowed Double time = 10.0 %** | 0.0 a 100.0 % | **Note:** this parameter, from version 4.3.0, is [G4.15.3.5-Allowed Double time ]. | Yes |
| **SG1.3.a14-Modulat. delay FPGA = 200 µs** | 10 to 2000 µs | **Note:** this parameter, from version 4.3.0, is [G4.15.2.3.3-Anti-Glitch delay]. | Yes |

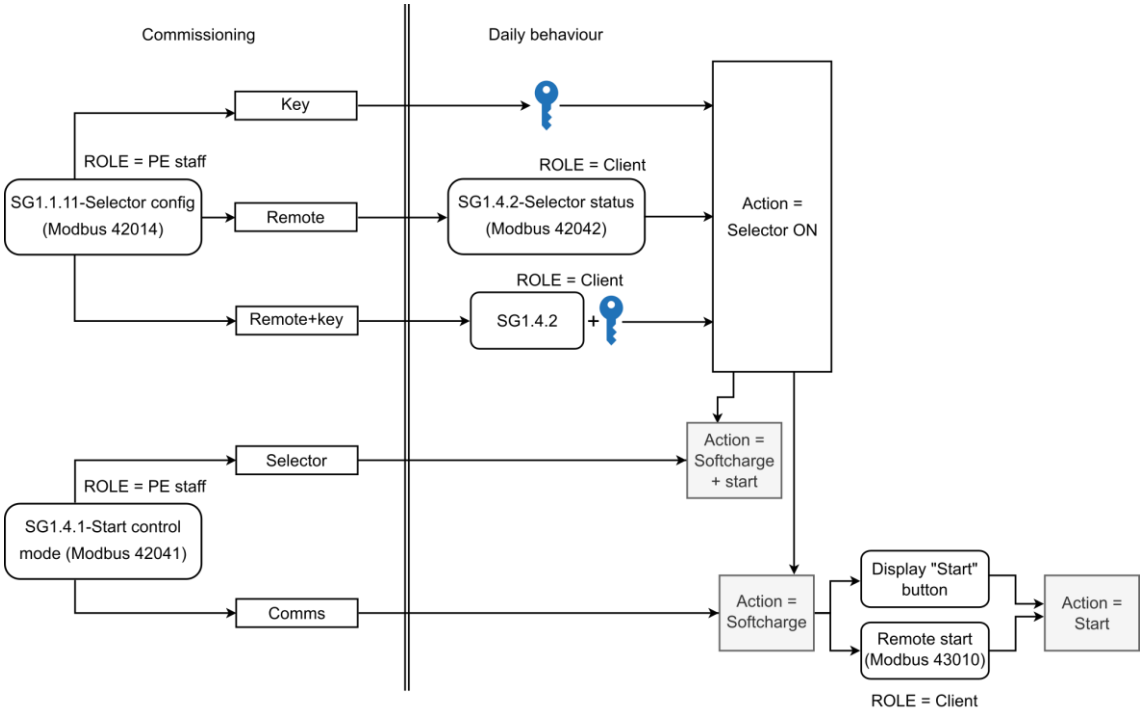
## Subgroup SG1.4: Selector status

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>SG1.4.1-Start control mode = Selector</td><td>Selector<br>Comms</td><td>Selects the mode for start control (selector or communications) and will execute the softcharge.<br><br><table><tr><th>Option</th><th>Description</th></tr><tr><td>Selector</td><td>The inverter will start automatically once start conditions are fulfilled.<br>This is the automatic start mode (only by switching SG1.4.2 to ON, if the start conditions are met (G2.X) the inverter would reach RUN or ON status.</td></tr><tr><td>Comms</td><td>The inverter will start if start conditions are met and once it has received the START command either from the display or from communications. This is the manual mode, once the selector (SG1.4.2=ON) is activated, the inverter performs only the soft load. Then, in order to start injecting power, the StartLocalControl command (register XXXX) or the green button on the display must be applied.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG1.4.2-Selector status = Off</td><td>Off<br>On</td><td>Sets the selector status (Off/On).<br><br>Note:<br>• This parameter will only be displayed if SG1.1.11 is set to ‘remote’ or ‘Key + remote’.<br>• If SG1.4.3 = Yes, Selector Status will be saved in EEPROM so that the inverter attempts to start automatically after a severe grid event.</td><td>Yes</td></tr>
<tr><td>SG1.4.3-Selector status EEPR = No</td><td>No<br>Yes</td><td>Defines whether the selector value should be stored in the EEPROM or not. If it is stored, the inverter will attempt to start automatically after a severe dip.<br><br>Note: This parameter will only be displayed if SG1.1.11 is set to ‘Remote’ or ‘Key + remote’.</td><td>Yes</td></tr>
</table>

**Note:** Parameters SG1.4.2 and SG1.4.3 are accessible to the client via Modbus (registers 42042 and 42047).

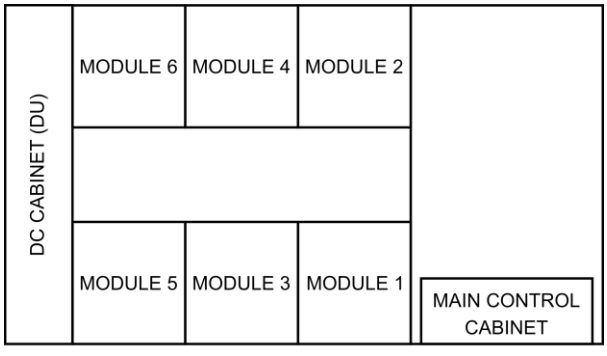
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The following diagram shows the types of equipment startup indicating how the start-up operation should be performed on the HEM inverter.




## Subgroup SG1.5: Enabled modules

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG1.5-Enabled modules = XXXXXX** | 000000 to XXXXXX | Shows which modules are enabled.<br>Each bit represents a module, being Module 1 the first from the right (0 = disabled, X = active).<br><br>**Note:** Phisically, Module 1 is the closest to the Main Control Cabinet. See figure below | Yes |




*Module numbering in HEM and HEMK inverters*

## Subgroup SG1.6: Selected Normative

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>SG1.6-Normative = UL 1741</td><td>0 to 10</td><td>It is possible to configure voltage and frequency protection ranges according to the normative that applies to each country.<br><br>Supported normative, plus an extra option are shown below:<br><br>
<table>
<tr><th>OPT.</th><th>NORMATIVE</th></tr>
<tr><td>0</td><td>UL 1741</td></tr>
<tr><td>1</td><td>Brazil</td></tr>
<tr><td>2</td><td>ERCOT</td></tr>
<tr><td>3</td><td>HECO 1</td></tr>
<tr><td>4</td><td>HECO 2</td></tr>
<tr><td>5</td><td>Rule 21</td></tr>
<tr><td>6</td><td>Jordan</td></tr>
<tr><td>7</td><td>Mexico</td></tr>
<tr><td>8</td><td>NERC Eastern</td></tr>
<tr><td>9</td><td>NERC Western</td></tr>
<tr><td>10</td><td>NERC Quebec</td></tr>
<tr><td>11</td><td>NERC ERCOT</td></tr>
<tr><td>12</td><td>Japan 50Hz</td></tr>
<tr><td>13</td><td>Japan 60Hz</td></tr>
<tr><td>14</td><td>SD Card (*)</td></tr>
</table>
(*) With this option, the file “Normative.txt” will be read from an SD card. This is useful in case none of the supported normative applies and a specific configuration needs to be created. Should this be the case, ask for Power Electronics’ support on creating the file.</td><td>Yes</td></tr>
</table>

## Subgroup SG1.7: Standby inverter

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG1.7-Standby inverter = No** | No<br>Yes | This is an option in case the regulation requires providing a power reserve, defined in percentage with respect to the pre-disturbance active power, to respond to low frequency events.<br><br>If SG1.7 = Yes, the inverter will operate in standby status (started but without injecting to the grid), acting as a power reserve. Whenever the PPC detects a low frequency event, it will command the inverter to start injecting active power. | Yes |

# Group SG2: Debug

## Subgroup SG2.1: DSP debug

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG2.1.1-Config DSP 1 = 0** | 0 to 65535 | DSP debug variables for testing and programming. | Yes |
| **SG2.1.2-Config DSP 2 = 0** | 0 to 65535 | | Yes |
| **SG2.1.3-Config DSP 3 = 0** | 0 to 65535 | | Yes |
| **SG2.1.4-Config DSP 4 = 0** | 0 to 65535 | | Yes |
| **SG2.1.5-Config DSP 5 = 0** | 0 to 65535 | | Yes |
| **SG2.1.6-Config DSP 6 = 0.00 %** | -100.00 to 100.00 % | | Yes |
| **SG2.1.7-Config DSP 7 = 0.00 %** | -100.00 to 100.00 % | | Yes |
| **SG2.1.8-Config DSP 8 = 0.00 %** | -100.00 to 100.00 % | | Yes |
| **SG2.1.9-Visualization DSP 1 = 0** | 0 to 65535 | Displays DSP debug variables. | Yes |
| **SG2.1.10-Visualization DSP 2 = 0** | 0 to 65535 | | Yes |
| **SG2.1.11-Visualization DSP 3 = 0** | 0 to 65535 | | Yes |
| **SG2.1.12-Visualization DSP 4 = 0** | 0 to 65535 | | Yes |
| **SG2.1.13-Visualization DSP 5 = 0** | 0 to 65535 | | Yes |

## Subgroup SG2.2: FPGA debug

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG2.2.1-Config FPGA 1 = 0** | 0 to 65535 | FPGA debug variables for testing and programming. | Yes |
| **SG2.2.2-Config FPGA 2 = 0** | 0 to 65535 | | Yes |
| **SG2.2.3-Config FPGA 3 = 0** | 0 to 65535 | | Yes |
| **SG2.2.4-Config FPGA 4 = 0** | 0 to 65535 | | Yes |
| **SG2.2.5-Config FPGA 5 = 0** | 0 to 65535 | | Yes |
| **SG2.2.6-Visualiz FPGA 1 = 0** | 0 to 65535 | Displays FPGA debug variables. | Yes |
| **SG2.2.7-Visualiz FPGA 2 = 0** | 0 to 65535 | | Yes |
| **SG2.2.8-Visualiz FPGA 3 = 0** | 0 to 65535 | | Yes |
| **SG2.2.9-Visualiz FPGA 4 = 0** | 0 to 65535 | | Yes |
| **SG2.2.10-Visualiz FPGA 5 = 0** | 0 to 65535 | | Yes |

## Subgroup SG2.3: uP debug

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>SG2.3.1-Init micro mode 1 = No data</td><td rowspan="3">No data<br>Watchdog<br>Power on<br>Sw reset</td><td rowspan="3">Displays the initial mode of the microcontroller, according to the following table:<br><table><tr><th>Option</th><th>Description</th></tr><tr><td>No data</td><td>There is no information.</td></tr><tr><td>Watchdog</td><td>Start by watchdog.</td></tr><tr><td>Power on</td><td>Start by power on.</td></tr><tr><td>Sw reset</td><td>Start by software reset.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG2.3.2-Init micro mode 2 = No data</td><td>Yes</td></tr>
<tr><td>SG2.3.3-Init micro mode 3 = No data</td><td>Yes</td></tr>
<tr><td>SG2.3.4-Hardfault LR = 0</td><td>0 to 65535</td><td rowspan="2">Displays LR (Link Register) register allocation. For internal use.</td><td>Yes</td></tr>
<tr><td>SG2.3.5-Hardfault LR = 0</td><td>0 to 65535</td><td>Yes</td></tr>
<tr><td>SG2.3.6-Hardfault PC = 0</td><td>0 to 65535</td><td rowspan="2">Displays PC (Program Counter) register allocation. For internal use.</td><td>Yes</td></tr>
<tr><td>SG2.3.7-Hardfault PC = 0</td><td>0 to 65535</td><td>Yes</td></tr>
<tr><td>SG2.3.8-Micro debug 1 = 0</td><td>0 to 65535</td><td rowspan="5">Microcontroller debug variables for testing and programming.</td><td>Yes</td></tr>
<tr><td>SG2.3.9-Micro debug 2 = 0</td><td>0 to 65535</td><td>Yes</td></tr>
<tr><td>SG2.3.10-Micro debug 3 = 0</td><td>0 to 65535</td><td>Yes</td></tr>
<tr><td>SG2.3.11-Micro debug 4 = 0</td><td>0 to 65535</td><td>Yes</td></tr>
<tr><td>SG2.3.12-Micro debug 5 = 0</td><td>0 to 65535</td><td>Yes</td></tr>
<tr><td>SG2.3.13-Debug logging level = WARNING log level</td><td>Disabled<br>DEBUG log level<br>INFO log level<br>WARNING log level<br>ERROR log level<br>CRITICAL log level</td><td>Allows to select the quantity of depuration frames stored on the SD card.<br><table><tr><th>Option</th><th>Description</th></tr><tr><td>Disabled</td><td>This option does not store frames.</td></tr><tr><td>DEBUG log level</td><td>Maximum quantity of frames stored.</td></tr><tr><td>INFO log level</td><td>Frames from the CRITICAL, ERROR, WARNING and this log level are stored.</td></tr><tr><td>WARNING log level</td><td>Frames from the CRITICAL, ERROR and this log level are stored.</td></tr><tr><td>ERROR log level</td><td>Frames from the CRITICAL and this log level are stored.</td></tr><tr><td>CRITICAL log level</td><td>Minimum quantity of frames stored.</td></tr></table>A higher quantity of stored frames means a higher overload of the microcontroller, this may affect the operation of the inverter.</td><td>Yes</td></tr>
</table>

EN

# Group SG3: SD card

In this group the storage of data in the SD can be configured.

**Note:** Recommended SG3.1 configuration (for SAT personnel) can be found in section "P2G Card Monitoring Configuration".

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>SG3.1.1-Enable = Yes</td><td>No<br>Yes</td><td>Enable or disable storing a configurable set of parameters (SG3.1.2 – SG3.1.16) in the SD card.</td><td>Yes</td></tr>
<tr><td>SG3.1.2-Modbus address 1 = 1192<br>SG3.1.3-Modbus address 2 = 1193<br>SG3.1.4-Modbus address 3 = 1194<br>SG3.1.5-Modbus address 4 = 1124<br>SG3.1.6-Modbus address 5 = 11248<br>SG3.1.7-Modbus address 6 = Off<br>SG3.1.8-Modbus address 7 = Off<br>SG3.1.9-Modbus address 8 = Off<br>SG3.1.10-Modbus address 9 = Off<br>SG3.1.11-Modbus address 10 = Off<br>SG3.1.12-Modbus address 11 = Off<br>SG3.1.13-Modbus address 12 = Off<br>SG3.1.14-Modbus address 13 = Off<br>SG3.1.15-Modbus address 14 = Off<br>SG3.1.16-Modbus address 15 = Off</td><td>Off = 0<br>1 to 9999</td><td>If [SG3.1.1] has been enabled, user can create a customized log that will be stored in the SD card. This log can contain up to 15 variables.<br><br>Each variable is represented in two columns: one for the Modbus register and one for the Content.<br><br>User must enter in parameters [SG3.1.2] to [SG3.1.16] the Modbus register (Modbus address – 40001) of each parameter to be stored.<br><br>If the register exists, its content will be stored in the log. Otherwise, if the register is not used, neither the register nor any content will be stored, and columns will be rearranged.<br><br>Example:<br><br>SG3.1.7 = 192.<br><br>Register 192 corresponds to parameter G4.2.1.3. Thus, the priority (P or Q) of OVRT algorithm will be stored in the SD card.</td><td>Yes</td></tr>
<tr><td>SG3.2-SD card status = OK</td><td>OK<br>Error<br>Not formatted<br>Not mounted<br>Not initialized<br>Settings loaded<br>Settings partially loaded<br>Error loading settings<br>Formatted SD</td><td>Displays the status of the SD card.<br><br><table><tr><th>Option</th><th>Description</th></tr><tr><td>OK</td><td>SD card is ready</td></tr><tr><td>Error</td><td>SD card has an error</td></tr><tr><td>Not formatted</td><td>SD card is not formatted</td></tr><tr><td>Not mounted</td><td>SD card is not mounted</td></tr><tr><td>Not initalized</td><td>SD card has not been initialized</td></tr><tr><td>Settings loaded</td><td>SD card settings are not loaded</td></tr><tr><td>Settings partially loaded</td><td>SD card settings are partially loaded</td></tr><tr><td>Error loading settings</td><td>An error occurred while loading settings from the SD card</td></tr><tr><td>Formatted SD</td><td>SD card has been formatted</td></tr></table></td><td>Yes</td></tr>
</table>

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG3.3-SD operation = Init SD card** | Init SD card<br>Format SD card<br>Load normative<br>Load settings<br>Update status<br>Backup parameters<br>Restore parameters | Sets the task to execute with the SD card.<br><br>Start SD card: Attempts to mount the SD card partition and access its file system.<br>Format SD card: Formats the SD card partition<br>Load standard: Loads the file Normative.txt that has been saved in the SD.<br>Load Settings: It registers in modbus addresses contained in a file that will be called txt settings.<br>Update status: Updates the status of the card; if it is mounted and accessible, OK will appear.<br>Save Parameters: Saves the parameters in Backup.txt<br>Restore Parameters: Load the Backup.txt parameters | Yes |
| **SG3.4-Apply changes = No** | No<br>Yes | Allows applying or discarding the modifications made to the SD card. | Yes |
| **SG3.5-Logger enable = Yes** | No<br>Yes | Besides having the possibility to store a customized set of parameters, the SD card will always keep a record of the following information:<br>• Full list of faults.<br>• Full list of warnings.<br>• Input event in OVRT mode.<br>• Insulation resistance:<br>  ○ For grounded pole: Value before starting.<br>  ○ For isolated poles: Registered value in case of fault; registered value in case of an emergency stop (fire mode or bus short-circuit). Maximum and minimum daily value. | Yes |
| **SG3.6-SD test stress = No** | No<br>Yes | This parameter is used for development tests. | Yes |
| **SG3.7-Blackbox download = No** | No<br>All<br>Module 1<br>Module 2<br>Module 3<br>Module 4<br>Module 5<br>Module 6<br>Module 7<br>Module 8<br>Module 9<br>Module 10 | Blackbox functionality consists on sending part of the FPGA information to the microprocessor in order to restore it in case of fault.<br><br>The FPGA stores the data in the F-RAM memory every 620ms. In case of failure, it will save in the SD card the additional 620ms + 100ms before and after the fault.<br><br>To consult this information, the user must download the csv file that is stored in the SD.<br>Configuring the following options defines the data download in the Blackbox:<br>**Option** / **Description**<br>No — No download.<br>All — Total download.<br>Module 1 … Module 10 — Select module to download. | No |
| **SG3.7a-AutoBbox download = No** | No<br>All Faults<br>Special Faults | Allows to configure in which situations the automatic download of the Black box must be executed. This information is stored in the SD card.<br>**Option** / **Description**<br>No — It is downloaded depending on the configuration of the SG3.7-Blackbox download.<br>All Faults — It is downloaded automatically if any failure occurs.<br>Special Faults — It is downloaded automatically if a special failure occurs.<br><br>When a system reboot is requested, it is required to wait until the data acquisition of the FPGA and its subsequent storage in the SD are finished. | Yes |
| **SG3.8-Used Capacity = 0.00 %** | 0.00 to 100.00 % | This parameter indicates the use percentage of the SD card. If it is zero, it could be because the SD is not being used or because the capacity could not be read.<br>**Note:** this is a read only parameter. | Yes |

EN

# Group SG4: Test mode

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG4.1-Force start conditions = No** | No<br>Yes | When set to Yes, forces start conditions | Yes |
| **SG4.2-Start delays = Yes** | No<br>Yes | When set to Yes, allows entering the start delay according to the normative. | Yes |
| **SG4.4-Open loop = No** | No<br>Yes | Configure open loop control. If this mode is enabled, parameters SG4.10 and SG4.11 must be configured. | Yes |
| **SG4.5-Short circuit = No** | No<br>Yes | Enables short circuit mode. Not used currently. | Yes |
| **SG4.6-AC source = No** | No<br>Yes | Enables or disables AC source mode. If enabled, parameters SG4.12, SG4.13 and SG4.14 must be configured. Not used currently. | Yes |
| **SG4.7-Cold commissioning = No** | No<br>Yes | Enables cold commissioning. If this mode is selected, modules that will work as AC source must be configured in SG4.15. | No |
| **SG4.8-High frequency = No** | No<br>Yes | Enables high frequency. Not used currently. | Yes |
| **SG4.9-Resistors mode = No** | No<br>Yes | Enables resistors mode. Not used currently. | Yes |
| **SG4.10-Ref Vd = 1.000** | 0.000 to 1.000 | Sets active power reference for open loop control.<br>**Note:** This parameter will only appear if SG4.4 = Yes. | Yes |
| **SG4.11-Ref Vq = 0.000** | -1.000 to 1.000 | Sets reactive power reference for open loop control.<br>**Note:** This parameter will only appear if SG4.4 = Yes. | Yes |
| **SG4.12-Ref Vac = 690 V** | 50 to 900 V | Sets the voltage reference.<br>**Note:** This parameter will only appear if SG4.6 = Yes. | Yes |
| **SG4.13-Grid = Yes Hz** | Yes = 399<br>Yes to 70.0 Hz | If this parameter is set to Yes, the inverter uses the grid normally (this should always be the setting on field).<br>Otherwise, a grid of 40 to 70 Hz can be simulated, which is useful for testing (e.g. set as AC source, cold commissioning…). Ensure to set SG4.13 back to 'Yes' once tests have been finished. | Yes |
| **SG4.14-Sequence = Positive** | Negative<br>Positive | If SG4.13 is other than Yes, use this parameter to configure the grid sequence as positive or negative. | Yes |
| **SG4.15-AC source mod select = 0000000000** | 0000000000 to XXXXXXXXXX | Sets which modules will work as AC source. | No |
| **SG4.16-Grid protections = No** | No<br>Yes | Enables grid protections. Used in Production testing. | Yes |
| **SG4.17-Heating test = No** | No<br>Yes | Enables heating test. Used in Production testing. | Yes |
| **SG4.18-Module DC source = No** | No<br>Yes | Enables DC source behavior. Only one module can be used. **This functionality should ONLY BE USED for testing**, never on field. | Yes |
| **SG4.19-Black start = No** | No<br>Yes | Test MPPt against passive charge (resistors) regulating Vac. Used sometimes in commissioning. | Yes |
| **SG4.20-Ref Vd Inv = 0.000** [2] | -1.000 to 1.000 | Similar to SG4.10 but for inverse sequence. | Yes |
| **SG4.21-Ref Vq Inv = 0.000** [2] | -1.000 to 1.000 | Similar to SG4.11 but for inverse sequence. | Yes |
| **SG4.22-Stop open loop = Ramp** [2] | Ramp<br>Instant | **OPT.** / **DESCRIPTION**<br>Ramp: Stop according to the configured ramp.<br>Instant: Stop immediately. | Yes |
| **SG4.23-AC Source DC comp = No** | No<br>Yes | Allows enabling the DC component in AC source, to compensate transformer DC component.<br>**Note: This functionality should ONLY be used for testing**, never on field. | Yes |

## NOTICE

For cold commissioning / off grid test, please refer to documents HEMIC001 and HEMDTI01, where the full procedure is described in detail.

# Group SG5: Remote upgrade

## Subgroup SG5.1: FTP Config

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG5.1.1-Ftp Address = 84.127.240.141** | 1 to 255 | Sets the IP address of the FTP server.<br><br>IP address format is: A.B.C.D.<br><br>To adjust this address, introduce a value in each of the four parameters that compose it. | Yes |
| **SG5.1.2-Ftp Port = 21** | 0 to 65535 | Sets the port to establish the FTP connection. Default port is 21. | Yes |
| **SG5.1.3-Ftp User = sat** | 10 to 10 | Sets the username for the FTP connection.<br><br>Enter the ASCII code for each character separately (up to 10 characters).<br><br>**Example**: SAT → 83 = 'S', 65 = 'A', 85 0 'T'.<br><br>**Note:** the Modbus addresses for this parameter are:<br>42216 – Size<br>42217 to 42226 – User | Yes |
| **SG5.1.4-Ftp password = sat** | 10 to 10 | Sets the password for the FTP connection.<br><br>Enter the ASCII code for each character separately (up to 10 characters).<br><br>**Note**: the Modbus addresses for this parameter are:<br>42227 – Size<br>42228 to 42237 –Password | Yes |
| **SG5.1.5-Ftp filename = HE2G.puf** | 32 to 32 | Enter the filename of the firmware file to download using ASCII codes for each character separately (up to 32 characters).<br><br>**Example**: to enter the filename "HE2G.puf", write the following ASCII codes: 72, 69, 50, 71, 46, 112, 117, 102.<br><br>**Note:** the Modbus addresses for this parameter are:<br>42238 – Size<br>42239 to 42270 – Filename | Yes |

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## Others

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>SG5.2.1-Upgrade = Off</td><td>Off<br>Full update<br>Download firmware<br>Install update<br>Erase downloaded file</td><td>Sets the remote upgrade function.</td><td>No</td></tr>
<tr><td>SG5.2.2-Status = Off</td><td>0 to 230</td><td>Displays the status of the remote upgrade, according to the following table:<br><br>
<table>
<tr><th>COD.</th><th>OPTION</th><th>COD.</th><th>OPTION</th></tr>
<tr><td>0</td><td>Off</td><td>210</td><td>Error upgrade</td></tr>
<tr><td>10</td><td>Downloading file</td><td>211</td><td>Firmware file error</td></tr>
<tr><td>11</td><td>Login failed error</td><td>212</td><td>Firmware read error</td></tr>
<tr><td>12</td><td>Access denied error</td><td>213</td><td>File format error</td></tr>
<tr><td>13</td><td>File not found error</td><td>214</td><td>File CRC error</td></tr>
<tr><td>14</td><td>Invalid path error</td><td>215</td><td>Section CRC error</td></tr>
<tr><td>15</td><td>Local file error</td><td>220</td><td>Upgrade OK but error at exit</td></tr>
<tr><td>16</td><td>Connecting error</td><td>221</td><td>Device not available</td></tr>
<tr><td>17</td><td>Timeout error</td><td>130</td><td>Upgrading DU</td></tr>
<tr><td>18</td><td>Unknown error</td><td>140</td><td>Upgrading DU ACQ</td></tr>
<tr><td>19</td><td>Download OK</td><td>150</td><td>Updating DU master</td></tr>
<tr><td>20</td><td>Upgrading uP</td><td>151</td><td>Updating the DU slave</td></tr>
<tr><td>21</td><td>Error SD full</td><td>155</td><td>Updating DU HEM/K</td></tr>
<tr><td>30</td><td>Upgrading MCF</td><td>156</td><td>Updating DU HEM/K I Meas</td></tr>
<tr><td>40</td><td>Upgrading Webserver</td><td>157</td><td>Updating DU HEM/K VI Meas</td></tr>
<tr><td>50</td><td>Upgrading PowerPLC</td><td>158</td><td>Updating MV Control</td></tr>
<tr><td>60</td><td>Upgrading DSP</td><td>160</td><td>Updating Isopro DC</td></tr>
<tr><td>80</td><td>Upgrading FPGA</td><td>161</td><td>Updating Isopro AC</td></tr>
<tr><td>90</td><td>Upgrading Cell DSP</td><td>170</td><td>Updating BFPS</td></tr>
<tr><td>100</td><td>Upgrading Cell FPGA</td><td>180</td><td>Upgrading GDULC MCF</td></tr>
<tr><td>110</td><td>Upgrading GDULC</td><td>230</td><td>Erase OK</td></tr>
<tr><td>200</td><td>Successfully upgrade</td><td></td><td></td></tr>
</table></td><td>Yes</td></tr>
<tr><td>SG5.2.3-Percentage = 0 %</td><td>0 to 100 %</td><td>Displays the remote upgrade progress.</td><td>Yes</td></tr>
<tr><td>SG5.2.4-Mod Error list = 0</td><td>0 to 65535</td><td>Displays the number of modules that have triggered an error when attempting to upgrade remotely.</td><td>Yes</td></tr>
<tr><td>SG5.3.1-Upgradable modules = XXXXXX</td><td>000000 to XXXXXX</td><td>Set which modules, can be upgraded remotely.<br>Each bit represents a module, being Module 1 the first from the right (0 = not upgradable, 1 = upgradable).<br><br>Note: Phisically, module 1 is the closest to the Main Control Cabinet. See figure “Module numbering in HEM and HEMK inverters” in case of doubt.</td><td>No</td></tr>
<tr><td>SG5.4.1-Upgrade du_acq 1 = XXXXXXXXXXXXXXX</td><td>0000000000000000 to XXXXXXXXXXXXXXX</td><td rowspan="3">From version 1.1.0, parameters SG5.4.1, SG5.4.2 and SG5.4.3 will allow selecting the number of boards of the measure channels. In general, we will have one board per channel.<br><br>In HEM and HEMK inverters, channels can be distributed in 3 groups of 12 or 4 groups of 9, depending on the configuration. Consult Power Electronics in case further information is required.<br><br>Each bit represents a channel. The lowest channel number corresponds to the bit further to the right.<br><br>Note: These parameters will not be shown if G10.1-DU Type = Standard</td><td>No</td></tr>
<tr><td>SG5.4.2-Upgrade du_acq 2 = XXXXXXXXXXXXXXX</td><td>0000000000000000 to XXXXXXXXXXXXXXX</td><td>No</td></tr>
<tr><td>SG5.4.3-Upgrade du_acq 3 = 00000000XXXXXXXX</td><td>0000000000000000 to 00000000XXXXXXXX</td><td>No</td></tr>
</table>

# Group SG6: Calibration

## Subgroup SG6.1: DU calibration

The calibration of the current and voltage measurements of the acquisition DU is carried out in the following way:

- Configure user "Power Electronics" or higher.
- Access group "SG6.1: DU calibration".
- Select the value to calibrate, either high (maximum) or low (minimum) value (SG6.1.1-Mode).
- Select the channel to calibrate, either voltage channel or current channel (SG6.1-3 – Mode V/I).
- Select the channel to be calibrated with SG6.1.2-Channel, in the case of wanting to calibrate the lower (minimum) value, it will be given the option to calibrate all channels simultaneously by placing in channel 1 and pressing down.
- Read the voltage/current with an external measuring instrument and write the value read in parameter SG6.1.5: Calibration value.
- Send calibration with SG6.16: Send calibration to yes.

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<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>SG6.1.1-Mode = Min calibration</td><td>Min calibration<br>Max calibration</td><td>Set the Disconnecting Unit calibration mode.</td><td>Yes</td></tr>
<tr><td>SG6.1.2-Channel = 1</td><td>0 to 40</td><td>Select channels to calibrate.</td><td>Yes</td></tr>
<tr><td>SG6.1.3-V/I mode = V mode</td><td>V mode<br>I mode</td><td>Set the calibration mode.<br>V = Voltage calibration. Reference value is set in SG6.1.5.1.<br>I = Current calibration. Reference value is set in SG6.1.5.2.</td><td>Yes</td></tr>
<tr><td>SG6.1.4-Channel status = Unknown</td><td>0 to 136</td><td>Shows the calibration status of the selected channel. By default, each channel has a theoretical calibration that is used as long as there is no complete calibration (maximum and minimum value of voltage or current). The possible states are as follows:<ul><li>No: There is no calibration parameter.</li><li>Min: Minimum value calibrated, partial calibration therefore continues to use theoretical calibration.</li><li>Max: Maximum value calibrated, partial calibration therefore continues to use theoretical calibration.</li><li>Ok: Minimum and maximum value calibrated, it will use the actual calibration entered.</li></ul><table><tr><th>COD.</th><th>OPTION</th><th>COD.</th><th>OPTION</th></tr><tr><td>0</td><td>Unknown</td><td>65</td><td>V:Max I:No</td></tr><tr><td>17</td><td>V:No I:No</td><td>66</td><td>V:Max I:Min</td></tr><tr><td>18</td><td>V:No I:Min</td><td>68</td><td>V:Max I:Max</td></tr><tr><td>20</td><td>V:No I:Max</td><td>72</td><td>V:Max I:Ok</td></tr><tr><td>24</td><td>V:No I:Ok</td><td>129</td><td>V:Ok I:No</td></tr><tr><td>33</td><td>V:Min I:No</td><td>130</td><td>V:Ok I:Min</td></tr><tr><td>34</td><td>V:Min I:Min</td><td>132</td><td>V:Ok I:Max</td></tr><tr><td>36</td><td>V:Min I:Max</td><td>136</td><td>V:Ok I:Ok</td></tr><tr><td>40</td><td>V:Min I:Ok</td><td></td><td></td></tr></table></td><td>Yes</td></tr>
<tr><td>SG6.1.5-Calibration value = 0.0V</td><td>0.0 to 1600.0 V</td><td>Sets the voltage calibration reference and current calibration reference, respectively.</td><td>Yes</td></tr>
</table>

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG6.1.5-Calibration value = 0.0A** | 0.0 to 300.0 A | **Note:** These parameters will be shown or hidden according to the calibration mode set in SG6.1.2. If V mode is chosen, the calibration value in Volts is shown. On the contrary, if I mode was chosen, calibration in Amperes is shown. | Yes |
| **SG6.1.6-Send calibration = No** | No<br>Yes | Send calibration reference. | Yes |

**Note:** The calibrations, once carried out, are stored in the EEPROM of the acquisition DU itself, therefore these values will remain even if the equipment parameters are initialized or the control board is changed.

## Subgroup SG6.2: AC calibration

The AC calibration will only have one parameter for a complete calibration, this is the **Gain.**

$$IAC_real = GAIN * IAC_measured\text{-}byelectronics$$

The steps for the calibration of the AC voltage will be as follows (Example for the 660Vac system):

1. See status of **SG6.2.1.1**, it will be “Not calibrated”.
2. The parameters **SG6.2.1.6-Measure Value Vrs, SG6.2.1.7-Measure Value Vst,** and **SG6.2.1.8-Measure Value Vtr** show a value of 645Vac (RMS value). What the PE STAFF is reading by the AC voltage measuring instrument is 655Vrms in all phases
3. Therefore, PE personnel have to configure **SG6.2.1.2-Cali Value Vrs, SG6.2.1.3-Cali Value Vst** and **SG6.2.1.4-Cali Value Vtr** to 655Vrms. Send calibration using **SG6.2.1.5 Send Cali/reset = Calibration**, then parameter **SG6.2.1.1-Status** will change to “Calibration sent”.
4. Once the gain has been calculated, parameter SG6.2.1.1 will change to “Calibration complete”.
5. Only by specific recommendation, a small gain adjustment can be made using **SG6.2.1.0.x.**

**SG6.2.1.0 Gain Vac adj**

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG6.2.1.0.1-Gain Vrs = 0.000** | 0.000 to 65.535 | Minimum gain setting Vrs. | Yes |
| **SG6.2.1.0.2-Gain Vst = 0.000** | 0.000 to 65.535 | Minimum gain setting Vst. | Yes |
| **SG6.2.1.0.3-Gain Vtr = 0.000** | 0.000 to 65.535 | Minimum gain setting Vtr. | Yes |
| **SG6.2.1.0.4-Measure Value Vrs = 0 V** | 0 to 65535 V | Measurement value Vrs. | Yes |
| **SG6.2.1.0.5-Measure Value Vst = 0 V** | 0 to 65535 V | Measurement value Vst. | Yes |
| **SG6.2.1.0.6-Measure Value Vtr = 0 V** | 0 to 65535 V | Measurement value Vtr. | Yes |

### SG6.2.1 Vac Calibration

Vrs, Vst and Vtr are calibrated at the same time by SG6.2.1

The possible states are: Unknown, Calibration sent, Not calibrated, Calibration complete.

The first status update serves to calibrate the offset of the DC measurement.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG6.2.1.1-Status = Not calibrated** | Unknown<br>Calibration sended<br>Not calibrated<br>Calibration complete | Displays the calibration status:<br>**OPTION** / **DESCRIPTION**<br>Unknown – Calibration not recognized.<br>Calibration sended – Offset has been calibrated.<br>Not calibrated – No calibration has been made.<br>Calibration complete – Calibration completed successfully. | Yes |
| **SG6.2.1.2-Cali Value Vrs = 0 V** | 0 to 65535 V | Calibration value adjustment Vrs. | Yes |
| **SG6.2.1.3-Cali Value Vst = 0 V** | 0 to 65535 V | Calibration value adjustment Vst. | Yes |
| **SG6.2.1.4-Cali Value Vtr = 0 V** | 0 to 65535 V | Calibration value adjustment Vtr. | Yes |
| **SG6.2.1.5-Send cali/reset = Idle** | Idle<br>Calibration<br>Reset Cal | Muestra el envío de calibración:<br>**OPTION** / **DESCRIPTION**<br>Idle – Calibration is not performed.<br>Calibration – Calibration values are sent.<br>Reset Cal – Reset calibration values | Yes |
| **SG6.2.1.6-Measure Value Vrs = 0 V** | 0 to 65535 V | Measurement value adjustment Vrs. | Yes |
| **SG6.2.1.7-Measure Value Vst = 0 V** | 0 to 65535 V | Measurement value adjustment Vst. | Yes |
| **SG6.2.1.8-Measure Value Vtr = 0 V** | 0 to 65535 V | Measurement value adjustment Vtr. | Yes |

### SG6.2.2.0-Gain Iac adj

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG6.2.2.0.1-Gain Ir = 0.000** | 0.000 to 65.535 | Minimum gain Ir setting. | Yes |
| **SG6.2.2.0.2-Gain Is = 0.000** | 0.000 to 65.535 | Minimum gain Is setting. | Yes |
| **SG6.2.2.0.3-Gain It = 0.000** | 0.000 to 65.535 | Minimum gain It setting. | Yes |
| **SG6.2.2.0.4-Measure Value Ir = 0 A** | 0 to 65535 A | Measurement value Ir. | Yes |
| **SG6.2.2.0.5-Measure Value Is = 0 A** | 0 to 65535 A | Measurement value Is. | Yes |
| **SG6.2.2.0.6-Measure Value It = 0 A** | 0 to 65535 A | Measurement value It. | Yes |

EN

### SG6.2.2 Iac Calibration

Ir, Is and It are calibrated at the same time by SG6.2.2. These are the overall measurements of the equipment (sum of the module currents), not those of the modules.

Current calibration is done in one step.

The possible states are: Unknown, Calibration sent, Not calibrated, Calibration complete.

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>SG6.2.2.1-Status = Not calibrated</td><td>Unknown<br>Calibration sended<br>Not calibrated<br>Calibration complete</td><td>Displays the calibration status:<br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Unknown</td><td>Calibration not recognized</td></tr><tr><td>Calibration sended</td><td>Offset has been calibrated.</td></tr><tr><td>Not calibrated</td><td>No calibration has been made.</td></tr><tr><td>Calibration complete</td><td>Calibration completed successfully.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG6.2.2.2-Cali Value Ir = 0 A</td><td>0 to 65535 A</td><td>Calibration value adjustment Ir</td><td>Yes</td></tr>
<tr><td>SG6.2.2.3-Cali Value Is = 0 A</td><td>0 to 65535 A</td><td>Calibration value adjustment Is</td><td>Yes</td></tr>
<tr><td>SG6.2.2.4-Cali Value It = 0 A</td><td>0 to 65535 A</td><td>Calibration value adjustment It</td><td>Yes</td></tr>
<tr><td>SG6.2.2.5-Send cali/reset = Idle</td><td>Idle = 0<br>Calibration = 1<br>Reset Cal = 2</td><td>Muestra el envío de calibración:<br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Idle</td><td>Calibration is not performed</td></tr><tr><td>Calibration</td><td>Calibration values are sent</td></tr><tr><td>Reset Cal</td><td>Reset Calibration values</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG6.2.2.6-Measure Value Ir = 0 A</td><td>0 to 65535 A</td><td>Measurement value adjustment Irs</td><td>Yes</td></tr>
<tr><td>SG6.2.2.7-Measure Value Is = 0 A</td><td>0 to 65535 A</td><td>Measurement value adjustment Ist</td><td>Yes</td></tr>
<tr><td>SG6.2.2.8-Measure Value It = 0 A</td><td>0 to 65535 A</td><td>Measurement value adjustment It</td><td>Yes</td></tr>
</table>

## Subgroup SG6.3: DC calibration

In general, DC calibrations (DC Voltage and DC Current) will have two parameters for a full calibration, those are offset and gain. Therefore, two measurements have to be made (minimum value and maximum value).

$$IDC_real = GAIN * IDC_measured\text{-}byelectronics + OFFSET$$

### SG6.3.0 Fine Tuning

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG6.3.0.1 Gain Vdc ch1 adj = 0.000 V** | 0.000 to 65.535 V | Vdc Gain Adjustment. | Yes |
| **SG6.3.0.2 Offset Vdc ch 1 adj = 0.00 V** | -327.67 to 327.67 V | Vdc offset adjustment. | Yes |
| **SG6.3.0.3-MeasureValueVdc ch1 = 0 V** | 0 to 65535 V | Measured value adjustment of Vdc. | Yes |
| **SG6.3.0.1 Gain I dc ch 1 adj = 0.000 A** | 0.000 to 65.535 A | Idc Gain Adjustment. | Yes |
| **SG6.3.0.2 Offset I dc ch1 adj = 0.00 A** | -327.67 to 327.67 A | Idc offset adjustment:<br>• When the calibration is restarted the Gain = 1, offset = 0.<br>• When not calibrated, the range is limited to +-2V (voltage case) and +2A (current case).<br>• When calibrated, the range is limited to 0.8*Calibrated_Offset and 1.2*Calibrated_Offset.<br>• The default value of SG6.3.0.2 is equal to the calibrated offset. | Yes |
| **SG6.3.0.3-MeasureValueIdc ch1 = 0 A** | -32768 to 32767 A | Measured value adjustment of Idc. | Yes |

### SG6.3-DC Calibration

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG6.3.1-Select channel = 1** | 1 to 1 | Channel selection for calibration. | Yes |
| **SG6.3.2-Select min/max = Minimum** | Minimum<br>Maximum | Minimum and maximum calibration selection. | Yes |
| **SG6.3.3-Select V/I = V mode** | V mode<br>I mode | Selection of Voltage (V) / Current (I) calibration mode. | Yes |
| **SG6.3.4-Satus channel/BUS 1 = Max: No,Min: No** | Max: No,Min: No<br>Max: No,Min: Yes<br>Max: Yes,Min: No<br>Max: Yes,Min: Yes<br>Calibration complete | It shows the channel / bus1 status according to the following table:<br>**OPT.** / **DESCRIPTION**<br>0 Max: No,Min: No<br>1 Max: No,Min: Yes<br>2 Max: Si,Min: No<br>3 Max: Si,Min: Yes<br>4 Calibración complete | Yes |
| **SG6.3.5-Cali Value Vdc = 0 V** | 0 to 65535 V | Adjusting the Vdc calibration value. | Yes |
| **SG6.3.5-Cali Value Idc = 0 A** | 0 to 65535 A | Calibration value adjustment Idc. | Yes |

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| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG6.3.6-Send cal./reset = Idle** | Idle<br>Calibration<br>Reset Cal<br>Reset Point | It shows the calibration status according to the values in the table below:<br><br>**OPT.** / **DESCRIPTION**<br>0 Idle<br>1 Calibration<br>2 Reset Cal<br>3 Reset Point | Yes |
| **SG6.3.7-Measure value Vdc ch1 = 0 V** | 0 to 65535 V | Measured value adjustment of Vdc. | Yes |
| **SG6.3.7-MeasureValue Idc ch1 = 0 A** | -32768 to 32767 A | Measured value adjustment of Idc. | Yes |
| **SG6.3.8-Point max V dc Ch1 = 0 V** | 0 to 65535 V | Measured value adjustment of the V. | Yes |
| **SG6.3.8-Point max I dc Ch1 = 0 A** | -32768 to 32767 A | Measured value adjustment of the maximum Idc point. | Yes |
| **SG6.3.9-Point max DSP Vdc Ch1 = 0 V** | 0 to 65535 V | Measured value adjustment of the DSP high point Vdc. | Yes |
| **SG6.3.9-Point max DSP Idc Ch1 = 0 A** | 0 to 65535 A | Measured value adjustment of the DSP high point Idc. | Yes |
| **SG6.3.10-Point min V dc Ch1 = 0 V** | 0 to 65535 V | Measured value adjustment of the lowest Vdc point. | Yes |
| **SG6.3.10-Point min I dc Ch1 = 0 A** | -32768 to 32767 A | Measured value adjustment of the lowest point of Idc. | Yes |
| **SG6.3.11-Point minDSP Vdc Ch1 = 0 V** | 0 to 65535 V | Measured value adjustment of the DSP minimum point Vdc. | Yes |
| **SG6.3.11-Point minDSP Idc Ch1 = 0 A** | 0 to 65535 A | Measured value setting of the minimum DSP Idc point. | Yes |

## Subgroup SG6.5: Power calibration

The inverter behavior can be modified with the following parameters:

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG6.5.1-Gain P/Q = 1.000** | 0.500 to 1.500 | Set the P/Q gain value.<br>It modifies the real power of the inverter. It mainly affects current loops, power and the maximum powers that the equipment delivers. A bad adjustment can be dangerous. | Yes |
| **SG6.5.2-Gain Q LCL = 1.000** | 0.500 to 1.500 | Set the Q LCL gain value. It modifies the compensation of the inverter filter. A bad adjustment can generate an unwanted reactive offset. | Yes |
| **SG6.5.3-Thermal gain = 1.020** | 0.800 to 1.500 | Set the thermal gain value. It works the same as the P/Q gain, but as a function of the thermal temperature (a drift was detected in the current measurement with the temperature). A bad adjustment can generate an unwanted reactive offset. | Yes |
| **SG6.5.4-P/Q visualization = 100.00 %** | 50.00 to 150.00% | Set the P/Q visualization percentage Modifies the display of the inverter without modifying its operation. | Yes |
| **SG6.5.5-Q Offset Visualiz = 0.00 kVAr** | -300.00 to 300.00 kVArh | Set an offset value for Q visualization. | Yes |
| **SG6.5.8-Current angle = 0** | -100 to 100 | Allows to set a delay to the current angle, due to measure acquisition time.<br>The calibration of this angle is done in terms of counts, 360º is equivalent to 1024 beads. E.g.; SG6.5.8 = 7 angle of 2,46º.<br>**Note: The use of this parameter is reserved for testing.** Do not modify it unless express authorization from Power Electronics. | Yes |

## Subgroup SG6.7 V MV calibration

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>SG6.7.1-Status = Not calibrated</td><td>Not calibrated<br>Positive seq calibrated<br>Negative seq calibrated<br>Calibration complete</td><td>Medium volage calibration is essential for the premagnetization process. The equipment must be correctly calibrated in, at least, the voltage sequence in which transformer premagnetization will be carried out.<br><br>This parameter shows the calibration status of the MV voltages (for HEM inverters).<br><br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Not calibrated</td><td>Equipment has not been calibrated.</td></tr><tr><td>Positive seq calibrated</td><td>Equipment is calibrated in positive sequence.</td></tr><tr><td>Negative seq calibrated</td><td>Equipment is calibrated in negative sequence.</td></tr><tr><td>Calibration complete</td><td>Equipment is calibrated in both sequences.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG6.7.2-Update = No</td><td>No<br>Yes<br>Updating</td><td>To force the calibration process, this parameter must be set to Yes. Value “Updating” will be shown.</td><td>Yes</td></tr>
<tr><td>SG6.7.3-Reset = No</td><td>No = 0<br>Yes = 1</td><td>Allows resetting the medium voltages calibration.</td><td>Yes</td></tr>
<tr><td>SG6.7.4-Gain MVr pos = 2047</td><td>1 to 4095</td><td rowspan="8">Shows the values obtained with the calibration. These parameters cannot be modified (read only parameters).</td><td>Yes</td></tr>
<tr><td>SG6.7.5-Gain MVs pos = 2047</td><td>1 to 4095</td><td>Yes</td></tr>
<tr><td>SG6.7.6-Gain MVt pos = 2047</td><td>1 to 4095</td><td>Yes</td></tr>
<tr><td>SG6.7.7-Gain MVr neg = 2047</td><td>1 to 4095</td><td>Yes</td></tr>
<tr><td>SG6.7.8-Gain MVs neg = 2047</td><td>1 to 4095</td><td>Yes</td></tr>
<tr><td>SG6.7.9-Gain MVt neg = 2047</td><td>1 to 4095</td><td>Yes</td></tr>
<tr><td>SG6.7.10-Seq pos angle = 3109</td><td>1 to 4095</td><td>Yes</td></tr>
<tr><td>SG6.7.11-Seq neg angle = 1721</td><td>1 to 4095</td><td>Yes</td></tr>
<tr><td>SG6.7.12-Estado interno = 0x00</td><td>0x00 to 0xFFFF</td><td>Shows the calibrations that have been performed:<br><br>0x00 = Not calibrated<br>0x01 = Positive sequence calibrated<br>0x02 = Negative sequence calibrated<br>0x03 = Calibration complete</td><td>Yes</td></tr>
</table>

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## Subgroup SG6.8 I MV calibration

**Note:** This group only applies to HEM inverters.

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG6.8.1-Status = Not calibrated** | Not calibrated<br>Calibration complete | Shows the calibration status of the medium voltage currents (for HEM inverters).<br><br>**OPTION** / **DESCRIPTION**<br>Not calibrated – Equipment has not been calibrated.<br>Calibration complete – Equipment is calibrated. | Yes |
| **SG6.8.1b-Visualization MV-I= MV** | MV<br>LV | Shows the measurement of medium voltage currents.<br>The average currents are measured with measuring transformers in the low voltage part. | Yes |
| **SG6.8.2-Update = No** | No<br>Yes<br>Updating | To force the calibration process, this parameter must be set to Yes. Value "Updating" will be shown. | Yes |
| **SG6.8.3-Reset = No** | No<br>Yes | Allows resetting the medium voltage currents calibration. | Yes |
| **SG6.8.4-Value Ir = 0.0 A** | 0.0 to 100.0 A | Shows the values obtained with the calibration. These parameters cannot be modified (read only parameters). | Yes |
| **SG6.8.5-Value Is = 0.0 A** | 0.0 to 100.0 A | | Yes |
| **SG6.8.6-Value It = 0.0 A** | 0.0 to 100.0 A | | Yes |
| **SG6.8.7-Gain Ir MV = 2047** | 1 to 4095 | | Yes |
| **SG6.8.8-Gain Is MV = 2047** | 1 to 4095 | | Yes |
| **SG6.8.9-Gain It MV = 2047** | 1 to 4095 | | Yes |

## Subgroup SG6.9: Auxiliary Calibration

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG6.9.1-Calibration 1 = 3** | 0 to 10 | Configurable offset to NTC measurements.<br>Serves to compensate the degradation of the IGBTs during their aging. | Yes |
| **SG6.9.2-Calibration 2 = 3** | 0 to 5 | Adjusts the percentage reduction of the current limit according to the number of modules in RUN state. | Yes |
| **SG6.9.3-Calibration PQ = 0.000** | -0.030 to 0.030 | Defines the adjustment on the calibration of the PQ visualization. | Yes |

# Group SG7: Reset counters

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG7.1-Total run time = No** | No<br>Yes | Reset total run time counters (number of years, days and hours). | Yes |
| **SG7.2-Partial run time = No** | No<br>Yes | Reset partial run time counters (years, days and hours). | Yes |
| **SG7.3-Total grid conns = No** | No<br>Yes | Reset the total number of grid connections. | Yes |
| **SG7.4-Partial grid conns = No** | No<br>Yes | Reset the partial number of grid connections. | Yes |
| **SG7.5-Faults = No** | No<br>Yes | Clear all fault counters. | Yes |
| **SG7.6-Warnings = No** | No<br>Yes | Clear all warning counters. | Yes |
| **SG7.7-Events = No** | No<br>Yes | Clear all event counters. | Yes |
| **SG7.8-Reset energy counters = No** | No<br>Yes | Reset energy (active and reactive) counters. | Yes |
| **SG7.9-Rst Premagn Closure = No** | No<br>Yes | Reset the MV switchgear counters without pre-magnetization. | Yes |
| **SG7.10-Rst Not Premag Clos = No** | No<br>Yes | Reset the MV switchgear counters with pre-magnetization. | Yes |
| **SG7.11-Rst Openings = No** | No<br>Yes | Reset the MV switchgear opening counters. | Yes |
| **SG7.12-Rst special faults = No** | No<br>Yes | Reset the special faults counters. | Yes |

EN

# Group SG8: DU diagnosis

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>SG8.1-Algorithm status = 0 Idle</td><td>0 to 65535</td><td>Displays the algorithm status, according to the following table:<br><table><tr><th>COD.</th><th>OPTION</th><th>COD.</th><th>OPTION</th></tr><tr><td>0</td><td>Idle</td><td>8</td><td>Meas closed contactor</td></tr><tr><td>1</td><td>Wait meas open GFDI</td><td>9</td><td>Process FB close issue</td></tr><tr><td>2</td><td>Wait until no current</td><td>10</td><td>Wait meas closed cont</td></tr><tr><td>3</td><td>Meas all contactors closed</td><td>11</td><td>Check FB acq open</td></tr><tr><td>4</td><td>Open all acqs</td><td>12</td><td>Check isolation status</td></tr><tr><td>5</td><td>Wait FB all acqs</td><td>13</td><td>All acqs processed</td></tr><tr><td>6</td><td>Close ind contactor</td><td>14</td><td>Wait meas closed GFDI</td></tr><tr><td>7</td><td>Wait FB contactor</td><td>15</td><td>Algorithm finished</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG8.2-Force event = Off</td><td>0 to 18</td><td>Set debug mode, according to the following table:<br><table><tr><th>COD.</th><th>OPTION</th><th>COD.</th><th>OPTION</th></tr><tr><td>0</td><td>Off</td><td>10</td><td>Enable check input miswired</td></tr><tr><td>3</td><td>New measure (No fault)</td><td>13</td><td>Reboot Central DU</td></tr><tr><td>4</td><td>New measure (Fault)</td><td>14</td><td>Reboot Acqs DU</td></tr><tr><td>5</td><td>Start conditions OK</td><td>15</td><td>Enable algorithm</td></tr><tr><td>6</td><td>Start conditions Off</td><td>16</td><td>Disable algorithm</td></tr><tr><td>7</td><td>Reset contactors status</td><td>17</td><td>Close soft charge contactor</td></tr><tr><td>8</td><td>Algorithm stop</td><td>18</td><td>Open soft charge contactor</td></tr><tr><td>9</td><td>Disable check input miswired</td><td></td><td></td></tr></table><br>Note: This parameter is for developers, do not modify it without authorization.</td><td>No</td></tr>
<tr><td>SG8.3-Contactors mode = Automatic</td><td>Automatic<br>Manual</td><td>Set contactors mode, between automatic and manual.</td><td>No</td></tr>
<tr><td>SG8.4-Reset channels status = No</td><td>No<br>Yes</td><td>Reset status for all channels.</td><td>Yes</td></tr>
<tr><td>SG8.5-Version = 0.0.0-0</td><td>0 to 65535</td><td>Displays the Disconnecting Unit firmware version.<br>Format: xx.xx.xx-xx Example: 1.0.0-1.<br><br>The value of this parameter is shown as a 32-bits DWord, two 16-bits words (MSB and LSB). The high byte (MSB) corresponds with the lowest Modbus address and the low byte (LSB) corresponds with the highest Modbus address. Thus, to correctly read the value, perform the calculation:<br>V = MSB * 2^16 + LSB</td><td>Yes</td></tr>
<tr><td>SG8.6-Status = 0000000000000000</td><td>0000000000000000 to XXXXXXXXXXXXXXXX</td><td>Displays the Disconnecting Unit status. Each bit (from right to left) represents a status (0 = Not active; 1 or X = Active):<br><table><tr><th>BIT</th><th>STATUS</th><th>BIT</th><th>STATUS</th></tr><tr><td>0</td><td>Online</td><td>8</td><td>Smoke detector</td></tr><tr><td>1</td><td>Diagnostic active</td><td>9</td><td>GFDI fuse</td></tr><tr><td>2</td><td>Contactor locked opening</td><td>10</td><td>Overvoltage DC</td></tr><tr><td>3</td><td>Contactor locked closing</td><td>11</td><td>Pole connection error</td></tr><tr><td>4</td><td>Isolation fault</td><td>12</td><td>Contactor</td></tr><tr><td>5</td><td>Acq comms error</td><td>13</td><td>No acq channels available</td></tr><tr><td>6</td><td>Diagnostic finished</td><td>14</td><td>Toggling contactor</td></tr><tr><td>7</td><td>Pole contactor</td><td>15</td><td>Ready</td></tr></table><br>Note: To represent all possible Disconnecting Unit status, 32 bits are needed. Thus, two variables, SG8.6 and SG8.7, are used.</td><td>Yes</td></tr>
</table>

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>SG8.7-Status 2 = 00000000000000</td><td>00000000000000 to XXXXXXXXXXXXX</td><td>Displays the Disconnecting Unit status 2.<br>Each bit (from right to left) represents a status (0 = Not active; 1 or X = Active):<br><table><tr><th>BIT</th><th>STATUS</th><th>BIT</th><th>STATUS</th></tr><tr><td>0</td><td>Acqs version mismatch</td><td>7</td><td>Initial isolation error</td></tr><tr><td>1</td><td>Poles contactor mismatch</td><td>8</td><td>Input miswired detection</td></tr><tr><td>2</td><td>Inverter comms lost</td><td>9</td><td>Algorithm enabled</td></tr><tr><td>3</td><td>Isolated channel</td><td>10</td><td>Softcharge contactor</td></tr><tr><td>4</td><td>Isolation fault grid</td><td>11</td><td>Softcharge error</td></tr><tr><td>5</td><td>Current present in acq</td><td>12</td><td>NEC 2017 mode</td></tr><tr><td>6</td><td>Channel miswired</td><td>13</td><td>Spring loaded error</td></tr></table><br>Note: To represent all possible Disconnecting Unit status, 32 bits are needed. Thus, two variables, SG8.6 and SG8.7, are used.</td><td>Yes</td></tr>
<tr><td>SG8.8-Acq selected = 1</td><td>1 to [G10.4]</td><td>Selects a channel (x = channel number), from 1 to [G10.4 = Number of DU channels].<br><br>Once a channel has been selected, its related information will be shown in parameters [SG8.9-Acq Version] , [SG8.10-Acq calibration] and [SG8.11-Acq Status].</td><td>Yes</td></tr>
<tr><td>SG8.9-Acq Version = 0.0.0-0</td><td>0 to 65535</td><td>Displays the firmware version of the channel selected in SG8.8.<br>Format: xx.xx.xx-xx Example: 1.0.0-1.<br><br>The value of this parameter is shown as a 32-bits DWord, two 16-bits words (MSB and LSB). The high byte (MSB) corresponds with the lowest Modbus address and the low byte (LSB) corresponds with the highest Modbus address. Thus, to correctly read the value, perform the calculation:<br>V = MSB * 2^16 + LSB</td><td>Yes</td></tr>
<tr><td>SG8.10-Acq calibration = Unknown</td><td>0 to 136</td><td>Shows the status calibration of the channel selected in SG8.8:<br><table><tr><th>COD.</th><th>OPTION</th><th>COD.</th><th>OPTION</th><th>COD.</th><th>OPTION</th></tr><tr><td>0</td><td>Unknown</td><td>34</td><td>V:MinI:Min</td><td>72</td><td>V:MaxI:Ok</td></tr><tr><td>17</td><td>V:NoI:No</td><td>36</td><td>V:MinI:Max</td><td>129</td><td>V:OkI:No</td></tr><tr><td>18</td><td>V:NoI:Min</td><td>40</td><td>V:MinI:Ok</td><td>130</td><td>V:OkI:Min</td></tr><tr><td>20</td><td>V:NoI:Max</td><td>65</td><td>V:MaxI:No</td><td>132</td><td>V:OkI:Max</td></tr><tr><td>24</td><td>V:NoI:Ok</td><td>66</td><td>V:MaxI:Min</td><td>136</td><td>V:OkI:Ok</td></tr><tr><td>33</td><td>V:MinI:No</td><td>68</td><td>V:MaxI:Max</td><td></td><td></td></tr></table></td><td>Yes</td></tr>
<tr><td>SG8.11-Acq Status = 0000000000000</td><td>0000000000000 to XXXXXXXXXXXXX</td><td>Displays the status of the channel selected in SG8.8. Each bit (from right to left) represents a status (0 = Not active; 1 or X = Active):<br><table><tr><th>BIT</th><th>STATUS</th><th>BIT</th><th>STATUS</th></tr><tr><td>1</td><td>Online</td><td>8</td><td>Negative pole</td></tr><tr><td>2</td><td>Contactor</td><td>9</td><td>Negative pole online</td></tr><tr><td>3</td><td>Contactor feedback</td><td>10</td><td>Channel isolated</td></tr><tr><td>4</td><td>Current present</td><td>11</td><td>Channel miswired</td></tr><tr><td>5</td><td>Contactor locked opening</td><td>12</td><td>Negative pole contactor</td></tr><tr><td>6</td><td>Contactor locked closing</td><td>13</td><td>Negative pole feedback</td></tr><tr><td>7</td><td>Comms error</td><td></td><td></td></tr></table></td><td>Yes</td></tr>
<tr><td>SG8.12-Contactors status = 000</td><td>000 to XXX</td><td>Set contactors status. Each bit (from right to left) represents a status (0 = Not active; 1 or X = Active):<br><table><tr><th>BIT</th><th>DESCRIPTION</th></tr><tr><td>1</td><td>DSP status. DU contactors status order sent to the uP by the DSP.</td></tr><tr><td>2</td><td>uP status. DU contactor status sent to the DU by the uP.</td></tr><tr><td>3</td><td>DU status. Current DU contactors status.</td></tr></table></td><td>Yes</td></tr>
</table>

EN

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>SG8.13-DU input status = 00000000</td><td>00000000 to XXXXXXX0</td><td>Shows the status of DU inputs. Each bit (from right to left) represents an input (0 = Not active; 1 or X = Active):<br><table><tr><th>BIT</th><th>ENTRADA</th></tr><tr><td>1</td><td>Aux (not used currently)</td></tr><tr><td>2</td><td>DC overvoltage</td></tr><tr><td>3</td><td>Energized spring breaker 2</td></tr><tr><td>4</td><td>Breaker 2 closed</td></tr><tr><td>5</td><td>Breaker 2 open</td></tr><tr><td>6</td><td>Energized spring breaker 1</td></tr><tr><td>7</td><td>Breaker 1 closed</td></tr><tr><td>8</td><td>Breaker 1 open</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG8.14-DU output status = 00000</td><td>00000 to XXXXX</td><td>Shows the status of DU outputs. Each bit (from right to left) represents an output (0 = Not active; 1 or X = Active.:<br><table><tr><th>BIT</th><th>ENTRADA</th></tr><tr><td>1</td><td>GFDI contctor</td></tr><tr><td>2</td><td>Breaker 1 close command</td></tr><tr><td>3</td><td>Breaker 1 open command</td></tr><tr><td>4</td><td>Breaker 2 close command</td></tr><tr><td>5</td><td>Breaker 2 open command</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG8.15-GFDI relay status = 00</td><td>00 to XX</td><td>It is used to manage the retries of the open or close command, depending on the DU and uP bits. If they do not match, a retry will be made. Bit 0 (the rightmost bit) is the microprocessor command and bit 1 is the status DU status.<br><br>Note: this parameter will only be shown if [G10.2-Working mode] = GFDI + IMI.</td><td>Yes</td></tr>
</table>

# Group SG9: Logger modules

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG9.1-Select module = Module 1** | 0 to 15 | Select the module (from 1 to number of available modules) for which its associated information will be displayed. | Yes |
| **SG9.2-Last status = 0** | 0 to 65535 | Displays a list of the last ten status, in chronological order, and coded as shown in section 3.<br><br>The most recent status is shown first and stored in parameter SG9.2. Whenever a new status occurs, it will be placed in the top of the list and older status messages will drop a position (the oldest status message will be discarded and registry SG9.11 replaced with the value that was stored in SG9.10). | Yes |
| **SG9.3-Nineth status = 0** | 0 to 65535 | | Yes |
| **SG9.4-Eighth status = 0** | 0 to 65535 | | Yes |
| **SG9.5-Seventh status = 0** | 0 to 65535 | | Yes |
| **SG9.6-Sixth status = 0** | 0 to 65535 | | Yes |
| **SG9.7-Fifth status = 0** | 0 to 65535 | | Yes |
| **SG9.8-Fourth status = 0** | 0 to 65535 | | Yes |
| **SG9.9-Third status = 0** | 0 to 65535 | | Yes |
| **SG9.10-Second status = 0** | 0 to 65535 | | Yes |
| **SG9.11-First status = 0** | 0 to 65535 | | Yes |

# Group SG10: Drive select

## Subgroup SG10.1: General

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG10.1.1-Password = 0** | 0 to 65535 | Password to enable drive select configuration. | No |
| **SG10.1.2-Force defaults = No** | No<br>Yes | Sets all values to factory default values. | No |
| **SG10.1.3-Product id = HEM** | HE V1500<br>HEMK<br>HEM | Sets the product ID. Currently, the following two options are supported:<br>VALUE / PRODUCT<br>46 / HEMK<br>47 / HEM | No |
| **SG10.1.3a-Size = 1500** | Normal<br>PLUS<br>1500 | Enables selection of the HEM hardware type:<br>OPTION / PRODUCT<br>Normal / HEM repowered inverter V1.<br>PLUS / PLUS repowered inverter V2.<br>1500 / V1500 repowered inverter. | No |
| **SG10.1.3b-Power Stage = Hybrid** | Full Mitsubishi<br>Hybrid | Enables selection of the HEM power stage:<br>OPTION / PRODUCT<br>Full Mitsubishi / Full Mitsubishi: Includes Mitsubishi IGTBs.<br>Hybrid / Hybrid: Includes Mitsubishi and Infineon IGTBs.<br>**Note:** this parameter is only shown if [SG10.1.3a-Size] ≠ 1500. | No |
| **SG10.1.4-Inverter type = Solar inverter** | Solar inverter<br>PCS<br>Statcom<br>M-PCS | Selects the type of inverter from the following options:<br>• Solar inverter.<br>• PCS (inverter for batteries).<br>• Statcom.<br>• M-PCS (each module is connected to a battery). | No |
| **SG10.1.4a-Bus Plus = No** | No<br>1 x 500 KW<br>2 x 500 KW<br>3 x 500 KW<br>5 x 500 KW<br>6 x 500 KW | Sets the maximum power at which the Bus Plus will work. | No |
| **SG10.1.5-Nominal Vac = 660V** | 50 to 750 | Sets the nominal Vac. | No |
| **SG10.1.5b-Nominal Frequency = 60.00 Hz** | 40.00 to 70.00 Hz | Sets the nominal Frecuency. | No |
| **SG10.1.6-Nominal S = 605.00kVA** | 20.00 to 650.00 kVA | Sets the nominal S. | No |
| **SG10.1.7-Number of modules = 6** | 1 to 10 | Sets the number of modules that compose the inverter. | No |
| **SG10.1.8-Number of Mpps = 1** | 1 to 5 | Sets the number of MPPs. | No |
| **SG10.1.9-Max Vbus = 1500V** | 70 to 1550 V | Sets the maximum bus voltage. | No |
| **SG10.1.10-Max Iac = 529A** | 30 to 600 A | Sets the maximum AC current. | No |
| **SG10.1.11-Max Idc = 670A** | 300 to 700 A | Sets the maximum DC current. | No |
| **SG10.1.12-DU Bus Bar Type = Aluminium** | Aluminium<br>Copper | Enables selection of the type of DU Bus Bar<br>VALUE / PRODUCT<br>0 / Aluminium<br>1 / Copper | No |
| **SG10.1.13- DU current measures = No** | No<br>Yes | It can be enabled when the DU includes current measures. | No |

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The following tables show the values for different parameters of group SG10.1, according to the inverter model. Refer to **project documentation** and *Hardware and Installation manual* (HEMHW01 and HEMHW02 for HEM inverters and HEMKHW01 and HEMKHW02 for HEMK) **to check that there is no limitation regarding the parameters**, especially regarding maximum currents or temperatures. **If the possible limitations are not followed**, **the MV transformers could be seriously damaged**.

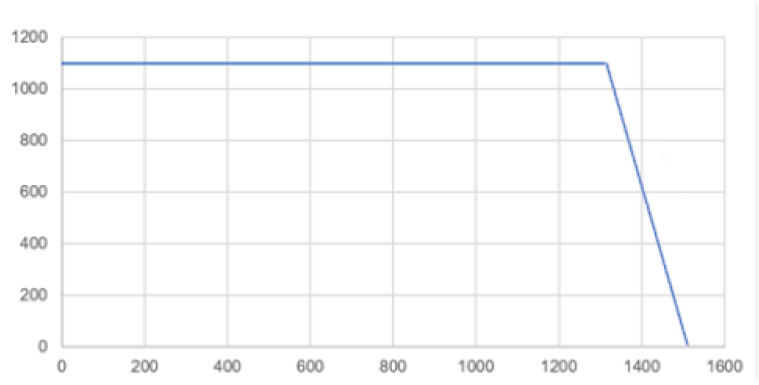
**Note**: The values shown are for a preset configuration of 40ºC. if the inverter is set to 50ºC, these values will change.

| SG10.1 | 690 V | | 660 V | | 645V | |
|---|---|---|---|---|---|---|
| | 4 mod. | 6 mod. | 4 mod. | 6 mod. | 4 mod. | 6 mod. |
| SG10.1.5-Nominal Vac | 690 | 690 | 660 | 660 | 645 | 645 |
| SG10.1.6-Nominal S | 633,0 | 633,0 | 605,0 | 605,0 | 591,0 | 591,0 |
| SG10.1.7-Number of modules | 4 | 6 | 4 | 6 | 4 | 6 |
| SG10.1.10-Max Iac (*) | 529 | 529 | 529 | 529 | 529 | 529 |
| SG10.1.11-Max Idc (*) | 670,0 | 670,0 | 670,0 | 670,0 | 670,0 | 670,0 |

| SG10.1 | 630 V | | 615 V | | 600 V | |
|---|---|---|---|---|---|---|
| | 4 mod. | 6 mod. | 4 mod. | 6 mod. | 4 mod. | 6 mod. |
| SG10.1.5-Nominal Vac | 630 | 630 | 615 | 615 | 600 | 600 |
| SG10.1.6-Nominal S | 578,0 | 578,0 | 564,0 | 564,0 | 550,0 | 550,0 |
| SG10.1.7-Number of modules | 4 | 6 | 4 | 6 | 4 | 6 |
| SG10.1.10-Max Iac (*) | 529 | 529 | 529 | 529 | 529 | 529 |
| SG10.1.11-Max Idc (*) | 670,0 | 670,0 | 670,0 | 670,0 | 670,0 | 670,0 |

## Subgroup SG10.2: Protections

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG10.2.1-Overcurr prot low = 1100A** | 0 to 1100 A | Set the low overcurrent protection threshold at Vdc[V] = 1050 V | No |
| **SG10.2.2-Overcurr prot mid 1 = 1100A** | 0 to 1100 A | Set the middle 1 overcurrent protection threshold at Vdc[V] = 1200 V | No |
| **SG10.2.3-Overcurr prot mid 2 = 1100A** | 0 to 1100 A | Set the middle 2 overcurrent protection threshold at Vdc[V] = 1310 V | No |
| **SG10.2.4-Overcurr prot high = 0A** | 0 to 600A | Set the high overcurrent protection threshold at Vdc[V] = 1500 V | No |
| **SG10.2.5-Igbts temp prot = 110ºC** | 20 to 600 ºC | Set IGBTs temperature protection threshold. | No |
| **SG10.2.6-Bus unbalance max = 250 V** | 200 to 400 V | Sets the bus maximum voltage unbalance.<br>**Note**: this parameter is only displayed for users with role Developer. | No |




The following table shows the usual values for the IGBTs temperature protection threshold in case of derating by IGBT temperature. In this case, the values to be defined varies according to the type of power stage:

| Power stage | Full Mitsubishi | Hybrid (New Mit 800A / Inf. 900A) |
|---|---|---|
| **Inverter configuration** | **Parameter SG10.2.5-Igbts temp prot (ºC)** | |
| **Normal** | 107 | 114 |
| **PLUS** | 110 | 128 |

**NOTE:** Please, refer to section G3.3: Derating T to find out indications to check the type of power stage installed in the inverter.

## Subgroup SG10.3: Limits

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG10.3.1-Inductive I limit = 492A** | 0 to 600 A | Maximum inductive current per module. | No |
| **SG10.3.2-Capacitive I limit = 440A** | 0 to 600 A | Maximum capacitive current per module. | No |
| **SG10.3.3-Max P limit = 100.00 %** | 0.00 to 100.00 % | Parameters SG10.3.3, SG10.3.4 and SG10.3.5 set the maximum P, Q and S limits, respectively, with respect to the inverter's nominal value.<br><br>They also limit the maximum value that parameters G3.1.1 (P limit), G3.1.2 (Q limit) and G3.1.4 (S limit) can take. This means that if a user attempts to set in G3 a higher value than the one set in the drive select limit, it will be rejected and overridden by the value of the drive select.<br><br>Example: If SG10.3.3 = 90% and user tries to set G3.1.1 to 95%, the resulting configuration will be G3.1.1 = 90%.<br><br>**Note:** P, Q and S may not have the expected value by default. Ensure they are correctly set, or output power may not be as expected. | No |
| **SG10.3.4-Max Q limit = 100.00 %** | 0.00 to 150.00 % | | No |
| **SG10.3.5-Max S limit = 100.00 %** | 0.00 to 200.00 % | | No |

## Subgroup SG10.4: Transformer

This group allows configuring parameters related to the medium voltage transformer. **In case modifications are required, please contact Power Electronics before making any changes.**

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG10.4.1-Cell type = V** | V<br>LV<br>LVL | Allows setting the medium voltage cell topology. | No |
| **SG10.4.2-MV Voltage = 34500 V** | 0 to 65535 V | Allows defining the medium voltage value. | No |
| **SG10.4.3-Pn = 3300 kW** | 0 to 5000 kW | Allows setting the medium voltage transformer nominal power. | No |
| **SG10.4.4-Xcc = 7.00 %** | 0.00 to 20.00 % | Allows setting the medium voltage transformer short-circuit reactance, in percentage. | No |
| **SG10.4.5-Pfe = 7.0 kW** | 0.0 to 100.0 kW | Allows setting the medium voltage transformer core losses for subsequent calculations. | No |
| **SG10.4.6-Pcu = 23.0 kW** | 0.0 to 100.0 kW | Allows setting the medium voltage transformer copper losses for subsequent calculations. | No |
| **SG10.4.7-Test frequency = 60 Hz** | 40 to 70 Hz | Allows setting the medium voltage transformer manufactured frequency. | No |
| **SG10.4.8-Open current = 0.500 %** | 0.000 to 10.000 % | Allows setting the medium voltage transformer open current in percentage. | No |

## Subgroup SG10.5: Filter

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG10.5.1-R (mOhm) = 5.00** | 0.00 to 200.00 | Sets the value of the LCL filter main resistance, in miliohms. | No |
| **SG10.5.2-L (uH) = 140** | 0 to 20000 | Sets the value of the LCL filter main inductor, in microhenries. | No |
| **SG10.5.3-C (uF) = 61.50** | 0.00 to 200.00 | Sets the value of the capacitance of each module of the LCL filter, in microfarads. | No |
| **SG10.5.4-Lc (uH) = 2.70** | 0.00 to 200.00 | Sets the value of the notch inductor, in microhenries. | No |

### Subgroup SG10.6: Econmode

| Screen | Range | Function | Set on run |
|---|---|---|---|
| **SG10.6.1 Econmode Ready = No** | No<br>Yes | Parameter to indicate that the HEM inverter is provided with the Econmode MV transformer. | Yes |

## Group SG11: Self-diagnosis modules

### CAUTION

The following group is supplementary to the normal hardware testing procedure performed by the qualified personnel and in no case will replace it.

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When active, the self diagnose will execute every time the inverter starts, before closing the DC and AC contactors. The self diagnose will be normally activated.

Once the system has checked that it is correctly isolated from DC and AC sides, it will proceed to check the power status for each module.

The self diagnose is executed module by module on all active modules, meaning that on each iteration, one module is checked at a time:

1. The DC bus soft charge is performed until reaching the maximum DC bus voltage, or a certain percentage of it, defined in SG11.2.
2. Once this voltage is reached, it will wait 10 seconds before starting the bus discharge.
3. The bus is actively discharged using just one inverter module.
    - If a module fault occurs: the process will be stopped, and the module disabled. The process will then start the self-diagnosis for the next module.
    - If an emergency special fault is triggered: the process will be interrupted, the inverter will stop, and the module will be disabled. **The AC contactor will NOT open.** The inverter will not start until it gets restarted by introducing the “power electronics” password.

When any special fault occurs (see G5.10.1) this causes the AC breaker to open. The fault will be saved even if the supply is disconnected and it cannot be reset until the Power Electronics user password is introduced.

After an emergency fault, the self diagnose will always be executed, even if it is not enabled. When this happens, the process will execute normally, except that if during the diagnose a module fault occurs (even not emergency fault), the system will report a new fault F32, that will not be resettable, and the inverter will stop as a consequence.

Before doing so, Power Electronics authorized user must check the inverter logs and parts to make sure that it is working properly, and then execute the inverter self-diagnosis in SG11.1.

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>SG11.1-Enable selfdiagnosis = Enabled</td><td>Disabled<br>Enabled</td><td>Selfdiagnosis will always be done before starting in the case where it is enabled, in case it is disabled it will only perform self-test after a special fault has been activated.<br>If it is disabled, it will only self-diagnose if it opens DU (F41, F45,F53).<br>Allows enabling the self-diagnosis process after a special fault (G5.10.1) has been triggered.<br>Note: Check the system before enabling the self-diagnosis process.<br>Note: This should always be performed by an authorized user.</td><td>Yes</td></tr>
<tr><td>SG11.2-Voltage threshold = 100.00</td><td>75.00 to 100.00 %</td><td>Set the voltage limit value that will be reached in the loading process, as a % of the rectified grid voltage.</td><td>Yes</td></tr>
<tr><td>SG11.3-State Selfdiagnosis = Init</td><td>Init<br>Open DU<br>Select module<br>SC Contactor On<br>SC Check Vdc<br>SC Contactor Off<br>Wait<br>Discharge<br>Error<br>Done</td><td>Shows the status of the self-diagnosis process, according to the following options:<br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Init</td><td>Initialization</td></tr><tr><td>Open DU</td><td>Forces the DU to open.</td></tr><tr><td>Select module</td><td>Checks for the next available module.</td></tr><tr><td>SC Contactor On</td><td>Opening soft charge contactor.</td></tr><tr><td>SC Check Vdc</td><td>Loading DC bus.</td></tr><tr><td>SC Contactor Off</td><td>Closing soft charge contactor.</td></tr><tr><td>Wait</td><td>Checking for faults after charge.</td></tr><tr><td>Discharge</td><td>Active discharge of the bus.</td></tr><tr><td>Error</td><td>Self diagnose couldn’t execute completely due to charging time or low voltage.</td></tr><tr><td>Done</td><td>Self diagnose complete.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG11.4-Ena disc deep 1st mod = Yes</td><td>No<br>Yes</td><td>Enables the deep discharge functionality to check the disconnection of the solar panels on the first module.<br>The inverter will always perform a first deep discharge (reduce the voltage to 250Vdc disconnecting the bus.)<br>Afterwards, the functionality will remain the same by performing the discharge sequence on the rest of the modules<br>If the deep discharge lasts longer than 300s, then the F193 is triggered.</td><td>Yes</td></tr>
</table>

# Group SG12: MV Cell service

These parameters allow configuring the maneuver of the medium voltage (MV) switchgear, as well as visualizing the status of each switch / interrupter.

**Note:** This group only applies to HEM inverters.

<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>SG12.1-Action Gen. Switch = Idle</td><td>Open<br>Idle<br>Close<br>Premagnetize</td><td>Allows to command the general switch maneuver of the MV switchgear.<br><br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Open</td><td>Open switch.</td></tr><tr><td>Idle</td><td>This state indicates that the general switch is ready to receive the order to open, close, or premagnetize. Once the action has been completed, the general switch returns to this state automatically. This is a visualization status, which means that choosing it will not affect the switch.</td></tr><tr><td>Close</td><td>Close switch.</td></tr><tr><td>Premagnetize</td><td>Performs the switchgear premagnetization. To do it, user must have previously carried out the medium voltage calibrations (group SG6.7) in both direct and reverse sequence and the inverter must be in “Ready” status.<br>After finishing the premagnetization, the main switch closes automatically.</td></tr></table></td><td>No</td></tr>
<tr><td>SG12.2-Action Timeout = 10 s</td><td>1 to 30 s</td><td>Allows setting a timeout in medium voltage switchgear automatic operation feedback failure. The failure in the operation process will trigger the fault once the timeout has elapsed with no cell operation.</td><td>No</td></tr>
<tr><td>SG12.3-Status Gen. Switch = Unknown</td><td>Unknown<br>Opened<br>Closed</td><td>Shows the MV switchgear general switch status:<br><br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Unknown</td><td>The status of the switch cannot be determined.</td></tr><tr><td>Opened</td><td>The general switch is open.</td></tr><tr><td>Closed</td><td>The general switch is closed.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG12.4-Status Earth Switch = Unknown</td><td>Unknown<br>Opened<br>Closed</td><td>Shows the MV switchgear earth switch status:<br><br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Unknown</td><td>The status of the switch cannot be determined.</td></tr><tr><td>Opened</td><td>The earth switch is open.</td></tr><tr><td>Closed</td><td>The earth switch is closed.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG12.5-Status Auto Int. = Unknown</td><td>Unknown<br>Opened<br>Closed</td><td>Shows the MV switchgear automatic interrupter status:<br><br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Unknown</td><td>The status of the automatic interrupter cannot be determined.</td></tr><tr><td>Opened</td><td>The automatic interrupter is open.</td></tr><tr><td>Closed</td><td>The automatic interrupter is closed.</td></tr></table></td><td>Yes</td></tr>
</table>

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<table>
<tr><th>Screen</th><th>Range</th><th>Function</th><th>Set on run</th></tr>
<tr><td>SG12.6-Premagn. Status = Off</td><td>Off<br>Checks MV cell condit.<br>Checks frequency condit.<br>Checks voltage condit.<br>Closes LC contactor<br>Starts 2 modules<br>Increases voltage to ref<br>Checks voltages<br>Closes MV switch<br>Done<br>Failed</td><td>Shows the current status of the premagnetization process. This parameter duplicates SV4.5.9, but in this case is shown in the SG subgroup that allows starting the premagnetization.<br><br><table><tr><th>VALUE</th><th>DESCRIPTION</th></tr><tr><td>0</td><td>Off</td></tr><tr><td>1</td><td>Checking MV cell conditions</td></tr><tr><td>2</td><td>Checking frequency condititions</td></tr><tr><td>3</td><td>Checking voltage conditions</td></tr><tr><td>4</td><td>Closes LC contactor</td></tr><tr><td>5</td><td>Starts 2 modules</td></tr><tr><td>6</td><td>Increases voltage to reference</td></tr><tr><td>7</td><td>Checks voltages</td></tr><tr><td>8</td><td>Closes MV switch</td></tr><tr><td>9</td><td>Done</td></tr><tr><td>10</td><td>Failed</td></tr></table></td><td>Yes</td></tr>
</table>

# TYPICAL CONFIGURATIONS AND USE CASES

9

## Parameters Initialization

When a new software update is performed, the following steps must be performed:

1. In the first software update, download the configuration file and the Excel file.
2. Perform software upload.
3. G1.1 (G.1.1 = 'Yes'). Initialize parameters. Sets all parameters to their default value except:
    - MAC address
    - RTC
    - Group 6: Communications
    - Group 9: Leak protection
    - Group 10: DU configuration
    - Group 12: BFPS configuration
    - Service Group (to initialize this group use "Init. service group")
    - Energy counters
    - Run time and grid connections
    - Faults history
    - Warnings history
    - Events history
    - EEPROM calibration
    - Custom Modbus map
4. SG1.1.4 (SG1.1.4 = 'Yes'). Init service group. Sets all Service Group parameters back to their default value, except for:
    - SG1.4.3-Selector status EEPR
    - Group SG6 "Calibration"
    - SG1.6-Normative
5. SG1.1.3, Init factory defaults: Sets all parameters back to their default value, except for the MAC address. **Please notice that this option deletes all data**, including calibration values. Thus, it should be avoided unless extraordinary circumstances. Consult Power Electronics in case of doubt.
6. Make a complete reboot if you are in field, if you are working remotely, perform a reset using password 1903.

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7. Load the configuration file of the inverter you downloaded previously, compare the parameters with the excel file to make sure everything was properly loaded.
8. Initialize drive select SG10.1.1 = '60989': Enter Drive Select password. To allow changes in Drive Select set the parameter to the value 'OK'.
9. SG10.1.2= 'Yes': This parameter sets the value of the Drive Select parameters to their default value.
10. Manually compare the parameters set with the Excel file.

**Note:** Initializing the SW is critical, not only because the parameters must be correct, but also because some memory areas and internal values need to be "formatted" for new versions. Unexpected" behaviors could be generated.

In addition, there are parameters that must be checked after all operations can be carried out:

- SG2.2.1 Config FPGA1= 0 Deactivation of homopolar fault for version 1.7.1 Assign a value of 2000 if 1.5.0 or higher version is configured.
- SG2.2.2 Config FPGA2= 312.
- SG4.1 Force Cond start = NO (This parameter is changed with functional testl).
- SG10.2.4-Umb. high overcurrent = 350 A.
- G8.1.1 Control Mode P= MPPT (This parameter is sometimes in power mode).
- G8.2.10 Kp Evening= 5%.
- G8.2.10 Kp Evening= 5%.
- G8.2.11 KI Night = 10%.

Once these steps have been carried out, check that the parameters appear with their predefined values.

# Internal Registers

The following Modbus register is not accessible by the user. It should only be controlled by Power Electronics' staff.

<table>
<tr><th>Function</th><th>Modbus Register</th><th>Range</th><th>Description</th></tr>
<tr><td>PPC Almeyda Unit Control Enable</td><td>4256</td><td>0 to 1</td><td>It allows to select a series of inverters in order not to interpret the information coming from the PPC by UDP frames.<br><br><table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>0</td><td>If the register is set to 0 the inverter will read the UDP emission frames coming from the PPC</td></tr><tr><td>1</td><td>If the register is set to 1, the functionality is activated and the inverter will not detect the UDP emission frames of the PPC</td></tr></table><br>Exception 1: When the "Unit Controller" function is activated (register 4256=1), the inverter will listen to UDP frames if the inverter is in NIGHT MODE. If in NIGHT MODE, the inverter will follow the UDP frames from the PPC (Qref in night mode).<br><br>Exception 2: When the Unit Controller function is activated (Record 4256=1), the HEM still listens to UDP frames from the PPC as far as writing is concerned G4.7.4 Night Mode Activation</td></tr>
</table>

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